Final research report

Repository: georgeredo, branch `claude/lucid-heisenberg-bpyq7k`. **Built:** 2026-10-23 by `research_notes/FINAL REPORT/build/` (`build.py`, then `render.py` for the PDF); **re-scored 2026-10-24 at Robinhood's fees**.

Fees: Robinhood is primary. From 2026-10-24 the research's default cost model is Robinhood's options fee schedule (`docs/FEES_ROBINHOOD.md`, read 2026-10-24). In short: $0 commission; ETF options about $0.04 a contract to buy and $0.06 to sell (a 1-lot); SPXW $1.04 a contract a side under $1.00 of premium and $1.13 at $1.00 and up (non-Gold); nothing at expiry or settlement. The studies charged $0.65 a contract each way, or $0.68748 with pass-throughs, or $0.70 for the older Compass studies; SPXW was $1.18748 / $1.27748.

Every trade is re-priced on its own fills: buys at the ask and sales at the bid, exactly as the study filled them. The study's modelled fee is added back and Robinhood's is charged. Every figure is printed at Robinhood's fees first, with the study's own beside it on page 1 and in each section's fee table.

Where a trade file lacks a leg's price, side or exit, Robinhood's fee is inferred from the rule. Those rows are marked ‡ and labelled "Robinhood fees inferred from the rule" in their section. Vendor-recorded rows (§) carry no fee in either column.

Every verdict stays exactly as pre-registered, on the study's own fees. The column "would the verdict differ at Robinhood fees?" only runs the frozen pass rule on the Robinhood-fee returns. It is descriptive, not a new verdict. **Inventory of every study:** `INVENTORY.md`, the file beside this one.

This report is read-only with respect to the research. It adds no test, variant or fetch, opens no holdout and changes no rule. Every metric is recomputed from a study's committed trade-level output, or from the scratchpad copy that INVENTORY.md names, with one exception: two studies (#31 and #40) were rerun deterministically from their committed scripts at their own commits, on the same stored inputs. Step 6's rule membership (#17) is recomputed by its committed `step6/trade.py choose()`, on the scratchpad population it was built from. Where a figure cannot be produced, the report writes "not available — ". GeorgeFi's own recorded card returns appear only in sections marked **vendor-recorded, not re-marked** (#1, and the grade check in #29f).

Units. Return per trade is given as:

- % of premium paid, for long options;
- % of max risk, for spreads and butterflies;
- % of the credit received, for naked short straddles and strangles;
- % of cash set aside, for cash-secured puts;
- % of notional, for stock trades. Their "$" column is $ per 100 shares, because every stock trade here is 100 shares.

$ per trade is one lot, after every fee the study charged.

Page 1. Every strategy, sorted by mean trade at Robinhood fees

One row per strategy, on the period that carries its verdict, sorted by mean trade at Robinhood fees (% in each row's own unit), with the mean at the study's own fees beside it. The table pools all names when a study judged them together, and shows one row per name or variant when it did not. The interval is the 20-session moving-block bootstrap of the mean % a trade (`audit_lib.session_stats`, 2,000 draws). It is "n/a" when the period has fewer than 60 sessions. "Biggest win/loss" are $ for one trade, with its entry date. The verdict is the study's own, as pre-registered. The last column is descriptive. It runs the study's frozen pass rule on the Robinhood-fee returns: through the study's own code where the trade list allows (the Compass features' Δ, the direction screen); otherwise through the rule's computable part, with the report's block-20 interval, and it says so. ‡ marks a row whose Robinhood fees are partly inferred from the rule. § marks a vendor-recorded row.

Robinhood against the study's fees, on page 1. 75 rows. The five largest changes in the mean trade:

- #40a Compass EMA ladder: PREVIEW 2025-03, 13 versions (rerun) (V1, preview 2025-03): −9.20 → **−4.89** % of premium (−2.76 → −1.48 $ a trade).

- #7f S12 Intraday momentum: first-half-hour direction → 10:00–15:00 long option (all names, 2022-11→2026-10): −15.07 → **−11.94** % of premium (−4.36 → −3.06 $ a trade).
- #21a Phase 2 hyp C: 15:00 SPY/QQQ P1 iron ±2 butterfly, held to settlement (all names, OOS 2025–26): +4.49 → **+7.46** % of max risk (+3.41 → +5.81 $ a trade).
- #21b Phase 2 hyp C: 15:00 SPY/QQQ P2 call ±2 butterfly, held to settlement (all names, OOS 2025–26): +2.09 → **+5.02** % of max risk (+1.34 → +3.75 $ a trade).
- #30e Compass features BX3e: the 14:00 exit on the physically settled names only … (0DTE→60 DTE, 15 names, holdout (opened)): −11.03 → **−8.12** % of premium (−14.09 → −12.83 $ a trade).

Rows whose mean trade changes sign: **2**: #4b all names ledger E2 (−0.13 → +1.50); #7c all names 2017-06→2026-10 (−0.08 → +0.83).

Rows where the frozen pass rule, run on Robinhood-fee returns, would read differently: **2**: #21a all names; #21c all names. This is descriptive; every verdict stays as pre-registered.

#	strategy	name	period	trades	mean trade at Robinhood fees	mean trade at the study's fees	win % Robinhood / study	biggest win	biggest loss	20-session block 95% (mean %, Robinhood)	verdict (as pre-registered, at the study's fees)	would the verdict differ at Robinhood fees? (descriptive)
6e	H15 carry lane: the 14:05 SPXW short strangle at ±0.5 expected move, held to …	SPXW	TEST	407	**+17.98 % credit** / +150.91 ‡	+18.02 / +151.17	70.8 / 70.8	+4,578 (2025-04-10)	−7,131 (2026-03-09)	[+3.49, +29.85]	Open, NOT promotable	Its computable part (mean > 0, interval clear of zero): No — passes at both (Robinhood: mean +17.98%, lower end +3.49); the reselection and the forward T-G are not in the trade list
1	GeorgeFi vendor cards (the vendor's own recorded P&L)	all names	Epoch 1	546	**+14.82 % prem** / +79.21 §	+14.82 / +79.21	78.8 / 78.8	+4,020 (2026-02-06)	−1,375 (2026-02-18)	n/a (< 60 sessions)	Not evidence	— (no pass rule: descriptive)
41a	EMA system v0: PREVIEW on 2026-10-12 → 16 (5 sessions)	Config A X1	preview 2026-10-12→16	57	**+11.68 % prem** / +43.87	+10.67 / +43.03	61.4 / 61.4	+718 (2026-10-14)	−772 (2026-10-15)	n/a (< 60 sessions)	PREVIEW	— (no pass rule: a preview)
6d	T-E: the 13:00 SPXW ATM straddle held to PM settlement, ex FOMC/CPI/NFP (H13)	SPXW	TEST	388	**+8.74 % credit** / +85.30 ‡	+8.75 / +85.56	67.0 / 67.0	+4,100 (2025-03-06)	−34,022 (2025-04-09)	[+1.85, +14.75]	FAIL	No — fails at both (Robinhood: mean +85.30, lower end −115.97)
21a	Phase 2 hyp C: 15:00 SPY/QQQ P1 iron ±2 butterfly, held to settlement	all names	OOS 2025–26	878	**+7.46 % risk** / +5.81	+4.49 / +3.41	53.3 / 52.3	+156 (2026-01-28)	−130 (2026-01-07)	[+0.15, +15.30]	Not confirmed	**Yes** — fails at the study's fees, passes at Robinhood's (all four parts; Robinhood: mean +7.46%, lower end +0.15, without the best five +3,788 $, years 2025 +5.17, 2026 +6.62)
27a	After-hours exercise study: fly5 iron ±5, case (b) the post-market decision	all names	ii	987	**+6.72 % risk** / +24.19	+6.25 / +22.52	55.8 / 55.7	+2,199 (2026-06-22)	−2,193 (2026-04-07)	[−5.36, +15.74]	DROP	Keep-or-drop rule: No — dropped at both (Robinhood: mean +24.19, lower end −17.22)
26c	Paper tracker: condor_d20 (SPY 2-wide 20-delta iron condor, 13:00–15:00)	SPY	ii	2,090	**+6.11 % risk** / +10.07	+4.68 / +7.67	82.8 / 82.1	+63 (2025-04-10)	−187 (2026-04-13)	[+3.87, +8.01]	Positive in ii	— (verdict at the forward N only; the historical rows are descriptive)
6c	T-C: the 13:00 SPXW short straddle bought back 15:55 (H11)	SPXW	2025-06→2026-10	339	**+5.07 % credit** / +71.91 ‡	+5.11 / +72.43	63.4 / 63.4	+3,370 (2025-09-17)	−8,500 (2026-06-11)	[−0.09, +8.97]	FAIL	No — fails at both (Robinhood: mean +71.91, lower end −56.51)
21b	Phase 2 hyp C: 15:00 SPY/QQQ P2 call ±2 butterfly, held to settlement	all names	OOS 2025–26	880	**+5.02 % risk** / +3.75	+2.09 / +1.34	52.3 / 51.2	+154 (2026-01-28)	−131 (2026-01-07)	[−2.27, +12.91]	Not confirmed	No — fails at both (all four parts; Robinhood: mean +5.02%, lower end −2.27, without the best five +2,002 $, years 2025 +2.90, 2026 +4.83)

#	strategy	name	period	trades	mean trade at Robinhood fees	mean trade at the study's fees	win % Robinhood / study	biggest win	biggest loss	20-session block 95% (mean %, Robinhood)	verdict (as pre-registered, at the study's fees)	would the verdict differ at Robinhood fees? (descriptive)
20b	Phase 2 hyp B: the midday SPY iron condor (em1, mid-VIX), the midday pick	SPY	OOS 2025–26	1,202	**+4.67 % risk** / +7.95	+3.25 / +5.55	81.9 / 81.6	+59 (2026-04-29)	−176 (2026-07-24)	[+1.62, +7.46]	"Needs X"	Its computable part (mean > 0, interval clear of zero): No — passes at both (Robinhood: mean +4.67%, lower end +1.62); the rule picks the cell on fit; read here on OOS
20a	Phase 2 hyp B: boundary touch — SPY 2-wide 25-delta credit spread, held (the …	SPY	OOS 2025–26	2,600	**+4.59 % risk** / +8.04	+3.91 / +6.84	88.2 / 88.1	+46 (2025-11-14)	−190 (2026-04-13)	[+2.24, +6.37]	No edge	Its computable part (mean > 0, interval clear of zero): No — passes at both (Robinhood: mean +4.59%, lower end +2.24); the rule picks the cell on fit; read here on OOS
25	Phase 3 hyp F: the 15:30 SPY/QQQ long call ±5 butterfly (cash account)	all names	OOS 2025–26	879	**+4.50 % risk** / +12.31	+3.85 / +9.89	63.0 / 62.2	+262 (2026-04-29)	−332 (2025-03-04)	[+2.14, +6.84]	Passes all four, but the cell was named after C's OOS was printed	No — passes at both (all four parts; Robinhood: mean +4.50%, lower end +2.14, at $0.01 a leg +3.43%, without the best five +8,588 $, names QQQ +9.11, SPY +15.50)
26b	Paper tracker: fly5 (SPY/QQQ ±5 butterflies at 15:00 and 15:30: iron, call, put)	all names	ii	5,271	**+4.29 % risk** / +12.05	+3.62 / +9.64	61.4 / 60.5	+283 (2026-04-29)	−468 (2026-05-12)	[+2.45, +6.13]	Iron clear of zero in i and ii	— (verdict at the forward N only; the historical rows are descriptive)
26a	Paper tracker: condor_em1 (SPY 2-wide iron condor at ± the ATM straddle, …	SPY	ii	2,068	**+3.90 % risk** / +6.50	+2.46 / +4.10	80.7 / 80.4	+66 (2026-07-29)	−185 (2025-04-09)	[+0.90, +6.46]	Historical periods through zero	— (verdict at the forward N only; the historical rows are descriptive)
27b	After-hours exercise study: condor_d20, case (b)	SPY	ii	1,176	**+3.70 % risk** / +6.07	+2.37 / +3.84	83.3 / 83.3	+868 (2026-09-16)	−1,796 (2026-04-07)	[−11.28, +15.87]	DROP	Keep-or-drop rule: No — dropped at both (Robinhood: mean +6.07, lower end −17.97)
41b	EMA system v1: PREVIEW on three drawn weeks (up, flat, down)	Config A X1	three weeks pooled	64	**+3.29 % prem** / +3.91	+2.23 / +2.98	54.7 / 54.7	+1,358 (2026-02-03)	−872 (2026-02-03)	[−13.43, +29.47]	PREVIEW	— (no pass rule: a preview)
1	GeorgeFi vendor cards (the vendor's own recorded P&L)	all names	Epoch 2	603	**+3.13 % prem** / −6.23 §	+3.13 / −6.23	48.1 / 48.1	+3,313 (2026-07-13)	−3,400 (2026-07-13)	[−0.74, +9.76]	Not evidence	— (no pass rule: descriptive)
21c	Phase 2 hyp C: 15:00 SPY/QQQ P3 call ±5 butterfly, held to settlement	all names	OOS 2025–26	878	**+2.84 % risk** / +6.94	+2.15 / +4.52	59.2 / 58.5	+277 (2026-04-29)	−468 (2026-05-12)	[+0.10, +5.46]	Not confirmed	**Yes** — fails at the study's fees, passes at Robinhood's (all four parts; Robinhood: mean +2.84%, lower end +0.10, without the best five +3,769 $, years 2025 +4.46, 2026 +10.13)
6a	T-A: the 13:00 short ATM 0DTE straddle, SPY/QQQ/IWM, bought back 15:55 (H9)	all names	2025-06→2026-10	1,017	**+2.44 % credit** / +5.17	+0.62 / +2.57	63.1 / 62.2	+346 (2026-06-12)	−1,184 (2026-06-11)	[−2.53, +6.28]	FAIL	No — fails at both (Robinhood: mean +5.17, lower end −6.16)
21d	Phase 2 hyp C: debit vertical 0.70/0.30, universe B, held (the vertical pick)	all names	OOS 2025–26	1,761	**+2.09 % prem** / +5.61 ‡	+1.25 / +4.95	52.0 / 51.6	+6,048 (2025-04-11)	−7,092 (2025-04-10)	[−2.01, +5.45]	Loses less than the naked contract but still loses	— (no pass rule: the pre-registered expectation was a negative OOS)

#	strategy	name	period	trades	mean trade at Robinhood fees	mean trade at the study's fees	win % Robinhood / study	biggest win	biggest loss	20-session block 95% (mean %, Robinhood)	verdict (as pre-registered, at the study's fees)	would the verdict differ at Robinhood fees? (descriptive)
28b	SPXW analogue B: 20-delta iron condor, wings 2×ratio, 13:00–15:00 (condor_d20)	SPXW	selection	2,157	**+1.53 % risk** / +25.28	+1.49 / +24.69	76.8 / 76.8	+605 (2026-07-29)	−1,849 (2026-04-13)	[−1.08, +3.50]	FAIL	No — fails at both (three parts; Robinhood: selection mean +25.28 $, lower end −18.27, at +$0.05 +20.28)
4b	GeorgeFi entries with the trail engine (reconstruction C, G2)	all names	ledger E2	516	**+1.50 % prem** / +2.46	−0.13 / +2.27	40.1 / 39.0	+1,752 (2026-07-06)	−778 (2026-03-16)	[−2.81, +3.24]	No robust entry timing	— (no pass rule: descriptive)
7c	S03 Quick Scalps with weekly expirations (1–7 days)	all names	2017-06→2026-10	818	**+0.83 % prem** / +2.57	−0.08 / +1.27	37.9 / 37.7	+464 (2020-05-15)	−226 (2020-03-10)	[−0.91, +2.53]	Nothing promoted	No — fails at both (Robinhood: mean +2.57, lower end −1.49)
7j-atm_put	S17 Index put-writing: monthly ATM cash-secured put	all names	2012-06→2026-10	334	**+0.66 % capital** / +200.60	+0.65 / +199.27	81.4 / 81.4	+2,118 (2026-07-17)	−9,870 (2020-02-21)	[+0.17, +1.08]	PASSES A7 for QQQ ATM, 0.30Δ and 0.15Δ and SPY ATM puts; alpha 0 …	No — passes at both (A7, per cell; Robinhood: mean +200.60 $, lower end +42.50)
7j-put_30d	S17 Index put-writing: monthly 0.30-delta cash-secured put	all names	2012-06→2026-10	334	**+0.46 % capital** / +142.99	+0.46 / +141.65	85.6 / 85.6	+1,428 (2026-05-15)	−9,370 (2020-02-21)	[+0.03, +0.79]	PASSES A7 for QQQ ATM, 0.30Δ and 0.15Δ and SPY ATM puts; alpha 0 …	No — passes at both (A7, per cell; Robinhood: mean +142.99 $, lower end +16.36)
37b	Edge H8: turn of the month (SPY, QQQ stock)	QQQ	single test	172	**+0.397 % notional** / +111.66	+0.387 / +110.22	59.3 / 59.3	+3,373 (2026-07-30)	−2,369 (2026-01-29)	[+0.016, +0.705]	SPY FAIL; QQQ PASS at 1× only	No — fails at both (each name read alone, at 1× (2× not re-read); Robinhood: mean +111.66 $, lower end −9.31)
37b	Edge H8: turn of the month (SPY, QQQ stock)	SPY	single test	117	**+0.315 % notional** / +135.10	+0.311 / +134.00	60.7 / 60.7	+2,808 (2026-07-30)	−2,463 (2024-07-30)	[−0.049, +0.646]	SPY FAIL; QQQ PASS at 1× only	No — fails at both (each name read alone, at 1× (2× not re-read); Robinhood: mean +135.10 $, lower end −19.75)
7j-put_15d	S17 Index put-writing: monthly 0.15-delta cash-secured put	all names	2012-06→2026-10	334	**+0.27 % capital** / +83.86	+0.26 / +82.52	94.0 / 94.0	+720 (2026-05-15)	−8,445 (2020-02-21)	[−0.09, +0.48]	PASSES A7 for QQQ ATM, 0.30Δ and 0.15Δ and SPY ATM puts; alpha 0 …	No — fails at both (A7, per cell; Robinhood: mean +83.86 $, lower end −14.32)
37a	Edge H7: the pre-FOMC drift (SPY, QQQ stock) — PASSED	QQQ	single test	108	**+0.214 % notional** / +70.01	+0.210 / +69.59	61.1 / 61.1	+1,162 (2024-07-30)	−543 (2026-07-28)	[+0.053, +0.358]	PASS both names	No — passes at both (each name read alone, at 1× (Bonferroni and 2× not re-read); Robinhood: mean +70.01 $, lower end +18.16)
37a	Edge H7: the pre-FOMC drift (SPY, QQQ stock) — PASSED	SPY	single test	77	**+0.191 % notional** / +70.54	+0.190 / +70.43	55.8 / 55.8	+823 (2024-07-30)	−620 (2021-01-26)	[+0.060, +0.355]	PASS both names	No — passes at both (each name read alone, at 1× (Bonferroni and 2× not re-read); Robinhood: mean +70.54 $, lower end +11.25)
32	Compass H1 / edge H1: overnight reversal of the late-day move (SPY, QQQ stock)	QQQ	single test	3,299	**+0.038 % notional** / +10.92	+0.027 / +9.42	51.6 / 50.5	+2,402 (2024-08-02)	−1,448 (2025-04-07)	[+0.005, +0.074]	FAIL	No — fails at both (both names must pass; at Robinhood's fees QQQ passes (changed) (mean +10.92, lower end +1.33); SPY fails (mean +12.26, lower end −3.39))

#	strategy	name	period	trades	mean trade at Robinhood fees	mean trade at the study's fees	win % Robinhood / study	biggest win	biggest loss	20-session block 95% (mean %, Robinhood)	verdict (as pre-registered, at the study's fees)	would the verdict differ at Robinhood fees? (descriptive)
32	Compass H1 / edge H1: overnight reversal of the late-day move (SPY, QQQ stock)	SPY	single test	2,177	**+0.035 % notional** / +12.26	+0.032 / +11.09	51.2 / 50.8	+3,028 (2020-03-13)	−1,749 (2025-04-07)	[−0.009, +0.088]	FAIL	No — fails at both (both names must pass; at Robinhood's fees QQQ passes (changed) (mean +10.92, lower end +1.33); SPY fails (mean +12.26, lower end −3.39))
35	Edge H4: market intraday momentum (SPY, QQQ stock)	QQQ	test 2025-10-13→2026-10-22	257	**+0.013 % notional** / +9.92	+0.014 / +10.32	49.8 / 49.8	+902 (2026-07-29)	−378 (2025-11-20)	[−0.017, +0.048]	FAIL	No — fails at both (each name read alone; Robinhood: mean +9.92 $, lower end −9.77)
9a	Lifecycle P1: Quick Scalps → cash-secured put sale (portfolio)	all names	hold-out 2025-10→2026-10	122	**+0.01 % risk** / +9.39 ‡	+0.01 / +8.09	63.1 / 63.1	+137 (2026-08-03)	−197 (2025-10-14)	[−0.02, +0.04]	Rejected	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean +0.01%, lower end −0.02); R1–R6 (Holm, the random-day control) need more than this list
35	Edge H4: market intraday momentum (SPY, QQQ stock)	SPY	test 2025-10-13→2026-10-22	257	**−0.005 % notional** / −3.45	−0.004 / −2.95	45.5 / 45.5	+573 (2026-07-29)	−373 (2026-09-30)	[−0.029, +0.019]	FAIL	No — fails at both (each name read alone; Robinhood: mean −3.45 $, lower end −20.01)
6b	T-B: gamma-conditioned last-30-minute momentum in SPY shares (H10)	SPY	negative-gamma days 2024-05→2026-10	299	**−0.029 % notional** / −18.38 ‡	−0.031 / −19.59	46.5 / 46.2	+523 (2025-04-04)	−698 (2025-03-04)	[−0.054, −0.001]	FAIL	No — fails at both (Robinhood: mean −0.03%, lower end −0.05)
28a	SPXW analogue A: iron fly ±5×ratio at 15:00 and 15:30 (fly5 iron)	SPXW	selection	882	**−0.40 % risk** / −19.23	−0.41 / −19.82	58.8 / 58.8	+2,569 (2026-09-16)	−3,549 (2026-03-09)	[−2.27, +1.43]	FAIL	No — fails at both (three parts; Robinhood: selection mean −19.23 $, lower end −86.71, at +$0.05 −24.23)
43	Compass ICT: sweep → displacement/MSS → FVG on SPXW 0DTE	v0	exploration 2022-05→2026-10-09	142	**−0.73 % prem** / +48.84	−0.76 / +48.26	37.3 / 37.3	+9,735 (2026-07-27)	−3,284 (2025-05-05)	[−7.74, +7.57]	Exploration only	— (no pass rule: exploration only; no holdout read)
40b	Compass EMA ladder: "take profits are the nearest EMAs" PREVIEW (E1 ATM, E2 …	E1	preview 2025-02 (fresh)	1,761	**−1.25 % prem** / −1.22	−2.75 / −2.49	31.8 / 30.8	+177 (2025-02-03)	−85 (2025-02-07)	n/a (< 60 sessions)	PREVIEW	— (no pass rule: a preview)
15	Step 4: turn-up buy (1.5× the low) + 20% trailing stop, near the money	all names	check 2025–26	26,258	**−1.37 % prem** / −2.06	−2.02 / −2.26	29.5 / 29.0	+21,608 (2025-04-07)	−39,942 (2025-04-09)	[−2.96, +0.56]	A trailing stop beats holding but still loses.	— (no pass rule: no pass rule pre-registered)
19	Phase 2 hyp A: the low-moc_mag closing pin (SPXW short call spread, 15:30, held)	SPXW	OOS 2025–26	177	**−1.43 % risk** / −33.83	−1.39 / −33.03	87.6 / 87.6	+173 (2025-04-14)	−2,312 (2025-04-30)	[−3.57, +0.26]	NO EDGE	No — fails at both (Robinhood: mean −33.83, lower end −85.45)
7e	S10 Short premium: fade the Quick Scalps signal (single contract, 2× stop, to …	all names	2022-11→2026-10	2,886	**−1.68 % credit** / +1.82	−3.63 / +0.52	55.5 / 55.4	+457 (2025-04-09)	−425 (2026-06-09)	[−7.80, +4.77]	Nothing promoted	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean −1.68%, lower end −7.80); the matched-random comparison is not recomputed
2b	George rebuild: the best of 144 variants (sweep / ict+gex / morning / …	all names	out-of-sample 2026-03→10	385	**−1.87 % prem** / −0.48	−3.04 / −1.78	40.3 / 40.3	+123 (2026-06-25)	−102 (2026-06-29)	[−3.37, −0.33]	Rejected	— (no pass rule: no pass rule pre-registered)

#	strategy	name	period	trades	mean trade at Robinhood fees	mean trade at the study's fees	win % Robinhood / study	biggest win	biggest loss	20-session block 95% (mean %, Robinhood)	verdict (as pre-registered, at the study's fees)	would the verdict differ at Robinhood fees? (descriptive)
4	GeorgeFi's own trades re-priced at the NBBO (G1)	all roots	ledger E2	594	**−2.02 % prem** / −41.59	−2.54 / −42.07	45.8 / 45.1	+3,018 (2026-09-21)	−3,476 (2026-07-13)	[−5.93, +2.73]	Not supported after execution	— (no pass rule: an execution check)
7g	S12b 10:00 long option in the first-half-hour direction	all names	2022-11→2026-10	2,499	**−2.27 % prem** / −3.58	−3.60 / −4.87	33.6 / 33.4	+2,479 (2025-04-09)	−412 (2026-06-25)	[−8.13, +4.26]	Nothing promoted	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean −2.27%, lower end −8.13); its two-part rule A1 also reads the underlying
2	George rebuild: the GeorgeFi-like reference (sweep / ict+gex / all / none / …	all names	out-of-sample 2026-03→10	1,399	**−2.33 % prem** / −1.69	−3.84 / −2.99	37.3 / 37.2	+143 (2026-06-26)	−127 (2026-06-25)	[−3.41, −1.58]	Rejected	— (no pass rule: no pass rule pre-registered)
5	Deep research round 1: flow rule F1 on SPY (H2 family)	SPY	2025-07→2026-10	638	**−2.48 % prem** / −2.82	−3.45 / −4.12	35.9 / 35.9	+177 (2025-12-18)	−82 (2026-07-29)	[−3.55, −1.17]	Rejected	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean −2.48%, lower end −3.55); its 62-cell multiple-testing correction needs the other cells
7a	S01 Quick Scalps (owner baseline) on index 0DTE	all names	2022-11→2026-10	2,886	**−3.01 % prem** / −2.89	−4.92 / −4.19	35.9 / 34.3	+347 (2026-06-09)	−156 (2025-04-07)	[−4.41, −1.79]	REJECTED	No — fails at both (Robinhood: mean −2.89, lower end −4.49)
29c	Picker, S2: "5 minutes" as a hard exit	all names	whole span (verdict)	2,943	**−3.03 % prem** / −12.07	−4.18 / −13.04	40.1 / 38.3	+908 (2025-04-09)	−992 (2026-02-05)	[−4.12, −2.02]	Labelled	— (no pass rule: a sensitivity)
7b	S02 Quick Scalps: 41-variant grid, walk-forward OOS (lower-bound selection)	all names	walk-forward OOS 2024-01→2026-10	3,200	**−3.09 % prem** / −2.51	−5.31 / −3.81	36.4 / 34.8	+347 (2026-06-09)	−162 (2025-04-07)	[−4.63, −1.82]	Nothing promoted	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean −3.09%, lower end −4.63); the matched-random comparison is not recomputed
9b	Lifecycle P2: Quick Scalps → credit put spread to expiry (portfolio)	all names	hold-out 2025-10→2026-10	196	**−3.13 % risk** / −30.07 ‡	−3.65 / −34.72	87.8 / 87.8	+1,951 (2025-10-14)	−5,189 (2026-01-16)	[−15.66, +4.66]	Rejected	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean −3.13%, lower end −15.66); R1–R6 (Holm, the random-day control) need more than this list
7d	S09 Quick Scalps on SPXW 0DTE	SPXW	2022-05→2026-10	1,229	**−3.93 % prem** / −47.04	−3.81 / −46.18	33.8 / 33.8	+1,483 (2025-04-09)	−1,002 (2026-03-09)	[−5.60, −2.27]	REJECTED	No — fails at both (Robinhood: mean −47.04, lower end −63.75)
17	Step 6: the six pre-registered runner rules × three exits	B1a	check 2025→2026-10-09	6,350	**−3.96 % prem** / −89.19	−3.97 / −89.48	41.6 / 41.6	+1,958 (2026-06-17)	−2,891 (2025-07-30)	[−4.78, −3.40]	FAIL	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean −3.96%, lower end −4.78); the 97.5th percentile of 500 cell-matched random draws is not recomputed
29e	Picker, IMPROVED (picked on the cards): PDH sweep → put, 10:00–12:00, +25% or …	all names	whole span (verdict)	516	**−4.37 % prem** / −17.87	−5.52 / −18.73	39.5 / 38.8	+478 (2024-08-07)	−1,172 (2026-02-20)	[−7.23, −1.84]	FAIL on every name	— (no pass rule: labelled, read only)

#	strategy	name	period	trades	mean trade at Robinhood fees	mean trade at the study's fees	win % Robinhood / study	biggest win	biggest loss	20-session block 95% (mean %, Robinhood)	verdict (as pre-registered, at the study's fees)	would the verdict differ at Robinhood fees? (descriptive)
7h	S15 VRP-conditioned 10:00 short ATM 0DTE straddle (unconditional row)	all names	2022-11→2026-10	1,941	**−4.50 % credit** / −12.86	−5.47 / −15.46	58.2 / 58.0	+663 (2024-08-05)	−3,321 (2025-04-09)	[−8.99, −0.44]	Nothing promoted	— (no pass rule: the rule reads the conditioned cells; this row is the unconditional one)
40a	Compass EMA ladder: PREVIEW 2025-03, 13 versions (rerun)	V1	preview 2025-03	2,781	**−4.89 % prem** / −1.48	−9.20 / −2.76	18.6 / 15.5	+149 (2025-03-28)	−30 (2025-03-05)	n/a (< 60 sessions)	PREVIEW	— (no pass rule: a preview)
29a	Picker (GeorgeFi rule), PRIMARY: opening-range fade, stop 0.68, target 1.25, …	all names	whole span (verdict)	2,943	**−4.89 % prem** / −21.65	−6.05 / −22.62	36.8 / 35.9	+1,388 (2025-04-09)	−2,952 (2025-04-07)	[−6.21, −3.68]	FAIL on every name	No — fails at both (every name must pass; at Robinhood's fees IWM fails (mean −6.54, lower end −9.08); QQQ fails (mean −3.99, lower end −5.98); SPXW fails (mean −4.93, lower end −6.70); SPY fails (mean −4.87, lower end −6.71))
29d	Picker, S3: SPXW rides to the official close when in the money at 15:58	SPXW	whole span (verdict)	908	**−4.93 % prem** / −59.13	−4.96 / −59.43	37.0 / 37.0	+1,388 (2025-04-09)	−2,952 (2025-04-07)	[−6.70, −3.46]	Labelled	— (no pass rule: a sensitivity)
29b	Picker, SECONDARY: the narrative's +6% breakeven lock and VIX trail	all names	whole span (verdict)	2,943	**−5.04 % prem** / −20.30	−6.20 / −21.27	41.9 / 39.8	+1,388 (2025-04-09)	−1,182 (2025-04-23)	[−6.20, −3.90]	Labelled	— (no pass rule: no verdict of its own)
7i	S16b Prior-day fade with a weekly contract (time exit)	all names	2017-06→2026-10	1,041	**−5.93 % prem** / −8.61	−6.78 / −9.91	38.0 / 37.8	+1,276 (2022-10-13)	−444 (2026-06-12)	[−10.19, −2.31]	Nothing promoted	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean −5.93%, lower end −10.19); its two-part rule A1 also reads the underlying
30d	Compass features BV1b: the V1 veto at 3% (PASSED, Phase B)	0DTE→60 DTE, 15 names	holdout (opened)	251,808	**−5.93 % prem** / −25.68	−6.73 / −26.77	32.0 / 31.9	+45,478 (2026-03-30)	−26,112 (2026-03-30)	[−8.57, −3.08]	PASS	No — passes at both (the study's own stats.delta on the holdout; Δ +4.30 → +3.46 points, Robinhood interval [+1.82, +4.95])
17	Step 6: the six pre-registered runner rules × three exits	A1a	check 2025→2026-10-09	1,035	**−6.72 % prem** / −23.79	−7.54 / −24.69	36.6 / 36.6	+4,162 (2025-04-09)	−3,532 (2025-04-04)	[−14.64, +0.78]	FAIL	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean −6.72%, lower end −14.64); the 97.5th percentile of 500 cell-matched random draws is not recomputed
30c	Compass features X3: sell a same-day long option at the 14:00 bid (PASSED)	0DTE→60 DTE, 15 names	holdout (opened)	233,454	**−6.97 % prem** / −10.58	−9.51 / −11.72	29.6 / 29.0	+19,168 (2025-11-20)	−4,252 (2025-11-21)	[−11.22, −4.91]	PASS	No — passes at both (the study's own stats.delta on the holdout; Δ +7.78 → +8.21 points, Robinhood interval [+4.53, +11.38])
30b	Compass features V1: veto a spread over 10% of mid (PASSED)	0DTE→60 DTE, 15 names	holdout (opened)	474,526	**−7.14 % prem** / −26.08	−8.43 / −27.16	29.7 / 29.5	+45,478 (2026-03-30)	−29,000 (2026-10-19)	[−9.97, −3.99]	PASS	No — passes at both (the study's own stats.delta on the holdout; Δ +2.61 → +2.25 points, Robinhood interval [+1.41, +3.07])

#	strategy	name	period	trades	mean trade at Robinhood fees	mean trade at the study's fees	win % Robinhood / study	biggest win	biggest loss	20-session block 95% (mean %, Robinhood)	verdict (as pre-registered, at the study's fees)	would the verdict differ at Robinhood fees? (descriptive)
30e	Compass features BX3e: the 14:00 exit on the physically settled names only ...	0DTE→60 DTE, 15 names	holdout (opened)	203,289	**−8.12 % prem** / −12.83	−11.03 / −14.09	29.1 / 28.4	+7,780 (2026-07-02)	−3,457 (2021-12-17)	[−12.22, −6.21]	PASS	No — passes at both (the study's own stats.delta on the holdout; Δ +7.67 → +8.15 points, Robinhood interval [+4.68, +11.18])
8d	V3a: single-name 0DTE short ATM straddle 10:00 → 15:50	all names	hold-out	1,622	**−8.50 % prem** / −22.24	−9.47 / −24.84	57.2 / 56.4	+2,455 (2026-06-12)	−7,237 (2026-07-02)	[−18.41, −1.91]	Rejected	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean −8.50%, lower end −18.41); rule A10 is not restated in the trade list
8e	V8: month-end 13:00 long straddle on unread chains	all names	2017-06→2022 (unread chains)	105	**−8.71 % prem** / −85.91	−9.14 / −86.06	39.0 / 39.0	+3,221 (2022-03-31)	−3,553 (2020-02-28)	[−23.53, +11.40]	Rejected	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean −8.71%, lower end −23.53); rule A15 is not restated in the trade list
30a	Compass features: the base trades (every feature is judged against them)	0DTE→60 DTE, 15 names	holdout (opened)	580,234	**−9.39 % prem** / −30.17	−11.03 / −31.26	28.3 / 28.2	+45,478 (2026-03-30)	−32,491 (2026-10-19)	[−12.26, −6.22]	Descriptive	— (no pass rule: the reference set)
7f	S12 Intraday momentum: first-half-hour direction → 10:00–15:00 long option	all names	2022-11→2026-10	2,214	**−11.94 % prem** / −3.06	−15.07 / −4.36	31.1 / 30.4	+1,189 (2026-07-29)	−221 (2025-04-08)	[−17.27, −6.14]	Nothing promoted	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean −11.94%, lower end −17.27); its two-part rule A1 also reads the underlying
13	Step 2: buy near the money, sell at +100% or hold (within 0.5% of the price)	all names	check 2025–26	227,851	**−17.61 % prem** / −54.14	−18.66 / −54.46	40.2 / 40.1	+9,038 (2025-04-09)	−18,091 (2025-11-28)	[−19.37, −16.02]	Every group loses before and after fees; near the money 29% double, ...	— (no pass rule: a population study)
31	Compass direction: post-earnings drift in 31–60 DTE options (rerun)	single names	exploration (holdout unopened)	12,114	**−23.52 % prem** / −107.74	−24.29 / −108.97	26.2 / 26.1	+18,029 (2023-10-30)	−13,865 (2026-06-26)	[−27.45, −19.03]	FAIL	No — fails at both (the study's own analyse(); Δ −0.90 → −0.90 pp, Robinhood interval [−3.54, +1.71])
8a	V1 T0: short every monthly ATM straddle across the optionable universe	all names	hold-out	12,484	**−24.13 % prem** / −241.82	−24.41 / −243.08	44.5 / 44.4	+30,841 (2026-07-20)	−65,572 (2026-01-20)	[−31.78, −16.70]	Rejected	— (no pass rule: A8 reads T0's sign and the sorts' long-short, not this list alone)
34	Edge H3: earnings IV-crush short iron condor (single names)	single names	exploration (holdout unopened)	2,209	**−27.65 % risk** / −156.97 ‡	−30.24 / −161.54	31.6 / 30.3	+1,565 (2022-04-26)	−6,536 (2022-07-26)	[−29.75, −25.26]	FAIL	No — fails at both (Robinhood: mean −156.97, lower end −176.19)
8c	V2b: long ATM straddle into the earnings report	single names	hold-out	3,543	**−30.57 % prem** / −339.77	−31.05 / −342.34	10.2 / 10.1	+19,714 (2026-05-21)	−11,371 (2026-02-10)	[−34.38, −25.36]	Rejected	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean −30.57%, lower end −34.38); rule A9 is not restated in the trade list

#	strategy	name	period	trades	mean trade at Robinhood fees	mean trade at the study's fees	win % Robinhood / study	biggest win	biggest loss	20-session block 95% (mean %, Robinhood)	verdict (as pre-registered, at the study's fees)	would the verdict differ at Robinhood fees? (descriptive)
8b	V2a: short ATM straddle across the earnings report	single names	hold-out	3,682	**−35.40 % prem** / −378.11	−35.95 / −380.58	26.2 / 25.6	+5,019 (2026-08-05)	−22,300 (2026-05-04)	[−38.38, −29.66]	Rejected	Its computable part (mean > 0, interval clear of zero): No — fails at both (Robinhood: mean −35.40%, lower end −38.38); rule A9 is not restated in the trade list

Studies with no trade-level result (each has its own short section below):

#	study	why no metrics	verdict
3	George engine audit: the phase 1 reproduction audit of #1–#2	the audit produced no trades of its own; its trade files are #2's, byte for byte (see #2).	Reproduced
9c	Lifecycle Tier I, daily tier, L-A3, H-FLO (by configuration)	committed as configuration-level cells (grid_by_setup.csv, cells.csv); the trade rows are in the scratchpad (lifecycle/archive, 281 MB) and were not re-run for	0 promoted
10	ATM straddle samples and the independent short-premium prereg (branch compass-research-2026-10-14)	price samples only (no trade with an exit); the pre-registered study never ran.	No result
11	Price-action catalogue and atlas (43 names)	descriptive tables, no trades.	Descriptive; V6 tested its levels
12	Step 1: every 0DTE contract broken down	a contract inventory, no trades.	Descriptive.
14	Step 3: 108 exits × 9 conditions	committed as per-rule summaries (exits.csv, exits_best.csv); the per-trade paths are committed but the exit grid was not re-run here.	Nothing made money in a way that held up; a −50% stop shrank …
16	Step 5: the runner model (model A/B) and its top 1% picks	the picks are committed as daily sums (top1_daily.csv), not trades.	Runners partly separable
18	Compass engine audit (six module groups)	the audit committed row-level paths and events (paths_*.parquet, events_*.parquet) and per-cell tables; its exit stacks were scored by cell, and no single trade	No directional rule survives after fees. What holds
22	Phase 2 hyp D: volatility breakout (straddles on the model's top decile)	committed as cells by scan minute and decile (cells_*.csv, minutes_*.parquet); no trade list of the pre-registered cell.	No edge
24	Phase 3 hyp E: short-premium lift of the vol model's quiet deciles	committed as by-scan cells and pooled tables; the candidate was set after the OOS read (see #26 condor_d20 for its trades).	No edge established — not pre-registrable
26d	Paper tracker: fly_spxw (SPX ±5 / ±25 flies)	a one-week sanity run outside the repository; no rows were kept.	Run only as a one-week sanity run outside the repository; replaced …
29f	Picker: flow gate and grade check	one session of flow (5 trades) and the vendor's own recorded returns (see #1, vendor-recorded).	Flow gate
30f	Compass features VN1 (venue SPY vs SPXW) and BC3 (V1 + X3 + D6)	VN1 pairs venues and BC3 needs the D6 direction feature; both are computed inside the study's screen code, and their per-trade arms are not on disk.	PASS
36	Edge H5 (SPY − SPXW late straddle) and H6 (monthly put spreads), forward only	forward-only: no trade exists. Every metric (trades, win rate, mean, interval, drawdown, by year) is "not available" because no forward session has been counted	No verdict.
38	Edge H9: overnight drift (forward only, withdrawn)	forward-only: no trade exists. Every metric (trades, win rate, mean, interval, drawdown, by year) is "not available" because no forward session has been counted	No verdict.
39	Compass scanner (score ≥ 80, SPY/QQQ/IWM 0DTE)	signal-only: the gate failed before any option was priced, so trades, win rate, average win/loss, biggest win/loss, profit factor, drawdown, top-5 share and by-	FAIL at the gate
42	George A+ entry + EMA targets	the run was stopped before its prereg was committed; no file exists on disk or in history, so every metric is not available.	Never run.
44	Compass EMA: the owner's three EMA / ORB models	the run was stopped before computing: a partial engine and a session list exist in the scratchpad (compass_ema_partial/), no signal, trade or P&L.	Never computed.

#	study	why no metrics	verdict
45	DR-GCE gates 1 and 2 (Mode 1 Gamma Vacuum Squeeze)	signal-only: touch outcomes (signals.csv, gate2_pairs.csv), never traded to an exit; no P&L metric exists.	Gate 1 PASS
46	Trend-day test	never run (the trend-day share is only described in step 6).	Never registered, never run.

The rules that passed, with their numbers

Seven rules passed a pre-registered holdout or single test. None of them makes a losing base set profitable, except H7, which is a stock trade.

rule	what it does	the test	result (Δ = with − without, points of the debit a trade; H7 = $ a trade)	forward N (committed)
V1 spread veto	skip a base trade whose (ask − bid) ÷ mid > 10% at entry	Compass features Phase A holdout (a2e14e4, 2a49049)	Δ **+2.61 [+1.76, +3.46]**; −8.43% with vs −11.03% without	23,960 sessions (out of reach)
X3 14:00 exit	sell a 0DTE long entered before 14:00 at the 14:00 bid	same	Δ **+7.78 [+4.14, +10.95]**; −9.51% vs −17.29%	369 sessions
VN1 venue	keep the SPXW/SPY venue with the lower ((ask − mid) + fee ÷ 100) ÷ mid	same (failed on the first, duplicated, read)	Δ **+1.18 [+0.50, +2.20]**	587
BV1b	V1 at 3%	Phase B holdout (b6737c7, e5dd8c6), not virgin	Δ **+4.30 [+2.66, +5.85]**; −6.73% vs −11.03%	481
BX3e	X3 on the physically settled names only	same	Δ **+7.67 [+4.23, +10.69]**; −11.03% vs −18.71%	265
BC3	V1 veto + X3 exit + D6 direction, against all base trades	same	Δ **+6.77 [+4.82, +8.19]**; −4.26% vs −11.03%	341
H7 pre-FOMC drift	100 shares from the close before a 2:00 pm FOMC statement to the 13:58 bar	single test (edca472, 6c94bb8)	SPY **+$70.43 [+14.64, +129.31]**, QQQ **+$69.59 [+23.10, +122.14]**; QQQ clears Bonferroni at m = 13	366 SPY / 483 QQQ trades (fcc1f4d)

These rules are frozen as written:

- Compass's decision layer uses V1, X3 and VN1 exactly as frozen (`FORWARD_RULE.md` b101ae1, `PREREG_PICKER.md` §4 fees).
- The daily forward job counts every rule here against its N. No verdict is read before a rule reaches its N.

Also positive on a single historical read, but not passed:

- S17's index put-writing passes its carry-lane rule (#7j). It is a carry with alpha 0, not an edge.
- H8 passes for QQQ at 1× only.
- Phase 3 hyp F (#25) passes its four-part rule on the OOS that chose it.

The same rules at Robinhood fees (descriptive; the pre-registered results above stand). The Compass features' Δ is re-run by the study's own `stats.delta` on the holdout with each trade re-priced; H7 is its single-test row.

rule	Δ or mean at the study's fees	at Robinhood fees	95% interval at Robinhood fees	the frozen pass rule at Robinhood fees
V1	Δ +2.61	Δ **+2.25**	[+1.41, +3.07]	passes
X3	Δ +7.78	Δ **+8.21**	[+4.53, +11.38]	passes
BV1b	Δ +4.30	Δ **+3.46**	[+1.82, +4.95]	passes
BX3e	Δ +7.67	Δ **+8.15**	[+4.68, +11.18]	passes

rule	Δ or mean at the study's fees	at Robinhood fees	95% interval at Robinhood fees	the frozen pass rule at Robinhood fees
VN1, BC3	not re-run: their per-trade arms are not on disk (and VN1's choice itself prices the fee)			
H7 QQQ	+69.59 $	**+70.01 $**	[+18.16, +115.30]	passes (report interval, 1×)
H7 SPY	+70.43 $	**+70.54 $**	[+11.25, +131.20]	passes (report interval, 1×)

The holdout ledger (the report's copy)

This is a copy of `research_notes/HOLDOUT_LEDGER.md` (9af93fc) with three rows added after it. The committed ledger is unchanged: nothing in it is wrong, it only predates DR-GCE.

Holdout A = index direction data 2017-06-01 → 2022-05-12. **Holdout B** = the 28 drawn months.

study (commit)	data and span	rules read	A	B
S04 (`b037899`)	SPY/QQQ/IWM bars 2017-06-01 → 2026-10-09	Quick Scalps signal, forward returns (+ ablations, not counted)	1	1
S06 (`4aba507`)	same	7 triggers × 4 filter sets	28	28
S12 (`be991de`)	bars 2017-06 → 2026-10-09	first-half-hour momentum, 3 windows	3	3
S12b (`be991de`)	0DTE 2022-11-11 → 2026-10-09	2 option expressions	0	2
S16a	bars 2017-06 → 2026-10-09	108 cells ÷ 3 names	36	36
S03, S16b (`3c8d6cb`)	weekly chains, sample 2017-06 → 2026-10-09	Quick Scalps weekly; prior-day fade 2 cells × 2 exits	5	5
H10 (registry)	SPY bars 2017 → 2026-10-09	last-30-minute momentum	1	1
H17 (registry)	SPX 2017 → 2026	4 day filters	4	4
S17	SPY/QQQ monthly chains 2012-06 → 2026-10	5 structures	5	5
Lifecycle daily tier (`50de62a`)	SPY/QQQ/IWM option EOD 2012-06 → 2026-10	≥ 19 walk-forward finalists	≥ 19	≥ 19
V6 / atlas (`0d706b6`, `1660366`)	43 names incl. SPY/QQQ/IWM bars 2017-06 → 2026-10	PDH/PDL, round numbers	2	2
V8 (registry)	SPXW 2017-06 → 2022-04, SPY 2017-06 → 2022-10 month-ends	month-end 13:00 straddle	1	1
Compass H1 (`4a02144`)	SPY 2017-01 → 2025-10-10, QQQ 2012-06 →	overnight reversal	1	1
Edge H4 (`dfa575d`)	SPY/QQQ 2012/2017 → 2026-10-22 (exploration context + test)	intraday momentum	1	1
Edge H7, H8 (`6c94bb8`)	SPY 2017 → 2026-09, QQQ 2012/2013 →	pre-FOMC drift; turn of month	2	2
Compass features (`a2e14e4`, `2a49049`, `e5dd8c6`)	ETFs 2012-06 →; SPXW/SPY/QQQ 0DTE 2022 → 2026-10-21; its own 43-month holdout read (twice)	23 features screened	23	23
Compass scanner signal check (`142fcf4`, `9e3888e`)	SPY/QQQ/IWM bars, Pick 2017-07-05 → 2022-11-18 (its non-held-out months; the 28 never opened)	the scanner score (RSI/VWAP/RVOL/structure family)	1	0
S01, S02 (`5e90201`, `6bf5934`)	SPY/QQQ/IWM 0DTE 2022-11-11 → 2026-10-09	Quick Scalps + 41 variants	0	42
S09, S10 (`7167541`, `3c8d6cb`)	SPXW 0DTE 2022-05 →; SPY/QQQ/IWM	QS on SPXW; QS fade (≥ 2 cells)	0	≥ 3

study (commit)	data and span	rules read	A	B
S15 (registry)	SPY/QQQ 0DTE 2022-11-11 → 2026-10-09	10:00 straddle, 3 conditions	0	3
H9, H11, H13, H15, H16 (registry)	SPY/QQQ/SPXW 0DTE 2022-05 / 2025-06 → 2026-10	straddles/strangles; grids of 1,890 + 2,736 configs	0	≥ 4,630
Steps 1–5 (`2185eb9` → `491a0c7`)	every 0DTE contract 2022-05-13 → 2026-10-09	buy/sell-and-hold, times, exits, turn-ups, volume (≥ 1 a step)	0	≥ 5
Lifecycle Tier I (`425c7f4`)	0DTE SPY/QQQ/IWM/SPXW 2022-05 → 2026-10-09	12 setups × 8 structures × 116 policies	0	11,136
V5, V7 (registry)	same-day chains 2022-05/11 →	calibration; event afternoons	0	2
Engine audit, phase 2 (4), phase 3 (E, F), backtest audit, SPXW analogues (`2820bb4`)	0DTE 2022 → 2026-10-09	≥ 1 each	0	≥ 9
Picker (`c502122`, `1e10053`)	SPY/QQQ/IWM/SPXW 0DTE 2022 → 2026-10-21	opening-range fade + improved	0	2
EMA system v1 preview	bars of 60 sessions before each week (2025-11, 2026-04 bars as warm-up only, no outcome)	—	0	0
Compass EMA ladder (previews 2025-02/03)	not held out; holdout not read	—	0	0
Step 6 (`30babaa`, `44b5547`)	0DTE 2022-05-13 → 2026-10-09, the 28 months never opened	18 rules	0	0
Compass ICT (`9d6c37e`, `0943030`)	exploration only; holdouts never opened	—	0	0
Compass EMA (this study, prereg)	not yet read	—	0	0
Compass Close (`research_notes/Compass Close/PREREG_COMPASS_CLOSE.md`, row added with the prereg, before any fetch or computation)	SPXW same-day PM chains 15:45–16:00, SPX prints, Cboe SPX closes: 470 new sessions 2017-06-02 → 2022-05-12 (gate = odd months 236, P&L = even months 234; 15 of B's 28 months inside); previously read 2022-05-13 → 2026-10-09 reported, not judged	1 judged rule (15:55 ATM straddle to settlement) + 8 reported entry/exit books (call, put, move side, narrower leg, 15:50, 15:58, 15:56 ask, one-strike-OTM strangle); fee and event-day lines not counted	9	9
Compass EMA (`9af93fc`)	prereg only; the run was stopped before computing	—	0	0
DR-GCE gates 1, 2 (`01d5c8e`, `041baba`, `b20eb9c`; `294d4f2`, `00eb3d5`)	SPXW 0DTE exploration 2023-12-01 → 2026-10-09, the 28 months never opened (guard selftest 5/5)	Mode 1 / Mode 2 touch gate; direction vs volatility	0	0
This final report (2026-10-23)	reads only spans already read; reruns #31 (exploration X only) and #40 (2025-02/03 only)	—	0	0

Running count with Compass Close's row (2026-10-24): holdout A ≥ 142 rules (≥ 133 before it); holdout B ≥ 15,975 rules (≥ 15,966 before it); Compass EMA's prereg has read nothing.

What has read each holdout, and what remains unopened

holdout	read by	unopened for
A (2017-06 → 2022-05-12)	≥ 133 rules (S04, S06, S12, S16a, S03/S16b, H10, H17, S17, the daily tier, V6, V8, Compass H1, H4, H7/H8, Compass features, the scanner's signal check)	every rule from the scanner on: Compass ICT (#43), Compass EMA (#44), the EMA ladder (#40), DR-GCE (#45), step 6 (#17)

holdout	read by	unopened for
B (28 months)	≥ 15,966 rules, before the holdout was defined (steps 1–5, S01/S02, S09/S10, S15, the short-premium grids, Tier I, V5/V7, the engine audit, phases 2–3, the picker)	the rules defined after it: #17, #39, #40, #43, #44, #45 (“unopened for this rule only”)
Compass features’ own (43 months + AVGO, MU, QQQ)	opened: Phase A twice (the first on duplicated rows), Phase B once	—
Compass direction: 1,600 names 2019-01 → 2022-12	—	**unopened** (the screen failed)
Edge H3: 1,600 names	—	**unopened** (not fetched)
The scanner’s held-out tickers AMD, AVGO	—	**unopened**
Lifecycle hold-out 2025-10-01 → 2026-10-09	opened once per group (Tier I, L-A3, H-FLO, the daily tier, P1/P2)	—
V1–V8 hold-outs	each opened once	—
ThetaData lanes TEST (2025 → 2026-10)	read (T-A, T-B, T-C, T-E, H15)	—
Phase 2/3 OOS 2025-01 → 2026-10-09	read once each; this is the tracker’s and the SPXW test’s “selection” period	—
Forward from 2026-10-12	the paper tracker (9 sessions to 10-22)	features (0 sessions counted), H5 (none), H6 (from 2026-11-02), H7 (from 2026-10-28)

The data inventory

All market data is stored files. Nothing was fetched for this report.

data	span	where
Same-day option chains, 1-minute NBBO (every contract)	SPY 989 sessions 2022-11-11 → 2026-10-22; QQQ 989, same span; IWM 629, 2024-04-19 → 2026-10-21; SPXW 1,115, 2022-05-13 → 2026-10-22	scratchpad `theta/data_full/chains/` (forward sessions 2026-10-12 → 21 also committed in `data/theta_forward/`)
Underlying 1-minute bars	SPY 3,687 sessions 2012-06-01 → 2026-10-22; QQQ 3,619; IWM 3,618	scratchpad `theta/data/underlying/`
SPX prints and VIX, 1-minute	SPX 2,362 sessions, VIX 2,395, 2017-06-01 → 2026-10-22	scratchpad `theta/data/index/`
Official closes	Cboe’s SPX history 1975 → 2026-10-21; UW daily bars	`data/official_closes/`
Option EOD and open interest, every contract	the account’s bulk files 2026-10-06 → 12; index-ETF option EOD 2012-06 → 2026-10	`data_lake/thetadata/flat/`, scratchpad `lifecycle/eod_by_exp`
Stock daily bars, every symbol	2012-06 → 2026-10-13 (26,855 symbols listed)	`data_lake/thetadata/stock_eod/`
Unusual Whales per-ticker histories	daily bars, greek exposure, IV rank, realised volatility, VRP, earnings; about one year to 2026-10-13	`data_lake/unusualwhales/`
Unusual Whales recorder (flow, dark pool, net-prem ticks, minute bars, OI by strike)	from 2026-10-21	`data_recorder/` (git-ignored)
Post-market 1-minute bars (UW)	16 name-sessions fetched once for the after-hours study, plus the backtest cache 2025-05 → 2026-10	`data/postmarket_1m/`, scratchpad `bt/cache`
Macro calendar (FOMC, CPI, payrolls)	2022 → 2027	`data/macro_events_2022_2027.csv`
GeorgeFi evidence (cards, books, API captures)	2026-01-20 → 2026-10-13	`evidence/` at 7dfea57
Trade-level outputs used here and not committed (kept in the scratchpad)	tracker `data_recorder/paper_backtest_official` (316 MB); step 6 population `step6/` (39 MB); features `features_work/{expl,hold}`; direction rerun `dir/work/expl_trades.parquet`; EMA ladder rerun `ema_ladder_out/`	as listed

Forward tracking

The daily job. `scripts/daily_forward.py` runs after each close, in this order, stopping at the first failure:

1. fetch the session's seven files: the Theta Terminal on the owner's machine, or one SDK login inside the owner's window;
2. verify them;
3. merge them;
4. read the official close: UW's daily bar, or Cboe for SPX;
5. read the post-market decision price;
6. replay the session through the live paper tracker (condor_em1, fly5, condor_d20), settled on the official close;
7. append it to `research_notes/Compass phase 3/forward_cumulative.md`;
8. run the Compass features forward test (V1, X3, VN1, BV1b, BX3e, BC3 under `FORWARD_RULE.md` b101ae1);
9. run the edge-search forward tests (H5, H6 and H7).

A session is appended only whole. No verdict is read before a model reaches its committed N. Interim figures are logged without early stopping.

Where it stands (2026-10-23).

- **The paper tracker's period iii:** 9 sessions, 2026-10-12 → 22. Results in that window:
  - condor_em1: 44 trades, +$9.99 a trade.
  - fly5 iron: 36 trades, +$52.93.
  - condor_d20: 41 trades, +$23.38.

  This is an execution and data-quality record. The thresholds are 60 → 120 sessions for condor_em1, 150 for condor_d20's D1 re-test, and 250 a name for the flies.
- **The Compass features:** 0 forward sessions counted (from 2026-10-22).
- **H5:** forward from 2026-10-23.
- **H6:** from 2026-11-02.

H7 live. The pre-FOMC drift is wired into step 8 of the job (712b0ad). It trades 100 SPY / 100 QQQ from the official close before each scheduled 2:00 pm statement to the 13:58 bar's close. It runs on the official closes and minute bars the job already keeps, with the macro calendar and H7's dividend table. The first forward trade is the statement of 2026-10-28. N is 366 trades for SPY and 483 for QQQ (`FORWARD_RULE_H7.md` fcc1f4d), about 46 and 60 years. The verdict is read at N only.

The H9 removal. H9 (overnight drift) was registered forward-only (edca472) and wired in 712b0ad. The owner withdrew it on 2026-10-23, and it was removed from the job (c68f389) before any H9 forward row was written. The registered rule is unchanged.

Method (every section)

- **Fills.** As each study ran them: ask in, bid out, at the 1-minute NBBO unless the section says otherwise. Older studies (#2, #4b, #5) used Unusual Whales bars with a modelled spread. Each section states its fee schedule.
- **Fees (from 2026-10-24).** Every trade is re-priced at Robinhood's fees: `pnl_rh = pnl + the study's fee − Robinhood's fee`, on the study's own fills (`scripts/fees_robinhood.py`, `docs/FEES_ROBINHOOD.md`).
  - The study's fee is taken from its own fee column when the file has one, and otherwise from its code's formula.
  - Robinhood's fee is figured from each leg's contracts, side and price.
  - A contract held to expiry or settlement pays no exit fee. Where a study charged a sale on a contract held in the money to the close (steps 2 and 4), Robinhood's fee is $0.
  - Where only the structure's total price is kept, a leg's price is taken as its share of that total. It is used only for the SEC fee's cent and SPXW's $1.00 tier. A row that needed any other guess is labelled "inferred".
  - Stock trades: the studies' $0.01 a share on an auction fill is their fee allowance, so it is replaced. Their $0.005 a share on a bar fill is half the spread, so it stays.

- **Win and loss.** A win is a trade whose P&L after fees is above zero. A breakeven counts as a loss.
- **Session span.** The calendar is every stored SPY minute-file session in the period's span. A session with no trade counts as zero.
- **Interval.** The interval is `audit_lib.session_stats` with blocks of 20 sessions (2,000 draws, its own seed), for the mean $ and the mean % a trade. A period under 60 sessions gets no interval. Some studies used other intervals, such as iid session or trade bootstraps, so the numbers here can differ from a study's own; the verdict is always the study's.
- **Drawdown.** Max drawdown is taken on the cumulative $ of session sums. The longest drawdown is the longest run of sessions spent below a previous peak.
- **Top-5 share.** This is the five best sessions' share of a positive total. Where the total is ≤ 0, the five best sessions' $ and the total without them are printed instead.
- **Random baselines.** Each is quoted from the study that computed it. For S10, S12, S12b and S16b, the mean of the study's own matched-control rows (seeds 11–15) is taken from the same trade file. None was re-simulated.
- **By year.** All periods together, grouped by entry year.
- **Charts.** Cumulative $ per name and pooled, with each period shaded and labelled.

Part A. Studies before the fresh start (75c29c9), read from 7dfea57

#1 GeorgeFi vendor cards (the vendor’s own recorded P&L)

Folder: `evidence/public_api/, research/georgefi_full_history/normalized/ (@7dfea57)`. **Commits:** — / errata 5ed66df / cf555c9. **Dates:** 2026-10-13.

Hypothesis. GeorgeFi’s published 0DTE and swing calls carry a positive return.

Rule. The vendor’s own cards: ticker, contract, entry and exit premium, size and P&L as GeorgeFi recorded them (books v2.0.0-ict, trade_now_v1, singles_today_v1, spx_plan_v1).

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). The vendor’s recorded fills; no fee or spread applied by us.

Data. 1,149 closed cards, 2026-01-20 → 2026-10-13 (Epoch 1 2026-01-20 → 02-27; Epoch 2 from 2026-03-02).

Periods and holdouts. Epoch 1 / Epoch 2, descriptive. No holdout.

Pass rule (as pre-registered). None (descriptive). The grade check of the picker test (#29) reads the same cards.

Verdict. Not evidence: re-priced at the NBBO (#4) Epoch 1 falls from +$43,248 recorded to +$12,039 executable and Epoch 2 from −$3,757 to −$24,988 (G1).

> **Vendor-recorded, not re-marked.** Every figure below is the vendor’s own recorded card P&L, as published; none was re-priced on market quotes.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib’s 20-session moving-block bootstrap over the period’s calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
v2.0.0-ict 0DTE	Epoch 1	373	25 / 29	82.6	+25.5, +161.46	−23.6, −195.44	+14.5, +39.00	**+16.92**, +99.27	n/a (< 60 sessions); n/a (< 60 sessions)
v2.0.0-ict SWING	Epoch 1	173	10 / 11	70.5	+16.1, +95.18	−3.6, −105.67	+10.7, +29.00	**+10.29**, +35.97	n/a (< 60 sessions); n/a (< 60 sessions)
all names	Epoch 1	546	28 / 29	78.8	+22.8, +142.65	−14.8, −155.97	+13.0, +36.00	**+14.82**, +79.21	n/a (< 60 sessions); n/a (< 60 sessions)
singles_today _v1 0DTE	Epoch 2	29	6 / 7	20.7	+43.3, +46.33	−45.8, −61.30	−50.0, −24.00	**−27.35**, −39.03	n/a (< 60 sessions); n/a (< 60 sessions)
singles_today _v1 SWING	Epoch 2	19	4 / 7	57.9	+20.3, +40.18	−16.2, −19.13	+8.2, +4.00	**+4.94**, +15.21	n/a (< 60 sessions); n/a (< 60 sessions)
spx_plan_v1 0DTE	Epoch 2	11	7 / 7	54.5	+97.4, +902.50	−79.0, −480.00	+54.1, +200.00	**+17.23**, +274.09	n/a (< 60 sessions); n/a (< 60 sessions)
trade_now_v1 0DTE	Epoch 2	55	35 / 69	25.5	+90.5, +1,247.32	−35.0, −667.44	−13.9, −130.00	**−3.06**, −180.05	[−32.03, +25.54]; [−540.38, +201.06]
v2.0.0-ict 0DTE	Epoch 2	429	113 / 159	51.0	+28.5, +129.99	−19.3, −120.22	+0.9, +3.00	**+5.13**, +7.51	[−0.81, +12.76]; [−17.29, +30.78]
v2.0.0-ict SWING	Epoch 2	60	28 / 121	56.7	+25.9, +94.03	−19.9, −94.02	+6.1, +22.00	**+6.06**, +12.54	[−8.91, +14.87]; [−57.80, +24.30]
all names	Epoch 2	603	132 / 162	48.1	+32.6, +190.56	−24.2, −188.56	+0.0, +0.00	**+3.13**, −6.23	[−0.74, +9.76]; [−62.87, +13.75]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
v2.0.0-ict 0DTE	Epoch 1	+4,020.00, +1,057.9%, 2026-02-06, O:SPX260206C06930000	−1,375.00, −78.6%, 2026-02-18, O:SPXW260218P06860000	3.91	310	1	60%
v2.0.0-ict SWING	Epoch 1	+910.00, +59.6%, 2026-01-29, O:USAR260130P00023000	−810.00, −21.4%, 2026-01-21, O:COIN260123P00230000	2.15	579	3	109%
all names	Epoch 1	+4,020.00, +1,057.9%, 2026-02-06, O:SPX260206C06930000	−1,375.00, −78.6%, 2026-02-18, O:SPXW260218P06860000	3.39	310	1	62%
singles_today_v1 0DTE	Epoch 2	+104.00, +50.5%, 2026-10-07, O:MU261007C01085000	−216.00, −63.5%, 2026-10-09, O:MU261009P01005000	0.20	1,132	7	total ≤ 0 (best 5 days −671; without them −461)
singles_today_v1 SWING	Epoch 2	+144.00, +38.9%, 2026-10-08, O:GOOGL261009P00347500	−70.00, −53.8%, 2026-10-08, O:TQQQ261009C00084000	2.89	0	0	100%
spx_plan_v1 0DTE	Epoch 2	+1,750.00, +140.0%, 2026-10-09, O:SPXW261009C07800000	−935.00, −94.4%, 2026-10-06, O:SPXW261006P07815000	2.26	2,140	5	162%
trade_now_v1 0DTE	Epoch 2	+3,312.50, +160.6%, 2026-07-13, O:SPXW260713P07505000	−3,400.00, −60.7%, 2026-07-13, O:SPXW260713C07570000	0.64	15,740	67	total ≤ 0 (best 5 days +8,318; without them −18,220)
v2.0.0-ict 0DTE	Epoch 2	+2,055.00, +24.9%, 2026-03-04, O:SPXW260304C06815000	−2,015.00, −24.7%, 2026-03-05, O:SPXW260305P06840000	1.13	4,066	63	227%
v2.0.0-ict SWING	Epoch 2	+501.00, +68.1%, 2026-03-04, O:IWM260313P00261000	−475.00, −19.2%, 2026-03-17, O:COIN260320C00210000	1.31	1,445	112	320%
all names	Epoch 2	+3,312.50, +160.6%, 2026-07-13, O:SPXW260713P07505000	−3,400.00, −60.7%, 2026-07-13, O:SPXW260713C07570000	0.94	14,299	101	total ≤ 0 (best 5 days +13,434; without them −17,191)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $0.00 a trade) and Robinhood's charged instead (mean $0.00). The study's fees are given beside them below. Re-pricing: vendor-recorded: no fee in either column (1,149 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
v2.0.0-ict 0DTE	Epoch 1	373	**+16.92**, +99.27	+16.92, +99.27	82.6 / 82.6	n/a (< 60 sessions) / n/a (< 60 sessions)
v2.0.0-ict SWING	Epoch 1	173	**+10.29**, +35.97	+10.29, +35.97	70.5 / 70.5	n/a (< 60 sessions) / n/a (< 60 sessions)
all names	Epoch 1	546	**+14.82**, +79.21	+14.82, +79.21	78.8 / 78.8	n/a (< 60 sessions) / n/a (< 60 sessions)
singles_today_v1 0DTE	Epoch 2	29	**−27.35**, −39.03	−27.35, −39.03	20.7 / 20.7	n/a (< 60 sessions) / n/a (< 60 sessions)
singles_today_v1 SWING	Epoch 2	19	**+4.94**, +15.21	+4.94, +15.21	57.9 / 57.9	n/a (< 60 sessions) / n/a (< 60 sessions)
spx_plan_v1 0DTE	Epoch 2	11	**+17.23**, +274.09	+17.23, +274.09	54.5 / 54.5	n/a (< 60 sessions) / n/a (< 60 sessions)
trade_now_v1 0DTE	Epoch 2	55	**−3.06**, −180.05	−3.06, −180.05	25.5 / 25.5	[−32.03, +25.54] / [−32.03, +25.54]
v2.0.0-ict 0DTE	Epoch 2	429	**+5.13**, +7.51	+5.13, +7.51	51.0 / 51.0	[−0.81, +12.76] / [−0.81, +12.76]
v2.0.0-ict SWING	Epoch 2	60	**+6.06**, +12.54	+6.06, +12.54	56.7 / 56.7	[−8.91, +14.87] / [−8.91, +14.87]
all names	Epoch 2	603	**+3.13**, −6.23	+3.13, −6.23	48.1 / 48.1	[−0.74, +9.76] / [−0.74, +9.76]

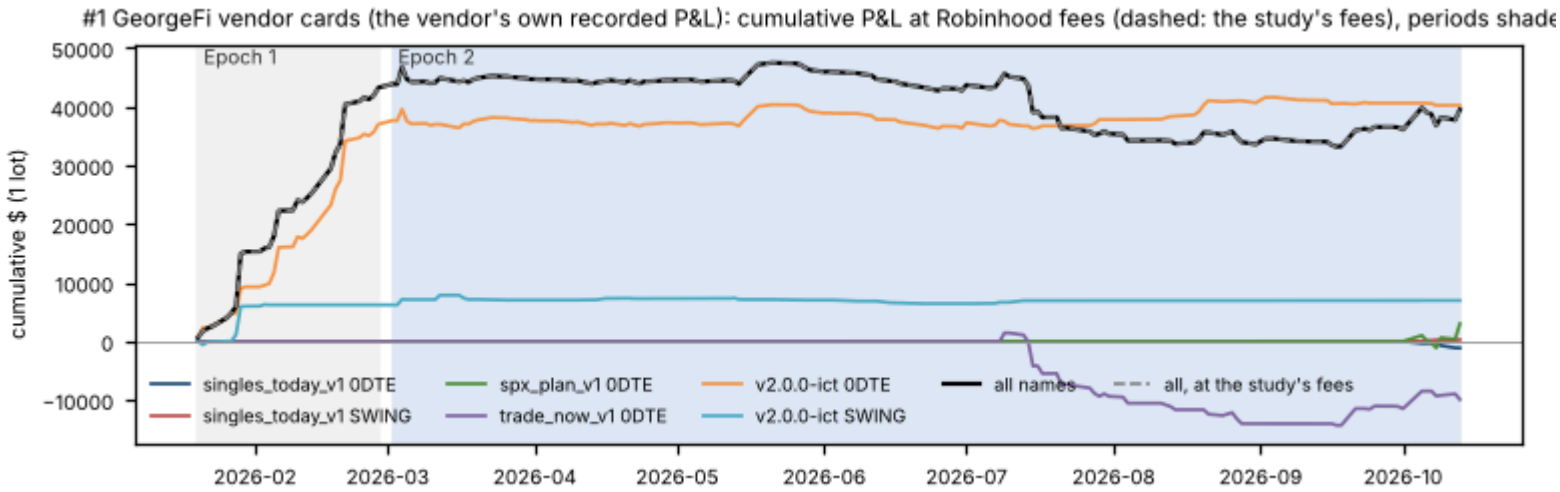
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- Epoch 1: ICT: Take profit at MAJOR liquidity pool 6.8%; TIER A +10% (1/5) 4.8%; EOD AUTO-CLOSE (3:58 PM) 4.6%; ICT TIER 1 PROFIT +11% (PARTIAL) 3.7%; ICT TIER 1 PROFIT +12% (PARTIAL) 3.3%; ICT TIER 1 PROFIT +13% (PARTIAL) 2.9%; TIER A +10% (1/4) 2.9%; TIER A +10% (2/8) 2.6%
- Epoch 2: ICT: Take profit at MAJOR liquidity pool 5.6%; SINGLES_TODAY_TRAIL 4.3%; STOP LOSS (-12%) 4.1%; GEX EXIT: Accelerator reversed against B 3.8%; SINGLES_TODAY_STOP 2.3%; STOP LOSS (0%) 2.0%; GEX REGIME SHIFT: NEG→POS while losing (1.8%; SWING PROFIT_TARGET PARTIAL EXIT (25%): 1.8%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2026	singles_today_v1 0DTE	29	20.7	−27.35	−39.03	−1,132
2026	singles_today_v1 SWING	19	57.9	+4.94	+15.21	+289
2026	spx_plan_v1 0DTE	11	54.5	+17.23	+274.09	+3,015
2026	trade_now_v1 0DTE	55	25.5	−3.06	−180.05	−9,902
2026	v2.0.0-ict 0DTE	802	65.7	+10.62	+50.18	+40,247
2026	v2.0.0-ict SWING	233	67.0	+9.20	+29.93	+6,975
2026	all names	1,149	62.7	+8.68	+34.37	+39,492



Amendments, deviations and integrity checks. pnl_reported is used as recorded; 24 cards carry a data_error flag and are kept, as the vendor kept them.

What it showed. The vendor’s record is positive in Epoch 1 and near zero in Epoch 2; the same rows re-priced at the market’s quotes are a loss in Epoch 2 (#4).

#2 George rebuild: the GeorgeFi-like reference (sweep / ict+gex / all / none / ladder)

Folder: `backtest/, BACKTEST_RESULTS.md (@7dfea57)`. **Commits:** — / — / 8f5b067. **Dates:** 2026-10-13.

Hypothesis. A level-sweep reversal into the nearest-strike 0DTE contract, managed by GeorgeFi's ladder, makes money after costs.

Rule. Levels PDH/PDL, PWH/PWL, PMH/PML, ORH/ORL and the prior-day call/put walls; a sweep within the last 3 bars closed back inside by max(0.25 ATR14, 0.02%) → buy the nearest-strike 0DTE call (low swept) or put (high swept) at the next bar's VWAP; ladder exit: stop −25%, T1 +25% half, T2 +45%, breakeven lock at +6%, trail at 65% of the high-water mark from +15%, 30-minute time stop, flat 15:55.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). The "spread" model: a half-spread of 1% (SPY, QQQ) or 2% (IWM) of premium each side plus $0.70 a contract each side; one contract. (Unusual Whales 1-minute bars, no NBBO.)

Data. SPY, QQQ, IWM 0DTE, 345 sessions 2025-06-02 → 2026-10-09 (SPX excluded: no index bars on the plan).

Periods and holdouts. In-sample < 2026-03-02, out-of-sample 2026-03-02 → 2026-10-09 (read). No holdout.

Pass rule (as pre-registered). None pre-registered (an exploratory grid of 144 variants).

Verdict. Rejected: 0 of 144 variants positive after costs (registry H1).

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	in-sample 2025-06→2026-02	504	175 / 192	35.5	+14.0, +8.71	−16.1, −9.98	−4.2, −2.30	**−5.41**, −3.34	[−7.13, −4.32]; [−4.70, −2.49]
QQQ	in-sample 2025-06→2026-02	526	165 / 192	35.0	+15.5, +20.22	−11.6, −15.06	−2.1, −2.14	**−2.11**, −2.72	[−3.48, −1.45]; [−4.68, −1.43]
SPY	in-sample 2025-06→2026-02	550	172 / 192	37.6	+14.9, +17.69	−11.3, −13.33	−2.1, −1.91	**−1.44**, −1.65	[−2.87, −0.20]; [−3.55, −0.06]
all names	in-sample 2025-06→2026-02	1,580	184 / 192	36.1	+14.8, +15.69	−13.0, −12.84	−2.1, −2.10	**−2.93**, −2.55	[−3.86, −2.52]; [−3.66, −1.86]
IWM	out-of-sample 2026-03→10	483	143 / 160	42.0	+13.3, +9.99	−14.6, −9.91	−4.1, −1.86	**−2.89**, −1.55	[−4.91, −1.54]; [−3.19, −0.58]
QQQ	out-of-sample 2026-03→10	461	142 / 160	36.9	+14.5, +26.57	−10.5, −17.33	−2.1, −2.70	**−1.29**, −1.14	[−3.21, +0.23]; [−3.38, +1.18]
SPY	out-of-sample 2026-03→10	455	144 / 160	32.7	+15.0, +21.32	−11.4, −13.96	−2.1, −2.40	**−2.78**, −2.41	[−3.73, −1.87]; [−3.44, −1.29]
all names	out-of-sample 2026-03→10	1,399	155 / 160	37.3	+14.2, +18.62	−12.1, −13.79	−2.1, −2.38	**−2.33**, −1.69	[−3.41, −1.58]; [−2.53, −1.01]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	in-sample 2025-06→2026-02	+29.27, +30.2%, 2026-01-14, IWM260114C00260000	−61.66, −28.5%, 2025-09-17, IWM250917P00240000	0.48	1,735	188	total ≤ 0 (best 5 days +198; without them −1,882)
QQQ	in-sample 2025-06→2026-02	+102.07, +32.6%, 2025-11-20, QQQ251120P00600000	−103.09, −26.8%, 2025-11-20, QQQ251120C00593000	0.72	1,658	177	total ≤ 0 (best 5 days +460; without them −1,890)
SPY	in-sample 2025-06→2026-02	+90.97, +32.6%, 2025-10-17, SPY251017C00658000	−83.29, −26.8%, 2025-11-20, SPY251120C00660000	0.80	1,071	180	total ≤ 0 (best 5 days +367; without them −1,275)
all names	in-sample 2025-06→2026-02	+102.07, +32.6%, 2025-11-20, QQQ251120P00600000	−103.09, −26.8%, 2025-11-20, QQQ251120C00593000	0.69	4,188	185	total ≤ 0 (best 5 days +582; without them −4,604)
IWM	out-of-sample 2026-03→10	+51.09, +30.2%, 2026-03-11, IWM260311C00251000	−43.42, −28.6%, 2026-06-11, IWM260611P00289000	0.73	949	141	total ≤ 0 (best 5 days +248; without them −994)
QQQ	out-of-sample 2026-03→10	+142.56, +32.6%, 2026-06-26, QQQ260626C00705000	−126.63, −26.8%, 2026-06-25, QQQ260625C00711000	0.90	675	137	total ≤ 0 (best 5 days +699; without them −1,226)
SPY	out-of-sample 2026-03→10	+105.99, +32.6%, 2026-06-25, SPY260625C00730000	−79.81, −26.8%, 2026-04-02, SPY260402P00655000	0.74	1,135	160	total ≤ 0 (best 5 days +326; without them −1,422)
all names	out-of-sample 2026-03→10	+142.56, +32.6%, 2026-06-26, QQQ260626C00705000	−126.63, −26.8%, 2026-06-25, QQQ260625C00711000	0.80	2,499	159	total ≤ 0 (best 5 days +701; without them −3,071)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (2,979 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	in-sample 2025-06→2026-02	504	**−5.41**, −3.34	−8.06, −4.64	35.5 / 35.1	[−7.13, −4.32] / [−9.87, −6.89]
QQQ	in-sample 2025-06→2026-02	526	**−2.11**, −2.72	−3.32, −4.02	35.0 / 35.0	[−3.48, −1.45] / [−4.73, −2.65]
SPY	in-sample 2025-06→2026-02	550	**−1.44**, −1.65	−2.79, −2.95	37.6 / 37.6	[−2.87, −0.20] / [−4.25, −1.51]
all names	in-sample 2025-06→2026-02	1,580	**−2.93**, −2.55	−4.65, −3.84	36.1 / 35.9	[−3.86, −2.52] / [−5.64, −4.20]
IWM	out-of-sample 2026-03→10	483	**−2.89**, −1.55	−5.30, −2.84	42.0 / 41.6	[−4.91, −1.54] / [−7.54, −3.82]
QQQ	out-of-sample 2026-03→10	461	**−1.29**, −1.14	−2.21, −2.44	36.9 / 36.9	[−3.21, +0.23] / [−4.23, −0.59]
SPY	out-of-sample 2026-03→10	455	**−2.78**, −2.41	−3.94, −3.71	32.7 / 32.7	[−3.73, −1.87] / [−4.92, −3.04]
all names	out-of-sample 2026-03→10	1,399	**−2.33**, −1.69	−3.84, −2.99	37.3 / 37.2	[−3.41, −1.58] / [−5.02, −3.01]

Random baseline (as the study computed it, at its own fees): random-time entries, same contract choice and ladder (9,896 trades): −$3.20 a trade [−3.70, −2.70] under the spread model; the reference is −$3.44 [−4.23, −2.70]. No percentile was computed.

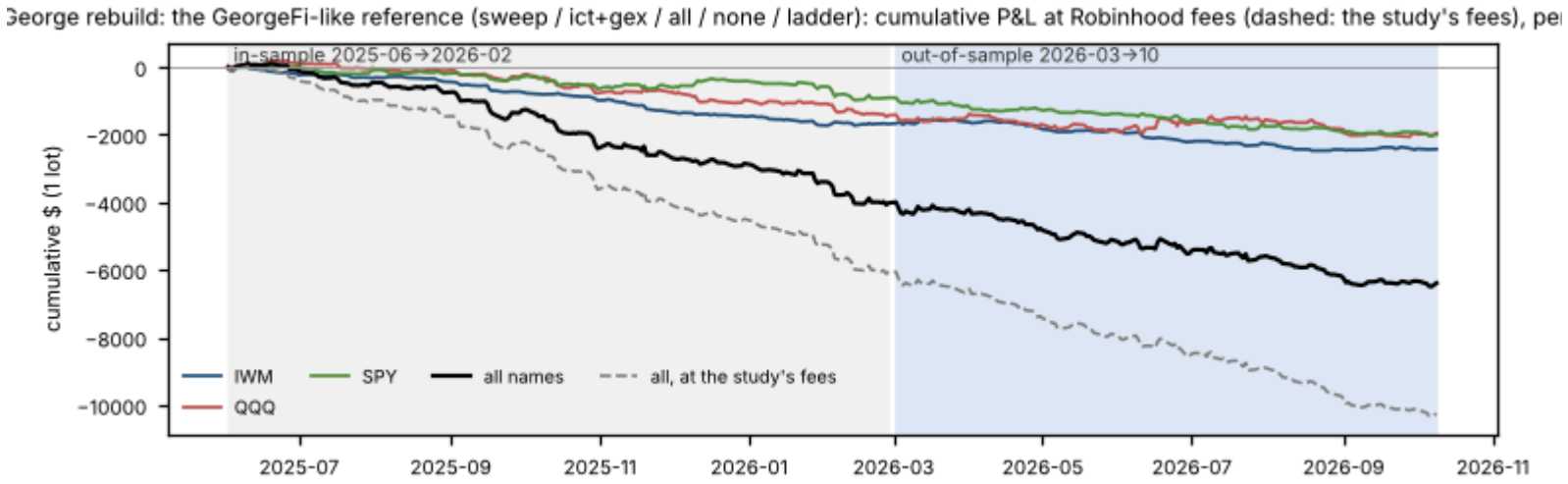
Exit reasons (share of trades):

- in-sample 2025-06→2026-02: BREAKEVEN 36.8%; STOP 26.1%; TRAIL 22.0%; T1+TRAIL 9.9%; T1+T2 4.2%; TIME_STOP 0.9%
- out-of-sample 2026-03→10: BREAKEVEN 38.5%; STOP 23.7%; TRAIL 23.6%; T1+TRAIL 10.4%; T1+T2 3.4%; TIME_STOP 0.6%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	IWM	399	34.3	−5.98	−3.69	−1,470
2026	IWM	588	41.7	−2.96	−1.63	−960

year	name	trades	win %	mean %	mean $	total $
2025	QQQ	425	35.5	−2.15	−2.32	−986
2026	QQQ	562	36.1	−1.40	−1.73	−971
2025	SPY	444	38.3	−0.91	−0.92	−411
2026	SPY	561	33.2	−2.95	−2.84	−1,594
2025	all names	1,268	36.1	−2.92	−2.26	−2,867
2026	all names	1,711	37.1	−2.44	−2.06	−3,524



Amendments, deviations and integrity checks. Reproduced byte for byte by the phase 1 audit (#3): trades.csv sha256 0e8c41c819e74eb0… on both runs.

What it showed. The reference variant loses $3.44 a trade, no better than random entry times with the same exits.

#2b George rebuild: the best of 144 variants (sweep / ict+gex / morning / posgamma_fade / trail)

Folder: `backtest/ (@7dfea57)`. **Commits:** — / — / 8f5b067. **Dates:** 2026-10-13.

Hypothesis. The best-ranked of the 144 variants (selected in sample) holds out of sample.

Rule. As #2 but 10:00–12:00 only, sweep reversals only when spot is above the prior-day gamma flip, and the trail engine (no profit targets).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). As #2 (spread model).

Data. As #2.

Periods and holdouts. As #2.

Pass rule (as pre-registered). None pre-registered.

Verdict. Rejected: −$2.41 a trade [−4.10, −0.47]; out of sample −$1.8 [−4.5, +0.9].

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	in-sample 2025-06→2026-02	134	91 / 187	29.9	+14.8, +9.73	−14.0, −9.82	−4.1, −2.76	**−5.40**, −3.99	[−7.91, −3.08]; [−5.60, −2.45]
QQQ	in-sample 2025-06→2026-02	161	98 / 189	32.3	+17.5, +27.01	−11.3, −14.72	−2.1, −2.34	**−2.02**, −1.24	[−5.84, +0.76]; [−5.87, +3.40]
SPY	in-sample 2025-06→2026-02	154	96 / 186	36.4	+18.0, +25.10	−11.4, −14.39	−2.1, −2.16	**−0.74**, −0.03	[−2.96, +1.88]; [−3.60, +4.91]
all names	in-sample 2025-06→2026-02	449	145 / 191	33.0	+17.0, +21.62	−12.2, −13.08	−2.1, −2.30	**−2.59**, −1.64	[−4.38, −1.28]; [−3.93, +0.49]
IWM	out-of-sample 2026-03→10	131	79 / 160	39.7	+12.0, +10.55	−15.6, −12.94	−4.1, −2.62	**−4.64**, −3.61	[−6.97, −2.75]; [−5.32, −2.17]
QQQ	out-of-sample 2026-03→10	122	76 / 157	45.9	+13.1, +29.78	−9.5, −17.21	−2.0, −2.77	**+0.88**, +4.36	[−1.20, +3.67]; [+0.16, +10.19]
SPY	out-of-sample 2026-03→10	132	76 / 160	35.6	+14.9, +22.59	−10.8, −15.36	−2.1, −2.64	**−1.66**, −1.85	[−4.04, +0.28]; [−5.66, +1.05]
all names	out-of-sample 2026-03→10	385	116 / 160	40.3	+13.3, +21.15	−12.1, −15.06	−2.1, −2.68	**−1.87**, −0.48	[−3.37, −0.33]; [−2.32, +1.51]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	in-sample 2025-06→2026-02	+38.51, +55.0%, 2026-01-02, IWM260102P00248000	−34.58, −28.6%, 2025-11-21, IWM251121P00231000	0.42	537	187	total ≤ 0 (best 5 days +124; without them −658)
QQQ	in-sample 2025-06→2026-02	+265.83, +84.9%, 2025-11-20, QQQ251120P00600000	−87.57, −26.8%, 2025-11-21, QQQ251121C00588000	0.88	499	182	total ≤ 0 (best 5 days +661; without them −861)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	in-sample 2025-06→2026-02	+200.88, +72.0%, 2025-10-17, SPY251017C00658000	−81.96, −26.8%, 2025-11-21, SPY251121C00655000	1.00	247	84	total ≤ 0 (best 5 days +481; without them −485)
all names	in-sample 2025-06→2026-02	+265.83, +84.9%, 2025-11-20, QQQ251120P00600000	−87.57, −26.8%, 2025-11-21, QQQ251121C00588000	0.81	938	181	total ≤ 0 (best 5 days +804; without them −1,542)
IWM	out-of-sample 2026-03→10	+47.29, +34.5%, 2026-03-25, IWM260325C00250000	−42.28, −28.6%, 2026-06-25, IWM260625P00300000	0.54	521	140	total ≤ 0 (best 5 days +146; without them −620)
QQQ	out-of-sample 2026-03→10	+123.00, +30.1%, 2026-06-25, QQQ260625C00711000	−102.02, −26.8%, 2026-06-29, QQQ260629C00712000	1.47	171	33	105%
SPY	out-of-sample 2026-03→10	+78.04, +43.6%, 2026-04-06, SPY260406P00658000	−79.81, −26.8%, 2026-04-02, SPY260402P00655000	0.81	393	99	total ≤ 0 (best 5 days +255; without them −499)
all names	out-of-sample 2026-03→10	+123.00, +30.1%, 2026-06-25, QQQ260625C00711000	−102.02, −26.8%, 2026-06-29, QQQ260629C00712000	0.95	520	87	total ≤ 0 (best 5 days +631; without them −816)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (834 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	in-sample 2025-06→2026-02	134	**−5.40**, −3.99	−7.51, −5.29	29.9 / 29.9	[−7.91, −3.08] / [−10.05, −5.20]
QQQ	in-sample 2025-06→2026-02	161	**−2.02**, −1.24	−3.11, −2.54	32.3 / 32.3	[−5.84, +0.76] / [−7.01, −0.26]
SPY	in-sample 2025-06→2026-02	154	**−0.74**, −0.03	−1.91, −1.33	36.4 / 36.4	[−2.96, +1.88] / [−4.22, +0.67]
all names	in-sample 2025-06→2026-02	449	**−2.59**, −1.64	−4.01, −2.94	33.0 / 33.0	[−4.38, −1.28] / [−5.87, −2.66]
IWM	out-of-sample 2026-03→10	131	**−4.64**, −3.61	−6.41, −4.91	39.7 / 39.7	[−6.97, −2.75] / [−8.92, −4.38]
QQQ	out-of-sample 2026-03→10	122	**+0.88**, +4.36	+0.17, +3.06	45.9 / 45.9	[−1.20, +3.67] / [−1.94, +2.98]
SPY	out-of-sample 2026-03→10	132	**−1.66**, −1.85	−2.65, −3.15	35.6 / 35.6	[−4.04, +0.28] / [−5.03, −0.65]
all names	out-of-sample 2026-03→10	385	**−1.87**, −0.48	−3.04, −1.78	40.3 / 40.3	[−3.37, −0.33] / [−4.57, −1.42]

Random baseline (as the study computed it, at its own fees): random-time baseline −$3.20 a trade (as #2); the best variant beats it by about $0.8, inside its interval.

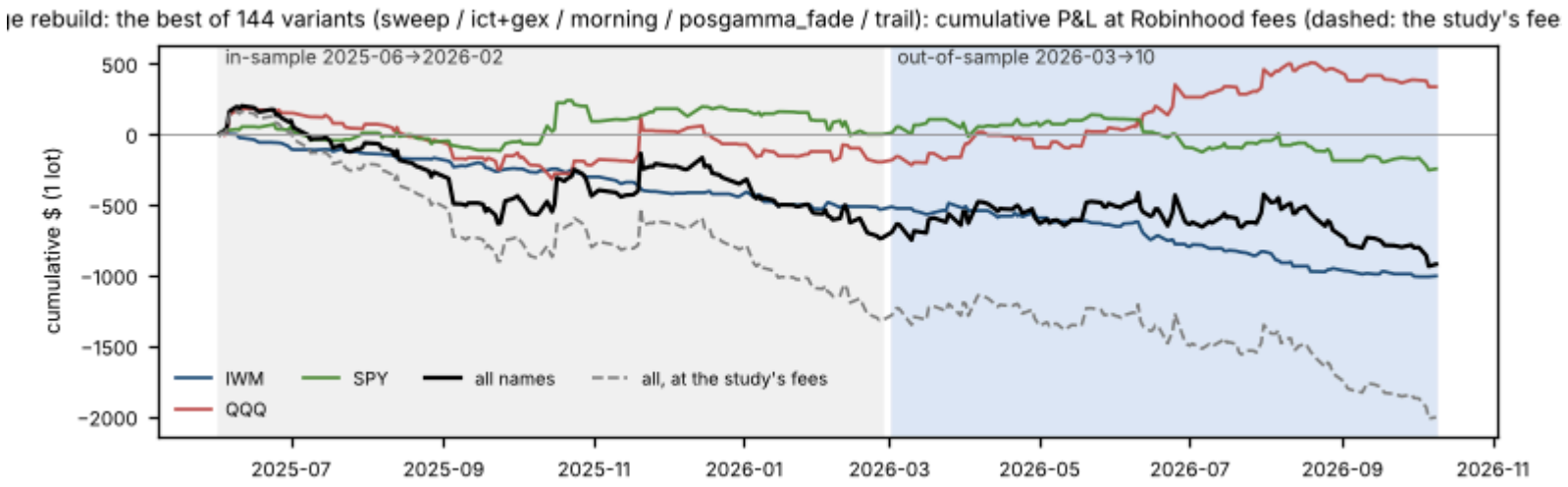
Exit reasons (share of trades):

- in-sample 2025-06→2026-02: BREAKEVEN 40.5%; TRAIL 33.0%; STOP 25.4%; TIME_STOP 1.1%
- out-of-sample 2026-03→10: TRAIL 40.3%; BREAKEVEN 36.9%; STOP 22.3%; TIME_STOP 0.5%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	IWM	98	30.6	−6.17	−4.54	−445
2026	IWM	167	37.1	−4.35	−3.37	−563
2025	QQQ	136	30.9	−1.73	−0.56	−76
2026	QQQ	147	44.9	+0.13	+2.77	+408
2025	SPY	124	37.1	−0.18	+1.33	+165
2026	SPY	162	35.2	−1.91	−2.55	−414
2025	all names	358	33.0	−2.41	−0.99	−355

year	name	trades	win %	mean %	mean $	total $
2026	all names	476	38.9	−2.14	−1.19	−569



Amendments, deviations and integrity checks. Selected as the best of 144 on the full sample (a selection effect the numbers include).

What it showed. Even the best of 144 variants loses after costs.

#3 George engine audit: the phase 1 reproduction audit of #1–#2

Folder: `audit/phase1/ (@7dfea57)` . **Commits:** — / — / 5ed66df (+ e95177c deterministic T5 rerun). **Dates:** 2026-10-13.

Hypothesis. The rebuild (#2) and the GeorgeFi validation tests T1–T7 reproduce from their artifacts.

Rule. Rerun of backtest/run.py and the validation scripts; byte comparison of outputs.

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). —

Data. As #2.

Periods and holdouts. —

Pass rule (as pre-registered). Reproduction (no trading rule).

Verdict. Reproduced: the regenerated trades and baseline files are byte-identical once decompressed.

Metrics. not available — the audit produced no trades of its own; its trade files are #2’s, byte for byte (see #2).

Amendments, deviations and integrity checks. Fixes the audit motivated were added, never overwritten (trading-days path, signals.json, deterministic T5 iteration).

What it showed. The rebuild is deterministic and reproducible; no new strategy was tested.

#4 GeorgeFi’s own trades re-priced at the NBBO (G1)

Folder: `research/georgefi_full_history/results/F_nbbo_\*.csv (@7dfea57)`. **Commits:** — / — / c27ff28. **Dates:** 2026-10-13.

Hypothesis. The vendor’s recorded trades were executable at the prices it recorded.

Rule. Each card’s own entry and exit second, filled at ThetaData’s NBBO (ask in, bid out), the card’s own size.

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). NBBO at the reported second; no fee added.

Data. 1,220 cards with both sides quoted, 2026-01-20 → 2026-10-13, every root the vendor traded.

Periods and holdouts. ledger Epoch 1, ledger Epoch 2, paper book. Descriptive.

Pass rule (as pre-registered). None (an execution check).

Verdict. Not supported after execution: Epoch 1 +$22.2 a trade [day CI −11.2, +66.2], Epoch 2 −$42.1 [−78.8, −9.6], paper −$73.9 (registry G1); 58% of Epoch 1 rows filled better than the market.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib’s 20-session moving-block bootstrap over the period’s calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
all roots	ledger E1	492	27 / 29	64.8	+15.7, +119.56	−22.4, −142.93	+7.0, +22.40	**+2.27**, +27.26	n/a (< 60 sessions); n/a (< 60 sessions)
all roots	ledger E2	594	132 / 162	45.8	+26.5, +154.61	−26.1, −207.32	−2.2, −6.25	**−2.02**, −41.59	[−5.93, +2.73]; [−122.29, −7.59]
all roots	paper book	84	38 / 61	36.9	+35.9, +131.73	−32.0, −196.34	−26.6, −79.43	**−6.94**, −75.27	[−13.24, −0.81]; [−97.75, −56.42]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days’ share of profit
all roots	ledger E1	+1,981.92, +38.1%, 2026-02-20, SPXW PUT y3bxx2s50sc0nh8	−1,677.02, −89.4%, 2026-02-18, SPXW PUT 2k5f36h32m9rqf7	1.54	4,914	12	109%
all roots	ledger E2	+3,017.74, +391.9%, 2026-09-21, SPXW CALL dni7it4jw9zjisg	−3,476.17, −62.1%, 2026-07-13, SPXW CALL vkmuy6n1j94hoee	0.63	31,395	159	total ≤ 0 (best 5 days +10,361; without them −35,063)
all roots	paper book	+597.74, +89.2%, 2026-10-01, SPXW PUT 4oduouxg1tp6xd2	−464.52, −36.3%, 2026-08-06, SPXW PUT qoxj9ry0ux3n0cz	0.39	6,770	61	total ≤ 0 (best 5 days +1,521; without them −7,843)

Fees. Every figure in this section is at **Robinhood’s fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study’s own fill prices, the study’s modelled fee added back (mean $3.63 a trade) and Robinhood’s charged instead (mean $2.31). The study’s fees are given beside them below. Re-pricing: exact (1,170 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study’s fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
all roots	ledger E1	492	**+2.27**, +27.26	+1.55, +24.46	64.8 / 64.4	n/a (< 60 sessions) / n/a (< 60 sessions)
all roots	ledger E2	594	**−2.02**, −41.59	−2.54, −42.07	45.8 / 45.1	[−5.93, +2.73] / [−6.22, +2.54]

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
all roots	paper book	84	**−6.94**, −75.27	−6.58, −73.92	36.9 / 36.9	[−13.24, −0.81] / [−12.81, −0.49]

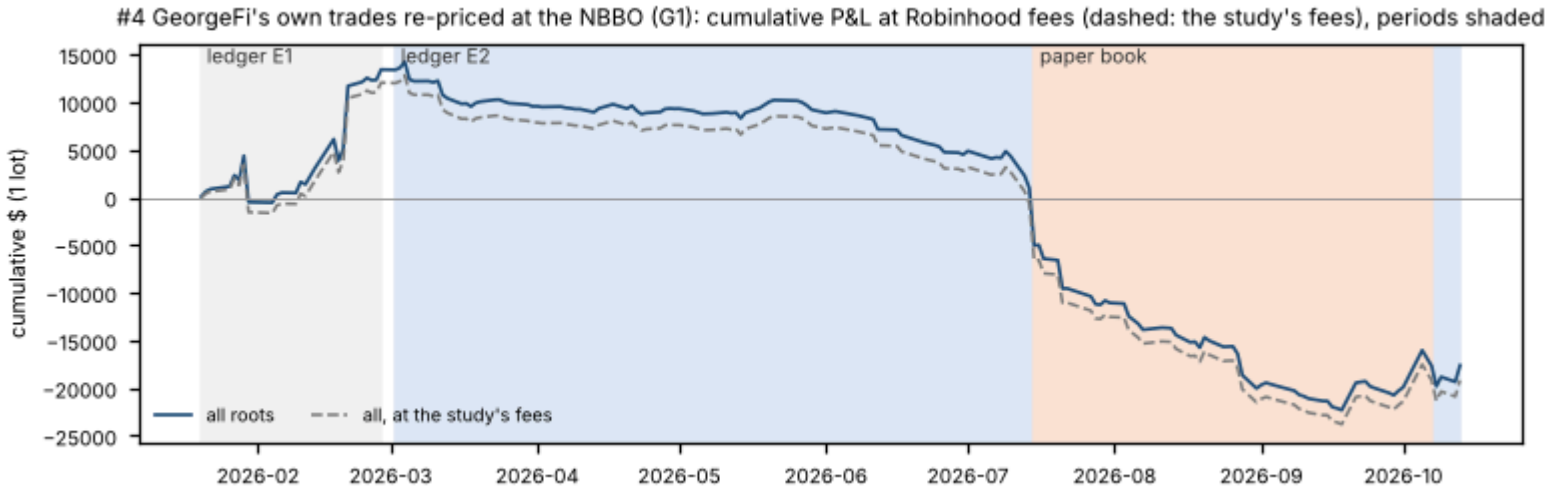
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- ledger E1: vendor exit time, filled at the NBBO 100.0%
- ledger E2: vendor exit time, filled at the NBBO 100.0%
- paper book: vendor exit time, filled at the NBBO 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2026	all roots	1,170	53.2	−0.57	−15.05	−17,614



Amendments, deviations and integrity checks. Cards with a side unquoted, and the stock rows, are left out (as F_nbbo_trades.csv flags them).

What it showed. At the market's own quotes the vendor's live epoch loses $42 a trade.

#4b GeorgeFi entries with the trail engine (reconstruction C, G2)

Folder: `research/georgefi_full_history/results/C_trades.csv (@7dfea57)`. **Commits:** — / — / 5ed66df. **Dates:** 2026-10-13.

Hypothesis. GeorgeFi's entry minutes carry timing information when managed by a fixed trail engine.

Rule. The vendor's entry minute, the reconstructed contract, entry at the bar VWAP, the paper-lane trail engine.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Spread model of #2 (half-spread + $0.70 a side).

Data. 972 reconstructed trades, 2026-01 → 2026-10.

Periods and holdouts. ledger Epoch 1 / Epoch 2 / paper.

Pass rule (as pre-registered). None (descriptive).

Verdict. No robust entry timing: Epoch 1 +$5.8 a contract [−5.7, +16.3], Epoch 2 +$2.3 [−10.5, +14.9]; MFE percentile against time-matched random minutes 53 and 48 (G2).

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
AAPL	ledger E1	1	1 / 1	100.0	+18.8, +4.00	—, —	+18.8, +4.00	**+18.81**, +4.00	n/a (< 60 sessions); n/a (< 60 sessions)
COIN	ledger E1	5	2 / 6	40.0	+27.0, +12.99	−21.7, −38.47	−12.6, −13.00	**−2.18**, −17.88	n/a (< 60 sessions); n/a (< 60 sessions)
CRCL	ledger E1	12	1 / 1	0.0	—, —	−27.2, −46.27	−27.2, −46.27	**−27.18**, −46.27	n/a (< 60 sessions); n/a (< 60 sessions)
CRM	ledger E1	1	1 / 1	100.0	+69.0, +54.79	—, —	+69.0, +54.79	**+69.05**, +54.79	n/a (< 60 sessions); n/a (< 60 sessions)
CRML	ledger E1	14	1 / 1	85.7	+26.0, +2.89	−21.3, −8.47	+26.7, +2.92	**+19.22**, +1.27	n/a (< 60 sessions); n/a (< 60 sessions)
CVNA	ledger E1	1	1 / 1	0.0	—, —	−27.1, −199.24	−27.1, −199.24	**−27.06**, −199.24	n/a (< 60 sessions); n/a (< 60 sessions)
GOOGL	ledger E1	3	1 / 1	0.0	—, —	−25.1, −19.46	−26.3, −20.34	**−25.13**, −19.46	n/a (< 60 sessions); n/a (< 60 sessions)
HOOD	ledger E1	3	1 / 1	66.7	+17.1, +9.94	−25.3, −38.22	+13.7, +7.97	**+2.95**, −6.11	n/a (< 60 sessions); n/a (< 60 sessions)
IWM	ledger E1	65	15 / 28	67.7	+27.2, +56.65	−7.5, −13.78	+7.8, +14.67	**+16.02**, +33.90	n/a (< 60 sessions); n/a (< 60 sessions)
NVDA	ledger E1	14	2 / 3	57.1	+30.6, +26.57	−2.7, −3.74	+30.2, +26.24	**+16.32**, +13.58	n/a (< 60 sessions); n/a (< 60 sessions)
OKLO	ledger E1	1	1 / 1	0.0	—, —	−24.3, −44.15	−24.3, −44.15	**−24.33**, −44.15	n/a (< 60 sessions); n/a (< 60 sessions)
QQQ	ledger E1	63	11 / 29	46.0	+11.7, +20.87	−6.5, −16.66	−0.6, −1.29	**+1.87**, +0.62	n/a (< 60 sessions); n/a (< 60 sessions)
RDDT	ledger E1	2	1 / 1	50.0	+60.0, +90.25	−25.1, −34.23	+17.5, +28.01	**+17.46**, +28.01	n/a (< 60 sessions); n/a (< 60 sessions)
RKLB	ledger E1	28	2 / 6	64.3	+24.3, +20.12	−3.2, −5.62	+7.1, +9.62	**+14.46**, +10.93	n/a (< 60 sessions); n/a (< 60 sessions)
RUTW	ledger E1	8	1 / 1	87.5	+8.2, +21.72	−3.2, −8.53	+8.2, +20.89	**+6.79**, +17.93	n/a (< 60 sessions); n/a (< 60 sessions)
SLV	ledger E1	3	1 / 1	0.0	—, —	−3.2, −5.66	−3.2, −5.66	**−3.19**, −5.66	n/a (< 60 sessions); n/a (< 60 sessions)
SPXW	ledger E1	59	11 / 25	54.2	+10.7, +142.36	−20.5, −150.77	+5.7, +5.62	**−3.57**, +8.22	n/a (< 60 sessions); n/a (< 60 sessions)
SPY	ledger E1	89	16 / 24	48.3	+13.9, +33.04	−7.3, −21.51	−0.3, −0.68	**+2.94**, +4.85	n/a (< 60 sessions); n/a (< 60 sessions)
all names	ledger E1	372	25 / 29	53.8	+18.8, +50.15	−11.3, −41.17	+4.5, +2.92	**+4.91**, +7.93	n/a (< 60 sessions); n/a (< 60 sessions)
AAPL	ledger E2	2	2 / 8	50.0	+14.5, +10.84	−2.9, −3.74	+5.8, +3.55	**+5.82**, +3.55	n/a (< 60 sessions); n/a (< 60 sessions)
AMD	ledger E2	1	1 / 1	0.0	—, —	−26.0, −34.41	−26.0, −34.41	**−25.96**, −34.41	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
AMZN	ledger E2	6	5 / 42	16.7	+22.0, +13.75	−6.8, −8.26	−2.1, −3.60	**−1.96**, −4.59	n/a (< 60 sessions); n/a (< 60 sessions)
AVGO	ledger E2	5	5 / 152	40.0	+9.7, +17.95	−16.0, −12.03	−1.0, −0.90	**−5.70**, −0.04	[−9.07, +7.77]; [−5.92, +23.51]
BAC	ledger E2	1	1 / 1	0.0	—, —	−11.8, −1.92	−11.8, −1.92	**−11.81**, −1.92	n/a (< 60 sessions); n/a (< 60 sessions)
COIN	ledger E2	1	1 / 1	100.0	+8.4, +22.90	—, —	+8.4, +22.90	**+8.39**, +22.90	n/a (< 60 sessions); n/a (< 60 sessions)
GLD	ledger E2	17	12 / 150	52.9	+137.9, +33.76	−10.1, −20.59	+0.6, +0.34	**+68.24**, +8.19	[+0.00, +61.34]; [+0.00, +38.22]
GOOGL	ledger E2	10	8 / 142	30.0	+11.6, +10.83	−8.9, −11.33	−1.7, −3.15	**−2.75**, −4.69	[−5.01, +10.11]; [−9.41, +10.23]
HOOD	ledger E2	1	1 / 1	0.0	—, —	−20.6, −7.33	−20.6, −7.33	**−20.59**, −7.33	n/a (< 60 sessions); n/a (< 60 sessions)
IBIT	ledger E2	2	2 / 6	0.0	—, —	−3.8, −2.11	−3.8, −2.11	**−3.83**, −2.11	n/a (< 60 sessions); n/a (< 60 sessions)
IWM	ledger E2	19	10 / 68	36.8	+14.1, +31.50	−12.6, −28.07	−1.9, −2.58	**−2.76**, −6.13	[−6.46, +17.44]; [−13.95, +30.20]
META	ledger E2	17	6 / 39	35.3	+7.3, +15.81	−9.5, −22.69	−3.6, −15.82	**−3.59**, −9.10	n/a (< 60 sessions); n/a (< 60 sessions)
MSFT	ledger E2	1	1 / 1	100.0	+9.8, +38.68	—, —	+9.8, +38.68	**+9.84**, +38.68	n/a (< 60 sessions); n/a (< 60 sessions)
MU	ledger E2	4	2 / 3	50.0	+17.8, +32.27	−14.4, −21.65	+4.0, +12.60	**+1.66**, +5.31	n/a (< 60 sessions); n/a (< 60 sessions)
NDXP	ledger E2	16	14 / 44	56.2	+48.8, +444.39	−24.3, −231.98	+15.7, +97.07	**+16.86**, +148.48	n/a (< 60 sessions); n/a (< 60 sessions)
QQQ	ledger E2	103	56 / 158	42.7	+14.6, +26.25	−12.4, −23.33	−0.7, −1.42	**−0.89**, −2.15	[−4.59, +4.08]; [−9.10, +7.45]
RUTW	ledger E2	23	17 / 77	56.5	+25.2, +156.97	−15.1, −91.73	+9.1, +53.40	**+7.66**, +48.84	[−1.21, +14.07]; [+5.82, +67.78]
SLV	ledger E2	3	3 / 29	100.0	+14.9, +5.38	—, —	+14.4, +5.03	**+14.88**, +5.38	n/a (< 60 sessions); n/a (< 60 sessions)
SMCI	ledger E2	1	1 / 1	0.0	—, —	−24.0, −16.17	−24.0, −16.17	**−24.05**, −16.17	n/a (< 60 sessions); n/a (< 60 sessions)
SMH	ledger E2	1	1 / 1	0.0	—, —	−2.2, −3.18	−2.2, −3.18	**−2.15**, −3.18	n/a (< 60 sessions); n/a (< 60 sessions)
SPXW	ledger E2	206	90 / 160	38.3	+19.4, +128.84	−15.0, −90.17	−3.1, −16.18	**−1.80**, −6.18	[−6.28, +2.53]; [−26.24, +7.53]
SPY	ledger E2	65	31 / 158	30.8	+11.0, +21.89	−9.2, −23.19	−1.4, −3.06	**−2.98**, −9.32	[−4.45, −1.04]; [−12.14, −2.43]
TSLA	ledger E2	8	6 / 126	62.5	+8.3, +8.99	−2.7, −8.01	+7.4, +3.99	**+4.19**, +2.61	[−2.97, +4.16]; [−9.47, +5.02]
UBER	ledger E2	1	1 / 1	0.0	—, —	−18.4, −5.05	−18.4, −5.05	**−18.40**, −5.05	n/a (< 60 sessions); n/a (< 60 sessions)
UCO	ledger E2	1	1 / 1	0.0	—, —	−26.2, −19.20	−26.2, −19.20	**−26.17**, −19.20	n/a (< 60 sessions); n/a (< 60 sessions)
USO	ledger E2	1	1 / 1	100.0	+28.4, +40.60	—, —	+28.4, +40.60	**+28.43**, +40.60	n/a (< 60 sessions); n/a (< 60 sessions)
all names	ledger E2	516	131 / 162	40.1	+23.2, +90.58	−13.0, −56.58	−1.5, −3.47	**+1.50**, +2.46	[−2.81, +3.24]; [−9.27, +19.34]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
AAPL	ledger E1	+4.00, +18.8%, 2026-02-06, AAPL PUT 955xs5861je0sz7	+4.00, +18.8%, 2026-02-06, AAPL PUT 955xs5861je0sz7	∞	0	0	100%
COIN	ledger E1	+19.57, +40.7%, 2026-01-23, COIN PUT j55nf3xhhgeu4hi	−74.94, −25.7%, 2026-01-30, COIN PUT 09x52w4963s144b	0.23	89	6	total ≤ 0 (best 5 days −89; without them +0)
CRCL	ledger E1	−46.27, −27.2%, 2026-01-30, CRCL PUT 0g311hmh7608g99	−46.27, −27.2%, 2026-01-30, CRCL PUT 0g311hmh7608g99	0.00	555	1	total ≤ 0 (best 5 days −555; without them +0)
CRM	ledger E1	+54.79, +69.0%, 2026-01-30, CRM PUT 78506oa5o61r701	+54.79, +69.0%, 2026-01-30, CRM PUT 78506oa5o61r701	∞	0	0	100%
CRML	ledger E1	+2.92, +26.7%, 2026-01-30, CRML PUT 163cd9t5w354tz2	−9.79, −24.7%, 2026-01-30, CRML PUT r2r64k78k862056	2.05	0	0	100%
CVNA	ledger E1	−199.24, −27.1%, 2026-01-30, CVNA PUT 3gris210t7m2b5w	−199.24, −27.1%, 2026-01-30, CVNA PUT 3gris210t7m2b5w	0.00	199	1	total ≤ 0 (best 5 days −199; without them +0)
GOOGL	ledger E1	−17.70, −22.9%, 2026-01-30, GOOGL PUT 3afgh5f7vn4ui47	−20.34, −26.3%, 2026-01-30, GOOGL PUT 95s51zbt917v348	0.00	58	1	total ≤ 0 (best 5 days −58; without them +0)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
HOOD	ledger E1	+11.92, +20.5%, 2026-01-30, HOOD PUT 5oox1vaao67k69q	−38.22, −25.3%, 2026-01-30, HOOD PUT 774ne079d7stl33	0.52	18	1	total ≤ 0 (best 5 days −18; without them +0)
IWM	ledger E1	+371.94, +164.1%, 2026-02-12, IWM PUT 9wo3d839ljqsl5l	−76.13, −26.6%, 2026-02-13, IWM PUT 960e22lsdv24ai6	8.61	164	11	105%
NVDA	ledger E1	+27.56, +31.7%, 2026-02-06, NVDA CALL u87y983118239z4	−4.18, −3.0%, 2026-02-04, NVDA PUT 1848jkd063c3950	9.47	22	2	100%
OKLO	ledger E1	−44.15, −24.3%, 2026-01-30, OKLO PUT 9xy7jas7kw9524i	−44.15, −24.3%, 2026-01-30, OKLO PUT 9xy7jas7kw9524i	0.00	44	1	total ≤ 0 (best 5 days −44; without them +0)
QQQ	ledger E1	+47.56, +21.4%, 2026-02-20, QQQ CALL x3tcpw8hmqi8w86	−119.17, −25.9%, 2026-02-18, QQQ CALL 402z467697wj3ie	1.07	366	22	1238%
RDDT	ledger E1	+90.25, +60.0%, 2026-01-30, RDDT PUT 5725tnis6i49604	−34.23, −25.1%, 2026-01-30, RDDT PUT 485pwz3e339pbdj	2.64	0	0	100%
RKLB	ledger E1	+33.88, +38.5%, 2026-01-30, RKLB PUT 0gi7xw49q6uthdz	−5.62, −3.2%, 2026-01-30, RKLB PUT 02rl903c51x3m7l	6.44	0	0	100%
RUTW	ledger E1	+26.77, +8.2%, 2026-01-27, RUTW CALL 1891h5gtbs11ue7	−8.53, −3.2%, 2026-01-27, RUTW CALL 592b74h07fy9i2j	17.82	0	0	100%
SLV	ledger E1	−5.66, −3.2%, 2026-02-06, SLV PUT ar9594rv59q1lpa	−5.66, −3.2%, 2026-02-06, SLV PUT ar9594rv59q1lpa	0.00	17	1	total ≤ 0 (best 5 days −17; without them +0)
SPXW	ledger E1	+639.34, +21.7%, 2026-02-17, SPXW PUT 0vsr61tk6e355f0	−855.63, −27.3%, 2026-02-19, SPXW PUT u3a2f8rxkb60t4x	1.12	811	7	396%
SPY	ledger E1	+162.30, +65.4%, 2026-02-24, SPY CALL 894j813el751a5q	−124.06, −17.0%, 2026-02-10, SPY CALL 9zf937w5328f71i	1.44	361	13	212%
all names	ledger E1	+639.34, +21.7%, 2026-02-17, SPXW PUT 0vsr61tk6e355f0	−855.63, −27.3%, 2026-02-19, SPXW PUT u3a2f8rxkb60t4x	1.42	765	7	127%
AAPL	ledger E2	+10.84, +14.5%, 2026-03-18, AAPL PUT yco9879mvrod1qa	−3.74, −2.9%, 2026-03-27, AAPL PUT lf35l8q98xq5m4i	2.90	4	1	100%
AMD	ledger E2	−34.41, −26.0%, 2026-10-09, AMD PUT i3b03ewpkryh99v	−34.41, −26.0%, 2026-10-09, AMD PUT i3b03ewpkryh99v	0.00	34	1	total ≤ 0 (best 5 days −34; without them +0)
AMZN	ledger E2	+13.75, +22.0%, 2026-09-23, AMZN PUT t5exm42pv39sbea	−27.28, −25.5%, 2026-08-17, AMZN PUT 9jcbom24ks2h5nl	0.33	35	42	total ≤ 0 (best 5 days −28; without them −0)
AVGO	ledger E2	+23.51, +7.8%, 2026-03-13, AVGO PUT rvd99u8867d6qku	−26.43, −25.4%, 2026-10-12, AVGO PUT ekylizkrfb48uzw	0.99	35	2	total ≤ 0 (best 5 days −0; without them +0)
BAC	ledger E2	−1.92, −11.8%, 2026-10-09, BAC CALL x0z4k4fwb8gos4x	−1.92, −11.8%, 2026-10-09, BAC CALL x0z4k4fwb8gos4x	0.00	2	1	total ≤ 0 (best 5 days −2; without them +0)
COIN	ledger E2	+22.90, +8.4%, 2026-10-09, COIN CALL 9oyzh2v0xe3d9bs	+22.90, +8.4%, 2026-10-09, COIN CALL 9oyzh2v0xe3d9bs	∞	0	0	100%
GLD	ledger E2	+142.92, +93.7%, 2026-09-14, GLD CALL kw39iz49kt101o8	−74.71, −26.5%, 2026-03-18, GLD CALL 4hc062it0yxcvu4	1.85	75	128	198%
GOOGL	ledger E2	+25.80, +22.9%, 2026-08-21, GOOGL CALL xk0nrxpvlxtv0rk	−41.71, −27.1%, 2026-03-27, GOOGL PUT i83o4578nx3v13m	0.41	47	142	total ≤ 0 (best 5 days +14; without them −61)
HOOD	ledger E2	−7.33, −20.6%, 2026-10-09, HOOD CALL 5dewoq9m3lm21bz	−7.33, −20.6%, 2026-10-09, HOOD CALL 5dewoq9m3lm21bz	0.00	7	1	total ≤ 0 (best 5 days −7; without them +0)
IBIT	ledger E2	−1.06, −1.1%, 2026-10-05, IBIT PUT mz572iaej6prqxd	−3.16, −6.5%, 2026-10-12, IBIT PUT ens8uikr9j4xd0u	0.00	4	6	total ≤ 0 (best 5 days −4; without them +0)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	ledger E2	+82.22, +27.4%, 2026-03-19, IWM CALL 1794163he39p5c3	−77.73, −27.0%, 2026-03-09, IWM CALL 9417g6bc21he872	0.65	213	67	total ≤ 0 (best 5 days +129; without them −245)
META	ledger E2	+20.17, +10.9%, 2026-09-04, META CALL kmr87mdhotjnbzb	−65.76, −26.9%, 2026-09-16, META CALL u8myue00egjv6ch	0.38	157	39	total ≤ 0 (best 5 days +2; without them −157)
MSFT	ledger E2	+38.68, +9.8%, 2026-03-27, MSFT PUT 8o6nv14p53b063h	+38.68, +9.8%, 2026-03-27, MSFT PUT 8o6nv14p53b063h	∞	0	0	100%
MU	ledger E2	+32.72, +24.7%, 2026-10-07, MU CALL m88pqldq0ixb2wd	−36.69, −26.1%, 2026-10-09, MU PUT cmzm0ywaiod9jxm	1.49	37	1	100%
NDXP	ledger E2	+1,752.21, +122.3%, 2026-07-06, NDXP PUT fugmhxmitbyc9jl	−590.99, −27.3%, 2026-06-12, NDXP PUT 744n57ni3vvbc5v	2.46	790	23	142%
QQQ	ledger E2	+151.46, +105.1%, 2026-05-11, QQQ PUT n65jjs8fech846i	−99.61, −26.1%, 2026-07-10, QQQ CALL g8rhxc7ftrxf8ox	0.84	655	154	total ≤ 0 (best 5 days +558; without them −779)
RUTW	ledger E2	+355.71, +39.1%, 2026-07-13, RUTW PUT hh5gls5egcdlfbb	−301.17, −27.2%, 2026-07-28, RUTW PUT kekqioeqylgb4kl	2.22	301	14	115%
SLV	ledger E2	+11.01, +30.1%, 2026-09-11, SLV CALL m2eeuqdcs6jl3xr	+0.10, +0.2%, 2026-10-07, SLV CALL 2ti9rbrixeb0oze	∞	0	0	100%
SMCI	ledger E2	−16.17, −24.0%, 2026-10-09, SMCI PUT u0lsesrgx6ikwsj	−16.17, −24.0%, 2026-10-09, SMCI PUT u0lsesrgx6ikwsj	0.00	16	1	total ≤ 0 (best 5 days −16; without them +0)
SMH	ledger E2	−3.18, −2.2%, 2026-10-12, SMH PUT c4i07emq4fnytre	−3.18, −2.2%, 2026-10-12, SMH PUT c4i07emq4fnytre	0.00	3	1	total ≤ 0 (best 5 days −3; without them +0)
SPXW	ledger E2	+823.27, +56.3%, 2026-09-29, SPXW CALL 4cwxg2xbhvss79h	−777.97, −27.2%, 2026-03-16, SPXW CALL r98xgo27bdx4g5f	0.89	3,530	159	total ≤ 0 (best 5 days +3,270; without them −4,544)
SPY	ledger E2	+40.40, +21.2%, 2026-08-18, SPY PUT kgrcrt2va87cvcl	−156.86, −26.3%, 2026-03-27, SPY PUT 5zy0jk44ra6yh18	0.42	606	158	total ≤ 0 (best 5 days +128; without them −734)
TSLA	ledger E2	+17.84, +6.7%, 2026-09-04, TSLA PUT mpxpmm8qv0ehcv0	−13.51, −3.3%, 2026-08-21, TSLA CALL v71km1xj03uovgs	1.87	24	122	165%
UBER	ledger E2	−5.05, −18.4%, 2026-10-09, UBER CALL lk1w0emiuxfqtk7	−5.05, −18.4%, 2026-10-09, UBER CALL lk1w0emiuxfqtk7	0.00	5	1	total ≤ 0 (best 5 days −5; without them +0)
UCO	ledger E2	−19.20, −26.2%, 2026-03-20, UCO CALL 0c63wm0dox7wy65	−19.20, −26.2%, 2026-03-20, UCO CALL 0c63wm0dox7wy65	0.00	19	1	total ≤ 0 (best 5 days −19; without them +0)
USO	ledger E2	+40.60, +28.4%, 2026-03-20, USO CALL 4ug6m04x120lnp8	+40.60, +28.4%, 2026-03-20, USO CALL 4ug6m04x120lnp8	∞	0	0	100%
all names	ledger E2	+1,752.21, +122.3%, 2026-07-06, NDXP PUT fugmhxmitbyc9jl	−777.97, −27.2%, 2026-03-16, SPXW CALL r98xgo27bdx4g5f	1.07	3,638	87	391%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $3.62 a trade) and Robinhood's charged instead (mean $2.63). The study's fees are given beside them below. Re-pricing: exact (888 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
AAPL	ledger E1	1	**+18.81**, +4.00	+6.49, +1.38	100.0 / 100.0	n/a (< 60 sessions) / n/a (< 60 sessions)
COIN	ledger E1	5	**−2.18**, −17.88	−12.50, −25.75	40.0 / 40.0	n/a (< 60 sessions) / n/a (< 60 sessions)
CRCL	ledger E1	12	**−27.18**, −46.27	−27.94, −47.57	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
CRM	ledger E1	1	**+69.05**, +54.79	+62.43, +49.54	100.0 / 100.0	n/a (< 60 sessions) / n/a (< 60 sessions)
CRML	ledger E1	14	**+19.22**, +1.27	+5.65, −0.59	85.7 / 78.6	n/a (< 60 sessions) / n/a (< 60 sessions)

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
CVNA	ledger E1	1	**−27.06**, −199.24	−27.94, −205.75	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
GOOGL	ledger E1	3	**−25.13**, −19.46	−27.94, −21.64	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
HOOD	ledger E1	3	**+2.95**, −6.11	−1.67, −9.61	66.7 / 66.7	n/a (< 60 sessions) / n/a (< 60 sessions)
IWM	ledger E1	65	**+16.02**, +33.90	+13.74, +30.74	67.7 / 64.6	n/a (< 60 sessions) / n/a (< 60 sessions)
NVDA	ledger E1	14	**+16.32**, +13.58	+14.71, +11.90	57.1 / 57.1	n/a (< 60 sessions) / n/a (< 60 sessions)
OKLO	ledger E1	1	**−24.33**, −44.15	−27.94, −50.70	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
QQQ	ledger E1	63	**+1.87**, +0.62	−0.60, −4.21	46.0 / 30.2	n/a (< 60 sessions) / n/a (< 60 sessions)
RDDT	ledger E1	2	**+17.46**, +28.01	+13.84, +22.76	50.0 / 50.0	n/a (< 60 sessions) / n/a (< 60 sessions)
RKLB	ledger E1	28	**+14.46**, +10.93	+11.45, +9.16	64.3 / 64.3	n/a (< 60 sessions) / n/a (< 60 sessions)
RUTW	ledger E1	8	**+6.79**, +17.93	+6.86, +18.09	87.5 / 87.5	n/a (< 60 sessions) / n/a (< 60 sessions)
SLV	ledger E1	3	**−3.19**, −5.66	−3.92, −6.96	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
SPXW	ledger E1	59	**−3.57**, +8.22	−0.02, +12.25	54.2 / 55.9	n/a (< 60 sessions) / n/a (< 60 sessions)
SPY	ledger E1	89	**+2.94**, +4.85	+1.29, +1.41	48.3 / 42.7	n/a (< 60 sessions) / n/a (< 60 sessions)
all names	ledger E1	372	**+4.91**, +7.93	+3.15, +5.83	53.8 / 49.2	n/a (< 60 sessions) / n/a (< 60 sessions)
AAPL	ledger E2	2	**+5.82**, +3.55	+0.91, −0.38	50.0 / 50.0	n/a (< 60 sessions) / n/a (< 60 sessions)
AMD	ledger E2	1	**−25.96**, −34.41	−27.94, −37.02	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
AMZN	ledger E2	6	**−1.96**, −4.59	−4.65, −7.65	16.7 / 16.7	n/a (< 60 sessions) / n/a (< 60 sessions)
AVGO	ledger E2	5	**−5.70**, −0.04	−9.00, −3.44	40.0 / 40.0	[−9.07, +7.77] / [−12.65, +5.61]
BAC	ledger E2	1	**−11.81**, −1.92	−27.94, −4.54	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
COIN	ledger E2	1	**+8.39**, +22.90	+7.91, +21.60	100.0 / 100.0	n/a (< 60 sessions) / n/a (< 60 sessions)
GLD	ledger E2	17	**+68.24**, +8.19	+7.49, +5.49	52.9 / 47.1	[+0.00, +61.34] / [+0.00, +32.98]
GOOGL	ledger E2	10	**−2.75**, −4.69	−5.62, −7.44	30.0 / 20.0	[−5.01, +10.11] / [−8.57, +8.89]
HOOD	ledger E2	1	**−20.59**, −7.33	−27.94, −9.95	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
IBIT	ledger E2	2	**−3.83**, −2.11	−7.92, −4.73	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
IWM	ledger E2	19	**−2.76**, −6.13	−4.12, −8.60	36.8 / 36.8	[−6.46, +17.44] / [−8.20, +13.22]
META	ledger E2	17	**−3.59**, −9.10	−4.40, −10.79	35.3 / 35.3	n/a (< 60 sessions) / n/a (< 60 sessions)
MSFT	ledger E2	1	**+9.84**, +38.68	+9.51, +37.38	100.0 / 100.0	n/a (< 60 sessions) / n/a (< 60 sessions)
MU	ledger E2	4	**+1.66**, +5.31	+0.31, +3.02	50.0 / 50.0	n/a (< 60 sessions) / n/a (< 60 sessions)
NDXP	ledger E2	16	**+16.86**, +148.48	+17.18, +149.80	56.2 / 56.2	n/a (< 60 sessions) / n/a (< 60 sessions)
QQQ	ledger E2	103	**−0.89**, −2.15	−2.11, −4.31	42.7 / 36.9	[−4.59, +4.08] / [−5.66, +2.78]
RUTW	ledger E2	23	**+7.66**, +48.84	+7.67, +48.88	56.5 / 56.5	[−1.21, +14.07] / [−1.20, +14.07]
SLV	ledger E2	3	**+14.88**, +5.38	+8.70, +2.76	100.0 / 66.7	n/a (< 60 sessions) / n/a (< 60 sessions)
SMCI	ledger E2	1	**−24.05**, −16.17	−27.94, −18.78	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
SMH	ledger E2	1	**−2.15**, −3.18	−3.92, −5.80	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
SPXW	ledger E2	206	**−1.80**, −6.18	+1.03, −3.72	38.3 / 40.3	[−6.28, +2.53] / [−1.75, +4.20]
SPY	ledger E2	65	**−2.98**, −9.32	−4.05, −11.57	30.8 / 29.2	[−4.45, −1.04] / [−5.86, −1.98]
TSLA	ledger E2	8	**+4.19**, +2.61	+2.11, +0.00	62.5 / 62.5	[−2.97, +4.16] / [−3.92, +1.72]
UBER	ledger E2	1	**−18.40**, −5.05	−27.94, −7.67	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
UCO	ledger E2	1	**−26.17**, −19.20	−27.94, −20.49	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
USO	ledger E2	1	**+28.43**, +40.60	+27.52, +39.30	100.0 / 100.0	n/a (< 60 sessions) / n/a (< 60 sessions)

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
all names	ledger E2	516	**+1.50**, +2.46	−0.13, +2.27	40.1 / 39.0	[−2.81, +3.24] / [−2.08, +3.55]

Random baseline (as the study computed it, at its own fees): time-matched random minutes, trail engine: MFE percentile 53 [47.5, 59.2] (Epoch 1) and 48 (Epoch 2).

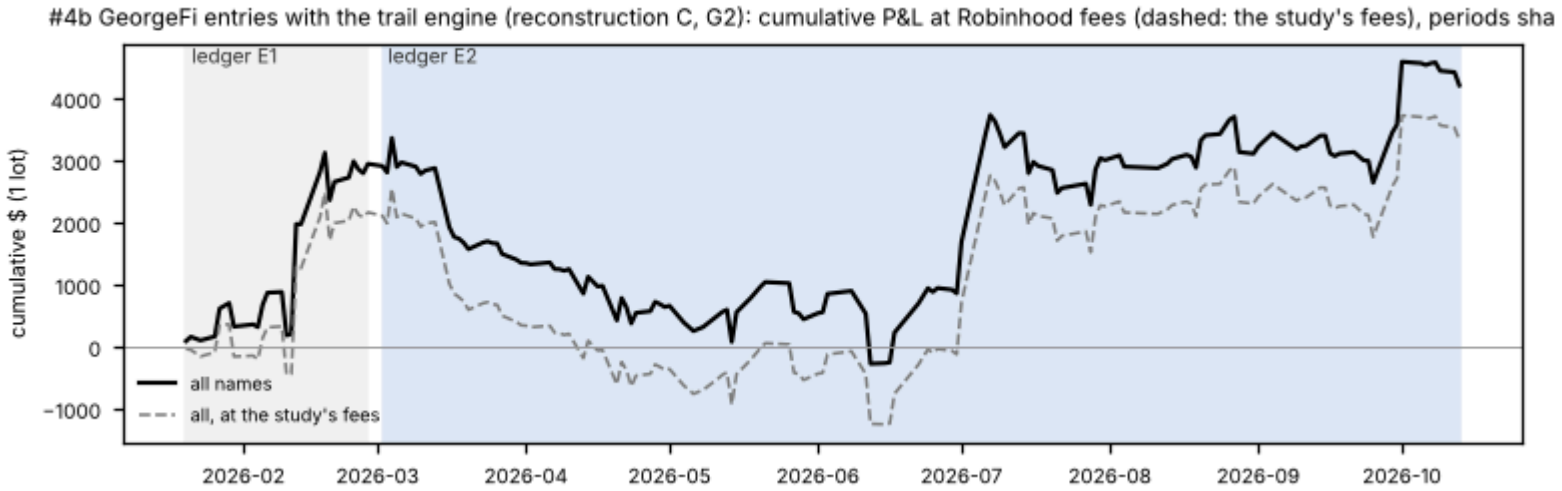
Exit reasons (share of trades):

- ledger E1: TRAIL 48.7%; BREAKEVEN 32.3%; STOP 16.4%; TIME_STOP 1.6%; CLOSE 1.1%
- ledger E2: TRAIL 39.0%; BREAKEVEN 37.0%; STOP 23.4%; TIME_STOP 0.6%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2026	AAPL	3	66.7	+10.15	+3.70	+11
2026	AMD	1	0.0	−25.96	−34.41	−34
2026	AMZN	6	16.7	−1.96	−4.59	−28
2026	AVGO	5	40.0	−5.70	−0.04	−0
2026	BAC	1	0.0	−11.81	−1.92	−2
2026	COIN	6	50.0	−0.42	−11.08	−67
2026	CRCL	12	0.0	−27.18	−46.27	−555
2026	CRM	1	100.0	+69.05	+54.79	+55
2026	CRML	14	85.7	+19.22	+1.27	+18
2026	CVNA	1	0.0	−27.06	−199.24	−199
2026	GLD	17	52.9	+68.24	+8.19	+139
2026	GOOGL	13	23.1	−7.91	−8.09	−105
2026	HOOD	4	50.0	−2.93	−6.42	−26
2026	IBIT	2	0.0	−3.83	−2.11	−4
2026	IWM	84	60.7	+11.77	+24.84	+2,087
2026	META	17	35.3	−3.59	−9.10	−155
2026	MSFT	1	100.0	+9.84	+38.68	+39
2026	MU	4	50.0	+1.66	+5.31	+21
2026	NDXP	16	56.2	+16.86	+148.48	+2,376
2026	NVDA	14	57.1	+16.32	+13.58	+190
2026	OKLO	1	0.0	−24.33	−44.15	−44
2026	QQQ	166	44.0	+0.16	−1.10	−183
2026	RDDT	2	50.0	+17.46	+28.01	+56
2026	RKLB	28	64.3	+14.46	+10.93	+306
2026	RUTW	31	64.5	+7.44	+40.87	+1,267
2026	SLV	6	50.0	+5.85	−0.14	−1
2026	SMCI	1	0.0	−24.05	−16.17	−16
2026	SMH	1	0.0	−2.15	−3.18	−3
2026	SPXW	265	41.9	−2.19	−2.98	−789
2026	SPY	154	40.9	+0.44	−1.13	−174

year	name	trades	win %	mean %	mean $	total $
2026	TSLA	8	62.5	+4.19	+2.61	+21
2026	UBER	1	0.0	−18.40	−5.05	−5
2026	UCO	1	0.0	−26.17	−19.20	−19
2026	USO	1	100.0	+28.43	+40.60	+41
2026	all names	888	45.8	+2.93	+4.75	+4,216



Amendments, deviations and integrity checks. —

What it showed. The entries are as good as random minutes once the exit is fixed.

#5 Deep research round 1: flow rule F1 on SPY (H2 family)

Folder: `research/flow_predictors/ (@7dfea57); report "Finding a 0DTE options edge"`. **Commits:** — / — / 96a2c09…fb4205f, 5208c8f. **Dates:** 2026-10-13.

Hypothesis. Options-flow imbalances (net premium, net delta, market tide, dark pool) lead the underlying.

Rule. F1: the flow predictor's signal minute on SPY → buy the 0DTE contract in its direction, trail engine (as #4b).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Spread model of #2 (1% half-spread + $0.70 a side), entry at the VWAP of the first option bar after the signal.

Data. SPY, Unusual Whales flow and bars 2025-07-01 → 2026-10-09; F1 = 15-minute net premium against its trailing-20-session 20th/80th percentiles, every 5 minutes 09:45–15:00, at most 2 a day.

Periods and holdouts. One span (descriptive test 5).

Pass rule (as pre-registered). Multiple-testing correction across 62 cells (exploratory).

Verdict. Rejected: 0 of 62 cells survive correction; forward ICs ≤ 0.02 (registry H2). The other round-1 notes (GEX regime and exit grid of 576 configurations, short-vol session T1–T5) kept results by configuration only; their verdicts: H4 rejected, H6 rejected, H7 0 of 576 exit configurations positive (White p = 1), H8 every option expression negative.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	2025-07→2026-10	638	319 / 331	35.9	+14.8, +23.24	−12.2, −17.41	−2.1, −2.60	**−2.48**, −2.82	[−3.55, −1.17]; [−4.47, −0.72]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	2025-07→2026-10	+177.21, +85.2%, 2025-12-18, SPY251218C00676000	−82.49, −26.8%, 2026-07-29, SPY260729C00736000	0.75	1,805	331	total ≤ 0 (best 5 days +647; without them −2,445)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (638 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPY	2025-07→2026-10	638	**−2.48**, −2.82	−3.45, −4.12	35.9 / 35.9	[−3.55, −1.17] / [−4.53, −2.12]

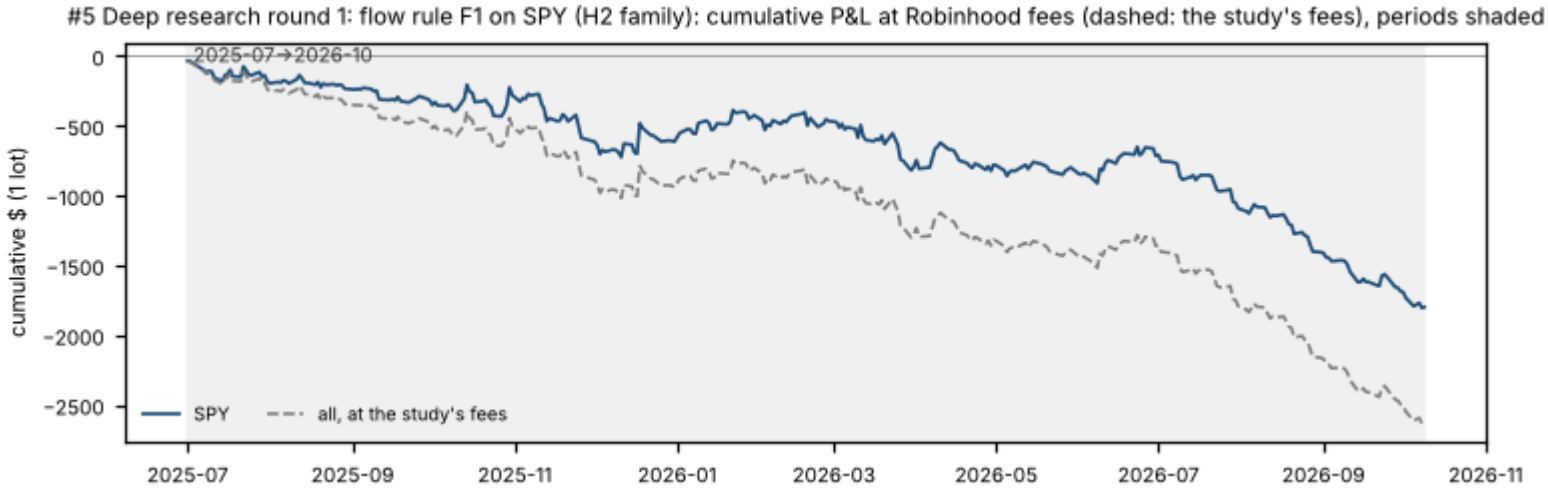
Random baseline (as the study computed it, at its own fees): random minutes matched per day, random side, same exits and costs (574 trades): −$3.34 a trade, −2.26% (the rule −$4.12, −2.38%).

Exit reasons (share of trades):

- 2025-07→2026-10: BREAKEVEN 37.8%; TRAIL 35.9%; STOP 26.0%; TIME_STOP 0.3%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	SPY	250	36.4	−2.93	−2.45	−614
2026	SPY	388	35.6	−2.18	−3.05	−1,185



Amendments, deviations and integrity checks. —

What it showed. The one flow rule with a trade list loses after costs.

#6a T-A: the 13:00 short ATM 0DTE straddle, SPY/QQQ/IWM, bought back 15:55 (H9)

Folder: `research/thetadata/straddle_nbbo/ (@7dfea57)`. **Commits:** — / gate v2 924aee9 / ebde6ae. **Dates:** 2026-10-13.

Hypothesis. Selling the 13:00 at-the-money straddle harvests intraday variance premium after measured costs.

Rule. Sell the same-day ATM straddle at the 13:00 bids, buy it back at the 15:55 asks.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Measured ThetaData NBBO both ways (the round trip $12.93 a SPY/QQQ straddle); return as % of the credit received.

Data. SPY, QQQ, IWM 0DTE, 339 sessions each, 2025-06-02 → 2026-10-09.

Periods and holdouts. One span; no holdout ("previously analysed").

Pass rule (as pre-registered). Gate v2: a positive lower bound at measured NBBO (T-A).

Verdict. FAIL: +$8.60 a straddle [−6.07, +22.45] SPY/QQQ (n 678); IWM −$9.50 [−18.69, −0.44].

Metrics at Robinhood fees (return = % of the credit received; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2025-06→2026-10	339	339 / 352	58.4	+40.1, +43.53	−71.3, −77.72	+17.3, +14.80	**−6.22**, −6.90	[−13.67, −1.11]; [−17.06, +0.58]
QQQ	2025-06→2026-10	339	339 / 352	65.8	+39.5, +90.29	−54.2, −133.74	+20.7, +47.80	**+7.41**, +13.63	[+2.39, +12.20]; [−0.70, +26.27]
SPY	2025-06→2026-10	339	339 / 352	65.2	+40.0, +77.36	−57.3, −119.68	+20.8, +38.80	**+6.12**, +8.77	[+1.16, +10.27]; [−4.18, +19.47]
all names	2025-06→2026-10	1,017	339 / 352	63.1	+39.8, +71.42	−61.6, −108.25	+20.0, +27.80	**+2.44**, +5.17	[−2.53, +6.28]; [−6.16, +14.35]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	2025-06→2026-10	+330.80, +81.5%, 2026-07-29, IWM 13:00 short straddle 289	−403.20, −238.6%, 2026-06-11, IWM 13:00 short straddle 285	0.79	3,358	346	total ≤ 0 (best 5 days +1,038; without them −3,378)
QQQ	2025-06→2026-10	+345.80, +77.2%, 2026-06-12, QQQ 13:00 short straddle 722	−1,184.20, −224.7%, 2026-06-11, QQQ 13:00 short straddle 700	1.30	2,434	125	34%
SPY	2025-06→2026-10	+321.80, +67.3%, 2025-09-17, SPY 13:00 short straddle 658	−764.20, −196.5%, 2026-06-11, SPY 13:00 short straddle 727	1.21	2,233	148	42%
all names	2025-06→2026-10	+345.80, +77.2%, 2026-06-12, QQQ 13:00 short straddle 722	−1,184.20, −224.7%, 2026-06-11, QQQ 13:00 short straddle 700	1.13	6,185	148	68%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.80 a trade) and Robinhood's charged instead (mean $0.20). The study's fees are given beside them below. Re-pricing: exact (1,017 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	2025-06→2026-10	339	**−6.22**, −6.90	−8.94, −9.50	58.4 / 57.2	[−13.67, −1.11] / [−16.30, −3.87]
QQQ	2025-06→2026-10	339	**+7.41**, +13.63	+6.15, +11.03	65.8 / 65.2	[+2.39, +12.20] / [+1.12, +10.94]
SPY	2025-06→2026-10	339	**+6.12**, +8.77	+4.64, +6.17	65.2 / 64.3	[+1.16, +10.27] / [−0.30, +8.72]
all names	2025-06→2026-10	1,017	**+2.44**, +5.17	+0.62, +2.57	63.1 / 62.2	[−2.53, +6.28] / [−4.33, +4.41]

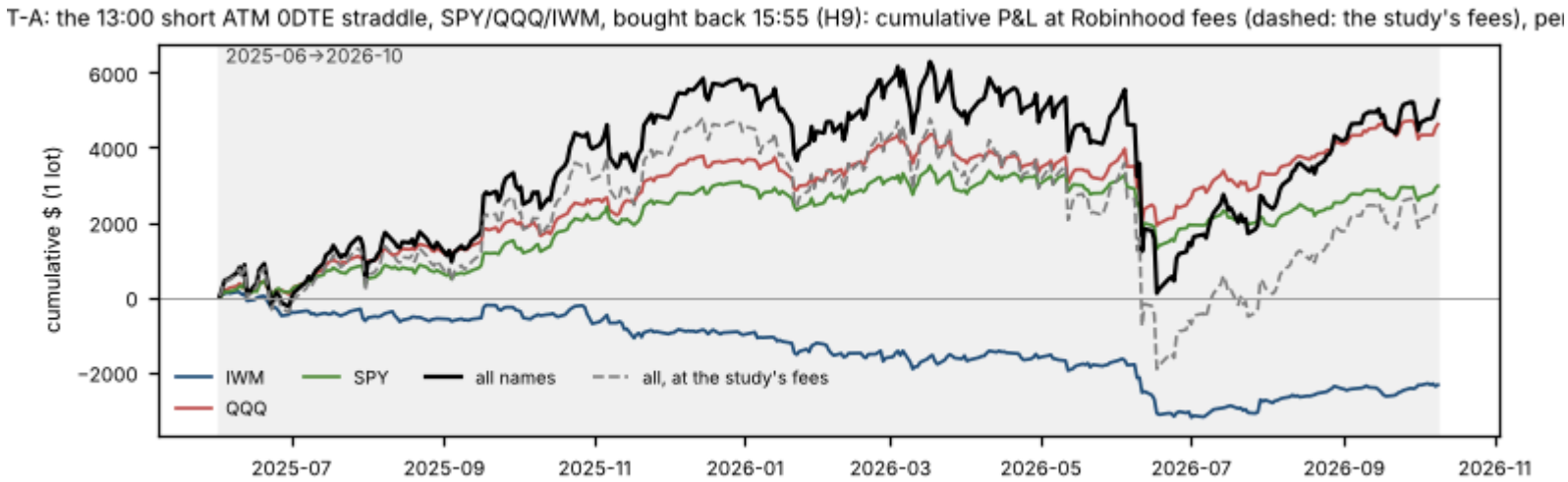
Random baseline (as the study computed it, at its own fees): random entry times: +$13.10 [−2.42, +29.14] against a $12.93 round trip (registry H9).

Exit reasons (share of trades):

- 2025-06→2026-10: bought back 15:55 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	IWM	145	57.9	−7.29	−6.93	−1,005
2026	IWM	194	58.8	−5.41	−6.88	−1,335
2025	QQQ	145	69.7	+12.48	+24.67	+3,577
2026	QQQ	194	62.9	+3.62	+5.38	+1,044
2025	SPY	145	68.3	+11.10	+20.50	+2,972
2026	SPY	194	62.9	+2.40	+0.00	+1
2025	all names	435	65.3	+5.43	+12.74	+5,543
2026	all names	582	61.5	+0.20	−0.50	−290



Amendments, deviations and integrity checks. The modelled-spread result (+$8.36) was optimistic by about $5 a straddle.

What it showed. The 13:00 short straddle is positive on SPY/QQQ with an interval through zero and negative on IWM.

#6b T-B: gamma-conditioned last-30-minute momentum in SPY shares (H10)

Folder: `research/thetadata/index_tests/tb_momentum/ (@7dfea57)`. **Commits:** — / — / b209f7b. **Dates:** 2026-10-13.

Hypothesis. On negative-gamma days the last 30 minutes continue the session's move.

Rule. On days with dealer gamma below the flip (prior-day UW levels): 100 SPY in the direction of the open→15:30 move, 15:30 → official close.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Net of the study's cost per day (0.4 bp).

Data. SPY, 299 negative-gamma days 2024-05-14 → 2026-10-07.

Periods and holdouts. One span.

Pass rule (as pre-registered). T-B: positive mean with a positive lower bound.

Verdict. FAIL (wrong sign): −3.07 bp a day [−5.73, −0.52], Sharpe −2.15.

Metrics at Robinhood fees (return = % of notional ($ = per 100 sh); win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	negative-gamma days 2024-05→2026-10	299	299 / 620	46.5	+0.128, +77.62	−0.165, −101.78	−0.012, −7.22	**−0.029**, −18.38	[−0.054, −0.001]; [−32.83, −2.86]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	negative-gamma days 2024-05→2026-10	+523.44, +1.0%, 2025-04-04, 100 SPY short 15:30→close	−698.04, −1.2%, 2025-03-04, 100 SPY long 15:30→close	0.66	5,964	620	total ≤ 0 (best 5 days +2,021; without them −7,517)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.55 a trade) and Robinhood's charged instead (mean $1.34). The study's fees are given beside them below. Re-pricing: Robinhood fees inferred from the rule: the study's 0.2 bp a side is a generic cost, taken as fees and replaced (299 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPY	negative-gamma days 2024-05→2026-10	299	**−0.029**, −18.38	−0.031, −19.59	46.5 / 46.2	[−0.054, −0.001] / [−0.056, −0.003]

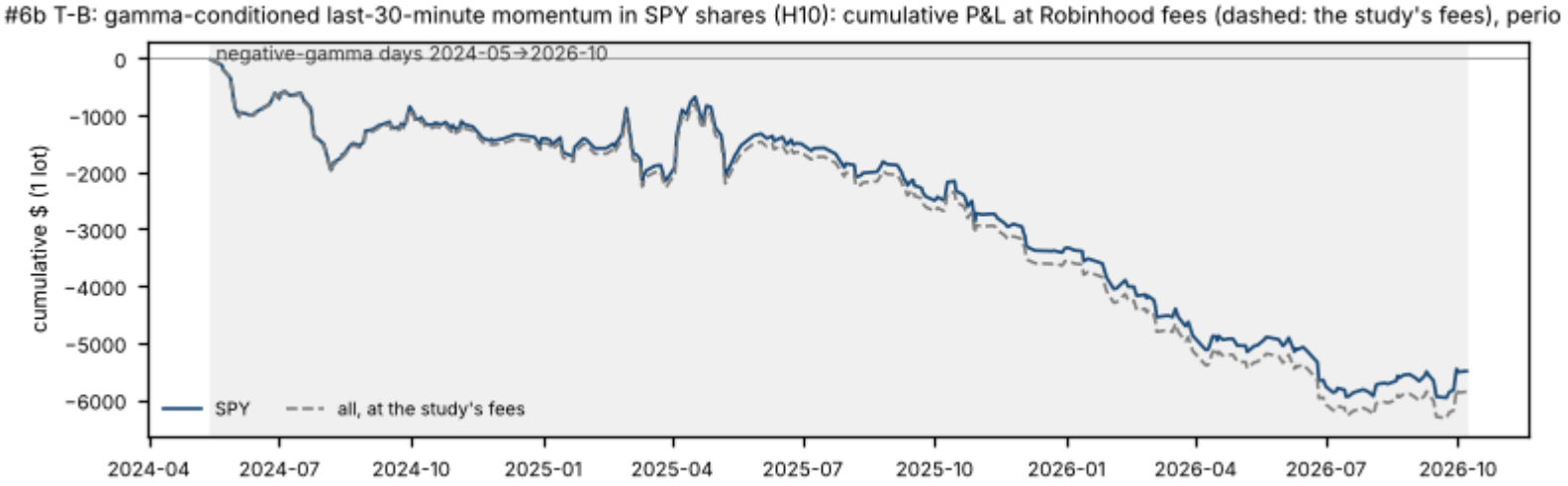
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- negative-gamma days 2024-05→2026-10: official close 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	SPY	77	50.6	−0.03	−18.29	−1,408
2025	SPY	125	44.0	−0.02	−15.41	−1,926
2026	SPY	97	46.4	−0.03	−22.28	−2,162



Amendments, deviations and integrity checks. The two gamma proxies agree on the regime 52% of days.

What it showed. The effect has the wrong sign.

#6c T-C: the 13:00 SPXW short straddle bought back 15:55 (H11)

Folder: `research/thetadata/index_tests/tc_spxw_straddle/ (@7dfea57)`. **Commits:** — / — / b209f7b. **Dates:** 2026-10-13.

Hypothesis. As T-A on SPXW (cash-settled).

Rule. Sell the SPXW ATM straddle at 13:00 (bids), buy back at 15:55 (asks).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Measured NBBO; return as % of the credit.

Data. SPXW 0DTE, 339 sessions 2025-06-02 → 2026-10-09.

Periods and holdouts. One span.

Pass rule (as pre-registered). T-C: positive lower bound.

Verdict. FAIL: +$72.4 a day [−75.8, +214.2], max drawdown −$21,180, worst day −$8,499.

Metrics at Robinhood fees (return = % of the credit received; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	2025-06→2026-10	339	339 / 352	63.4	+46.5, +832.04	−66.7, −1,246.05	+26.2, +395.48	**+5.07**, +71.91	[−0.09, +8.97]; [−56.51, +173.38]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	2025-06→2026-10	+3,370.48, +71.7%, 2025-09-17, SPXW 13:00 short straddle 6590.0	−8,499.52, −230.3%, 2026-06-11, SPXW 13:00 short straddle 7275.0	1.16	21,186	91	54%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $4.00 a trade) and Robinhood's charged instead (mean $4.52). The study's fees are given beside them below. Re-pricing: Robinhood fees inferred from the rule: the 15:55 buy-back legs' price tier (only the straddle's debit is kept) (339 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPXW	2025-06→2026-10	339	**+5.07**, +71.91	+5.11, +72.43	63.4 / 63.4	[−0.09, +8.97] / [−0.06, +9.01]

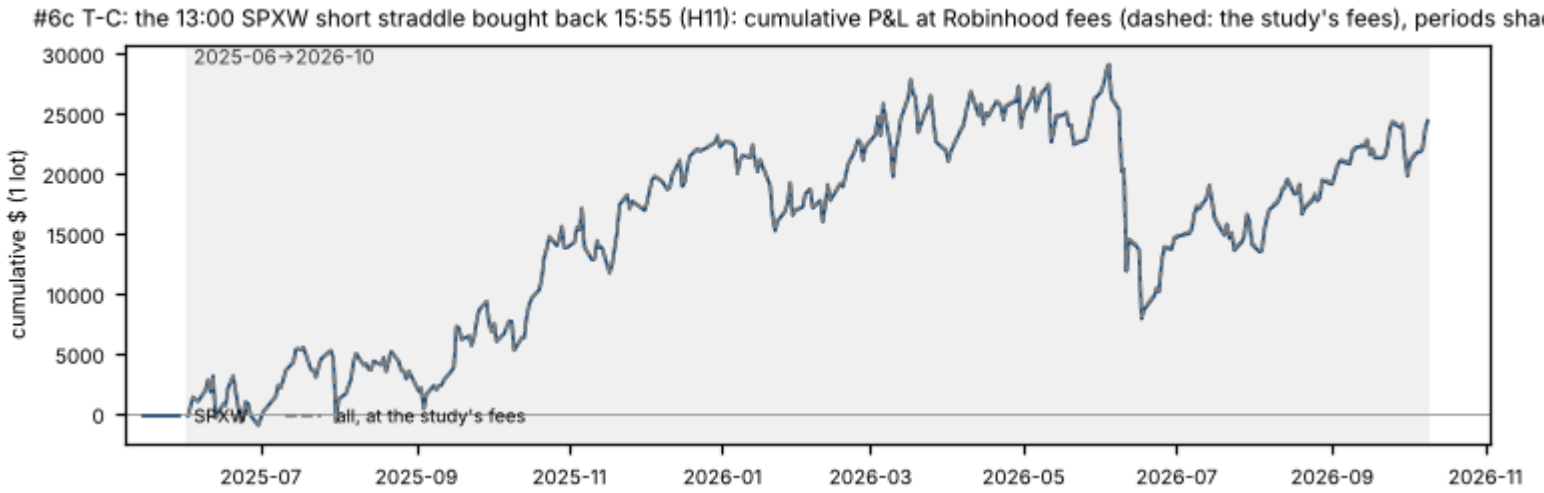
Random baseline (as the study computed it, at its own fees): random entry: +$40.6 [−73.4, +164.2] against a $68.9 cost.

Exit reasons (share of trades):

- 2025-06→2026-10: bought back 15:55 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	SPXW	145	63.4	+7.60	+153.24	+22,219
2026	SPXW	194	63.4	+3.19	+11.12	+2,158



Amendments, deviations and integrity checks. —

What it showed. Positive mean, interval through zero, large tail days.

#6d T-E: the 13:00 SPXW ATM straddle held to PM settlement, ex FOMC/CPI/NFP (H13)

Folder: `research/thetadata/hwr_grid/spxw_v2/ (@7dfea57)`. **Commits:** protocol sha256 in protocol_sha256.txt / — / 01ee3f3, 5f253c6 (full chains). **Dates:** 2026-10-14.

Hypothesis. Holding the short straddle to cash settlement adds the last hour's decay.

Rule. Sell the SPXW ATM straddle at 13:00, hold to the PM settlement (Cboe close), skip FOMC/CPI/NFP days.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Measured NBBO in; settlement value out.

Data. SPXW 0DTE 2022-05-13 → 2026-10-09 on complete chains.

Periods and holdouts. TRAIN 2022-05→2024 / TEST 2025→2026-10 (TEST read).

Pass rule (as pre-registered). T-E gate on TEST.

Verdict. FAIL: +$26.94 a day [−136.73, +170.75]; 2025-04-09 alone −$33,667 a lot; trails random entry in both windows.

Metrics at Robinhood fees (return = % of the credit received; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	TRAIN	575	575 / 683	60.2	+54.3, +881.09	−81.0, −1,229.08	+22.1, +353.74	**+0.40**, +40.69	[−6.13, +6.47]; [−56.80, +135.66]
SPXW	TEST	388	388 / 457	67.0	+52.8, +997.58	−80.7, −1,767.78	+28.0, +486.74	**+8.74**, +85.30	[+1.85, +14.75]; [−115.97, +252.47]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	TRAIN	+2,926.74, +96.3%, 2022-05-13, SPXW 13:00 4025P/4025C	−9,211.26, −860.9%, 2024-04-04, SPXW 13:00 5250P/5250C	1.08	22,502	526	58%
SPXW	TEST	+4,099.74, +96.5%, 2025-03-06, SPXW 13:00 5740P/5740C	−34,022.26, −392.4%, 2025-04-09, SPXW 13:00 5030P/5030C	1.15	38,495	175	53%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.00 a trade) and Robinhood's charged instead (mean $2.26). The study's fees are given beside them below. Re-pricing: Robinhood fees inferred from the rule: each leg's price tier (only the credit is kept; $1.00 and up unless the credit is under $1.00) (963 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPXW	TRAIN	575	**+0.40**, +40.69	+0.42, +40.95	60.2 / 60.2	[−6.13, +6.47] / [−6.12, +6.49]
SPXW	TEST	388	**+8.74**, +85.30	+8.75, +85.56	67.0 / 67.0	[+1.85, +14.75] / [+1.86, +14.77]

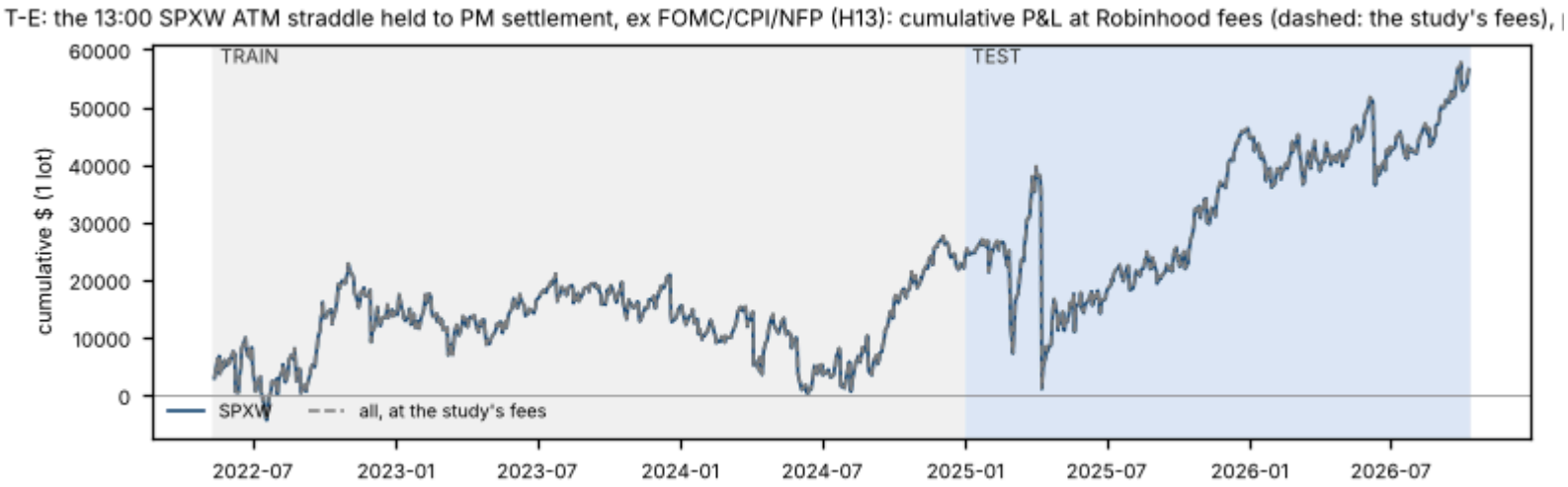
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- TRAIN: settle 100.0%
- TEST: settle 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	140	61.4	+4.58	+102.76	+14,387
2023	SPXW	217	58.5	−1.50	+1.62	+352
2024	SPXW	218	61.0	−0.40	+39.72	+8,659
2025	SPXW	218	69.3	+10.31	+97.72	+21,302
2026	SPXW	170	64.1	+6.72	+69.37	+11,793



Amendments, deviations and integrity checks. Re-run on complete chains (5f253c6, 8fb5bad): no TEST figure changed.

What it showed. One day can erase years; no edge.

#6e H15 carry lane: the 14:05 SPXW short strangle at ±0.5 expected move, held to settlement

Folder: `research/thetadata/hwr_grid/spxw_v2/ (@7dfea57); report "High win rate 0DTE lanes"`. **Commits:** protocol_sha256.txt / — / 01ee3f3, 97e685a, 6691368. **Dates:** 2026-10-14.

Hypothesis. A 70%+ win-rate short strangle with a positive cost-realistic lower bound exists.

Rule. 14:05 sell the SPXW strangle at ±0.5 × the 14:05 expected move, held to cash settlement, skip FOMC and month-end (selection S2 on TRAIN).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Measured NBBO in; settlement out.

Data. SPXW 0DTE 2022-05-13 → 2026-10-09.

Periods and holdouts. TRAIN (selection) / TEST (read). The deciding test T-G (forward paper ≥ 150 sessions) was never started.

Pass rule (as pre-registered). Positive TEST lower bound after the two selections; then T-G forward.

Verdict. Open, NOT promotable: TEST +$151.17 a lot [+42.15, +251.73], win 70.8%; the pre-registered reselection on complete TRAIN picks a different S2 that fails (+$52.54 [−22.24, +122.49]); ±10% shock ≈ −$65k a lot.

Metrics at Robinhood fees (return = % of the credit received; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	TRAIN	606	606 / 682	72.1	+83.8, +528.16	−166.7, −922.10	+97.3, +347.74	**+13.98**, +123.71	[+1.57, +24.80]; [+60.00, +188.18]
SPXW	TEST	407	407 / 457	70.8	+86.4, +646.09	−147.7, −1,047.53	+95.8, +352.74	**+17.98**, +150.91	[+3.49, +29.85]; [+33.90, +263.67]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	TRAIN	+1,827.74, +99.9%, 2024-08-05, SPXW 14:05 5175P/5215C	−7,146.26, −1,125.4%, 2024-04-04, SPXW 14:05 5225P/5240C	1.48	10,109	91	10%
SPXW	TEST	+4,577.74, +100.0%, 2025-04-10, SPXW 14:05 5245P/5335C	−7,131.26, −735.2%, 2026-03-09, SPXW 14:05 6690P/6715C	1.49	13,025	157	28%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.00 a trade) and Robinhood's charged instead (mean $2.26). The study's fees are given beside them below. Re-pricing: Robinhood fees inferred from the rule: each leg's price tier (only the credit is kept; $1.00 and up unless the credit is under $1.00) (1,013 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPXW	TRAIN	606	**+13.98**, +123.71	+14.03, +123.98	72.1 / 72.1	[+1.57, +24.80] / [+1.62, +24.85]
SPXW	TEST	407	**+17.98**, +150.91	+18.02, +151.17	70.8 / 70.8	[+3.49, +29.85] / [+3.54, +29.89]

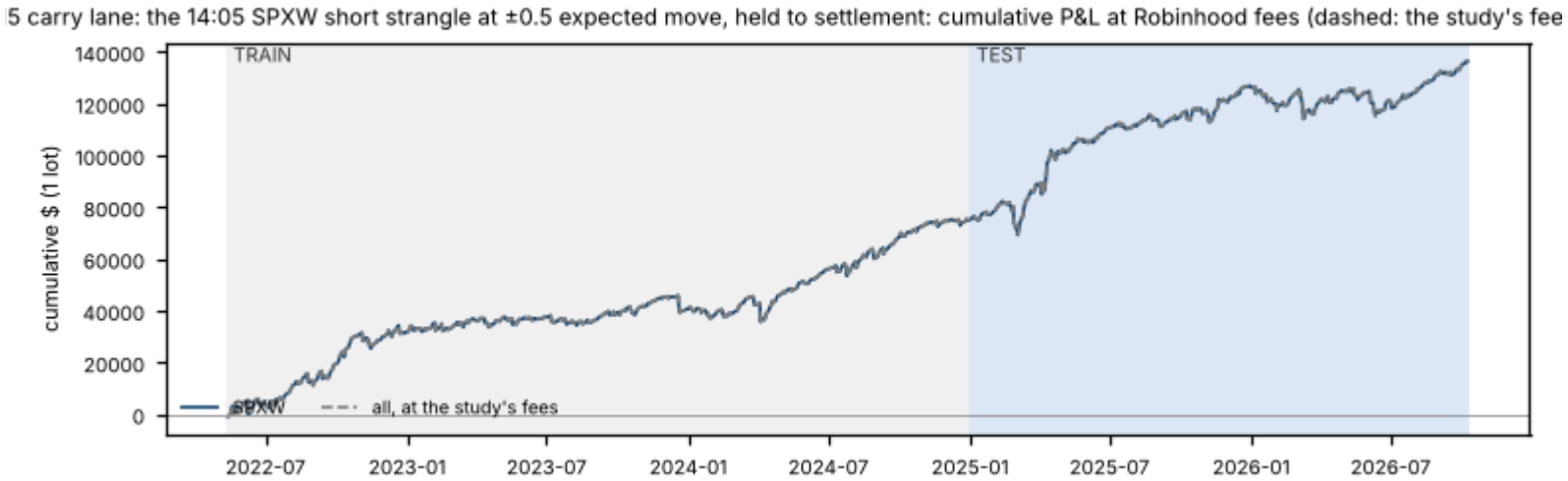
Random baseline (as the study computed it, at its own fees): matched random: +$22.60 [−147.61, +261.12].

Exit reasons (share of trades):

- TRAIN: settle 100.0%
- TEST: settle 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	146	73.3	+20.70	+217.14	+31,703
2023	SPXW	228	67.1	+5.37	+38.58	+8,797
2024	SPXW	232	76.3	+18.20	+148.58	+34,472
2025	SPXW	228	71.9	+22.48	+228.34	+52,062
2026	SPXW	179	69.3	+12.24	+52.28	+9,357



Amendments, deviations and integrity checks. All historical data has been used for selection; no further historical re-selection allowed (registry).

What it showed. Positive on TEST, but the reselection fails and the tail risk is unbounded.

#7a S01 Quick Scalps (owner baseline) on index 0DTE

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** protocol 113cad7 / — / 5e90201. **Dates:** 2026-10-14.

Hypothesis. The owner's Quick Scalps signal (RSI14 cross 55/45 + EMA9 lead, 5m EMA20 + VWAP confirm, 15m EMA alignment) predicts 0DTE moves.

Rule. Signal → buy the 0DTE contract in a delta band at the ask; exits: stop, two targets, VWAP loss, invalidation, time.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM 0DTE 2022-11-11 → 2026-10-08.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). Positive net expectancy with a positive lower bound (protocol).

Verdict. REJECTED: −$4.19 a contract [−5.68, −2.81].

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2022-11→2026-10	685	274 / 638	35.0	+37.4, +20.71	−26.2, −14.50	−25.2, −9.10	**−3.89**, −2.16	[−6.28, −1.52]; [−3.61, −0.83]
QQQ	2022-11→2026-10	1,075	421 / 1,009	36.8	+36.1, +39.24	−25.6, −26.85	−22.2, −14.10	**−2.85**, −2.50	[−4.78, −1.21]; [−4.52, −0.66]
SPY	2022-11→2026-10	1,126	444 / 1,009	35.5	+36.1, +32.76	−24.0, −23.77	−17.9, −12.10	**−2.63**, −3.69	[−4.52, −0.90]; [−6.03, −1.75]
all names	2022-11→2026-10	2,886	614 / 1,009	35.9	+36.4, +32.44	−25.1, −22.67	−21.6, −11.10	**−3.01**, −2.89	[−4.41, −1.79]; [−4.49, −1.58]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	2022-11→2026-10	+72.40, +51.0%, 2026-06-12, IWM 2026-06-12 292.0CALL	−59.10, −29.6%, 2025-04-07, IWM 2025-04-07 178.0PUT	0.77	1,610	571	total ≤ 0 (best 5 days +396; without them −1,879)
QQQ	2022-11→2026-10	+346.89, +112.3%, 2026-06-09, QQQ 2026-06-09 692.0PUT	−156.10, −34.2%, 2025-04-07, QQQ 2025-04-07 417.0PUT	0.85	3,381	1,007	total ≤ 0 (best 5 days +1,405; without them −4,095)
SPY	2022-11→2026-10	+145.90, +51.7%, 2025-03-11, SPY 2025-03-11 557.0PUT	−114.10, −24.8%, 2025-04-09, SPY 2025-04-09 495.0PUT	0.76	4,510	966	total ≤ 0 (best 5 days +785; without them −4,943)
all names	2022-11→2026-10	+346.89, +112.3%, 2026-06-09, QQQ 2026-06-09 692.0PUT	−156.10, −34.2%, 2025-04-07, QQQ 2025-04-07 417.0PUT	0.80	8,689	991	total ≤ 0 (best 5 days +1,575; without them −9,906)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (2,886 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	2022-11→2026-10	685	**−3.89**, −2.16	−6.74, −3.46	35.0 / 33.3	[−6.28, −1.52] / [−9.19, −4.39]
QQQ	2022-11→2026-10	1,075	**−2.85**, −2.50	−4.38, −3.80	36.8 / 35.0	[−4.78, −1.21] / [−6.34, −2.71]
SPY	2022-11→2026-10	1,126	**−2.63**, −3.69	−4.32, −4.99	35.5 / 34.3	[−4.52, −0.90] / [−6.19, −2.60]
all names	2022-11→2026-10	2,886	**−3.01**, −2.89	−4.92, −4.19	35.9 / 34.3	[−4.41, −1.79] / [−6.31, −3.69]

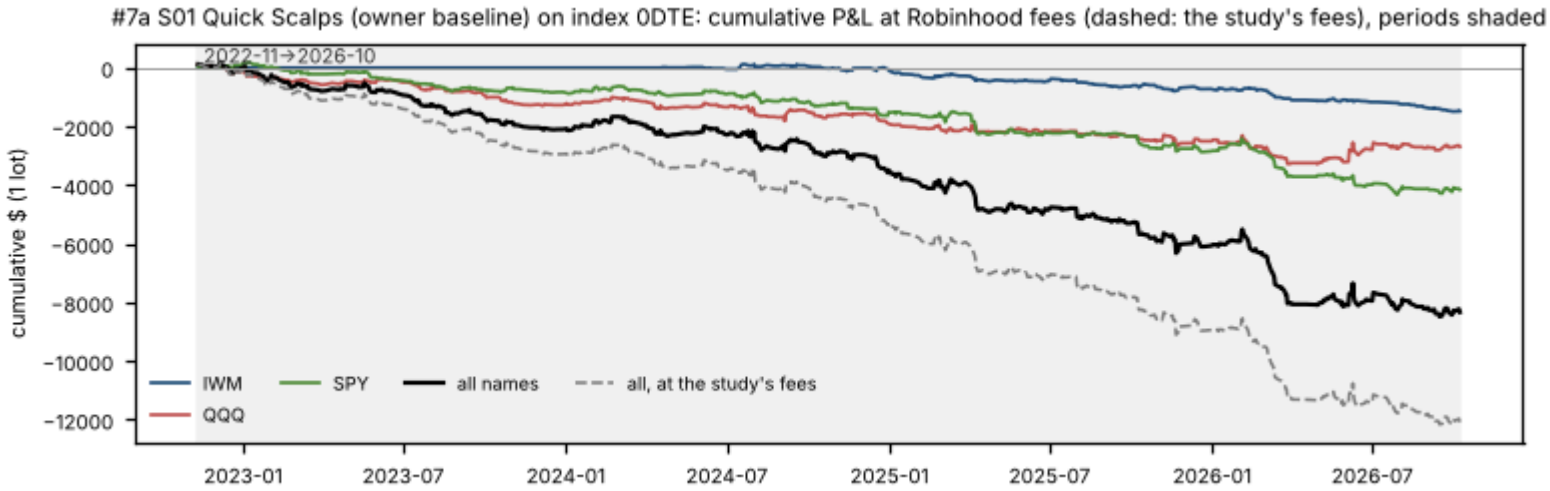
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- 2022-11→2026-10: stop 51.9%; target_2 26.3%; vwap_loss 15.3%; time 5.4%; invalidation 1.1%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	157	36.9	−2.73	−0.47	−73
2025	IWM	311	35.7	−2.94	−2.08	−647
2026	IWM	217	32.7	−6.08	−3.51	−762
2022	QQQ	34	35.3	−4.44	−3.66	−124
2023	QQQ	293	33.4	−5.58	−3.91	−1,146
2024	QQQ	287	38.0	−2.42	−2.20	−631
2025	QQQ	241	39.0	−1.29	−2.50	−602
2026	QQQ	220	37.7	−1.21	−0.85	−187
2022	SPY	40	45.0	+3.25	+2.29	+91
2023	SPY	285	32.6	−4.51	−3.22	−919
2024	SPY	281	35.9	−1.98	−2.44	−685
2025	SPY	282	37.2	−2.44	−4.72	−1,331
2026	SPY	238	34.9	−2.39	−5.52	−1,315
2022	all names	74	40.5	−0.28	−0.45	−33
2023	all names	578	33.0	−5.05	−3.57	−2,065
2024	all names	725	37.0	−2.32	−1.92	−1,389
2025	all names	834	37.2	−2.29	−3.09	−2,580
2026	all names	675	35.1	−3.20	−3.35	−2,264



Amendments, deviations and integrity checks. Status ≠ filled rows (no chain, in position, no expiry) are not trades.

What it showed. The baseline loses $4 a contract after costs.

#7b S02 Quick Scalps: 41-variant grid, walk-forward OOS (lower-bound selection)

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** 113cad7 / — / 6bf5934. **Dates:** 2026-10-14.

Hypothesis. A single-factor variant of the baseline holds out of sample.

Rule. Each year, the variant with the best lower bound on the years before is traded the next year (walk-forward).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM 0DTE, OOS 2024-01 → 2026-10.

Periods and holdouts. Walk-forward OOS years 2024–2026.

Pass rule (as pre-registered). Positive OOS lower bound, better than matched random.

Verdict. Nothing promoted: OOS −$3.81 [−5.23, −2.35]; no variant positive or better than random.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	walk-forward OOS 2024-01→2026-10	935	314 / 638	34.5	+35.9, +18.92	−26.1, −13.42	−25.3, −7.10	**−4.67**, −2.25	[−6.71, −2.72]; [−3.43, −1.16]
QQQ	walk-forward OOS 2024-01→2026-10	1,117	352 / 715	38.0	+35.7, +39.64	−25.4, −26.27	−21.3, −10.10	**−2.22**, −1.25	[−4.12, −0.57]; [−3.55, +1.03]
SPY	walk-forward OOS 2024-01→2026-10	1,148	373 / 715	36.4	+35.6, +31.63	−24.5, −24.30	−18.2, −10.10	**−2.65**, −3.93	[−4.92, −0.91]; [−6.75, −1.69]
all names	walk-forward OOS 2024-01→2026-10	3,200	488 / 715	36.4	+35.7, +31.02	−25.3, −21.70	−22.0, −9.10	**−3.09**, −2.51	[−4.63, −1.82]; [−4.16, −1.16]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	walk-forward OOS 2024-01→2026-10	+109.90, +199.8%, 2025-02-07, IWM 227.5PUT no_15m	−119.10, −70.5%, 2025-09-17, IWM 239.0PUT no_15m	0.74	2,158	571	total ≤ 0 (best 5 days +561; without them −2,661)
QQQ	walk-forward OOS 2024-01→2026-10	+346.89, +112.3%, 2026-06-09, QQQ 692.0PUT no_15m	−156.10, −34.2%, 2025-04-07, QQQ 417.0PUT no_15m	0.92	1,615	668	total ≤ 0 (best 5 days +1,714; without them −3,116)
SPY	walk-forward OOS 2024-01→2026-10	+173.39, +40.3%, 2025-04-10, SPY 516.0PUT no_15m	−162.11, −23.5%, 2025-04-07, SPY 496.0PUT no_15m	0.75	4,899	692	total ≤ 0 (best 5 days +956; without them −5,472)
all names	walk-forward OOS 2024-01→2026-10	+346.89, +112.3%, 2026-06-09, QQQ 692.0PUT no_15m	−162.11, −23.5%, 2025-04-07, SPY 496.0PUT no_15m	0.82	8,361	668	total ≤ 0 (best 5 days +2,083; without them −10,101)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (3,200 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	walk-forward OOS 2024-01→2026-10	935	**−4.67**, −2.25	−7.85, −3.55	34.5 / 32.2	[−6.71, −2.72] / [−10.10, −5.79]
QQQ	walk-forward OOS 2024-01→2026-10	1,117	**−2.22**, −1.25	−3.97, −2.55	38.0 / 36.6	[−4.12, −0.57] / [−5.94, −2.21]
SPY	walk-forward OOS 2024-01→2026-10	1,148	**−2.65**, −3.93	−4.55, −5.23	36.4 / 35.1	[−4.92, −0.91] / [−6.76, −2.77]
all names	walk-forward OOS 2024-01→2026-10	3,200	**−3.09**, −2.51	−5.31, −3.81	36.4 / 34.8	[−4.63, −1.82] / [−6.87, −3.98]

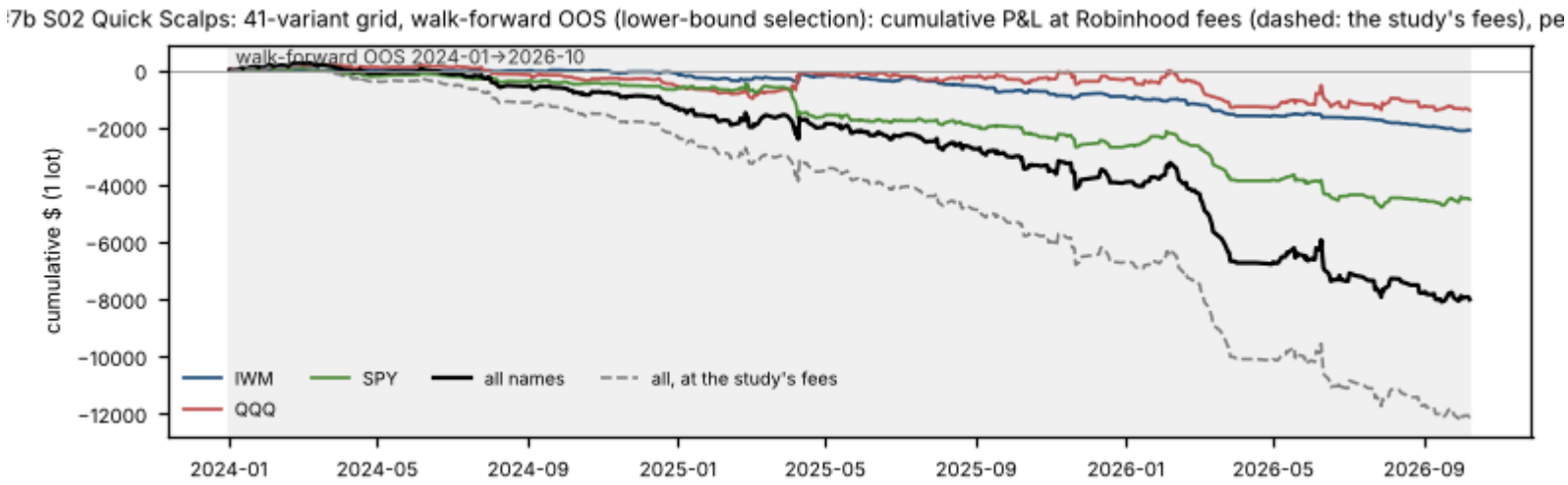
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- walk-forward OOS 2024-01→2026-10: stop 52.8%; target_2 24.4%; vwap_loss 14.3%; ema_target 4.2%; time 3.6%; invalidation 0.8%; settlement 0.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	150	32.7	−5.10	−0.79	−118
2025	IWM	475	36.4	−3.21	−1.68	−796
2026	IWM	310	32.6	−6.70	−3.83	−1,186
2024	QQQ	317	34.7	−3.70	−1.73	−548
2025	QQQ	429	42.2	−0.46	+0.49	+211
2026	QQQ	371	35.8	−2.98	−2.87	−1,064
2024	SPY	300	33.0	−3.66	−2.06	−618
2025	SPY	475	39.6	−2.20	−4.32	−2,050
2026	SPY	373	35.1	−2.40	−4.96	−1,849
2024	all names	767	33.6	−3.96	−1.67	−1,284
2025	all names	1,379	39.3	−2.01	−1.91	−2,635
2026	all names	1,054	34.6	−3.87	−3.89	−4,099



Amendments, deviations and integrity checks. —

What it showed. Variant selection does not rescue the signal.

#7c S03 Quick Scalps with weekly expirations (1–7 days)

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** 113cad7 / — / 3c8d6cb. **Dates:** 2026-10-14.

Hypothesis. The signal works on weekly contracts.

Rule. Quick Scalps signal → nearest Friday contract, OI > 250.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM weekly chains, sampled sessions 2017-06 → 2026-10.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). As S01.

Verdict. Nothing promoted: +$1.27 a contract [−3.69, +6.25] (= random).

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2017-06→2026-10	239	141 / 2,394	41.4	+27.7, +40.24	−14.1, −22.66	−3.9, −6.10	**+3.21**, +3.39	[−0.34, +6.81]; [−1.55, +9.44]
QQQ	2017-06→2026-10	217	135 / 2,386	36.9	+23.7, +65.41	−15.5, −39.52	−6.6, −13.10	**−1.07**, −0.83	[−4.07, +2.56]; [−10.65, +10.52]
SPY	2017-06→2026-10	362	221 / 2,393	36.2	+29.2, +72.20	−15.9, −34.57	−5.9, −8.10	**+0.39**, +4.06	[−2.01, +2.90]; [−1.87, +11.12]
all names	2017-06→2026-10	818	376 / 2,399	37.9	+27.3, +60.24	−15.3, −32.62	−5.7, −8.10	**+0.83**, +2.57	[−0.91, +2.53]; [−1.49, +6.93]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	2017-06→2026-10	+168.89, +39.5%, 2025-04-08, IWM 2025-04-11 176.0PUT	−99.10, −21.8%, 2020-06-12, IWM 2020-06-19 134.5PUT	1.26	497	1,422	92%
QQQ	2017-06→2026-10	+416.87, +39.9%, 2025-04-08, QQQ 2025-04-11 423.0PUT	−167.10, −33.6%, 2026-02-05, QQQ 2026-02-06 594.0PUT	0.97	1,797	1,942	total ≤ 0 (best 5 days +1,661; without them −1,842)
SPY	2017-06→2026-10	+464.39, +120.3%, 2020-05-15, SPY 2020-05-22 286.0CALL	−226.11, −31.8%, 2020-03-10, SPY 2020-03-13 273.0PUT	1.18	854	739	121%
all names	2017-06→2026-10	+464.39, +120.3%, 2020-05-15, SPY 2020-05-22 286.0CALL	−226.11, −31.8%, 2020-03-10, SPY 2020-03-13 273.0PUT	1.13	902	726	118%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (818 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	2017-06→2026-10	239	**+3.21**, +3.39	+2.14, +2.09	41.4 / 40.6	[−0.34, +6.81] / [−1.42, +5.76]
QQQ	2017-06→2026-10	217	**−1.07**, −0.83	−1.82, −2.13	36.9 / 36.9	[−4.07, +2.56] / [−4.84, +1.90]
SPY	2017-06→2026-10	362	**+0.39**, +4.06	−0.50, +2.76	36.2 / 36.2	[−2.01, +2.90] / [−2.94, +2.05]
all names	2017-06→2026-10	818	**+0.83**, +2.57	−0.08, +1.27	37.9 / 37.7	[−0.91, +2.53] / [−1.87, +1.70]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

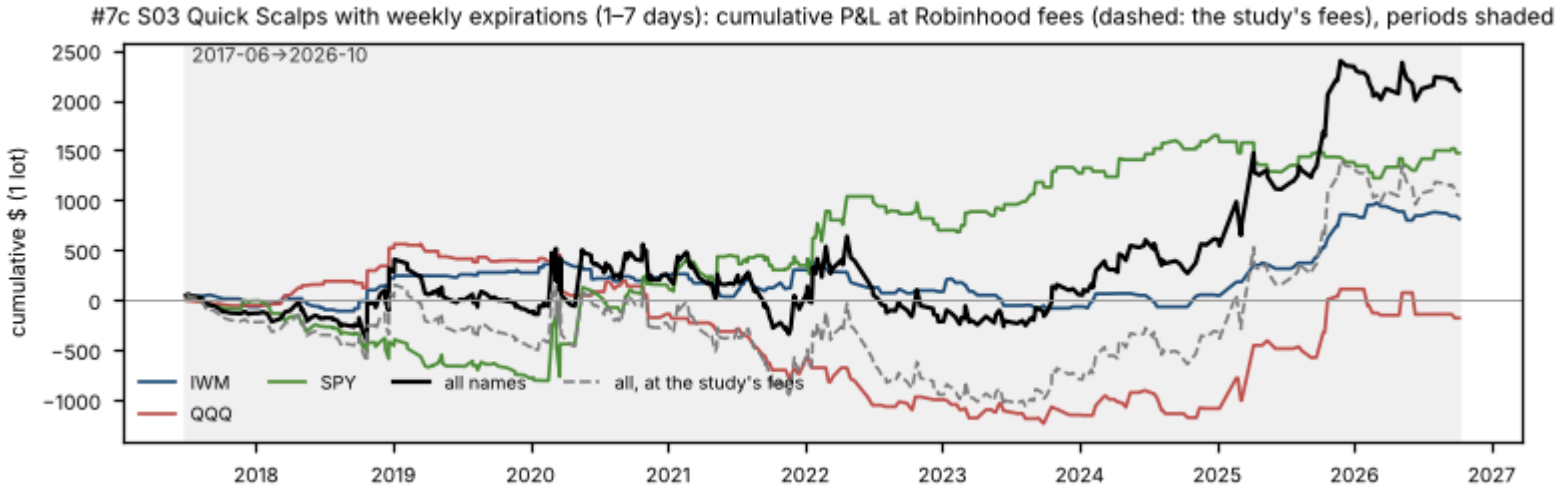
Exit reasons (share of trades):

- 2017-06→2026-10: vwap_loss 50.6%; data_end 17.6%; target_2 17.5%; stop 8.6%; invalidation 5.4%; time 0.4%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2017	IWM	20	35.0	−2.01	−3.43	−69
2018	IWM	30	53.3	+8.83	+9.48	+284
2019	IWM	15	46.7	+2.74	+3.87	+58
2020	IWM	34	35.3	+2.62	−0.79	−27
2021	IWM	29	37.9	+2.60	+1.88	+55
2022	IWM	20	20.0	−3.05	−10.88	−218
2023	IWM	31	35.5	−1.72	−4.86	−151
2024	IWM	17	41.2	+3.23	+6.99	+119
2025	IWM	24	66.7	+16.71	+33.11	+795
2026	IWM	19	42.1	−0.22	−1.89	−36
2017	QQQ	14	14.3	−6.61	−3.99	−56
2018	QQQ	21	76.2	+15.13	+27.45	+576
2019	QQQ	16	37.5	−8.39	−8.19	−131
2020	QQQ	32	37.5	−4.01	−16.56	−530
2021	QQQ	27	18.5	−11.97	−22.19	−599
2022	QQQ	26	30.8	−0.82	−9.93	−258
2023	QQQ	26	34.6	−0.72	−5.87	−153
2024	QQQ	23	30.4	+2.49	+2.81	+65
2025	QQQ	22	54.5	+8.04	+54.33	+1,195
2026	QQQ	10	30.0	−6.52	−29.05	−291
2017	SPY	32	25.0	−3.41	−0.32	−10
2018	SPY	75	29.3	−2.69	−5.44	−408
2019	SPY	46	21.7	−7.32	−7.95	−366
2020	SPY	39	41.0	+5.90	+24.03	+937
2021	SPY	51	41.2	−0.11	+3.79	+193
2022	SPY	36	38.9	+3.19	+9.77	+352
2023	SPY	30	53.3	+8.63	+19.10	+573
2024	SPY	21	57.1	+8.43	+17.90	+376

year	name	trades	win %	mean %	mean $	total $
2025	SPY	17	35.3	−2.76	−15.69	−267
2026	SPY	15	40.0	+4.01	+6.03	+90
2017	all names	66	25.8	−3.67	−2.04	−135
2018	all names	126	42.9	+3.02	+3.59	+453
2019	all names	77	29.9	−5.58	−5.70	−439
2020	all names	105	38.1	+1.82	+3.62	+380
2021	all names	107	34.6	−2.37	−3.28	−351
2022	all names	82	31.7	+0.40	−1.51	−124
2023	all names	87	41.4	+2.15	+3.10	+270
2024	all names	61	42.6	+4.74	+9.17	+559
2025	all names	63	54.0	+8.43	+27.35	+1,723
2026	all names	44	38.6	−0.21	−5.36	−236



Amendments, deviations and integrity checks. A cut-short pull is a uniform sample by strata (d706328).

What it showed. Indistinguishable from zero.

#7d S09 Quick Scalps on SPXW 0DTE

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** 113cad7 / — / 7167541. **Dates:** 2026-10-14.

Hypothesis. The baseline on SPXW.

Rule. As S01, signals on SPX values with SPY volume.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO; SPXW per its fees

Data. SPXW 0DTE 2022-05-19 → 2026-10-08, 1,229 fills on 481 days.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). As S01.

Verdict. REJECTED: −$46.18 a contract [−63.19, −30.02], −$24 against matched random.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	2022-05→2026-10	1,229	481 / 1,135	33.8	+37.6, +316.91	−25.1, −232.60	−22.2, −142.26	**−3.93**, −47.04	[−5.60, −2.27]; [−63.75, −31.69]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	2022-05→2026-10	+1,482.74, +41.0%, 2025-04-09, SPXW 2025-04-09 5380.0CALL	−1,002.26, −40.6%, 2026-03-09, SPXW 2026-03-09 6640.0PUT	0.69	58,726	1,135	total ≤ 0 (best 5 days +8,061; without them −65,879)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $2.26). The study's fees are given beside them below. Re-pricing: exact (1,229 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPXW	2022-05→2026-10	1,229	**−3.93**, −47.04	−3.81, −46.18	33.8 / 33.8	[−5.60, −2.27] / [−5.48, −2.15]

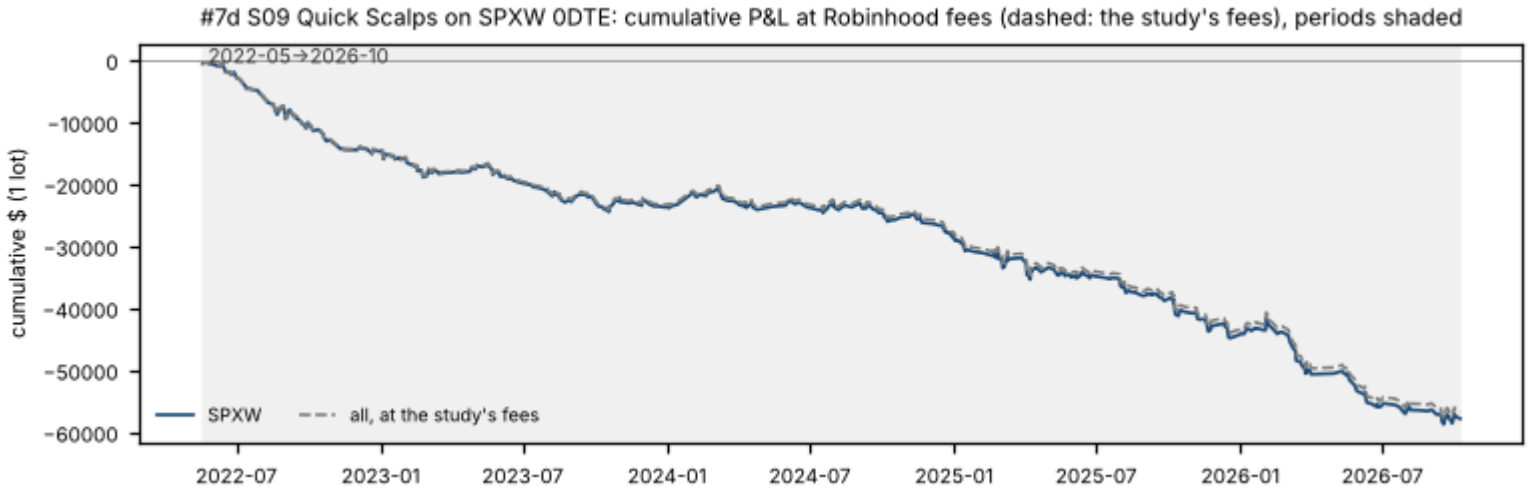
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- 2022-05→2026-10: stop 52.7%; target_2 26.2%; vwap_loss 15.3%; time 3.3%; invalidation 2.4%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	159	27.7	−7.55	−92.48	−14,704
2023	SPXW	300	34.3	−3.60	−30.08	−9,023
2024	SPXW	279	37.6	−1.33	−16.66	−4,648
2025	SPXW	266	33.8	−5.45	−59.79	−15,904
2026	SPXW	225	32.4	−3.24	−60.17	−13,539



Amendments, deviations and integrity checks. —

What it showed. Loses on SPXW too.

#7e S10 Short premium: fade the Quick Scalps signal (single contract, 2× stop, to 15:50)

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A4 912b415 / rerun 1ecdfd5 / cf124a0, 3c8d6cb. **Dates:** 2026-10-14.

Hypothesis. Selling the contract the signal would buy earns its premium.

Rule. Sell the signal's contract at the bid, buy back at 15:50 or at a 2× credit stop (the best cell).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM 0DTE 2022-11 → 2026-10.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). Positive lower bound and better than random.

Verdict. Nothing promoted: breakeven at best (+$0.52 a contract).

Metrics at Robinhood fees (return = % of the credit received; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2022-11→2026-10	685	274 / 638	51.7	+82.7, +49.31	−102.2, −52.69	+17.1, +8.90	**−6.67**, +0.02	[−17.08, +3.47]; [−5.57, +5.31]
QQQ	2022-11→2026-10	1,075	421 / 1,009	56.3	+73.9, +81.09	−97.0, −102.74	+26.2, +20.90	**−0.84**, +0.72	[−8.80, +8.60]; [−9.93, +12.06]
SPY	2022-11→2026-10	1,126	444 / 1,009	57.1	+74.7, +75.78	−98.2, −91.64	+34.6, +22.90	**+0.55**, +3.97	[−6.55, +7.56]; [−3.28, +11.71]
all names	2022-11→2026-10	2,886	614 / 1,009	55.5	+76.1, +71.94	−98.8, −85.66	+30.1, +18.90	**−1.68**, +1.82	[−7.80, +4.77]; [−4.24, +8.50]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	2022-11→2026-10	+186.90, +99.4%, 2025-04-09, IWM short 173.0PUT	−194.10, −106.6%, 2025-09-17, IWM short 241.0CALL	1.00	2,829	362	7604%
QQQ	2022-11→2026-10	+419.90, +92.9%, 2025-04-07, QQQ short 417.0PUT	−425.10, −138.5%, 2026-06-09, QQQ short 692.0PUT	1.02	7,141	244	515%
SPY	2022-11→2026-10	+456.90, +99.5%, 2025-04-09, SPY short 495.0PUT	−413.10, −101.0%, 2025-04-11, SPY short 527.0CALL	1.10	3,022	358	78%
all names	2022-11→2026-10	+456.90, +99.5%, 2025-04-09, SPY short 495.0PUT	−425.10, −138.5%, 2026-06-09, QQQ short 692.0PUT	1.05	8,737	358	135%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (2,886 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	2022-11→2026-10	685	**−6.67**, +0.02	−9.63, −1.27	51.7 / 51.4	[−17.08, +3.47] / [−20.09, +0.58]
QQQ	2022-11→2026-10	1,075	**−0.84**, +0.72	−2.40, −0.58	56.3 / 56.3	[−8.80, +8.60] / [−10.34, +7.08]
SPY	2022-11→2026-10	1,126	**+0.55**, +3.97	−1.17, +2.67	57.1 / 57.1	[−6.55, +7.56] / [−8.30, +5.93]
all names	2022-11→2026-10	2,886	**−1.68**, +1.82	−3.63, +0.52	55.5 / 55.4	[−7.80, +4.77] / [−9.79, +2.83]

Random baseline (as the study computed it, at its own fees): the study's matched random controls (5 seeds, same symbols, sessions and exits, 14,218 trades): mean −5.71 dollars and −10.27% a trade (no percentile was computed).

Exit reasons (share of trades):

- 2022-11→2026-10: session_cutoff 61.9%; stop 38.1%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	157	56.1	−0.78	+1.38	+217
2025	IWM	311	46.9	−14.49	−3.18	−988
2026	IWM	217	55.3	+0.28	+3.63	+788
2022	QQQ	34	67.6	+23.21	+20.99	+714
2023	QQQ	293	53.9	−0.91	+0.49	+144
2024	QQQ	287	59.6	−0.15	+4.20	+1,205
2025	QQQ	241	59.3	+3.89	+9.16	+2,209
2026	QQQ	220	50.0	−10.54	−15.91	−3,501
2022	SPY	40	50.0	−8.08	−9.88	−395
2023	SPY	285	58.9	+6.11	+6.91	+1,968
2024	SPY	281	57.7	−4.92	+2.18	+613
2025	SPY	282	58.2	+2.96	+7.35	+2,073
2026	SPY	238	54.2	−1.07	+0.87	+208
2022	all names	74	58.1	+6.30	+4.30	+319
2023	all names	578	56.4	+2.55	+3.65	+2,112
2024	all names	725	58.1	−2.13	+2.81	+2,035
2025	all names	834	54.3	−3.28	+3.95	+3,293
2026	all names	675	53.2	−3.72	−3.71	−2,505



Amendments, deviations and integrity checks. The vertical rows were superseded after the protective-leg fix; the single-contract rows (shown) are unchanged.

What it showed. The fade is about breakeven.

#7f S12 Intraday momentum: first-half-hour direction → 10:00–15:00 long option

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A5 fe93f87 / — / 2430c9b. **Dates:** 2026-10-14.

Hypothesis. The first half hour's direction continues (Gao et al.).

Rule. Buy the 0DTE contract in the first-half-hour direction at 10:00, exit 15:00.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM 2022-11 → 2026-10 (options).

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). Two-part promotion rule (A1).

Verdict. Nothing promoted (QQQ-only underlying effect, not batch-robust; option cell negative).

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2022-11→2026-10	388	388 / 638	28.9	+116.9, +42.41	−69.3, −22.27	−58.6, −13.60	**−15.58**, −3.60	[−26.74, −4.07]; [−7.74, +0.90]
QQQ	2022-11→2026-10	920	920 / 1,010	31.3	+96.5, +67.15	−61.8, −35.44	−42.3, −19.10	**−12.22**, −3.32	[−18.81, −6.00]; [−8.32, +1.98]
SPY	2022-11→2026-10	906	906 / 1,010	31.8	+98.5, +63.60	−60.7, −33.41	−40.0, −17.10	**−10.10**, −2.57	[−15.74, −2.87]; [−6.60, +2.60]
all names	2022-11→2026-10	2,214	961 / 1,010	31.1	+100.7, +61.64	−62.7, −32.24	−44.0, −17.10	**−11.94**, −3.06	[−17.27, −6.14]; [−6.83, +1.36]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	2022-11→2026-10	+371.90, +743.8%, 2026-07-29, IWM 293.0PUT	−108.10, −99.2%, 2024-09-18, IWM 222.0CALL	0.77	1,852	465	total ≤ 0 (best 5 days +1,190; without them −2,586)
QQQ	2022-11→2026-10	+1,188.88, +861.5%, 2026-07-29, QQQ 678.0PUT	−176.10, −77.9%, 2025-04-07, QQQ 424.0CALL	0.86	5,106	1,008	total ≤ 0 (best 5 days +3,418; without them −6,477)
SPY	2022-11→2026-10	+972.88, +299.3%, 2025-04-09, SPY 532.0CALL	−221.10, −97.4%, 2025-04-08, SPY 502.0CALL	0.89	3,997	1,010	total ≤ 0 (best 5 days +3,046; without them −5,377)
all names	2022-11→2026-10	+1,188.88, +861.5%, 2026-07-29, QQQ 678.0PUT	−221.10, −97.4%, 2025-04-08, SPY 502.0CALL	0.86	10,128	1,010	total ≤ 0 (best 5 days +7,114; without them −13,899)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (2,214 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	2022-11→2026-10	388	**−15.58**, −3.60	−20.62, −4.90	28.9 / 28.1	[−26.74, −4.07] / [−31.88, −9.17]
QQQ	2022-11→2026-10	920	**−12.22**, −3.32	−14.88, −4.62	31.3 / 30.9	[−18.81, −6.00] / [−21.50, −8.59]
SPY	2022-11→2026-10	906	**−10.10**, −2.57	−12.89, −3.87	31.8 / 30.8	[−15.74, −2.87] / [−18.58, −5.60]
all names	2022-11→2026-10	2,214	**−11.94**, −3.06	−15.07, −4.36	31.1 / 30.4	[−17.27, −6.14] / [−20.45, −9.22]

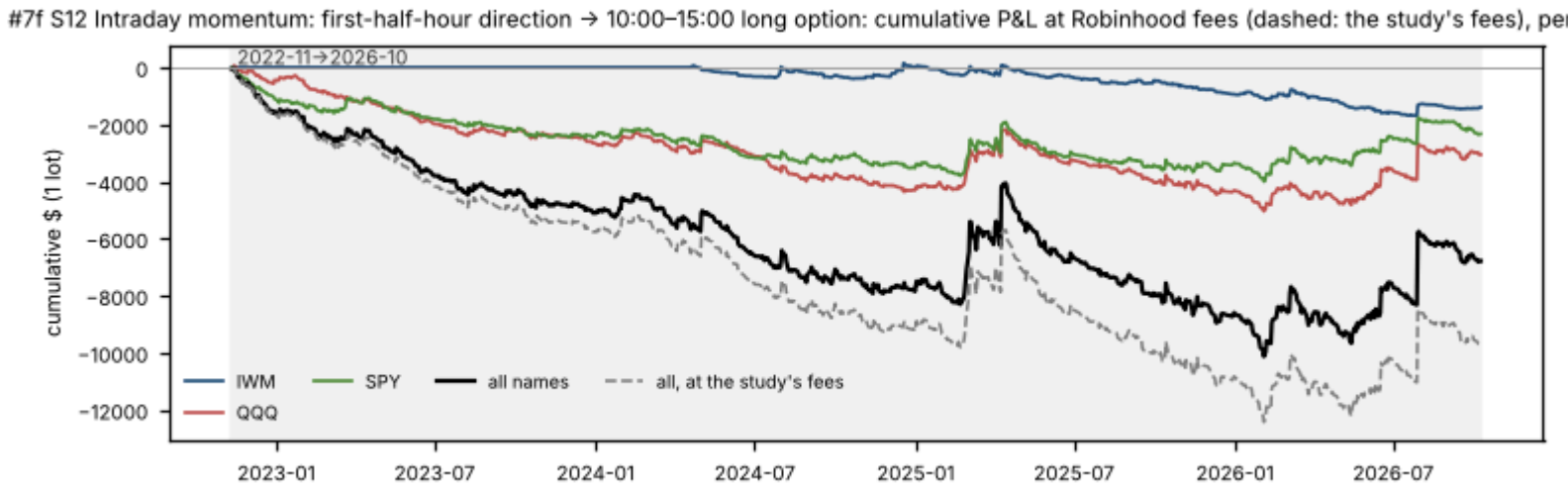
Random baseline (as the study computed it, at its own fees): the study's matched random controls (5 seeds, same symbols, sessions and exits, 10,984 trades): mean −5.53 dollars and −14.32% a trade (no percentile was computed).

Exit reasons (share of trades):

- 2022-11→2026-10: session_cutoff 98.5%; settlement 1.5%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	98	31.6	−8.90	−0.11	−11
2025	IWM	166	27.7	−21.21	−5.60	−930
2026	IWM	124	28.2	−13.31	−3.67	−455
2022	QQQ	31	16.1	−23.64	−14.29	−443
2023	QQQ	225	31.1	−22.55	−9.79	−2,204
2024	QQQ	236	28.4	−16.18	−7.05	−1,664
2025	QQQ	239	33.1	−7.83	−0.00	−1
2026	QQQ	189	35.4	+1.33	+6.63	+1,253
2022	SPY	33	9.1	−58.40	−33.49	−1,105
2023	SPY	217	27.6	−13.59	−6.22	−1,351
2024	SPY	233	32.2	−10.10	−4.62	−1,076
2025	SPY	236	32.6	−8.48	+0.79	+186
2026	SPY	187	39.0	+0.42	+5.43	+1,016
2022	all names	64	12.5	−41.56	−24.19	−1,548
2023	all names	442	29.4	−18.15	−8.04	−3,554
2024	all names	567	30.5	−12.42	−4.85	−2,751
2025	all names	641	31.5	−11.54	−1.16	−744
2026	all names	500	35.0	−2.64	+3.63	+1,814



Amendments, deviations and integrity checks. seed 0 = the signal; 11–15 random.

What it showed. The option expression loses.

#7g S12b 10:00 long option in the first-half-hour direction

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A5.1 b9b9206 / — / be991de. **Dates:** 2026-10-14.

Hypothesis. As S12 with the baseline rule's contract.

Rule. Buy the baseline-rule 0DTE contract at 10:00 in the first-half-hour direction; exit 15:00.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM 2022-11 → 2026-10.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). As S12.

Verdict. Nothing promoted: −$4.87 a contract [−14.18, +4.87].

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2022-11→2026-10	564	564 / 639	32.8	+145.0, +111.15	−81.1, −59.47	−73.8, −40.10	**−6.92**, −3.51	[−17.17, +6.37]; [−12.57, +7.74]
QQQ	2022-11→2026-10	964	964 / 1,010	33.0	+148.6, +207.30	−76.1, −108.99	−60.9, −70.60	**−2.01**, −4.66	[−8.95, +4.98]; [−14.84, +5.37]
SPY	2022-11→2026-10	971	971 / 1,010	34.6	+144.1, +169.16	−76.0, −93.39	−59.6, −57.10	**+0.18**, −2.54	[−6.40, +7.45]; [−11.49, +9.04]
all names	2022-11→2026-10	2,499	971 / 1,010	33.6	+146.0, +170.82	−77.2, −91.72	−63.1, −54.10	**−2.27**, −3.58	[−8.13, +4.26]; [−11.65, +5.47]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	2022-11→2026-10	+859.88, +398.1%, 2025-04-09, IWM 176.0CALL	−180.09, −100.1%, 2025-04-23, IWM 194.0CALL	0.91	3,136	249	total ≤ 0 (best 5 days +3,049; without them −5,027)
QQQ	2022-11→2026-10	+2,478.84, +482.3%, 2025-04-09, QQQ 423.0CALL	−412.10, −99.5%, 2026-06-25, QQQ 704.0PUT	0.94	6,199	980	total ≤ 0 (best 5 days +6,404; without them −10,894)
SPY	2022-11→2026-10	+2,331.85, +411.3%, 2025-04-09, SPY 502.0CALL	−376.09, −100.0%, 2025-04-08, SPY 524.0CALL	0.96	4,288	620	total ≤ 0 (best 5 days +5,599; without them −8,068)
all names	2022-11→2026-10	+2,478.84, +482.3%, 2025-04-09, QQQ 423.0CALL	−412.10, −99.5%, 2026-06-25, QQQ 704.0PUT	0.94	11,917	605	total ≤ 0 (best 5 days +13,086; without them −22,022)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (2,499 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	2022-11→2026-10	564	**−6.92**, −3.51	−8.94, −4.81	32.8 / 32.8	[−17.17, +6.37] / [−19.12, +4.50]
QQQ	2022-11→2026-10	964	**−2.01**, −4.66	−3.05, −5.96	33.0 / 32.7	[−8.95, +4.98] / [−9.98, +4.00]
SPY	2022-11→2026-10	971	**+0.18**, −2.54	−1.04, −3.84	34.6 / 34.4	[−6.40, +7.45] / [−7.66, +6.24]
all names	2022-11→2026-10	2,499	**−2.27**, −3.58	−3.60, −4.87	33.6 / 33.4	[−8.13, +4.26] / [−9.45, +2.93]

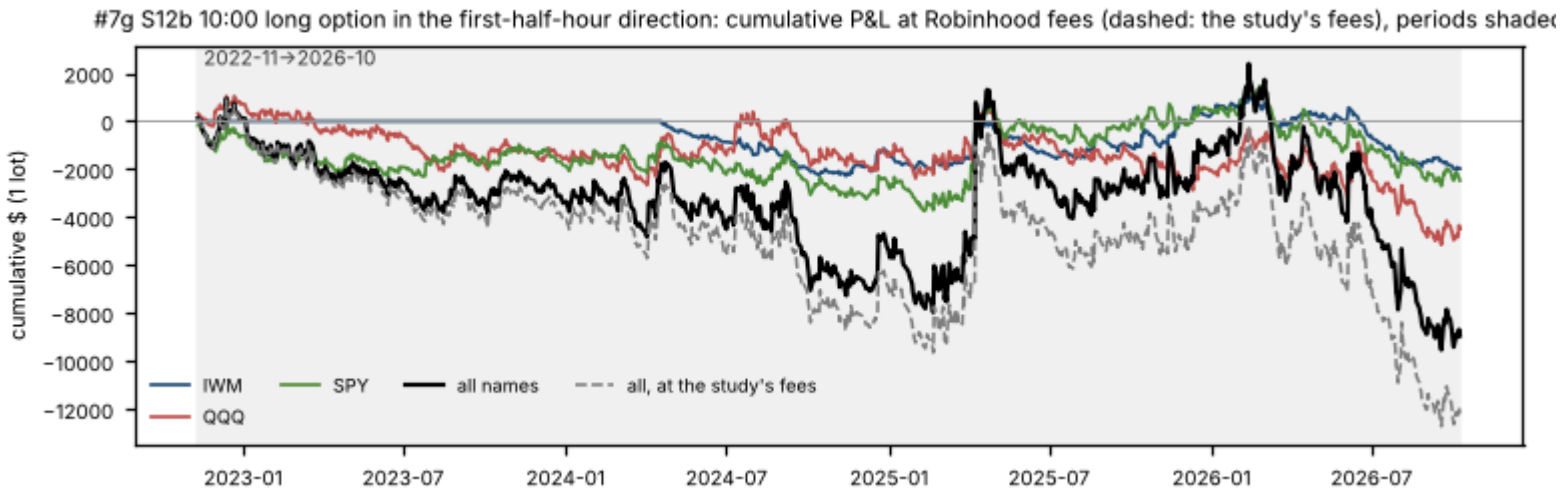
Random baseline (as the study computed it, at its own fees): the study's matched random controls (5 seeds, same symbols, sessions and exits, 12,446 trades): mean −11.99 dollars and −9.75% a trade (no percentile was computed).

Exit reasons (share of trades):

- 2022-11→2026-10: time 95.5%; settlement 4.5%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	144	29.9	−15.19	−9.30	−1,339
2025	IWM	227	35.2	+2.83	+7.08	+1,607
2026	IWM	193	32.1	−12.22	−11.63	−2,245
2022	QQQ	31	45.2	+16.56	+23.03	+714
2023	QQQ	247	30.8	−8.73	−9.31	−2,299
2024	QQQ	248	32.7	+3.88	+1.17	+290
2025	QQQ	246	33.7	−2.72	−2.96	−727
2026	QQQ	192	33.3	−3.05	−12.85	−2,467
2022	SPY	33	30.3	−4.33	−18.40	−607
2023	SPY	248	33.5	−5.80	−4.91	−1,218
2024	SPY	249	35.3	+1.08	−3.47	−863
2025	SPY	247	37.7	+6.98	+12.87	+3,178
2026	SPY	194	32.0	−1.19	−15.25	−2,959
2022	all names	64	37.5	+5.79	+1.67	+107
2023	all names	495	32.1	−7.26	−7.10	−3,517
2024	all names	641	33.1	−1.50	−2.98	−1,913
2025	all names	720	35.6	+2.35	+5.64	+4,058
2026	all names	579	32.5	−5.48	−13.25	−7,671



Amendments, deviations and integrity checks. —

What it showed. Negative.

#7h S15 VRP-conditioned 10:00 short ATM 0DTE straddle (unconditional row)

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A5 6dfba60 / — / 6ed12b6. **Dates:** 2026-10-14.

Hypothesis. Rich implied vs trailing realised volatility marks sellable straddles.

Rule. Sell the 10:00 ATM straddle at the bids, buy back at 15:50 at the asks.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ 0DTE, 970/971 sessions 2022-11 → 2026-10.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). A conditioned cell with a positive lower bound.

Verdict. Nothing promoted: unconditional −$15.46 a straddle [−32.68, +0.29]; rich implied forecasts large moves.

Metrics at Robinhood fees (return = % of the credit received; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	2022-11→2026-10	971	971 / 1,010	58.4	+44.8, +146.03	−73.9, −235.27	+14.9, +43.80	**−4.60**, −12.62	[−9.49, −0.07]; [−32.26, +3.85]
SPY	2022-11→2026-10	970	970 / 1,010	58.0	+44.0, +123.56	−71.4, −202.13	+14.9, +38.80	**−4.40**, −13.10	[−8.87, −0.60]; [−31.42, +0.76]
all names	2022-11→2026-10	1,941	971 / 1,010	58.2	+44.4, +134.83	−72.6, −218.64	+14.9, +41.80	**−4.50**, −12.86	[−8.99, −0.44]; [−31.36, +1.76]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	2022-11→2026-10	+662.80, +76.0%, 2024-08-05, QQQ 10:00 short straddle 433.0	−3,321.22, −278.9%, 2025-04-09, QQQ 10:00 short straddle 421.0	0.87	17,201	1,003	total ≤ 0 (best 5 days +2,665; without them −14,915)
SPY	2022-11→2026-10	+660.80, +76.5%, 2024-08-05, SPY 10:00 short straddle 515.0	−3,097.22, −230.6%, 2025-04-09, SPY 10:00 short straddle 500.0	0.85	14,929	999	total ≤ 0 (best 5 days +2,528; without them −15,233)
all names	2022-11→2026-10	+662.80, +76.0%, 2024-08-05, QQQ 10:00 short straddle 433.0	−3,321.22, −278.9%, 2025-04-09, QQQ 10:00 short straddle 421.0	0.86	31,025	1,003	total ≤ 0 (best 5 days +5,081; without them −30,037)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.80 a trade) and Robinhood's charged instead (mean $0.20). The study's fees are given beside them below. Re-pricing: exact (1,941 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	2022-11→2026-10	971	**−4.60**, −12.62	−5.49, −15.22	58.4 / 58.1	[−9.49, −0.07] / [−10.37, −0.97]

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPY	2022-11→2026-10	970	**−4.40**, −13.10	−5.44, −15.70	58.0 / 57.9	[−8.87, −0.60] / [−9.91, −1.65]
all names	2022-11→2026-10	1,941	**−4.50**, −12.86	−5.47, −15.46	58.2 / 58.0	[−8.99, −0.44] / [−9.93, −1.42]

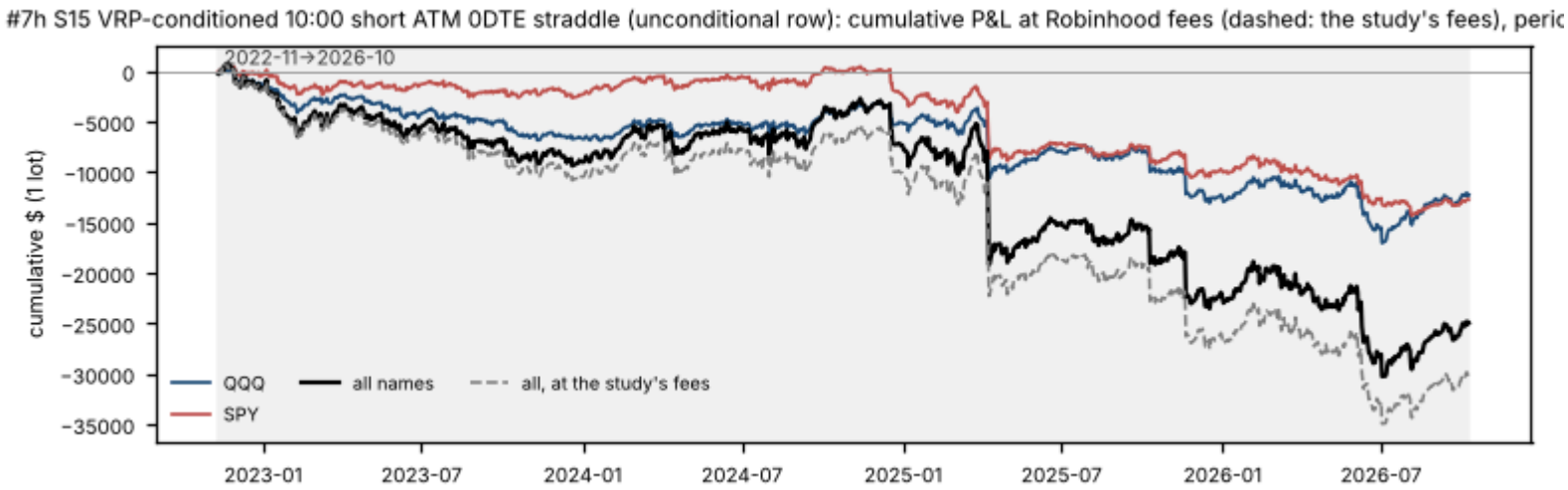
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- 2022-11→2026-10: bought back 15:50 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	QQQ	33	45.5	−14.07	−37.63	−1,242
2023	QQQ	248	52.0	−10.88	−21.56	−5,346
2024	QQQ	249	63.5	+1.61	+6.15	+1,531
2025	QQQ	247	59.1	−6.33	−29.53	−7,293
2026	QQQ	194	61.3	−0.73	+0.51	+99
2022	SPY	33	51.5	−5.42	−13.26	−438
2023	SPY	248	57.3	−5.28	−7.76	−1,924
2024	SPY	248	63.7	−0.01	+0.38	+94
2025	SPY	247	55.1	−6.93	−30.65	−7,570
2026	SPY	194	56.7	−5.48	−14.78	−2,868
2022	all names	66	48.5	−9.74	−25.44	−1,679
2023	all names	496	54.6	−8.08	−14.66	−7,270
2024	all names	497	63.6	+0.80	+3.27	+1,625
2025	all names	494	57.1	−6.63	−30.09	−14,863
2026	all names	388	59.0	−3.11	−7.14	−2,769



Amendments, deviations and integrity checks. —

What it showed. The 10:00 short straddle loses.

#7i S16b Prior-day fade with a weekly contract (time exit)

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A5 fe93f87 / — / 5cabc13, 3c8d6cb. **Dates:** 2026-10-14.

Hypothesis. The prior session's close-to-close move reverses.

Rule. Buy the weekly contract against the prior move at the 09:31 close; time exit.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM weekly chains 2017-06 → 2026-10.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). As S12.

Verdict. Nothing promoted: −$9.91 a contract [−19.31, −0.26].

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2017-06→2026-10	287	287 / 2,411	36.9	+48.1, +77.16	−41.2, −64.72	−16.8, −22.10	**−8.24**, −12.32	[−13.81, −2.14]; [−21.52, −2.43]
QQQ	2017-06→2026-10	290	290 / 2,411	38.6	+46.5, +134.39	−40.9, −104.20	−13.6, −27.60	**−7.16**, −12.06	[−12.80, −1.26]; [−29.31, +6.13]
SPY	2017-06→2026-10	464	464 / 2,419	38.4	+58.1, +132.32	−42.2, −89.11	−14.6, −23.10	**−3.74**, −4.17	[−10.96, +2.56]; [−19.85, +10.79]
all names	2017-06→2026-10	1,041	699 / 2,419	38.0	+52.1, +118.14	−41.6, −86.43	−14.9, −24.10	**−5.93**, −8.61	[−10.19, −2.31]; [−17.65, +0.82]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	2017-06→2026-10	+376.89, +221.7%, 2026-05-20, IWM 275.0CALL	−277.10, −72.5%, 2025-04-08, IWM 186.0CALL	0.70	3,902	1,453	total ≤ 0 (best 5 days +1,240; without them −4,775)
QQQ	2017-06→2026-10	+1,175.86, +123.8%, 2025-04-08, QQQ 435.0PUT	−444.10, −65.9%, 2026-06-12, QQQ 713.0PUT	0.81	3,518	2,362	total ≤ 0 (best 5 days +2,972; without them −6,469)
SPY	2017-06→2026-10	+1,275.87, +405.0%, 2022-10-13, SPY 350.0CALL	−415.10, −89.7%, 2022-05-04, SPY 415.0PUT	0.92	4,102	1,484	total ≤ 0 (best 5 days +4,195; without them −6,129)
all names	2017-06→2026-10	+1,275.87, +405.0%, 2022-10-13, SPY 350.0CALL	−444.10, −65.9%, 2026-06-12, QQQ 713.0PUT	0.84	9,949	2,388	total ≤ 0 (best 5 days +4,600; without them −13,565)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (1,041 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	2017-06→2026-10	287	**−8.24**, −12.32	−9.21, −13.62	36.9 / 36.6	[−13.81, −2.14] / [−14.76, −3.11]
QQQ	2017-06→2026-10	290	**−7.16**, −12.06	−7.85, −13.36	38.6 / 38.3	[−12.80, −1.26] / [−13.53, −1.97]
SPY	2017-06→2026-10	464	**−3.74**, −4.17	−4.60, −5.46	38.4 / 38.1	[−10.96, +2.56] / [−11.77, +1.69]
all names	2017-06→2026-10	1,041	**−5.93**, −8.61	−6.78, −9.91	38.0 / 37.8	[−10.19, −2.31] / [−11.05, −3.10]

Random baseline (as the study computed it, at its own fees): the study's matched random controls (5 seeds, same symbols, sessions and exits, 5,218 trades): mean −9.80 dollars and −4.48% a trade (no percentile was computed).

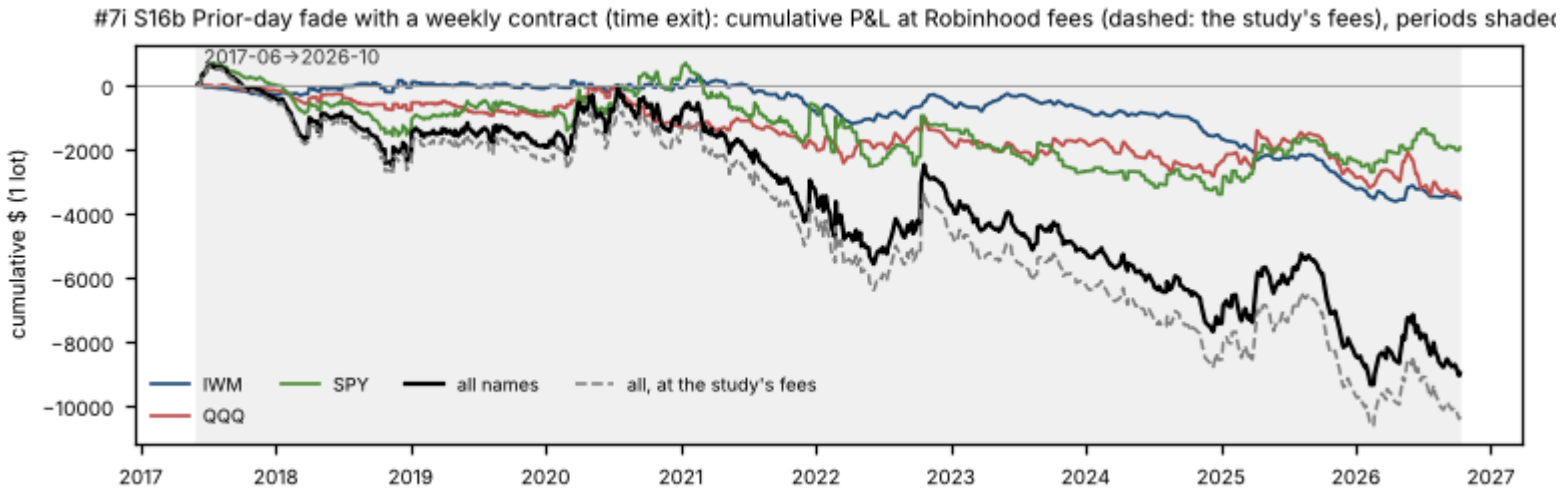
Exit reasons (share of trades):

- 2017-06→2026-10: data_end 88.1%; time 11.9%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2017	IWM	16	18.8	−27.60	−16.66	−267
2018	IWM	30	50.0	+9.17	+7.30	+219
2019	IWM	32	37.5	−2.93	−0.54	−17
2020	IWM	28	42.9	−1.08	+0.04	+1
2021	IWM	31	38.7	−14.71	−22.94	−711
2022	IWM	32	50.0	−0.39	+6.56	+210
2023	IWM	31	38.7	−1.68	−6.07	−188
2024	IWM	31	29.0	−15.08	−25.33	−785
2025	IWM	32	21.9	−30.49	−52.63	−1,684
2026	IWM	24	33.3	−4.60	−13.02	−312
2017	QQQ	17	23.5	−19.87	−8.04	−137
2018	QQQ	31	32.3	−17.10	−20.42	−633
2019	QQQ	31	32.3	−7.11	−5.58	−173
2020	QQQ	30	46.7	+1.84	−11.83	−355
2021	QQQ	31	38.7	−8.57	−16.88	−523
2022	QQQ	32	43.8	−1.23	+1.15	+37
2023	QQQ	31	38.7	−1.89	+1.51	+47
2024	QQQ	31	38.7	−7.38	−24.20	−750
2025	QQQ	32	40.6	−10.12	−12.01	−384
2026	QQQ	24	45.8	−5.25	−26.02	−625
2017	SPY	58	44.8	+2.02	+0.59	+34
2018	SPY	62	33.9	−20.01	−25.44	−1,577
2019	SPY	64	34.4	+6.00	+10.41	+667
2020	SPY	60	45.0	+1.73	+17.46	+1,048
2021	SPY	64	29.7	−10.06	−12.65	−809
2022	SPY	37	40.5	−3.28	−19.86	−735
2023	SPY	32	34.4	−14.53	−40.44	−1,294

year	name	trades	win %	mean %	mean $	total $
2024	SPY	31	35.5	−8.14	−23.36	−724
2025	SPY	31	48.4	+9.03	+34.12	+1,058
2026	SPY	25	44.0	+4.15	+16.02	+400
2017	all names	91	36.3	−7.27	−4.06	−369
2018	all names	123	37.4	−12.16	−16.19	−1,991
2019	all names	127	34.6	+0.55	+3.75	+476
2020	all names	118	44.9	+1.09	+5.88	+694
2021	all names	126	34.1	−10.84	−16.22	−2,044
2022	all names	101	44.6	−1.71	−4.84	−488
2023	all names	94	37.2	−6.13	−15.27	−1,435
2024	all names	93	34.4	−10.20	−24.29	−2,259
2025	all names	95	36.8	−10.73	−10.64	−1,011
2026	all names	73	41.1	−1.82	−7.35	−537



Amendments, deviations and integrity checks. —

What it showed. Negative.

#7j-atm_put S17 Index put-writing: monthly ATM cash-secured put

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A7 c81522a / wording 9ea4adf / 5e7e1bd. **Dates:** 2026-10-15.

Hypothesis. Index put-writing earns the variance risk premium (Cboe PUT).

Rule. Sell the standard monthly ATM put at the bid at expiry roll, hold to expiration, cash-secured.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Bid in; settlement out; $0.70 a contract.

Data. SPY and QQQ standard monthlies 2012-06 → 2026-10.

Periods and holdouts. One span (2012–2026); no holdout.

Pass rule (as pre-registered). A7 carry-lane rule (positive mean with a positive lower bound per cell).

Verdict. PASSES A7 for QQQ ATM, 0.30Δ and 0.15Δ and SPY ATM puts; alpha 0 against the index, crash risk stated (a carry lane, not an edge).

Metrics at Robinhood fees (return = % of the cash set aside; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	2012-06→2026-10	167	167 / 3,633	80.2	+1.9, +496.48	−4.0, −1,003.19	+1.6, +216.94	**+0.72**, +200.14	[+0.21, +1.16]; [+57.42, +322.89]
SPY	2012-06→2026-10	167	166 / 3,633	82.6	+1.5, +527.73	−3.9, −1,353.39	+1.2, +330.94	**+0.59**, +201.07	[+0.08, +1.01]; [+20.70, +347.85]
all names	2012-06→2026-10	334	167 / 3,633	81.4	+1.7, +512.34	−4.0, −1,166.99	+1.4, +267.94	**+0.66**, +200.60	[+0.17, +1.08]; [+42.50, +333.89]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	2012-06→2026-10	+2,117.90, +3.0%, 2026-07-17, QQQ 2026-08-21 695.0	−5,456.06, −23.7%, 2020-02-21, QQQ 2020-03-20 230.0	2.01	7,613	581	24%
SPY	2012-06→2026-10	+1,693.91, +2.3%, 2026-05-15, SPY 2026-07-17 735.0	−9,870.07, −29.6%, 2020-02-21, SPY 2020-03-20 333.0	1.86	9,870	586	22%
all names	2012-06→2026-10	+2,117.90, +3.0%, 2026-07-17, QQQ 2026-08-21 695.0	−9,870.07, −29.6%, 2020-02-21, SPY 2020-03-20 333.0	1.93	15,326	581	23%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.07). The study's fees are given beside them below. Re-pricing: exact (334 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	2012-06→2026-10	167	**+0.72**, +200.14	+0.72, +198.81	80.2 / 80.2	[+0.21, +1.16] / [+0.20, +1.16]
SPY	2012-06→2026-10	167	**+0.59**, +201.07	+0.59, +199.73	82.6 / 82.6	[+0.08, +1.01] / [+0.07, +1.01]
all names	2012-06→2026-10	334	**+0.66**, +200.60	+0.65, +199.27	81.4 / 81.4	[+0.17, +1.08] / [+0.16, +1.07]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

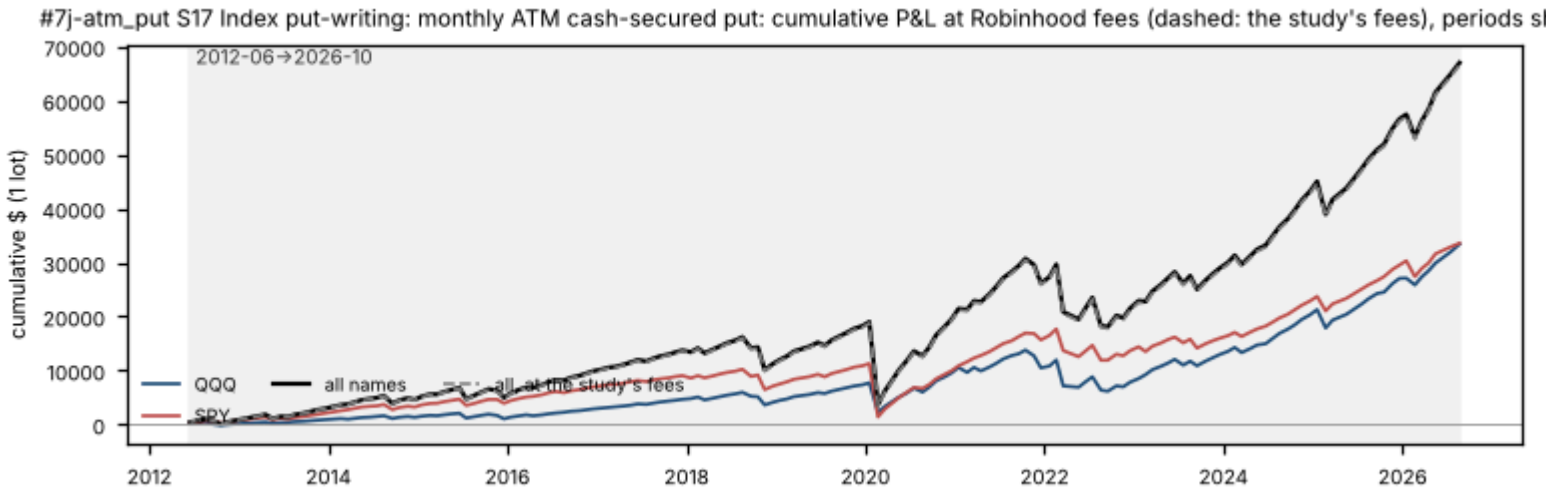
Exit reasons (share of trades):

- 2012-06→2026-10: held to expiration 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	QQQ	7	71.4	+0.33	+16.44	+115
2013	QQQ	12	91.7	+0.89	+65.65	+788
2014	QQQ	12	75.0	+0.45	+37.06	+445
2015	QQQ	12	66.7	−0.14	−23.69	−284
2016	QQQ	12	91.7	+1.41	+153.48	+1,842
2017	QQQ	12	91.7	+1.06	+147.61	+1,771
2018	QQQ	12	66.7	−0.20	−43.56	−523
2019	QQQ	11	90.9	+1.58	+292.21	+3,214
2020	QQQ	12	83.3	+0.72	+184.26	+2,211
2021	QQQ	12	66.7	+0.33	+76.59	+919
2022	QQQ	11	54.5	−0.51	−240.35	−2,644
2023	QQQ	12	83.3	+1.32	+424.18	+5,090
2024	QQQ	12	91.7	+1.31	+617.35	+7,408
2025	QQQ	11	90.9	+1.09	+611.29	+6,724
2026	QQQ	7	85.7	+1.33	+906.63	+6,346
2012	SPY	7	71.4	+0.75	+100.37	+703
2013	SPY	12	83.3	+0.71	+115.86	+1,390
2014	SPY	12	83.3	+0.53	+96.36	+1,156
2015	SPY	12	83.3	+0.30	+57.48	+690
2016	SPY	12	91.7	+1.25	+254.36	+3,052
2017	SPY	12	91.7	+0.71	+172.48	+2,070
2018	SPY	12	66.7	−0.53	−158.98	−1,908
2019	SPY	11	90.9	+1.18	+340.03	+3,740
2020	SPY	12	83.3	+0.02	−76.32	−916
2021	SPY	12	83.3	+1.18	+475.68	+5,708
2022	SPY	11	63.6	−0.32	−182.12	−2,003
2023	SPY	12	75.0	+0.48	+190.18	+2,282
2024	SPY	12	91.7	+1.05	+579.10	+6,949
2025	SPY	11	90.9	+0.95	+601.20	+6,613
2026	SPY	7	85.7	+0.80	+578.78	+4,051
2012	all names	14	71.4	+0.54	+58.40	+818
2013	all names	24	87.5	+0.80	+90.75	+2,178
2014	all names	24	79.2	+0.49	+66.71	+1,601
2015	all names	24	75.0	+0.08	+16.90	+406
2016	all names	24	91.7	+1.33	+203.92	+4,894
2017	all names	24	91.7	+0.88	+160.04	+3,841
2018	all names	24	66.7	−0.37	−101.27	−2,430

year	name	trades	win %	mean %	mean $	total $
2019	all names	22	90.9	+1.38	+316.12	+6,955
2020	all names	24	83.3	+0.37	+53.97	+1,295
2021	all names	24	75.0	+0.75	+276.14	+6,627
2022	all names	22	59.1	−0.42	−211.24	−4,647
2023	all names	24	79.2	+0.90	+307.18	+7,372
2024	all names	24	91.7	+1.18	+598.22	+14,357
2025	all names	22	90.9	+1.02	+606.24	+13,337
2026	all names	14	85.7	+1.07	+742.70	+10,398



Amendments, deviations and integrity checks. Covered calls not significantly negative (9ea4adf).

What it showed. Put-writing earned its premium like the index it is short; a carry, not an edge.

#7j-put_30d S17 Index put-writing: monthly 0.30-delta cash-secured put

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A7 c81522a / wording 9ea4adf / 5e7e1bd. **Dates:** 2026-10-15.

Hypothesis. Index put-writing earns the variance risk premium (Cboe PUT).

Rule. Sell the standard monthly 0.30-delta put at the bid at expiry roll, hold to expiration, cash-secured.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Bid in; settlement out; $0.70 a contract.

Data. SPY and QQQ standard monthlies 2012-06 → 2026-10.

Periods and holdouts. One span (2012–2026); no holdout.

Pass rule (as pre-registered). A7 carry-lane rule (positive mean with a positive lower bound per cell).

Verdict. PASSES A7 for QQQ ATM, 0.30Δ and 0.15Δ and SPY ATM puts; alpha 0 against the index, crash risk stated (a carry lane, not an edge).

Metrics at Robinhood fees (return = % of the cash set aside; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	2012-06→2026-10	167	167 / 3,633	85.6	+1.2, +318.38	−3.7, −833.96	+1.0, +156.94	**+0.52**, +152.78	[+0.12, +0.86]; [+44.84, +240.05]
SPY	2012-06→2026-10	167	166 / 3,633	85.6	+1.0, +338.50	−3.3, −1,090.06	+0.8, +223.94	**+0.40**, +133.20	[−0.07, +0.73]; [−21.54, +242.25]
all names	2012-06→2026-10	334	167 / 3,633	85.6	+1.1, +328.44	−3.5, −962.01	+0.9, +186.94	**+0.46**, +142.99	[+0.03, +0.79]; [+16.36, +239.59]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	2012-06→2026-10	+1,427.92, +2.1%, 2026-05-15, QQQ 2026-07-17 675.0	−5,030.06, −22.5%, 2020-02-21, QQQ 2020-03-20 224.0	2.27	5,030	368	21%
SPY	2012-06→2026-10	+1,116.92, +1.6%, 2026-05-15, SPY 2026-07-17 715.0	−9,370.06, −28.7%, 2020-02-21, SPY 2020-03-20 326.0	1.85	9,370	408	20%
all names	2012-06→2026-10	+1,427.92, +2.1%, 2026-05-15, QQQ 2026-07-17 675.0	−9,370.06, −28.7%, 2020-02-21, SPY 2020-03-20 326.0	2.03	14,400	343	21%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.06). The study's fees are given beside them below. Re-pricing: exact (334 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	2012-06→2026-10	167	**+0.52**, +152.78	+0.51, +151.44	85.6 / 85.6	[+0.12, +0.86] / [+0.11, +0.86]
SPY	2012-06→2026-10	167	**+0.40**, +133.20	+0.40, +131.86	85.6 / 85.6	[−0.07, +0.73] / [−0.08, +0.72]
all names	2012-06→2026-10	334	**+0.46**, +142.99	+0.46, +141.65	85.6 / 85.6	[+0.03, +0.79] / [+0.02, +0.79]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

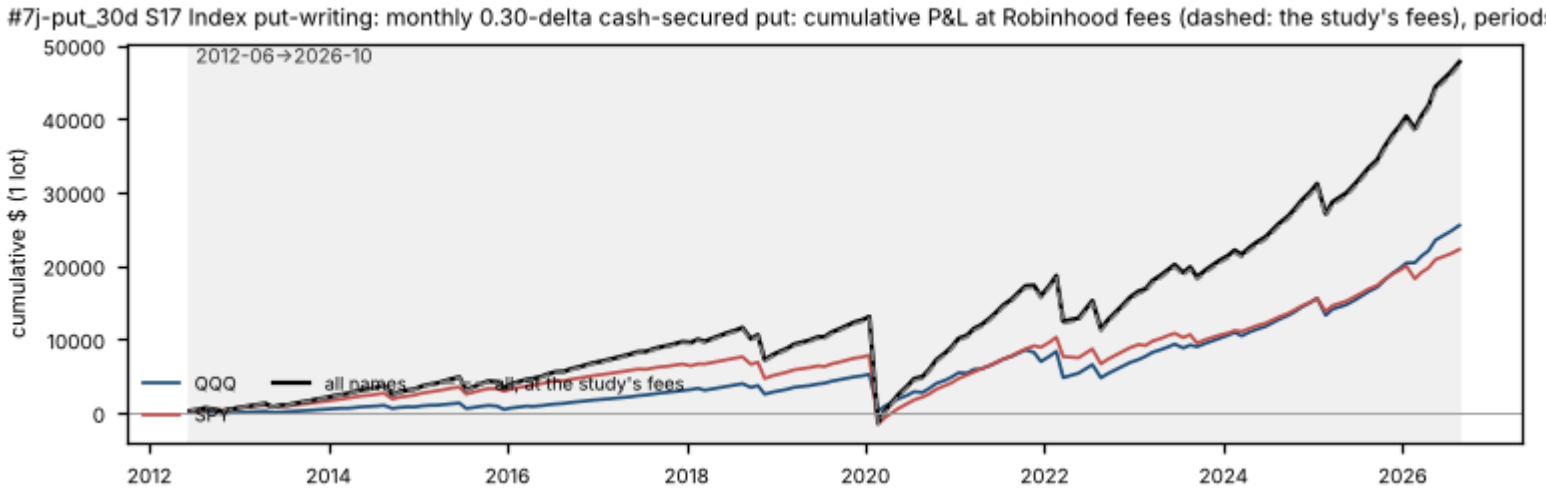
Exit reasons (share of trades):

- 2012-06→2026-10: held to expiration 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	QQQ	7	71.4	+0.11	+2.87	+20
2013	QQQ	12	91.7	+0.52	+37.81	+454
2014	QQQ	12	75.0	+0.33	+26.81	+322
2015	QQQ	12	75.0	−0.20	−27.69	−332
2016	QQQ	12	91.7	+0.99	+105.69	+1,268
2017	QQQ	12	100.0	+0.78	+107.86	+1,294
2018	QQQ	12	75.0	−0.06	−15.31	−184
2019	QQQ	11	100.0	+1.11	+199.94	+2,199
2020	QQQ	12	83.3	−0.09	−14.57	−175
2021	QQQ	12	75.0	+0.58	+176.10	+2,113
2022	QQQ	11	81.8	+0.18	−11.70	−129
2023	QQQ	12	83.3	+0.87	+275.69	+3,308
2024	QQQ	12	91.7	+0.88	+402.18	+4,826
2025	QQQ	11	90.9	+0.75	+420.93	+4,630
2026	QQQ	7	100.0	+1.32	+842.50	+5,897
2012	SPY	7	85.7	+0.66	+88.80	+622
2013	SPY	12	83.3	+0.49	+78.69	+944
2014	SPY	12	91.7	+0.39	+70.69	+848
2015	SPY	12	83.3	+0.23	+42.94	+515
2016	SPY	12	91.7	+0.88	+176.73	+2,121
2017	SPY	12	91.7	+0.55	+132.02	+1,584
2018	SPY	12	66.7	−0.45	−129.56	−1,555
2019	SPY	11	90.9	+0.81	+229.12	+2,520
2020	SPY	12	91.7	−0.71	−292.32	−3,508
2021	SPY	12	91.7	+0.99	+401.44	+4,817
2022	SPY	11	72.7	+0.09	−3.98	−44
2023	SPY	12	75.0	+0.36	+140.69	+1,688
2024	SPY	12	91.7	+0.70	+374.52	+4,494
2025	SPY	11	90.9	+0.63	+393.57	+4,329
2026	SPY	7	85.7	+0.58	+409.50	+2,866
2012	all names	14	78.6	+0.38	+45.83	+642
2013	all names	24	87.5	+0.50	+58.25	+1,398
2014	all names	24	83.3	+0.36	+48.75	+1,170
2015	all names	24	79.2	+0.01	+7.63	+183
2016	all names	24	91.7	+0.93	+141.21	+3,389
2017	all names	24	95.8	+0.67	+119.94	+2,879
2018	all names	24	70.8	−0.26	−72.44	−1,738

year	name	trades	win %	mean %	mean $	total $
2019	all names	22	95.5	+0.96	+214.53	+4,720
2020	all names	24	87.5	−0.40	−153.44	−3,683
2021	all names	24	83.3	+0.79	+288.77	+6,930
2022	all names	22	77.3	+0.13	−7.84	−173
2023	all names	24	79.2	+0.62	+208.19	+4,997
2024	all names	24	91.7	+0.79	+388.35	+9,320
2025	all names	22	90.9	+0.69	+407.25	+8,959
2026	all names	14	92.9	+0.95	+626.00	+8,764



Amendments, deviations and integrity checks. Covered calls not significantly negative (9ea4adf).

What it showed. Put-writing earned its premium like the index it is short; a carry, not an edge.

#7j-put_15d S17 Index put-writing: monthly 0.15-delta cash-secured put

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A7 c81522a / wording 9ea4adf / 5e7e1bd. **Dates:** 2026-10-15.

Hypothesis. Index put-writing earns the variance risk premium (Cboe PUT).

Rule. Sell the standard monthly 0.15-delta put at the bid at expiry roll, hold to expiration, cash-secured.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Bid in; settlement out; $0.70 a contract.

Data. SPY and QQQ standard monthlies 2012-06 → 2026-10.

Periods and holdouts. One span (2012–2026); no holdout.

Pass rule (as pre-registered). A7 carry-lane rule (positive mean with a positive lower bound per cell).

Verdict. PASSES A7 for QQQ ATM, 0.30Δ and 0.15Δ and SPY ATM puts; alpha 0 against the index, crash risk stated (a carry lane, not an edge).

Metrics at Robinhood fees (return = % of the cash set aside; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	2012-06→2026-10	167	167 / 3,633	93.4	+0.6, +151.92	−3.9, −780.51	+0.5, +86.94	**+0.31**, +90.50	[−0.01, +0.52]; [+20.03, +138.63]
SPY	2012-06→2026-10	167	166 / 3,633	94.6	+0.5, +160.77	−4.6, −1,389.73	+0.4, +114.94	**+0.22**, +77.21	[−0.18, +0.45]; [−47.55, +149.04]
all names	2012-06→2026-10	334	167 / 3,633	94.0	+0.5, +156.37	−4.2, −1,054.66	+0.5, +107.44	**+0.27**, +83.86	[−0.09, +0.48]; [−14.32, +142.48]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	2012-06→2026-10	+719.93, +1.1%, 2026-05-15, QQQ 2026-07-17 635.0	−4,190.06, −19.6%, 2020-02-21, QQQ 2020-03-20 214.0	2.76	4,190	388	17%
SPY	2012-06→2026-10	+528.93, +0.8%, 2026-05-15, SPY 2026-07-17 675.0	−8,445.06, −26.8%, 2020-02-21, SPY 2020-03-20 315.0	2.03	8,520	1,336	17%
all names	2012-06→2026-10	+719.93, +1.1%, 2026-05-15, QQQ 2026-07-17 635.0	−8,445.06, −26.8%, 2020-02-21, SPY 2020-03-20 315.0	2.33	12,635	776	17%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.06). The study's fees are given beside them below. Re-pricing: exact (334 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	2012-06→2026-10	167	**+0.31**, +90.50	+0.30, +89.16	93.4 / 93.4	[−0.01, +0.52] / [−0.01, +0.51]
SPY	2012-06→2026-10	167	**+0.22**, +77.21	+0.22, +75.87	94.6 / 94.6	[−0.18, +0.45] / [−0.19, +0.45]
all names	2012-06→2026-10	334	**+0.27**, +83.86	+0.26, +82.52	94.0 / 94.0	[−0.09, +0.48] / [−0.10, +0.47]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

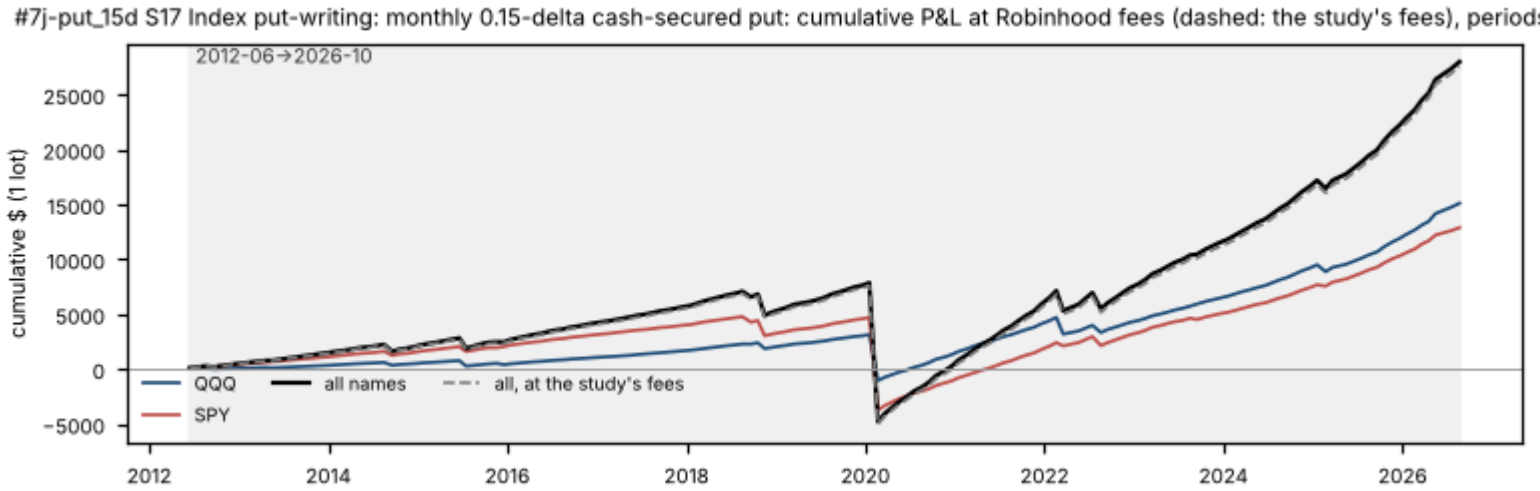
Exit reasons (share of trades):

- 2012-06→2026-10: held to expiration 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	QQQ	7	85.7	+0.18	+9.23	+65
2013	QQQ	12	91.7	+0.32	+22.65	+272
2014	QQQ	12	91.7	+0.18	+14.27	+171
2015	QQQ	12	83.3	−0.05	−8.19	−98
2016	QQQ	12	100.0	+0.52	+53.77	+645
2017	QQQ	12	100.0	+0.38	+51.44	+617
2018	QQQ	12	83.3	+0.18	+27.77	+333
2019	QQQ	11	100.0	+0.54	+94.67	+1,041
2020	QQQ	12	91.7	−0.68	−143.14	−1,718
2021	QQQ	12	100.0	+0.72	+232.52	+2,790
2022	QQQ	11	81.8	+0.14	+13.76	+151
2023	QQQ	12	100.0	+0.58	+184.11	+2,209
2024	QQQ	12	100.0	+0.52	+229.77	+2,757
2025	QQQ	11	90.9	+0.46	+243.03	+2,673
2026	QQQ	7	100.0	+0.76	+457.65	+3,204
2012	SPY	7	100.0	+0.46	+60.51	+424
2013	SPY	12	100.0	+0.36	+56.52	+678
2014	SPY	12	91.7	+0.21	+36.61	+439
2015	SPY	12	91.7	+0.24	+43.27	+519
2016	SPY	12	100.0	+0.45	+88.52	+1,062
2017	SPY	12	100.0	+0.31	+72.69	+872
2018	SPY	12	83.3	−0.23	−64.23	−771
2019	SPY	11	100.0	+0.45	+124.67	+1,371
2020	SPY	12	91.7	−1.37	−471.06	−5,653
2021	SPY	12	100.0	+0.60	+235.77	+2,829
2022	SPY	11	81.8	+0.37	+124.39	+1,368
2023	SPY	12	91.7	+0.40	+158.36	+1,900
2024	SPY	12	100.0	+0.39	+200.44	+2,405
2025	SPY	11	90.9	+0.44	+260.30	+2,863
2026	SPY	7	100.0	+0.56	+369.22	+2,585
2012	all names	14	92.9	+0.32	+34.87	+488
2013	all names	24	95.8	+0.34	+39.59	+950
2014	all names	24	91.7	+0.20	+25.44	+611
2015	all names	24	87.5	+0.10	+17.54	+421
2016	all names	24	100.0	+0.49	+71.15	+1,708
2017	all names	24	100.0	+0.35	+62.06	+1,490
2018	all names	24	83.3	−0.03	−18.23	−437

year	name	trades	win %	mean %	mean $	total $
2019	all names	22	100.0	+0.50	+109.67	+2,413
2020	all names	24	91.7	−1.02	−307.10	−7,370
2021	all names	24	100.0	+0.66	+234.15	+5,620
2022	all names	22	81.8	+0.25	+69.08	+1,520
2023	all names	24	95.8	+0.49	+171.23	+4,110
2024	all names	24	100.0	+0.45	+215.11	+5,163
2025	all names	22	90.9	+0.45	+251.67	+5,537
2026	all names	14	100.0	+0.66	+413.44	+5,788



Amendments, deviations and integrity checks. Covered calls not significantly negative (9ea4adf).

What it showed. Put-writing earned its premium like the index it is short; a carry, not an edge.

#8a V1 T0: short every monthly ATM straddle across the optionable universe

Folder: `research/compass/intel/, studies/v*.py, results/v1/ (@7dfea57)`. **Commits:** A8 29f20e3 (A8.1–A8.4) / — / aff15d9, df4fe32. **Dates:** 2026-10-15.

Hypothesis. Single-name monthly straddles are overpriced (volatility risk premium).

Rule. Sell the ATM straddle of every optionable name at the cycle's entry, hold to expiration (adverse fill: one tick worse).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side; mid ± one tick ("adv").

Data. The optionable universe 2023-11 → 2026-08, 44,447 name-cycles.

Periods and holdouts. development / validation / hold-out (each hold-out opened once).

Pass rule (as pre-registered). A8: negative T0 with sorts (T1 IV/RV, T2 term slope, T3 skew) adding a positive long-short.

Verdict. Rejected: T0 −13.2% of premium dev+val [−22.4, −5.0], −23.6% hold-out; T1–T3 add nothing.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
ADR	development	731	14 / 283	51.2	+46.9, +305.51	−77.4, −436.05	+1.3, +5.88	**−13.81**, −56.64	[−29.13, −3.90]; [−173.85, +26.14]
Common Stock	development	12,281	14 / 283	54.7	+47.5, +505.78	−73.6, −747.20	+8.5, +36.88	**−7.38**, −62.30	[−8.83, +0.77]; [−109.35, +10.32]
ETF	development	2,642	14 / 283	50.8	+44.8, +257.19	−61.0, −253.09	+1.2, +3.88	**−7.25**, +6.11	[−18.04, +8.70]; [−59.66, +84.56]
Structured Product	development	12	12 / 283	75.0	+51.1, +156.21	−42.9, −108.45	+35.2, +95.88	**+27.60**, +90.05	[−6.44, +50.40]; [−13.76, +129.89]
Unit	development	51	14 / 283	58.8	+42.2, +39.65	−94.1, −93.64	+15.2, +11.88	**−13.93**, −15.24	[−31.13, +23.27]; [−25.28, +24.29]
stock	development	14	14 / 283	57.1	+55.2, +105.38	−48.5, −77.29	+19.0, +26.88	**+10.77**, +27.09	[−17.45, +41.25]; [−26.68, +82.50]
all names	development	15,731	14 / 283	53.9	+47.0, +455.19	−71.5, −640.64	+7.1, +24.88	**−7.63**, −50.20	[−9.73, +0.78]; [−90.34, +12.98]
ADR	validation	761	12 / 239	50.6	+44.0, +355.98	−86.0, −584.97	+1.1, +4.82	**−20.23**, −108.93	[−32.01, −4.74]; [−189.57, +31.75]
Common Stock	validation	12,557	12 / 239	48.6	+45.8, +573.82	−82.4, −939.57	−2.7, −12.12	**−20.11**, −204.27	[−35.39, −5.24]; [−393.76, −25.91]
ETF	validation	2,832	12 / 239	45.8	+43.3, +287.12	−79.0, −499.64	−7.5, −20.12	**−22.94**, −139.04	[−51.85, −0.88]; [−393.80, +28.71]
Other	validation	2	2 / 21	0.0	—, —	−100.7, −335.62	−100.7, −335.62	**−100.69**, −335.62	n/a (< 60 sessions); n/a (< 60 sessions)
Structured Product	validation	12	12 / 239	58.3	+44.1, +241.74	−121.4, −746.72	+21.3, +107.38	**−24.86**, −170.12	[−139.67, +22.36]; [−756.56, +127.70]
Unit	validation	45	12 / 239	57.8	+51.3, +74.30	−75.4, −90.28	+26.0, +27.88	**−2.21**, +4.81	[−41.90, +38.57]; [−48.07, +54.95]
stock	validation	23	12 / 239	34.8	+45.0, +151.38	−103.5, −375.79	−33.8, −120.12	**−51.82**, −192.42	[−104.30, −6.97]; [−282.32, −12.45]
all names	validation	16,232	12 / 239	48.2	+45.3, +513.15	−82.0, −840.26	−3.5, −14.12	**−20.61**, −187.82	[−37.72, −4.97]; [−373.25, −17.85]
ADR	hold-out	588	8 / 155	46.3	+41.1, +557.43	−81.7, −791.98	−10.0, −33.12	**−24.91**, −167.77	[−46.04, −11.98]; [−503.04, +7.84]
Common Stock	hold-out	9,500	8 / 155	46.2	+43.9, +771.40	−78.3, −1,107.72	−7.9, −44.12	**−21.89**, −240.16	[−26.52, −13.80]; [−377.47, −37.68]

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
ETF	hold-out	2,296	8 / 155	37.2	+40.4, +375.41	−77.6, −662.56	−24.2, −77.62	**−33.69**, −276.03	[−53.03, −22.38]; [−495.65, −182.98]
Other	hold-out	11	7 / 155	45.5	+40.8, +348.67	−44.1, −418.63	−9.9, −169.14	**−5.51**, −69.86	[−53.68, +36.47]; [−459.63, +389.86]
Structured Product	hold-out	8	8 / 155	62.5	+45.2, +167.88	−24.8, −108.79	+16.6, +42.88	**+18.93**, +64.13	[−18.51, +45.31]; [−89.42, +132.74]
Unit	hold-out	34	8 / 155	44.1	+39.0, +61.01	−80.1, −104.49	−10.4, −18.12	**−27.57**, −31.47	[−53.97, +8.22]; [−68.41, +11.82]
stock	hold-out	47	8 / 155	42.6	+58.5, +660.32	−63.0, −621.76	−3.8, −40.13	**−11.31**, −76.19	[−51.78, +4.77]; [−610.34, +117.37]
all names	hold-out	12,484	8 / 155	44.5	+43.2, +696.78	−78.2, −995.02	−10.9, −49.12	**−24.13**, −241.82	[−31.78, −16.70]; [−367.78, −68.92]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
ADR	development	+4,425.75, +66.1%, 2024-07-22, ASML 2024-08-16 940.0 straddle	−11,414.21, −224.7%, 2024-01-22, ASML 2024-02-16 765.0 straddle	0.73	57,178	283	total ≤ 0 (best 5 days +31,201; without them −72,607)
Common Stock	development	+19,754.81, +36.9%, 2024-03-18, MSTR 2024-04-19 1500.0 straddle	−95,326.37, −739.7%, 2024-02-20, MSTR 2024-03-15 705.0 straddle	0.82	836,734	283	total ≤ 0 (best 5 days +268,994; without them −1,034,062)
ETF	development	+2,699.76, +41.9%, 2024-11-18, MSTX 2024-12-20 115.0 straddle	−5,420.16, −209.3%, 2024-05-20, NVDX 2024-06-21 105.0 straddle	1.05	116,495	282	765%
Structured Product	development	+490.88, +91.0%, 2024-12-23, VXX 2025-01-17 44.0 straddle	−181.12, −96.3%, 2024-04-22, VXX 2024-05-17 14.5 straddle	4.32	181	109	104%
Unit	development	+117.88, +73.7%, 2024-08-19, WES 2024-09-20 39.0 straddle	−443.12, −190.6%, 2023-11-20, NEP 2023-12-15 23.5 straddle	0.60	876	193	total ≤ 0 (best 5 days +691; without them −1,469)
stock	development	+191.88, +89.2%, 2024-10-21, SPLG 2024-11-15 69.0 straddle	−154.12, −83.3%, 2024-04-22, SPLG 2024-05-17 59.0 straddle	1.82	398	237	189%
all names	development	+19,754.81, +36.9%, 2024-03-18, MSTR 2024-04-19 1500.0 straddle	−95,326.37, −739.7%, 2024-02-20, MSTR 2024-03-15 705.0 straddle	0.83	918,409	283	total ≤ 0 (best 5 days +345,303; without them −1,134,960)
ADR	validation	+4,876.78, +87.2%, 2025-05-19, ASML 2025-06-20 750.0 straddle	−23,247.25, −315.6%, 2025-12-22, ASML 2026-01-16 1055.0 straddle	0.62	98,216	195	total ≤ 0 (best 5 days +26,071; without them −108,969)
Common Stock	validation	+21,042.30, +68.0%, 2025-05-19, BKNG 2025-06-20 5380.0 straddle	−45,597.98, −100.8%, 2025-04-21, BKNG 2025-05-16 4440.0 straddle	0.58	2,565,025	239	total ≤ 0 (best 5 days +64,256; without them −2,629,282)
ETF	validation	+4,382.79, +87.3%, 2025-08-18, ETHU 2025-09-19 155.0 straddle	−8,801.19, −210.3%, 2025-04-21, VGT 2025-05-16 490.0 straddle	0.49	416,912	239	total ≤ 0 (best 5 days +123,089; without them −516,862)
Other	validation	−198.12, −67.2%, 2025-06-23, RDW 2025-07-18 15.0 straddle	−473.12, −134.2%, 2025-07-21, RDW 2025-08-15 17.0 straddle	0.00	671	21	total ≤ 0 (best 5 days −671; without them +0)
Structured Product	validation	+533.88, +70.9%, 2025-05-19, VXX 2025-06-20 52.5 straddle	−2,173.12, −453.7%, 2025-03-24, VXX 2025-04-17 45.5 straddle	0.45	3,230	195	total ≤ 0 (best 5 days +1,477; without them −3,519)
Unit	validation	+202.88, +93.3%, 2025-07-21, MPLX 2025-08-15 50.0 straddle	−260.12, −125.4%, 2025-03-24, MPLX 2025-04-17 55.0 straddle	1.13	934	105	609%
stock	validation	+411.88, +56.8%, 2025-07-21, HIMZ 2025-08-15 19.0 straddle	−1,175.12, −162.1%, 2025-12-22, Q 2026-01-16 80.0 straddle	0.21	4,578	215	total ≤ 0 (best 5 days +717; without them −5,143)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
all names	validation	+21,042.30, +68.0%, 2025-05-19, BKNG 2025-06-20 5380.0 straddle	−45,597.98, −100.8%, 2025-04-21, BKNG 2025-05-16 4440.0 straddle	0.57	3,048,619	239	total ≤ 0 (best 5 days +121,340; without them −3,169,960)
ADR	hold-out	+21,363.40, +87.6%, 2026-07-20, ASML 2026-08-21 1740.0 straddle	−29,898.44, −178.8%, 2026-05-18, ASML 2026-06-18 1470.0 straddle	0.61	124,610	155	total ≤ 0 (best 5 days +22,101; without them −120,748)
Common Stock	hold-out	+30,840.83, +58.9%, 2026-07-20, SNDK 2026-08-21 1390.0 straddle	−65,571.76, −189.2%, 2026-01-20, BKNG 2026-02-20 5050.0 straddle	0.60	2,530,009	155	total ≤ 0 (best 5 days −790,140; without them −1,491,384)
ETF	hold-out	+7,362.74, +95.7%, 2026-07-20, SMH 2026-08-21 560.0 straddle	−47,328.44, −266.6%, 2026-05-18, MUU 2026-06-18 445.0 straddle	0.34	644,350	131	total ≤ 0 (best 5 days −166,531; without them −467,244)
Other	hold-out	+744.86, +43.1%, 2026-05-18, FCFS 2026-06-18 220.0 straddle	−1,184.14, −67.5%, 2026-04-20, FCFS 2026-05-15 200.0 straddle	0.69	1,579	111	total ≤ 0 (best 5 days +1,088; without them −1,857)
Structured Product	hold-out	+352.88, +60.8%, 2026-01-20, VXX 2026-02-20 29.5 straddle	−215.12, −43.3%, 2026-02-23, VXX 2026-03-20 29.5 straddle	2.57	215	85	164%
Unit	hold-out	+194.88, +74.2%, 2026-04-20, MPLX 2026-05-15 55.0 straddle	−296.12, −240.7%, 2026-01-20, EPD 2026-02-20 32.5 straddle	0.46	1,423	155	total ≤ 0 (best 5 days +329; without them −1,400)
stock	hold-out	+2,137.84, +74.9%, 2026-07-20, Q 2026-08-21 135.0 straddle	−1,752.13, −174.3%, 2026-06-22, SOLS 2026-07-17 85.0 straddle	0.79	9,304	155	total ≤ 0 (best 5 days +6,741; without them −10,323)
all names	hold-out	+30,840.83, +58.9%, 2026-07-20, SNDK 2026-08-21 1390.0 straddle	−65,571.76, −189.2%, 2026-01-20, BKNG 2026-02-20 5050.0 straddle	0.56	3,279,363	155	total ≤ 0 (best 5 days −1,248,558; without them −1,770,295)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.13). The study's fees are given beside them below. Re-pricing: exact (44,447 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
ADR	development	731	**−13.81**, −56.64	−14.31, −57.92	51.2 / 50.8	[−29.13, −3.90] / [−29.64, −4.41]
Common Stock	development	12,281	**−7.38**, −62.30	−7.69, −63.56	54.7 / 54.6	[−8.83, +0.77] / [−9.11, +0.46]
ETF	development	2,642	**−7.25**, +6.11	−7.86, +4.83	50.8 / 50.5	[−18.04, +8.70] / [−18.67, +8.12]
Structured Product	development	12	**+27.60**, +90.05	+27.04, +88.77	75.0 / 75.0	[−6.44, +50.40] / [−7.09, +49.70]
Unit	development	51	**−13.93**, −15.24	−15.55, −16.52	58.8 / 58.8	[−31.13, +23.27] / [−32.77, +21.79]
stock	development	14	**+10.77**, +27.09	+10.02, +25.81	57.1 / 57.1	[−17.45, +41.25] / [−18.20, +40.54]
all names	development	15,731	**−7.63**, −50.20	−8.01, −51.47	53.9 / 53.7	[−9.73, +0.78] / [−10.10, +0.41]
ADR	validation	761	**−20.23**, −108.93	−20.69, −110.21	50.6 / 50.2	[−32.01, −4.74] / [−32.48, −5.20]
Common Stock	validation	12,557	**−20.11**, −204.27	−20.38, −205.54	48.6 / 48.5	[−35.39, −5.24] / [−35.64, −5.51]
ETF	validation	2,832	**−22.94**, −139.04	−23.43, −140.32	45.8 / 45.4	[−51.85, −0.88] / [−52.31, −1.38]
Other	validation	2	**−100.69**, −335.62	−101.09, −336.90	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
Structured Product	validation	12	**−24.86**, −170.12	−25.09, −171.40	58.3 / 58.3	[−139.67, +22.36] / [−139.90, +22.16]

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
Unit	validation	45	**−2.21**, +4.81	−3.33, +3.53	57.8 / 57.8	[−41.90, +38.57] / [−43.01, +37.34]
stock	validation	23	**−51.82**, −192.42	−52.24, −193.70	34.8 / 34.8	[−104.30, −6.97] / [−104.78, −7.46]
all names	validation	16,232	**−20.61**, −187.82	−20.93, −189.08	48.2 / 48.0	[−37.72, −4.97] / [−38.05, −5.29]
ADR	hold-out	588	**−24.91**, −167.77	−25.31, −169.03	46.3 / 46.3	[−46.04, −11.98] / [−46.43, −12.36]
Common Stock	hold-out	9,500	**−21.89**, −240.16	−22.12, −241.42	46.2 / 46.0	[−26.52, −13.80] / [−26.76, −14.02]
ETF	hold-out	2,296	**−33.69**, −276.03	−34.11, −277.31	37.2 / 37.1	[−53.03, −22.38] / [−53.44, −22.80]
Other	hold-out	11	**−5.51**, −69.86	−5.76, −71.13	45.5 / 45.5	[−53.68, +36.47] / [−53.95, +36.37]
Structured Product	hold-out	8	**+18.93**, +64.13	+18.57, +62.85	62.5 / 62.5	[−18.51, +45.31] / [−18.81, +44.93]
Unit	hold-out	34	**−27.57**, −31.47	−28.55, −32.75	44.1 / 44.1	[−53.97, +8.22] / [−55.00, +7.31]
stock	hold-out	47	**−11.31**, −76.19	−11.47, −77.46	42.6 / 42.6	[−51.78, +4.77] / [−51.92, +4.60]
all names	hold-out	12,484	**−24.13**, −241.82	−24.41, −243.08	44.5 / 44.4	[−31.78, −16.70] / [−32.05, −16.96]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

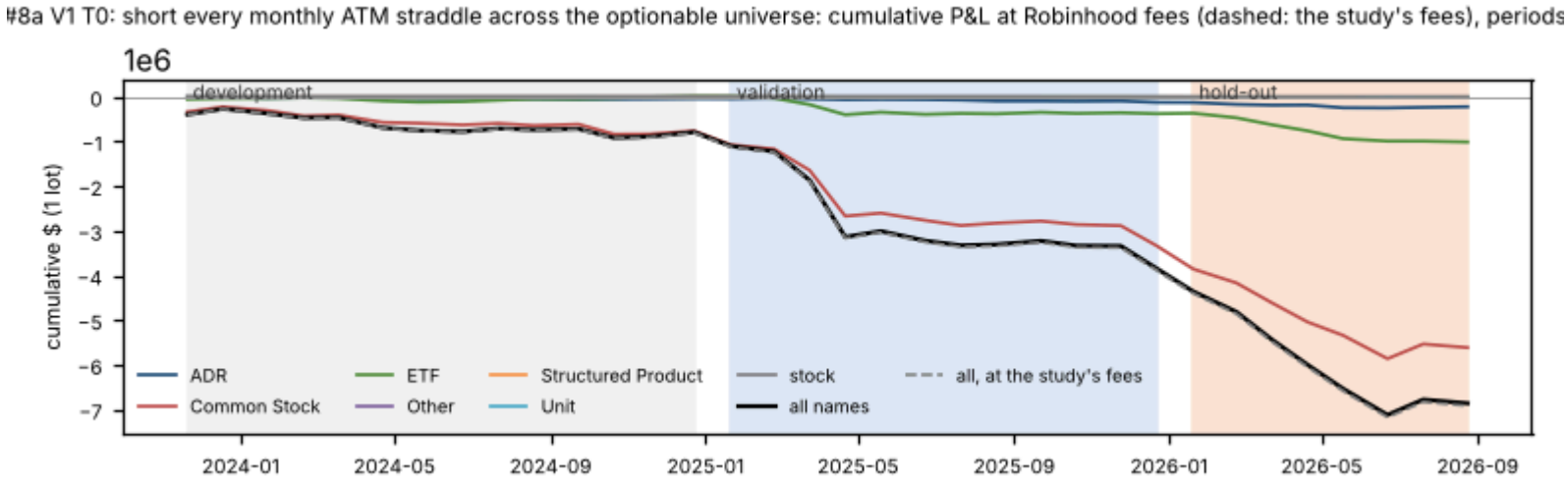
Exit reasons (share of trades):

- development: held to expiration 100.0%
- validation: held to expiration 100.0%
- hold-out: held to expiration 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2023	ADR	106	47.2	−13.65	−67.88	−7,195
2024	ADR	625	51.8	−13.84	−54.74	−34,211
2025	ADR	761	50.6	−20.23	−108.93	−82,898
2026	ADR	588	46.3	−24.91	−167.77	−98,647
2023	Common Stock	1,681	52.1	−22.75	−133.90	−225,093
2024	Common Stock	10,600	55.1	−4.94	−50.94	−539,975
2025	Common Stock	12,557	48.6	−20.11	−204.27	−2,565,025
2026	Common Stock	9,500	46.2	−21.89	−240.16	−2,281,524
2023	ETF	404	39.9	−32.49	−91.40	−36,924
2024	ETF	2,238	52.8	−2.70	+23.71	+53,059
2025	ETF	2,832	45.8	−22.94	−139.04	−393,774
2026	ETF	2,296	37.2	−33.69	−276.03	−633,775
2025	Other	2	0.0	−100.69	−335.62	−671
2026	Other	11	45.5	−5.51	−69.86	−768
2023	Structured Product	2	50.0	+16.90	+35.38	+71
2024	Structured Product	10	80.0	+29.74	+100.98	+1,010
2025	Structured Product	12	58.3	−24.86	−170.12	−2,041
2026	Structured Product	8	62.5	+18.93	+64.13	+513

year	name	trades	win %	mean %	mean $	total $
2023	Unit	7	71.4	+3.99	−43.69	−306
2024	Unit	44	56.8	−16.78	−10.71	−471
2025	Unit	45	57.8	−2.21	+4.81	+217
2026	Unit	34	44.1	−27.57	−31.47	−1,070
2023	stock	2	50.0	−23.70	−32.12	−64
2024	stock	12	58.3	+16.51	+36.96	+444
2025	stock	23	34.8	−51.82	−192.42	−4,426
2026	stock	47	42.6	−11.31	−76.19	−3,581
2023	all names	2,202	49.6	−23.98	−122.39	−269,511
2024	all names	13,529	54.6	−4.97	−38.45	−520,145
2025	all names	16,232	48.2	−20.61	−187.82	−3,048,619
2026	all names	12,484	44.5	−24.13	−241.82	−3,018,853



Amendments, deviations and integrity checks. Corporate actions detected from the options market (A8.1–A8.3c).

What it showed. Selling every straddle lost; the spread takes the premium.

#8b V2a: short ATM straddle across the earnings report

Folder: `research/compass/intel/, studies/v*.py, results/v1/ (@7dfea57)`. **Commits:** A9 47c2c2f / — / 4ed84d4, 152f786. **Dates:** 2026-10-15.

Hypothesis. Earnings straddles are overpriced.

Rule. Sell the ATM straddle at the close before the report, buy back after it (adverse fill).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side; adverse tick.

Data. 1,578 names, reports 2023-11 → 2026-09.

Periods and holdouts. development / validation / hold-out (each hold-out opened once).

Pass rule (as pre-registered). A9.

Verdict. Rejected: −26.1% dev+val [−28.5, −23.3], −34.2% hold-out.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
single names	development	3,558	234 / 276	40.2	+27.6, +301.83	−60.3, −622.70	−12.5, −52.20	**−25.03**, −251.38	[−27.70, −21.41]; [−288.01, −214.25]
single names	validation	4,244	218 / 246	33.0	+26.3, +355.73	−56.3, −553.12	−17.6, −86.20	**−29.02**, −252.89	[−33.20, −23.79]; [−315.21, −192.68]
single names	hold-out	3,682	165 / 191	26.2	+23.6, +339.98	−56.3, −632.44	−24.0, −146.71	**−35.40**, −378.11	[−38.38, −29.66]; [−411.00, −332.44]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
single names	development	+5,621.52, +31.6%, 2024-05-02, BKNG 2024-05-03 3475.0	−23,114.48, −129.2%, 2024-02-22, BKNG 2024-02-23 3900.0	0.33	898,664	266	total ≤ 0 (best 5 days +22,380; without them −916,788)
single names	validation	+26,833.24, +84.4%, 2025-02-20, BKNG 2025-02-21 5020.0	−18,091.28, −406.1%, 2025-09-09, SNPS 2025-09-12 605.0	0.32	1,074,808	244	total ≤ 0 (best 5 days +43,663; without them −1,116,914)
single names	hold-out	+5,019.46, +30.4%, 2026-08-05, SNDK 2026-08-07 1350.0	−22,300.32, −330.6%, 2026-05-04, STRL 2026-05-15 530.0	0.19	1,399,007	191	total ≤ 0 (best 5 days +19,525; without them −1,411,739)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.67 a trade) and Robinhood's charged instead (mean $0.21). The study's fees are given beside them below. Re-pricing: exact (11,484 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
single names	development	3,558	**−25.03**, −251.38	−25.67, −253.83	40.2 / 39.2	[−27.70, −21.41] / [−28.33, −22.08]
single names	validation	4,244	**−29.02**, −252.89	−29.62, −255.34	33.0 / 32.5	[−33.20, −23.79] / [−33.80, −24.38]

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
single names	hold-out	3,682	**−35.40**, −378.11	−35.95, −380.58	26.2 / 25.6	[−38.38, −29.66] / [−38.94, −30.24]

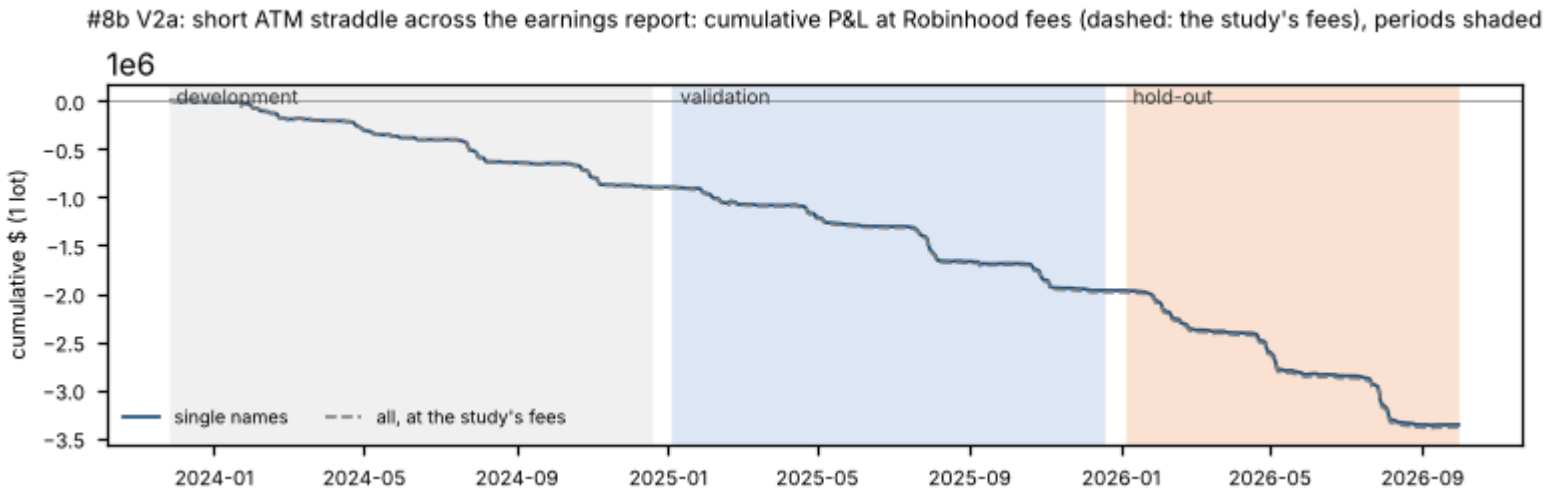
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- development: closed after the report 100.0%
- validation: closed after the report 100.0%
- hold-out: closed after the report 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2023	single names	71	47.9	−19.84	−99.28	−7,049
2024	single names	3,487	40.0	−25.13	−254.48	−887,359
2025	single names	4,244	33.0	−29.02	−252.89	−1,073,251
2026	single names	3,682	26.2	−35.40	−378.11	−1,392,215



Amendments, deviations and integrity checks. Friday-only expiry probe checked on all names (95c90f9: no change).

What it showed. Selling earnings straddles lost a quarter of the premium.

#8c V2b: long ATM straddle into the earnings report

Folder: `research/compass/intel/, studies/v*.py, results/v1/ (@7dfea57)`. **Commits:** A9 47c2c2f / — / 152f786. **Dates:** 2026-10-15.

Hypothesis. Buying into the report captures the run-up in implied volatility.

Rule. Buy the ATM straddle five sessions before the report, sell at the report.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). As V2a.

Data. As V2a.

Periods and holdouts. development / validation / hold-out (each hold-out opened once).

Pass rule (as pre-registered). A9.

Verdict. Rejected: −20.9% / −29.3%.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
single names	development	3,485	234 / 277	21.2	+18.0, +204.81	−22.5, −231.86	−13.2, −74.20	**−13.91**, −139.26	[−15.90, −11.09]; [−165.27, −101.68]
single names	validation	4,100	217 / 246	9.2	+20.2, +272.46	−34.2, −369.07	−26.4, −194.20	**−29.18**, −309.92	[−33.00, −24.91]; [−356.97, −263.79]
single names	hold-out	3,543	164 / 191	10.2	+24.0, +483.68	−36.8, −433.77	−28.6, −232.20	**−30.57**, −339.77	[−34.38, −25.36]; [−393.53, −267.97]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
single names	development	+8,234.62, +93.4%, 2024-10-30, TPL 2024-11-15 1150.0	−11,695.66, −50.7%, 2024-04-22, MSTR 2024-05-03 1320.0	0.24	485,792	277	total ≤ 0 (best 5 days +22,585; without them −507,907)
single names	validation	+10,494.60, +99.1%, 2025-02-18, AXON 2025-03-21 710.0	−10,930.90, −32.1%, 2025-07-22, BKNG 2025-08-01 5775.0	0.07	1,270,690	246	total ≤ 0 (best 5 days +3,917; without them −1,274,607)
single names	hold-out	+19,714.24, +71.8%, 2026-05-21, AZO 2026-06-18 3450.0	−11,370.84, −36.5%, 2026-02-10, BKNG 2026-02-20 4300.0	0.13	1,203,797	191	total ≤ 0 (best 5 days +27,607; without them −1,231,404)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.80 a trade) and Robinhood's charged instead (mean $0.22). The study's fees are given beside them below. Re-pricing: exact (11,128 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
single names	development	3,485	**−13.91**, −139.26	−14.51, −141.84	21.2 / 20.3	[−15.90, −11.09] / [−16.48, −11.70]
single names	validation	4,100	**−29.18**, −309.92	−29.71, −312.50	9.2 / 9.0	[−33.00, −24.91] / [−33.51, −25.43]

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
single names	hold-out	3,543	**−30.57**, −339.77	−31.05, −342.34	10.2 / 10.1	[−34.38, −25.36] / [−34.85, −25.85]

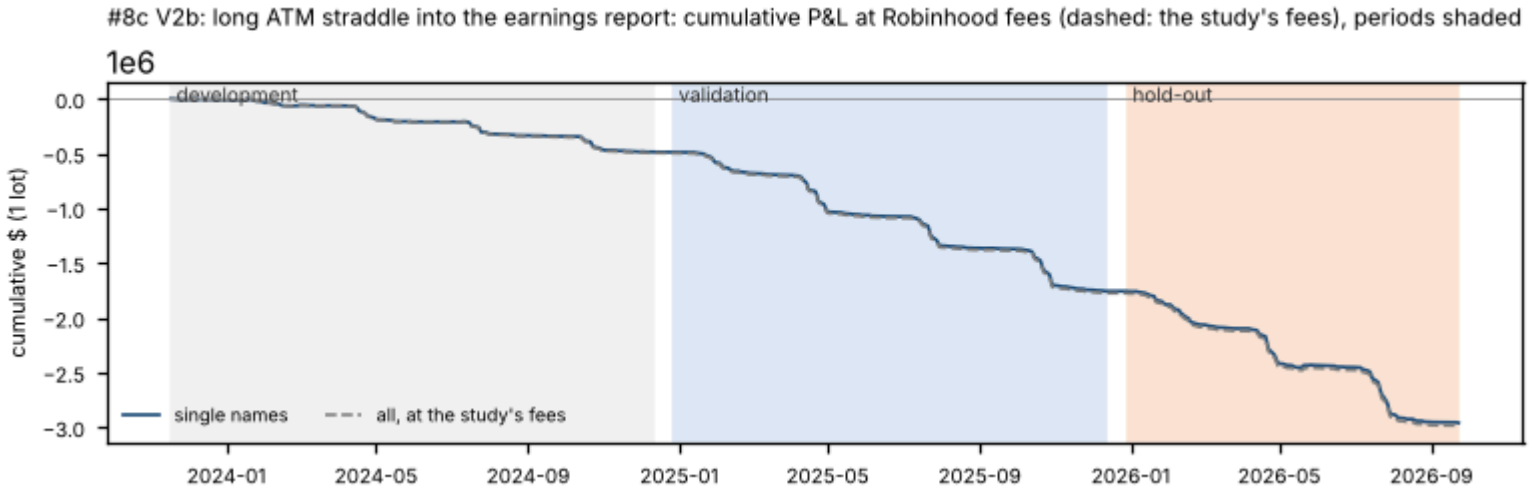
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- development: closed at the report 100.0%
- validation: closed at the report 100.0%
- hold-out: closed at the report 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2023	single names	76	17.1	−9.07	−95.74	−7,277
2024	single names	3,417	21.2	−14.06	−140.45	−479,930
2025	single names	4,101	9.2	−29.19	−310.06	−1,271,575
2026	single names	3,534	10.2	−30.55	−339.85	−1,201,028



Amendments, deviations and integrity checks. —

What it showed. Both sides of the earnings straddle lose.

#8d V3a: single-name 0DTE short ATM straddle 10:00 → 15:50

Folder: `research/compass/intel/, studies/v*.py, results/v1/ (@7dfea57)`. **Commits:** A10–A10.4 9c476c5, 048d33a / — / e4e4cea. **Dates:** 2026-10-16.

Hypothesis. Single-name same-day premium is overpriced.

Rule. Sell the ATM straddle at 10:00, buy back 15:50 (adverse).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side; adverse tick.

Data. 11 Mon/Wed/Fri names 2017-06 → 2026-10.

Periods and holdouts. development / validation / hold-out (each hold-out opened once).

Pass rule (as pre-registered). A10.

Verdict. Rejected: −19.8% dev+val [−25.5, −14.7], −7.4% hold-out.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
AAPL	development	283	283 / 1,446	49.8	+45.7, +84.21	−81.0, −152.69	−0.2, −0.20	**−17.87**, −34.66	[−26.74, −4.24]; [−56.36, −11.16]
AMD	development	284	284 / 1,446	50.4	+38.6, +44.78	−78.3, −75.02	+1.8, +0.80	**−19.44**, −14.70	[−28.06, −9.69]; [−26.95, −3.56]
AMZN	development	283	283 / 1,446	51.6	+40.5, +802.01	−81.0, −1,484.71	+4.1, +23.78	**−18.33**, −304.99	[−26.52, −7.14]; [−520.81, −95.64]
AVGO	development	284	284 / 1,446	31.7	+33.2, +129.67	−97.1, −370.79	−37.7, −126.20	**−55.78**, −212.20	[−66.69, −42.24]; [−262.59, −158.95]
GOOGL	development	283	283 / 1,446	41.7	+36.9, +473.30	−89.1, −1,069.53	−14.7, −104.20	**−36.58**, −426.23	[−46.05, −23.83]; [−583.44, −271.48]
INTC	development	283	283 / 1,446	41.0	+35.8, +19.76	−86.9, −42.88	−12.3, −6.20	**−36.59**, −17.20	[−48.71, −24.04]; [−23.72, −10.92]
META	development	284	284 / 1,446	52.5	+43.2, +108.30	−70.4, −167.28	+5.0, +12.80	**−10.81**, −22.69	[−17.22, −1.13]; [−42.29, −0.45]
MSFT	development	284	284 / 1,446	45.1	+38.2, +67.15	−92.7, −141.54	−10.5, −10.70	**−33.74**, −47.48	[−43.70, −22.16]; [−67.09, −29.27]
MU	development	284	284 / 1,446	51.1	+38.8, +32.45	−81.5, −66.99	+4.3, +2.80	**−20.08**, −16.22	[−29.24, −8.91]; [−24.51, −7.51]
NVDA	development	283	283 / 1,446	55.8	+41.0, +168.40	−71.8, −329.11	+11.1, +36.80	**−8.78**, −51.35	[−16.83, −0.31]; [−93.86, −7.94]
TSLA	development	282	282 / 1,446	57.1	+47.1, +482.48	−84.7, −858.84	+17.1, +99.80	**−9.48**, −93.05	[−18.33, −0.81]; [−215.17, +22.77]
all names	development	3,117	284 / 1,446	48.0	+40.4, +226.69	−84.0, −425.57	−3.8, −5.20	**−24.33**, −112.72	[−28.82, −17.16]; [−148.99, −72.62]
AAPL	validation	102	102 / 512	60.8	+49.3, +76.86	−63.3, −104.25	+23.2, +39.30	**+5.16**, +5.84	[−0.92, +21.03]; [−3.78, +31.69]
AMD	validation	102	102 / 512	58.8	+46.8, +99.75	−80.3, −166.80	+20.9, +32.80	**−5.52**, −10.01	[−15.28, +11.32]; [−38.48, +23.38]
AMZN	validation	102	102 / 512	52.9	+45.3, +76.87	−68.6, −107.16	+5.9, +10.30	**−8.31**, −9.73	[−16.57, +7.97]; [−22.77, +16.20]
AVGO	validation	101	101 / 512	48.5	+36.5, +319.04	−89.4, −815.52	−1.1, −11.22	**−28.33**, −265.09	[−45.56, −9.94]; [−446.57, −96.17]
GOOGL	validation	102	102 / 512	53.9	+42.0, +55.09	−88.1, −110.48	+8.7, +11.30	**−17.99**, −21.20	[−31.46, −1.42]; [−35.16, +0.48]
INTC	validation	102	102 / 512	62.7	+36.3, +17.13	−91.6, −40.10	+16.2, +6.30	**−11.37**, −4.19	[−20.89, +3.78]; [−9.40, +2.60]
META	validation	102	102 / 512	56.9	+49.2, +225.42	−97.7, −386.70	+18.0, +62.80	**−14.17**, −38.63	[−32.49, +1.89]; [−119.61, +33.39]
MSFT	validation	102	102 / 512	50.0	+39.5, +118.88	−71.8, −197.42	+1.1, +2.80	**−16.14**, −39.27	[−31.11, −2.06]; [−76.64, −0.82]
MU	validation	102	102 / 512	54.9	+38.3, +46.16	−68.0, −81.46	+9.4, +10.30	**−9.65**, −11.40	[−16.99, +4.15]; [−22.10, +5.98]

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
NVDA	validation	101	101 / 512	54.5	+47.4, +239.69	−90.3, −647.64	+4.1, +13.80	**−15.30**, −164.44	[−30.21, +7.22]; [−364.49, −5.92]
TSLA	validation	102	102 / 512	54.9	+51.7, +219.24	−71.3, −262.64	+14.3, +41.80	**−3.76**, +1.93	[−14.69, +13.33]; [−44.26, +68.41]
all names	validation	1,120	102 / 512	55.4	+43.9, +132.00	−80.0, −276.29	+10.3, +12.30	**−11.38**, −50.27	[−17.28, −0.43]; [−88.36, −9.27]
AAPL	hold-out	161	161 / 456	54.0	+44.8, +109.86	−80.5, −191.23	+6.5, +14.80	**−12.79**, −28.53	[−29.37, −0.27]; [−66.61, +2.54]
AMD	hold-out	116	116 / 456	60.3	+44.9, +255.78	−70.1, −323.03	+20.6, +78.80	**−0.73**, +26.25	[−14.94, +8.72]; [−39.74, +75.10]
AMZN	hold-out	161	161 / 456	60.9	+40.7, +101.80	−74.7, −182.01	+17.3, +36.80	**−4.44**, −9.26	[−17.22, +6.96]; [−37.65, +17.31]
AVGO	hold-out	159	159 / 456	53.5	+37.6, +186.18	−57.0, −318.46	+7.5, +34.80	**−6.44**, −48.69	[−15.07, +2.08]; [−103.74, +0.86]
GOOGL	hold-out	159	159 / 456	52.2	+45.7, +144.28	−73.7, −219.04	+5.3, +13.80	**−11.39**, −29.38	[−25.56, −1.45]; [−72.75, +2.67]
INTC	hold-out	116	116 / 456	54.3	+37.7, +57.70	−75.4, −97.03	+5.5, +4.80	**−13.97**, −12.99	[−31.20, −1.03]; [−35.43, +4.57]
META	hold-out	159	159 / 456	55.3	+37.4, +293.02	−71.4, −499.71	+3.7, +28.80	**−11.19**, −60.96	[−25.81, −3.35]; [−159.79, −8.66]
MSFT	hold-out	159	159 / 456	52.2	+44.3, +174.96	−79.3, −288.00	+7.3, +27.80	**−14.77**, −46.33	[−34.37, −2.24]; [−121.96, +3.72]
MU	hold-out	115	115 / 456	62.6	+40.0, +515.28	−94.5, −791.52	+11.9, +57.80	**−10.27**, +26.65	[−27.84, +6.42]; [−151.18, +235.33]
NVDA	hold-out	158	158 / 456	65.8	+44.3, +106.99	−86.9, −196.65	+22.8, +54.80	**−0.56**, +3.22	[−15.00, +12.08]; [−29.62, +32.92]
TSLA	hold-out	159	159 / 456	59.7	+47.5, +292.67	−87.3, −522.37	+21.9, +117.80	**−6.77**, −35.40	[−20.47, +5.64]; [−115.37, +43.42]
all names	hold-out	1,622	161 / 456	57.2	+42.4, +199.58	−76.6, −318.86	+10.9, +32.80	**−8.50**, −22.24	[−18.41, −1.91]; [−62.38, +8.26]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
AAPL	development	+438.80, +59.9%, 2020-08-28, AAPL 10:00 short straddle 505.0	−1,147.20, −146.1%, 2020-03-20, AAPL 10:00 short straddle 250.0	0.55	9,952	1,446	total ≤ 0 (best 5 days +1,622; without them −11,431)
AMD	development	+223.80, +74.1%, 2022-05-06, AMD 10:00 short straddle 95.0	−838.20, −262.3%, 2022-02-11, AMD 10:00 short straddle 123.0	0.61	4,832	1,446	total ≤ 0 (best 5 days +1,030; without them −5,205)
AMZN	development	+3,219.66, +46.9%, 2020-09-04, AMZN 10:00 short straddle 3275.0	−7,893.22, −413.3%, 2021-10-15, AMZN 10:00 short straddle 3310.0	0.58	90,721	1,446	total ≤ 0 (best 5 days +13,718; without them −100,029)
AVGO	development	+412.80, +55.4%, 2022-06-10, AVGO 10:00 short straddle 545.0	−1,891.20, −244.0%, 2022-08-26, AVGO 10:00 short straddle 547.5	0.16	60,878	1,446	total ≤ 0 (best 5 days +1,855; without them −62,119)
GOOGL	development	+2,033.74, +68.0%, 2022-07-15, GOOGL 10:00 short straddle 2225.0	−6,481.22, −498.6%, 2021-09-10, GOOGL 10:00 short straddle 2890.0	0.32	122,179	1,446	total ≤ 0 (best 5 days +8,920; without them −129,543)
INTC	development	+94.80, +62.2%, 2020-07-24, INTC 10:00 short straddle 50.0	−382.20, −534.5%, 2018-03-29, INTC 10:00 short straddle 50.0	0.32	4,868	1,446	total ≤ 0 (best 5 days +338; without them −5,206)
META	development	+324.80, +64.5%, 2021-12-03, META 10:00 short straddle 305.0	−891.20, −221.4%, 2020-06-26, META 10:00 short straddle 227.5	0.71	6,962	1,446	total ≤ 0 (best 5 days +1,533; without them −7,978)
MSFT	development	+267.80, +50.3%, 2022-01-21, MSFT 10:00 short straddle 300.0	−844.20, −226.6%, 2022-08-26, MSFT 10:00 short straddle 280.0	0.39	13,546	1,446	total ≤ 0 (best 5 days +1,200; without them −14,685)
MU	development	+120.80, +65.7%, 2022-01-21, MU 10:00 short straddle 82.5	−341.20, −300.6%, 2020-06-12, MU 10:00 short straddle 50.0	0.51	4,732	1,446	total ≤ 0 (best 5 days +445; without them −5,051)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
NVDA	development	+662.80, +76.2%, 2020-05-22, NVDA 10:00 short straddle 360.0	−2,157.20, −243.8%, 2020-11-06, NVDA 10:00 short straddle 557.5	0.65	14,965	1,441	total ≤ 0 (best 5 days +2,830; without them −17,362)
TSLA	development	+2,874.70, +50.6%, 2020-08-21, TSLA 10:00 short straddle 2060.0	−12,862.24, −442.8%, 2020-07-10, TSLA 10:00 short straddle 1385.0	0.75	27,343	1,441	total ≤ 0 (best 5 days +10,858; without them −37,099)
all names	development	+3,219.66, +46.9%, 2020-09-04, AMZN 10:00 short straddle 3275.0	−12,862.24, −442.8%, 2020-07-10, TSLA 10:00 short straddle 1385.0	0.49	353,850	1,446	total ≤ 0 (best 5 days +27,310; without them −378,672)
AAPL	validation	+198.80, +55.5%, 2024-08-02, AAPL 10:00 short straddle 222.5	−451.20, −213.3%, 2024-12-20, AAPL 10:00 short straddle 247.5	1.14	591	105	131%
AMD	validation	+260.80, +80.7%, 2024-01-26, AMD 10:00 short straddle 177.5	−873.20, −328.9%, 2024-01-19, AMD 10:00 short straddle 162.5	0.85	1,679	453	total ≤ 0 (best 5 days +1,186; without them −2,207)
AMZN	validation	+201.80, +75.3%, 2023-10-27, AMZN 10:00 short straddle 128.0	−337.20, −242.6%, 2024-02-09, AMZN 10:00 short straddle 170.0	0.81	1,252	512	total ≤ 0 (best 5 days +815; without them −1,808)
AVGO	validation	+1,858.76, +67.1%, 2024-06-21, AVGO 10:00 short straddle 1680.0	−4,831.22, −406.0%, 2023-05-26, AVGO 10:00 short straddle 755.0	0.37	27,246	512	total ≤ 0 (best 5 days +5,488; without them −32,262)
GOOGL	validation	+129.80, +78.4%, 2024-12-27, GOOGL 10:00 short straddle 192.5	−505.20, −204.9%, 2024-12-20, GOOGL 10:00 short straddle 185.0	0.58	2,292	512	total ≤ 0 (best 5 days +604; without them −2,767)
INTC	validation	+54.80, +52.4%, 2024-08-02, INTC 10:00 short straddle 21.0	−175.20, −216.3%, 2024-03-08, INTC 10:00 short straddle 46.0	0.72	520	512	total ≤ 0 (best 5 days +206; without them −634)
META	validation	+966.78, +85.0%, 2024-02-02, META 10:00 short straddle 475.0	−1,493.20, −323.2%, 2024-07-05, META 10:00 short straddle 522.5	0.77	4,914	512	total ≤ 0 (best 5 days +3,136; without them −7,077)
MSFT	validation	+336.80, +67.1%, 2024-08-02, MSFT 10:00 short straddle 407.5	−771.20, −357.0%, 2023-04-06, MSFT 10:00 short straddle 282.5	0.60	4,775	512	total ≤ 0 (best 5 days +1,278; without them −5,284)
MU	validation	+135.80, +68.8%, 2024-04-05, MU 10:00 short straddle 123.0	−356.20, −137.3%, 2024-08-02, MU 10:00 short straddle 96.0	0.69	1,260	512	total ≤ 0 (best 5 days +563; without them −1,726)
NVDA	validation	+1,068.78, +77.2%, 2024-05-03, NVDA 10:00 short straddle 887.5	−6,702.22, −541.6%, 2024-04-19, NVDA 10:00 short straddle 840.0	0.44	17,826	378	total ≤ 0 (best 5 days +4,328; without them −20,937)
TSLA	validation	+573.80, +69.1%, 2024-11-15, TSLA 10:00 short straddle 317.5	−1,069.20, −290.5%, 2023-01-27, TSLA 10:00 short straddle 165.0	1.02	2,458	343	1194%
all names	validation	+1,858.76, +67.1%, 2024-06-21, AVGO 10:00 short straddle 1680.0	−6,702.22, −541.6%, 2024-04-19, NVDA 10:00 short straddle 840.0	0.59	59,100	512	total ≤ 0 (best 5 days +9,405; without them −65,708)
AAPL	hold-out	+336.80, +61.2%, 2026-09-09, AAPL 10:00 short straddle 317.5	−733.20, −417.8%, 2025-10-10, AAPL 10:00 short straddle 255.0	0.68	6,414	443	total ≤ 0 (best 5 days +1,181; without them −5,774)
AMD	hold-out	+845.78, +58.7%, 2026-06-12, AMD 10:00 short straddle 510.0	−2,432.22, −199.4%, 2026-07-02, AMD 10:00 short straddle 545.0	1.20	3,520	252	118%
AMZN	hold-out	+303.80, +73.0%, 2026-08-03, AMZN 10:00 short straddle 285.0	−913.20, −493.6%, 2025-10-10, AMZN 10:00 short straddle 227.5	0.87	3,747	293	total ≤ 0 (best 5 days +1,386; without them −2,876)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
AVGO	hold-out	+438.80, +58.1%, 2026-05-15, AVGO 10:00 short straddle 425.0	−3,666.24, −137.1%, 2026-09-02, AVGO 10:00 short straddle 375.0	0.67	7,809	447	total ≤ 0 (best 5 days +2,057; without them −9,798)
GOOGL	hold-out	+397.80, +81.2%, 2026-02-06, GOOGL 10:00 short straddle 322.5	−1,528.20, −384.9%, 2026-08-05, GOOGL 10:00 short straddle 380.0	0.72	5,992	275	total ≤ 0 (best 5 days +1,622; without them −6,294)
INTC	hold-out	+238.80, +72.4%, 2026-04-24, INTC 10:00 short straddle 82.0	−895.20, −259.1%, 2026-07-02, INTC 10:00 short straddle 130.0	0.71	2,143	456	total ≤ 0 (best 5 days +980; without them −2,487)
META	hold-out	+1,095.78, +76.9%, 2025-04-11, META 10:00 short straddle 542.5	−2,417.22, −222.8%, 2026-09-21, META 10:00 short straddle 712.5	0.73	13,402	438	total ≤ 0 (best 5 days +3,825; without them −13,518)
MSFT	hold-out	+412.80, +63.8%, 2026-07-31, MSFT 10:00 short straddle 462.5	−952.20, −269.4%, 2026-06-17, MSFT 10:00 short straddle 390.0	0.66	10,826	438	total ≤ 0 (best 5 days +1,788; without them −9,155)
MU	hold-out	+2,454.74, +71.8%, 2026-06-12, MU 10:00 short straddle 985.0	−7,237.26, −236.9%, 2026-07-02, MU 10:00 short straddle 1060.0	1.09	9,813	361	288%
NVDA	hold-out	+271.80, +83.9%, 2026-06-08, NVDA 10:00 short straddle 207.5	−803.20, −378.9%, 2025-10-10, NVDA 10:00 short straddle 195.0	1.05	2,421	145	250%
TSLA	hold-out	+798.78, +77.4%, 2025-11-21, TSLA 10:00 short straddle 395.0	−1,992.20, −427.1%, 2026-04-15, TSLA 10:00 short straddle 370.0	0.83	9,938	456	total ≤ 0 (best 5 days +3,470; without them −9,098)
all names	hold-out	+2,454.74, +71.8%, 2026-06-12, MU 10:00 short straddle 985.0	−7,237.26, −236.9%, 2026-07-02, MU 10:00 short straddle 1060.0	0.84	57,068	443	total ≤ 0 (best 5 days +18,242; without them −54,316)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.80 a trade) and Robinhood's charged instead (mean $0.20). The study's fees are given beside them below. Re-pricing: exact (5,859 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
AAPL	development	283	**−17.87**, −34.66	−19.52, −37.26	49.8 / 49.1	[−26.74, −4.24] / [−28.35, −5.92]
AMD	development	284	**−19.44**, −14.70	−24.12, −17.30	50.4 / 48.2	[−28.06, −9.69] / [−32.87, −14.20]
AMZN	development	283	**−18.33**, −304.99	−18.63, −307.55	51.6 / 51.6	[−26.52, −7.14] / [−26.79, −7.45]
AVGO	development	284	**−55.78**, −212.20	−56.57, −214.80	31.7 / 31.0	[−66.69, −42.24] / [−67.43, −43.02]
GOOGL	development	283	**−36.58**, −426.23	−36.98, −428.81	41.7 / 41.7	[−46.05, −23.83] / [−46.44, −24.19]
INTC	development	283	**−36.59**, −17.20	−42.26, −19.80	41.0 / 39.9	[−48.71, −24.04] / [−54.48, −29.44]
META	development	284	**−10.81**, −22.69	−12.07, −25.29	52.5 / 52.1	[−17.22, −1.13] / [−18.45, −2.33]
MSFT	development	284	**−33.74**, −47.48	−36.16, −50.08	45.1 / 43.7	[−43.70, −22.16] / [−46.26, −24.48]
MU	development	284	**−20.08**, −16.22	−23.59, −18.82	51.1 / 50.7	[−29.24, −8.91] / [−32.65, −12.50]
NVDA	development	283	**−8.78**, −51.35	−9.58, −53.95	55.8 / 55.8	[−16.83, −0.31] / [−17.63, −1.10]
TSLA	development	282	**−9.48**, −93.05	−9.89, −95.64	57.1 / 57.1	[−18.33, −0.81] / [−18.72, −1.20]
all names	development	3,117	**−24.33**, −112.72	−26.32, −115.32	48.0 / 47.4	[−28.82, −17.16] / [−30.82, −19.20]
AAPL	validation	102	**+5.16**, +5.84	+3.45, +3.24	60.8 / 60.8	[−0.92, +21.03] / [−2.56, +19.29]
AMD	validation	102	**−5.52**, −10.01	−6.93, −12.60	58.8 / 57.8	[−15.28, +11.32] / [−16.69, +10.03]

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
AMZN	validation	102	**−8.31**, −9.73	−10.03, −12.33	52.9 / 52.9	[−16.57, +7.97] / [−18.30, +6.15]
AVGO	validation	101	**−28.33**, −265.09	−28.79, −267.68	48.5 / 47.5	[−45.56, −9.94] / [−46.02, −10.45]
GOOGL	validation	102	**−17.99**, −21.20	−20.12, −23.80	53.9 / 52.9	[−31.46, −1.42] / [−33.58, −3.41]
INTC	validation	102	**−11.37**, −4.19	−17.56, −6.79	62.7 / 59.8	[−20.89, +3.78] / [−27.04, −2.17]
META	validation	102	**−14.17**, −38.63	−14.87, −41.23	56.9 / 56.9	[−32.49, +1.89] / [−33.14, +1.23]
MSFT	validation	102	**−16.14**, −39.27	−17.09, −41.87	50.0 / 50.0	[−31.11, −2.06] / [−32.02, −2.96]
MU	validation	102	**−9.65**, −11.40	−12.21, −14.00	54.9 / 54.9	[−16.99, +4.15] / [−19.70, +1.67]
NVDA	validation	101	**−15.30**, −164.44	−16.01, −167.04	54.5 / 54.5	[−30.21, +7.22] / [−30.76, +6.52]
TSLA	validation	102	**−3.76**, +1.93	−4.47, −0.67	54.9 / 54.9	[−14.69, +13.33] / [−15.41, +12.65]
all names	validation	1,120	**−11.38**, −50.27	−13.13, −52.87	55.4 / 54.8	[−17.28, −0.43] / [−19.02, −2.17]
AAPL	hold-out	161	**−12.79**, −28.53	−13.89, −31.13	54.0 / 53.4	[−29.37, −0.27] / [−30.49, −1.41]
AMD	hold-out	116	**−0.73**, +26.25	−1.50, +23.66	60.3 / 60.3	[−14.94, +8.72] / [−15.86, +8.01]
AMZN	hold-out	161	**−4.44**, −9.26	−5.53, −11.86	60.9 / 60.9	[−17.22, +6.96] / [−18.28, +5.86]
AVGO	hold-out	159	**−6.44**, −48.69	−7.00, −51.28	53.5 / 53.5	[−15.07, +2.08] / [−15.63, +1.50]
GOOGL	hold-out	159	**−11.39**, −29.38	−12.32, −31.98	52.2 / 51.6	[−25.56, −1.45] / [−26.47, −2.39]
INTC	hold-out	116	**−13.97**, −12.99	−17.28, −15.59	54.3 / 51.7	[−31.20, −1.03] / [−34.85, −4.09]
META	hold-out	159	**−11.19**, −60.96	−11.56, −63.56	55.3 / 55.3	[−25.81, −3.35] / [−26.18, −3.73]
MSFT	hold-out	159	**−14.77**, −46.33	−15.47, −48.93	52.2 / 50.9	[−34.37, −2.24] / [−35.07, −2.94]
MU	hold-out	115	**−10.27**, +26.65	−10.96, +24.07	62.6 / 60.0	[−27.84, +6.42] / [−28.77, +5.92]
NVDA	hold-out	158	**−0.56**, +3.22	−1.71, +0.62	65.8 / 63.3	[−15.00, +12.08] / [−16.12, +10.93]
TSLA	hold-out	159	**−6.77**, −35.40	−7.21, −37.99	59.7 / 59.7	[−20.47, +5.64] / [−20.89, +5.22]
all names	hold-out	1,622	**−8.50**, −22.24	−9.47, −24.84	57.2 / 56.4	[−18.41, −1.91] / [−19.41, −2.80]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- development: bought back 15:50 100.0%
- validation: bought back 15:50 100.0%
- hold-out: bought back 15:50 100.0%

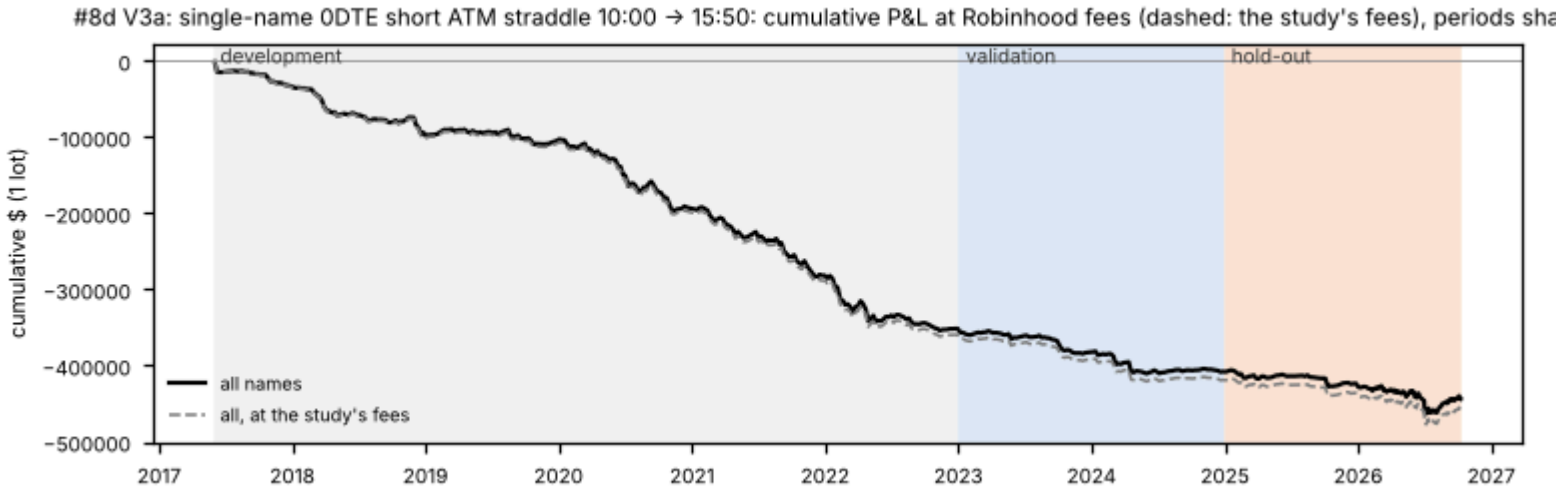
By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2017	AAPL	30	53.3	−38.65	−27.40	−822
2018	AAPL	51	47.1	−25.99	−37.99	−1,937
2019	AAPL	51	58.8	−5.75	−13.22	−674
2020	AAPL	49	42.9	−18.00	−84.67	−4,149
2021	AAPL	51	51.0	−13.29	−11.57	−590
2022	AAPL	51	47.1	−14.08	−32.08	−1,636
2023	AAPL	51	58.8	+6.16	+9.96	+508
2024	AAPL	51	62.7	+4.16	+1.72	+88

year	name	trades	win %	mean %	mean $	total $
2025	AAPL	50	50.0	−23.90	−52.76	−2,638
2026	AAPL	111	55.9	−7.78	−17.62	−1,955
2017	AMD	30	50.0	−30.95	−7.77	−233
2018	AMD	51	45.1	−40.00	−17.87	−911
2019	AMD	51	45.1	−15.26	−6.89	−351
2020	AMD	50	48.0	−21.31	−26.08	−1,304
2021	AMD	51	51.0	−15.00	−18.75	−956
2022	AMD	51	62.7	+1.11	−8.22	−419
2023	AMD	51	56.9	−9.40	−7.75	−395
2024	AMD	51	60.8	−1.64	−12.26	−625
2025	AMD	50	54.0	−13.31	−19.66	−983
2026	AMD	66	65.2	+8.80	+61.03	+4,028
2017	AMZN	30	46.7	−49.74	−322.50	−9,675
2018	AMZN	51	52.9	−21.37	−380.11	−19,386
2019	AMZN	51	64.7	−1.96	−30.20	−1,540
2020	AMZN	50	44.0	−19.52	−443.86	−22,193
2021	AMZN	51	51.0	−22.41	−438.78	−22,378
2022	AMZN	50	48.0	−7.71	−222.80	−11,140
2023	AMZN	51	52.9	−8.15	−6.67	−340
2024	AMZN	51	52.9	−8.47	−12.79	−652
2025	AMZN	50	70.0	−3.78	−0.70	−35
2026	AMZN	111	56.8	−4.74	−13.11	−1,455
2017	AVGO	30	33.3	−91.66	−203.83	−6,115
2018	AVGO	51	33.3	−32.53	−88.30	−4,503
2019	AVGO	51	51.0	−17.84	−37.79	−1,927
2020	AVGO	50	20.0	−68.55	−266.56	−13,328
2021	AVGO	51	15.7	−99.57	−400.65	−20,433
2022	AVGO	51	37.3	−39.53	−273.67	−13,957
2023	AVGO	51	43.1	−40.88	−348.99	−17,799
2024	AVGO	50	54.0	−15.53	−179.52	−8,976
2025	AVGO	50	56.0	−7.23	−35.74	−1,787
2026	AVGO	109	52.3	−6.07	−54.62	−5,954
2017	GOOGL	30	33.3	−61.32	−289.07	−8,672
2018	GOOGL	51	33.3	−46.95	−462.19	−23,572
2019	GOOGL	51	51.0	−15.02	−82.68	−4,216
2020	GOOGL	50	42.0	−34.67	−425.80	−21,290
2021	GOOGL	51	43.1	−45.53	−739.78	−37,729
2022	GOOGL	50	44.0	−25.94	−502.89	−25,145
2023	GOOGL	51	56.9	−17.54	−13.28	−677
2024	GOOGL	51	51.0	−18.44	−29.12	−1,485

year	name	trades	win %	mean %	mean $	total $
2025	GOOGL	50	60.0	−12.72	−11.36	−568
2026	GOOGL	109	48.6	−10.79	−37.65	−4,104
2017	INTC	29	24.1	−62.75	−15.96	−463
2018	INTC	51	29.4	−81.58	−40.28	−2,054
2019	INTC	51	54.9	−9.42	−3.73	−190
2020	INTC	50	42.0	−34.85	−19.80	−990
2021	INTC	51	45.1	−31.44	−19.24	−981
2022	INTC	51	43.1	−10.72	−3.71	−189
2023	INTC	51	56.9	−15.05	−5.99	−305
2024	INTC	51	68.6	−7.68	−2.40	−122
2025	INTC	50	48.0	−30.93	−16.36	−818
2026	INTC	66	59.1	−1.13	−10.44	−689
2017	META	30	43.3	−31.80	−29.70	−891
2018	META	51	41.2	−20.15	−38.53	−1,965
2019	META	51	66.7	+6.85	+13.09	+668
2020	META	50	50.0	−22.11	−60.86	−3,043
2021	META	51	56.9	−1.40	−4.79	−244
2022	META	51	52.9	−5.12	−19.01	−969
2023	META	51	49.0	−26.54	−82.20	−4,192
2024	META	51	64.7	−1.79	+4.94	+252
2025	META	50	38.0	−15.32	−78.88	−3,944
2026	META	109	63.3	−9.30	−52.74	−5,749
2017	MSFT	30	43.3	−80.99	−31.10	−933
2018	MSFT	51	43.1	−56.62	−46.63	−2,378
2019	MSFT	51	51.0	−10.56	−10.63	−542
2020	MSFT	50	40.0	−26.77	−49.10	−2,455
2021	MSFT	51	45.1	−19.05	−50.06	−2,553
2022	MSFT	51	47.1	−27.74	−90.65	−4,623
2023	MSFT	51	49.0	−24.22	−58.93	−3,005
2024	MSFT	51	51.0	−8.07	−19.61	−1,000
2025	MSFT	50	42.0	−17.86	−56.32	−2,816
2026	MSFT	109	56.9	−13.35	−41.75	−4,551
2017	MU	30	40.0	−55.95	−23.77	−713
2018	MU	51	47.1	−16.14	−11.83	−603
2019	MU	51	51.0	−6.40	−3.30	−168
2020	MU	50	54.0	−40.51	−34.58	−1,729
2021	MU	51	56.9	−13.95	−21.61	−1,102
2022	MU	51	52.9	−2.73	−5.69	−290
2023	MU	51	54.9	−8.19	−7.06	−360
2024	MU	51	54.9	−11.11	−15.73	−802

year	name	trades	win %	mean %	mean $	total $
2025	MU	50	50.0	−30.11	−49.04	−2,452
2026	MU	65	72.3	+5.00	+84.88	+5,517
2017	NVDA	30	50.0	−24.55	−70.90	−2,127
2018	NVDA	51	56.9	−7.20	−16.83	−858
2019	NVDA	51	68.6	+7.19	+17.76	+906
2020	NVDA	50	36.0	−36.47	−247.80	−12,390
2021	NVDA	50	68.0	+8.05	+30.50	+1,525
2022	NVDA	51	52.9	−6.43	−31.12	−1,587
2023	NVDA	51	54.9	−17.06	−79.63	−4,061
2024	NVDA	50	54.0	−13.50	−250.95	−12,547
2025	NVDA	50	72.0	+0.31	+9.04	+452
2026	NVDA	108	63.0	−0.97	+0.52	+56
2017	TSLA	30	63.3	−26.55	−86.40	−2,592
2018	TSLA	51	54.9	−8.34	−59.04	−3,011
2019	TSLA	51	58.8	−7.48	−45.12	−2,301
2020	TSLA	49	61.2	−1.05	−127.33	−6,239
2021	TSLA	51	54.9	−5.11	−44.42	−2,265
2022	TSLA	50	52.0	−15.14	−196.64	−9,832
2023	TSLA	51	54.9	−6.79	−4.14	−211
2024	TSLA	51	54.9	−0.74	+7.99	+408
2025	TSLA	50	66.0	−3.97	−2.58	−129
2026	TSLA	109	56.9	−8.05	−50.45	−5,499
2017	all names	329	43.8	−50.41	−101.02	−33,236
2018	all names	561	44.0	−32.44	−109.05	−61,180
2019	all names	561	56.5	−6.88	−18.43	−10,338
2020	all names	548	43.6	−29.51	−162.61	−89,111
2021	all names	560	48.9	−23.57	−156.62	−87,708
2022	all names	558	49.1	−13.99	−125.07	−69,789
2023	all names	561	53.5	−15.24	−54.97	−30,839
2024	all names	559	57.2	−7.50	−45.55	−25,464
2025	all names	550	55.1	−14.44	−28.58	−15,719
2026	all names	1,072	58.3	−5.45	−18.99	−20,355



Amendments, deviations and integrity checks. —

What it showed. Negative.

#8e V8: month-end 13:00 long straddle on unread chains

Folder: `research/compass/intel/, studies/v*.py, results/v1/ (@7dfea57)`. **Commits:** A15 8bc3baf / — / e4e4cea. **Dates:** 2026-10-16.

Hypothesis. Month-end afternoons move more than priced (from V7).

Rule. Buy the ATM straddle at 13:00 on the month's last session, hold to settlement.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Ask in (adverse); settlement out.

Data. SPXW 2017-06 → 2022-04, SPY 2017-06 → 2022-10 (chains no study had read).

Periods and holdouts. One unread span.

Pass rule (as pre-registered). A15.

Verdict. Rejected: SPXW −6.6% [−24.5, +12.6], SPY −8.4% [−27.2, +11.3].

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	2017-06→2022 (unread chains)	56	56 / 1,251	41.1	+61.9, +810.44	−57.6, −808.56	−24.7, −289.26	**−8.51**, −143.62	[−26.73, +11.41]; [−475.31, +118.60]
SPY	2017-06→2022 (unread chains)	49	49 / 1,382	36.7	+65.6, +109.64	−52.2, −95.19	−29.7, −39.08	**−8.94**, −19.95	[−23.93, +13.10]; [−52.12, +20.74]
all names	2017-06→2022 (unread chains)	105	62 / 1,382	39.0	+63.5, +502.77	−55.0, −463.02	−28.7, −54.58	**−8.71**, −85.91	[−23.53, +11.40]; [−258.98, +99.54]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	2017-06→2022 (unread chains)	+3,220.74, +154.1%, 2022-03-31, SPXW 13:00 long straddle 4585.0	−3,553.26, −78.9%, 2020-02-28, SPXW 13:00 long straddle 2945.0	0.70	15,944	971	total ≤ 0 (best 5 days +11,961; without them −20,003)
SPY	2017-06→2022 (unread chains)	+303.92, +133.9%, 2022-03-31, SPY 13:00 long straddle 457.0	−377.08, −78.2%, 2020-02-28, SPY 13:00 long straddle 295.0	0.67	1,823	1,190	total ≤ 0 (best 5 days +1,337; without them −2,314)
all names	2017-06→2022 (unread chains)	+3,220.74, +154.1%, 2022-03-31, SPXW 13:00 long straddle 4585.0	−3,553.26, −78.9%, 2020-02-28, SPXW 13:00 long straddle 2945.0	0.70	17,754	1,102	total ≤ 0 (best 5 days +13,098; without them −22,118)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $1.24). The study's fees are given beside them below. Re-pricing: exact (105 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPXW	2017-06→2022 (unread chains)	56	**−8.51**, −143.62	−8.42, −142.76	41.1 / 41.1	[−26.73, +11.41] / [−26.64, +11.50]
SPY	2017-06→2022 (unread chains)	49	**−8.94**, −19.95	−9.96, −21.27	36.7 / 36.7	[−23.93, +13.10] / [−24.94, +12.12]
all names	2017-06→2022 (unread chains)	105	**−8.71**, −85.91	−9.14, −86.06	39.0 / 39.0	[−23.53, +11.40] / [−23.88, +10.99]

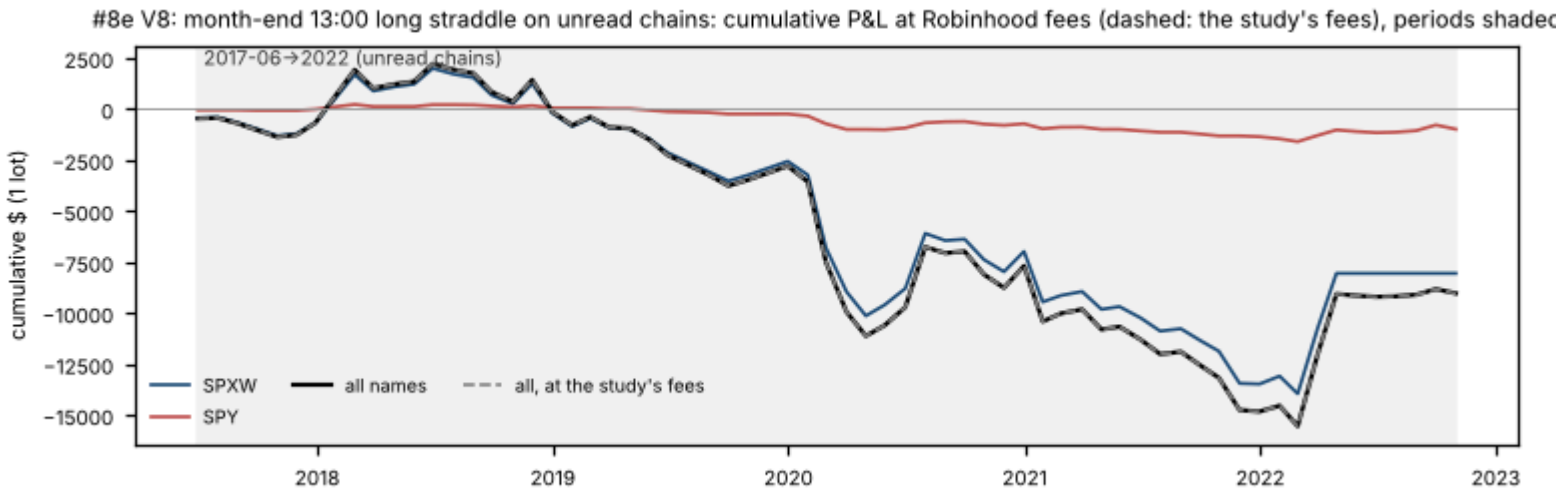
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- 2017-06→2022 (unread chains): held to settlement 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2017	SPXW	7	42.9	−25.25	−95.83	−671
2018	SPXW	11	45.5	+10.50	+44.19	+486
2019	SPXW	10	30.0	−23.62	−236.76	−2,368
2020	SPXW	12	41.7	−3.04	−368.01	−4,416
2021	SPXW	12	33.3	−31.32	−541.34	−6,496
2022	SPXW	4	75.0	+58.32	+1,355.49	+5,422
2017	SPY	3	33.3	−2.40	−0.08	−0
2018	SPY	9	33.3	+15.59	+6.70	+60
2019	SPY	6	16.7	−51.70	−48.58	−291
2020	SPY	11	45.5	+0.47	−43.63	−480
2021	SPY	10	30.0	−41.59	−63.48	−635
2022	SPY	10	50.0	+14.98	+36.87	+369
2017	all names	10	40.0	−18.40	−67.11	−671
2018	all names	20	40.0	+12.79	+27.32	+546
2019	all names	16	25.0	−34.15	−166.19	−2,659
2020	all names	23	43.5	−1.36	−212.87	−4,896
2021	all names	22	31.8	−35.99	−324.13	−7,131
2022	all names	14	57.1	+27.37	+413.62	+5,791



Amendments, deviations and integrity checks. Read once on unread chains.

What it showed. The month-end effect V7 suggested did not appear on unread data.

#9a Lifecycle P1: Quick Scalps → cash-secured put sale (portfolio)

Folder: `research/compass/lifecycle/ (@7dfea57)`. **Commits:** protocol 1f99d75 / L-A4 8bd322a / 425c7f4, b9d04b5. **Dates:** 2026-10-15.

Hypothesis. The Tier I finalists, run as a portfolio, earn after costs.

Rule. QS signal → sell a cash-secured put (s20 policy), one unit, the lifecycle position engine.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side, NBBO.

Data. SPY/QQQ/IWM 0DTE 2022-11 → 2026-10.

Periods and holdouts. development + validation / hold-out 2025-10-01 → 2026-10-09 (opened once).

Pass rule (as pre-registered). R1–R6 promotion rules (Holm; R2 random-day control).

Verdict. Rejected: hold-out +$987 (about 1% a year, collateral-bound).

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	development + validation	88	78 / 318	56.8	+0.1, +26.45	−0.2, −37.34	+0.0, +10.40	**−0.01**, −1.09	[−0.05, +0.03]; [−9.52, +6.92]
QQQ	development + validation	165	157 / 738	61.8	+0.1, +45.84	−0.1, −58.91	+0.1, +27.90	**+0.01**, +5.84	[−0.00, +0.03]; [−2.44, +13.34]
SPY	development + validation	155	148 / 742	62.6	+0.1, +46.81	−0.1, −59.34	+0.1, +27.90	**+0.01**, +7.09	[−0.00, +0.03]; [−1.56, +14.65]
all names	development + validation	408	269 / 742	61.0	+0.1, +42.32	−0.1, −53.91	+0.1, +22.40	**+0.01**, +4.82	[−0.00, +0.02]; [−0.06, +9.09]
IWM	hold-out 2025-10→2026-10	51	48 / 259	49.0	+0.1, +33.06	−0.2, −41.87	−0.0, −13.10	**−0.02**, −5.14	[−0.06, +0.02]; [−15.48, +5.30]
QQQ	hold-out 2025-10→2026-10	32	32 / 252	81.2	+0.1, +71.86	−0.2, −98.10	+0.1, +59.90	**+0.06**, +39.99	[+0.03, +0.10]; [+17.30, +69.30]
SPY	hold-out 2025-10→2026-10	39	35 / 240	66.7	+0.1, +51.78	−0.1, −93.72	+0.1, +34.90	**+0.00**, +3.28	[−0.04, +0.04]; [−24.04, +27.80]
all names	hold-out 2025-10→2026-10	122	89 / 262	63.1	+0.1, +52.48	−0.2, −64.35	+0.1, +30.40	**+0.01**, +9.39	[−0.02, +0.04]; [−5.59, +23.17]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	development + validation	+98.90, +0.4%, 2025-09-17, IWM CSP m85	−84.10, −0.4%, 2024-07-17, IWM CSP m16	0.93	464	240	total ≤ 0 (best 5 days +286; without them −383)
QQQ	development + validation	+171.90, +0.4%, 2025-04-11, QQQ CSP m181	−167.10, −0.4%, 2025-04-23, QQQ CSP m54	1.26	513	195	61%

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	development + validation	+243.90, +0.5%, 2025-04-11, SPY CSP m75	−197.10, −0.4%, 2025-04-10, SPY CSP m309	1.32	440	218	62%
all names	development + validation	+243.90, +0.5%, 2025-04-11, SPY CSP m75	−197.10, −0.4%, 2025-04-10, SPY CSP m309	1.23	923	195	52%
IWM	hold-out 2025-10→2026-10	+91.90, +0.3%, 2026-06-12, IWM CSP m21	−114.10, −0.5%, 2025-10-29, IWM CSP m139	0.76	301	248	total ≤ 0 (best 5 days +273; without them −536)
QQQ	hold-out 2025-10→2026-10	+136.90, +0.2%, 2026-08-03, QQQ CSP m15	−166.10, −0.3%, 2025-10-29, QQQ CSP m44	3.17	210	129	47%
SPY	hold-out 2025-10→2026-10	+91.90, +0.1%, 2026-04-01, SPY CSP m38	−197.10, −0.3%, 2025-10-14, SPY CSP m283	1.11	507	147	327%
all names	hold-out 2025-10→2026-10	+136.90, +0.2%, 2026-08-03, QQQ CSP m15	−197.10, −0.3%, 2025-10-14, SPY CSP m283	1.40	763	82	62%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.40 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: Robinhood fees inferred from the rule: the legs' prices and the expiry exercise fee are not in the trade file (530 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	development + validation	88	**−0.01**, −1.09	−0.01, −2.39	56.8 / 56.8	[−0.05, +0.03] / [−0.05, +0.02]
QQQ	development + validation	165	**+0.01**, +5.84	+0.01, +4.54	61.8 / 61.8	[−0.00, +0.03] / [−0.01, +0.03]
SPY	development + validation	155	**+0.01**, +7.09	+0.01, +5.79	62.6 / 62.6	[−0.00, +0.03] / [−0.00, +0.03]
all names	development + validation	408	**+0.01**, +4.82	+0.01, +3.52	61.0 / 61.0	[−0.00, +0.02] / [−0.01, +0.02]
IWM	hold-out 2025-10→2026-10	51	**−0.02**, −5.14	−0.02, −6.44	49.0 / 49.0	[−0.06, +0.02] / [−0.06, +0.02]
QQQ	hold-out 2025-10→2026-10	32	**+0.06**, +39.99	+0.06, +38.69	81.2 / 81.2	[+0.03, +0.10] / [+0.02, +0.10]
SPY	hold-out 2025-10→2026-10	39	**+0.00**, +3.28	+0.00, +1.98	66.7 / 66.7	[−0.04, +0.04] / [−0.04, +0.04]
all names	hold-out 2025-10→2026-10	122	**+0.01**, +9.39	+0.01, +8.09	63.1 / 63.1	[−0.02, +0.04] / [−0.02, +0.03]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

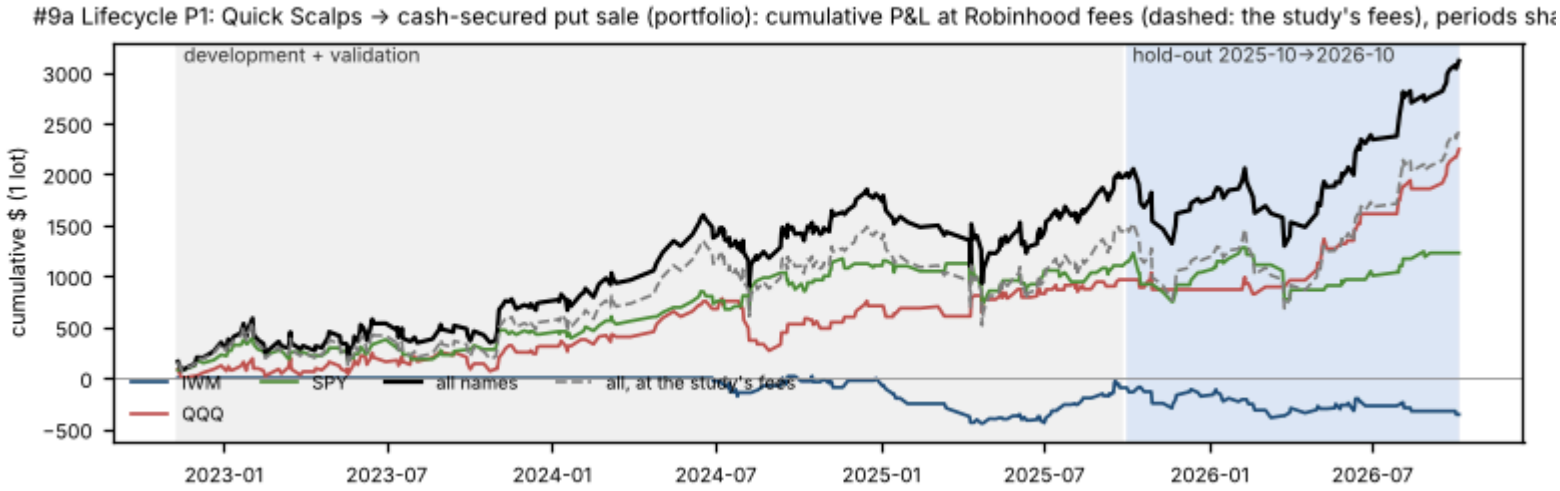
Exit reasons (share of trades):

- development + validation: policy 100.0%
- hold-out 2025-10→2026-10: policy 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	36	58.3	−0.00	−0.10	−4
2025	IWM	66	54.5	−0.01	−2.67	−176
2026	IWM	37	48.6	−0.02	−4.83	−179
2022	QQQ	6	83.3	+0.08	+20.23	+121
2023	QQQ	76	57.9	+0.01	+2.49	+189
2024	QQQ	48	62.5	+0.02	+8.25	+396

year	name	trades	win %	mean %	mean $	total $
2025	QQQ	40	65.0	+0.01	+4.02	+161
2026	QQQ	27	85.2	+0.07	+50.97	+1,376
2022	SPY	7	85.7	+0.08	+32.61	+228
2023	SPY	59	55.9	+0.01	+3.39	+200
2024	SPY	54	68.5	+0.02	+12.50	+675
2025	SPY	47	61.7	−0.00	−1.44	−68
2026	SPY	27	66.7	+0.01	+7.08	+191
2022	all names	13	84.6	+0.08	+26.90	+350
2023	all names	135	57.0	+0.01	+2.88	+389
2024	all names	138	63.8	+0.02	+7.74	+1,068
2025	all names	153	59.5	−0.00	−0.54	−83
2026	all names	91	64.8	+0.02	+15.26	+1,389



Amendments, deviations and integrity checks. A superseded hold-out file is kept separately.

What it showed. The put sale earns cents on the cash it ties up.

#9b Lifecycle P2: Quick Scalps → credit put spread to expiry (portfolio)

Folder: `research/compass/lifecycle/ (@7dfea57)`. **Commits:** 1f99d75 / L-A4 / d8a7ebb. **Dates:** 2026-10-15.

Hypothesis. As P1 with a defined-risk spread.

Rule. QS signal → credit put spread held to expiry.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). As P1.

Data. SPY/QQQ/IWM/SPXW 0DTE.

Periods and holdouts. As P1.

Pass rule (as pre-registered). As P1.

Verdict. Rejected: hold-out −$6,805.

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	development + validation	80	80 / 335	93.8	+12.9, +112.46	−67.6, −614.64	+8.1, +74.56	**+7.83**, +67.01	[+5.39, +13.17]; [+45.16, +114.00]
QQQ	development + validation	173	173 / 738	89.6	+14.0, +124.25	−108.0, −959.24	+9.8, +89.56	**+1.26**, +11.51	[−6.04, +7.97]; [−54.56, +71.24]
SPXW	development + validation	182	182 / 844	85.2	+12.0, +106.03	−79.8, −703.62	+11.5, +102.74	**−1.66**, −14.09	[−7.34, +3.03]; [−64.09, +27.31]
SPY	development + validation	169	169 / 742	92.3	+13.5, +120.52	−98.2, −830.68	+9.2, +84.56	**+4.91**, +47.35	[−1.92, +11.38]; [−12.08, +103.91]
all names	development + validation	604	299 / 844	89.6	+13.1, +116.32	−90.7, −795.81	+9.8, +89.56	**+2.27**, +21.18	[−2.26, +5.99]; [−18.54, +53.64]
IWM	hold-out 2025-10→2026-10	53	53 / 259	79.2	+7.7, +66.86	−87.2, −795.58	+6.3, +54.56	**−11.97**, −112.14	[−23.36, −3.21]; [−216.18, −31.72]
QQQ	hold-out 2025-10→2026-10	48	48 / 252	95.8	+16.8, +147.16	−100.8, −889.94	+10.4, +89.56	**+11.89**, +103.95	[−1.34, +21.64]; [−16.34, +191.82]
SPXW	hold-out 2025-10→2026-10	47	47 / 261	89.4	+11.1, +98.93	−99.8, −882.92	+10.9, +97.74	**−0.73**, −5.52	[−11.24, +8.66]; [−98.70, +77.67]
SPY	hold-out 2025-10→2026-10	48	48 / 261	87.5	+16.2, +149.39	−199.0, −1,825.69	+9.8, +89.56	**−10.73**, −97.50	[−45.02, +8.20]; [−415.90, +78.51]
all names	hold-out 2025-10→2026-10	196	91 / 262	87.8	+13.0, +116.32	−118.9, −1,079.17	+9.2, +84.56	**−3.13**, −30.07	[−15.66, +4.66]; [−146.03, +40.69]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	development + validation	+1,571.06, +179.5%, 2024-07-25, IWM CPS m163	−887.44, −100.8%, 2024-07-24, IWM CPS m140	2.74	887	44	62%
QQQ	development + validation	+1,766.06, +187.9%, 2024-03-06, QQQ CPS m226	−2,153.94, −238.0%, 2023-02-03, QQQ CPS m105	1.12	5,931	590	279%
SPXW	development + validation	+157.74, +18.8%, 2024-07-31, SPXW CPS m76	−943.66, −100.4%, 2024-07-12, SPXW CPS m311	0.87	5,356	416	total ≤ 0 (best 5 days +754; without them −3,318)
SPY	development + validation	+2,461.06, +284.5%, 2025-06-05, SPY CPS m85	−1,503.94, −168.0%, 2023-02-03, SPY CPS m92	1.74	4,686	304	65%
all names	development + validation	+2,461.06, +284.5%, 2025-06-05, SPY CPS m85	−2,153.94, −238.0%, 2023-02-03, QQQ CPS m105	1.26	9,144	496	68%
IWM	hold-out 2025-10→2026-10	+114.56, +12.9%, 2026-03-31, IWM CPS m225	−2,353.94, −255.9%, 2026-01-16, IWM CPS m187	0.32	6,493	241	total ≤ 0 (best 5 days +548; without them −6,491)
QQQ	hold-out 2025-10→2026-10	+1,781.06, +196.8%, 2026-02-10, QQQ CPS m61	−902.44, −100.8%, 2026-08-28, QQQ CPS m123	3.80	902	56	66%
SPXW	hold-out 2025-10→2026-10	+137.74, +16.0%, 2026-08-27, SPXW CPS m21	−898.66, −100.4%, 2025-10-14, SPXW CPS m283	0.94	1,835	203	total ≤ 0 (best 5 days +634; without them −893)
SPY	hold-out 2025-10→2026-10	+1,951.06, +204.3%, 2025-10-14, SPY CPS m283	−5,188.94, −558.0%, 2026-01-16, SPY CPS m316	0.57	8,609	250	total ≤ 0 (best 5 days +3,081; without them −7,761)
all names	hold-out 2025-10→2026-10	+1,951.06, +204.3%, 2025-10-14, SPY CPS m283	−5,188.94, −558.0%, 2026-01-16, SPY CPS m316	0.77	11,888	251	total ≤ 0 (best 5 days +5,214; without them −11,107)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $5.39 a trade) and Robinhood's charged instead (mean $0.97). The study's fees are given beside them below. Re-pricing: Robinhood fees inferred from the rule: the legs' prices and the expiry exercise fee are not in the trade file (800 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	development + validation	80	**+7.83**, +67.01	+7.11, +60.54	93.8 / 93.8	[+5.39, +13.17] / [+4.68, +12.45]
QQQ	development + validation	173	**+1.26**, +11.51	+0.52, +4.89	89.6 / 89.6	[−6.04, +7.97] / [−6.78, +7.22]
SPXW	development + validation	182	**−1.66**, −14.09	−1.56, −13.21	85.2 / 85.2	[−7.34, +3.03] / [−7.24, +3.13]
SPY	development + validation	169	**+4.91**, +47.35	+4.17, +40.71	92.3 / 92.3	[−1.92, +11.38] / [−2.65, +10.65]
all names	development + validation	604	**+2.27**, +21.18	+1.79, +16.83	89.6 / 89.6	[−2.26, +5.99] / [−2.76, +5.49]
IWM	hold-out 2025-10→2026-10	53	**−11.97**, −112.14	−12.68, −118.47	79.2 / 79.2	[−23.36, −3.21] / [−24.08, −3.92]
QQQ	hold-out 2025-10→2026-10	48	**+11.89**, +103.95	+11.17, +97.67	95.8 / 95.8	[−1.34, +21.64] / [−2.07, +20.92]
SPXW	hold-out 2025-10→2026-10	47	**−0.73**, −5.52	−0.64, −4.66	89.4 / 89.4	[−11.24, +8.66] / [−11.14, +8.75]
SPY	hold-out 2025-10→2026-10	48	**−10.73**, −97.50	−11.45, −104.05	87.5 / 87.5	[−45.02, +8.20] / [−45.74, +7.48]
all names	hold-out 2025-10→2026-10	196	**−3.13**, −30.07	−3.65, −34.72	87.8 / 87.8	[−15.66, +4.66] / [−16.19, +4.15]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

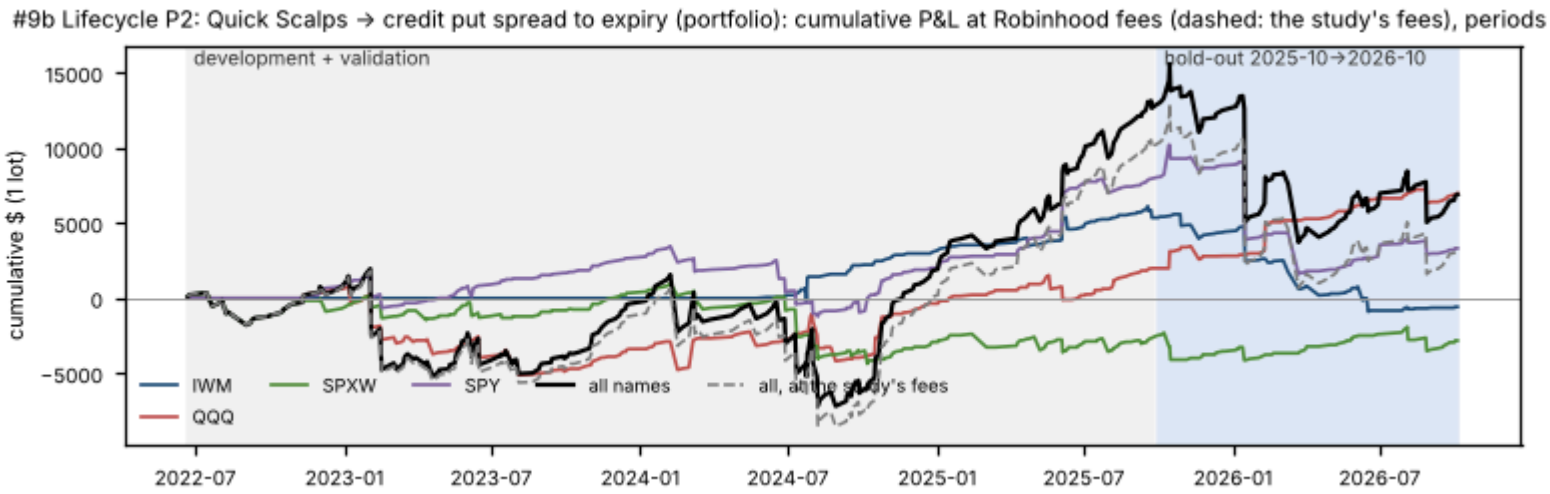
Exit reasons (share of trades):

- development + validation: policy 100.0%

- hold-out 2025-10→2026-10: policy 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	31	96.8	+12.09	+99.03	+3,070
2025	IWM	62	90.3	+2.54	+21.84	+1,354
2026	IWM	40	77.5	−13.50	−125.16	−5,006
2022	QQQ	6	100.0	+11.33	+101.70	+610
2023	QQQ	68	85.3	−6.27	−58.75	−3,995
2024	QQQ	53	88.7	+5.89	+57.16	+3,030
2025	QQQ	56	94.6	+6.39	+56.61	+3,170
2026	QQQ	38	97.4	+12.58	+109.64	+4,166
2022	SPXW	28	85.7	−2.86	−24.25	−679
2023	SPXW	51	86.3	+2.29	+20.11	+1,026
2024	SPXW	56	80.4	−6.60	−58.25	−3,262
2025	SPXW	57	87.7	−1.37	−10.86	−619
2026	SPXW	37	91.9	+2.01	+19.21	+711
2022	SPY	7	100.0	+14.56	+119.16	+834
2023	SPY	54	92.6	+3.66	+33.88	+1,829
2024	SPY	61	86.9	−2.38	−10.61	−647
2025	SPY	58	94.8	+12.99	+117.73	+6,828
2026	SPY	37	89.2	−16.06	−149.25	−5,522
2022	all names	41	90.2	+2.19	+18.67	+765
2023	all names	173	87.9	−0.65	−6.59	−1,140
2024	all names	201	87.1	+0.85	+10.90	+2,191
2025	all names	233	91.8	+5.11	+46.07	+10,734
2026	all names	152	88.8	−3.82	−37.18	−5,652



Amendments, deviations and integrity checks. —

What it showed. Loses on the hold-out.

#9c Lifecycle Tier I, daily tier, L-A3, H-FLO (by configuration)

Folder: `research/compass/lifecycle/ (@7dfea57)`. **Commits:** 1f99d75 / L-A2, L-A3 889691a, L-A4 8bd322a, L-A4.1 f815d27 / 425c7f4, f6d5a98, d8a7ebb, 66bc5c9, 50de62a. **Dates:** 2026-10-15.

Hypothesis. Some of 12 setups × 8 structures × 116 management policies (Tier I) or daily setups (Tier D) survive a hold-out.

Rule. Grids; finalists chosen on development + validation.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.70 a contract each side, NBBO.

Data. 0DTE SPY/QQQ/IWM/SPXW; index-ETF option EOD 2012-06 → 2026-10.

Periods and holdouts. hold-out 2025-10-01 → 2026-10-09, opened once per group.

Pass rule (as pre-registered). R1–R6.

Verdict. 0 promoted: Tier I no long-premium policy with a positive lower bound, finalists all put sales; L-A3 debit call spread +$64.37 [−18.16, +138.67] p 0.064 (fail); daily tier 19 finalists, all fail R2 (random day); H-FLO a regime descriptor, not an edge.

Metrics. not available — committed as configuration-level cells (grid_by_setup.csv, cells.csv); the trade rows are in the scratchpad (lifecycle/archive, 281 MB) and were not re-run for this report.

Amendments, deviations and integrity checks. Random-minute look-ahead corrected (L-A2) before the hold-out.

What it showed. No configuration survived the hold-out.

#10 ATM straddle samples and the independent short-premium prereg (branch compass-research-2026-10-14)

Folder: `research/compass_2026_10_14/ on origin/compass-research-2026-10-14`. **Commits:** short_premium PREREG 2f97a0c / — / BLOCKED. **Dates:** 2026-10-14 → 15.

Hypothesis. (Measurement) what short-dated ATM straddles cost on 10 names.

Rule. —

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). —

Data. 8–19 sessions a name.

Periods and holdouts. —

Pass rule (as pre-registered). —

Verdict. No result: the short-premium study was blocked (ThetaData session invalidated off-box).

Metrics. not available — price samples only (no trade with an exit); the pre-registered study never ran.

Amendments, deviations and integrity checks. —

What it showed. Medians of ATM straddle prices; no strategy.

#11 Price-action catalogue and atlas (43 names)

Folder: `research/compass/intel/04, 06, 07 (@7dfea57)`. **Commits:** — / — / 048d33a, 0d706b6, 98fa5f0. **Dates:** 2026-10-15.

Hypothesis. (Descriptive) how price action behaves around levels.

Rule. —

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). —

Data. 43 names 2017–2026 bars.

Periods and holdouts. —

Pass rule (as pre-registered). —

Verdict. Descriptive; V6 tested its levels (rejected).

Metrics. not available — descriptive tables, no trades.

Amendments, deviations and integrity checks. —

What it showed. No strategy.

Part B. Studies after the fresh start

#12 Step 1: every 0DTE contract broken down

Folder: `step1/`. **Commits:** — / — / 2185eb9. **Dates:** 2026-10-16.

Hypothesis. (Descriptive) the anatomy of 1,176,935 same-day contracts over 3,687 sessions.

Rule. —

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). —

Data. SPY, QQQ, IWM, SPX 0DTE.

Periods and holdouts. —

Pass rule (as pre-registered). —

Verdict. Descriptive.

Metrics. not available — a contract inventory, no trades.

Amendments, deviations and integrity checks. Checked against the raw quotes (verify.py).

What it showed. No strategy.

#13 Step 2: buy near the money, sell at +100% or hold (within 0.5% of the price)

Folder: `step2/`. **Commits:** — / — / b15b56b. **Dates:** 2026-10-16.

Hypothesis. A same-day contract that can double pays for those that do not.

Rule. Buy every contract within 0.5% of the underlying at the ask at 9:31, 10:00, 10:30, 11:00, 12:00, 1:00, 2:00, 3:00, 3:30; sell the first minute the bid reaches 2× the price paid (at exactly 2×), else hold to the close (worth intrinsic).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.65 a contract on the buy and on the sale; none when it expires worthless.

Data. SPY, QQQ, IWM, SPX 0DTE 2022-05-13 → 2026-10-09 (the near-the-money slice of 10,547,433 trades).

Periods and holdouts. pick 2022–24 / check 2025–26 (no holdout; holdout B months read).

Pass rule (as pre-registered). None pre-registered (a population study).

Verdict. Every group loses before and after fees; near the money 29% double, below the share needed to break even.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	pick 2022–24	6,960	177 / 182	38.0	+84.1, +54.63	−84.6, −49.19	−62.8, −13.04	**−20.47**, −9.75	[−24.27, −17.67]; [−11.94, −8.14]
QQQ	pick 2022–24	39,574	536 / 553	38.6	+83.2, +95.68	−84.8, −86.19	−59.7, −23.04	**−19.94**, −15.94	[−21.42, −18.71]; [−19.08, −13.54]
SPXW	pick 2022–24	109,721	662 / 683	41.3	+83.1, +914.13	−85.4, −810.56	−46.6, −121.13	**−15.83**, −98.21	[−17.16, −14.68]; [−117.17, −79.16]
SPY	pick 2022–24	46,425	536 / 553	39.2	+80.8, +88.58	−84.0, −77.77	−53.2, −16.04	**−19.43**, −12.61	[−20.88, −18.21]; [−14.81, −10.73]
all names	pick 2022–24	202,680	662 / 683	40.2	+82.6, +548.24	−85.0, −467.67	−51.3, −31.04	**−17.62**, −59.50	[−18.93, −16.54]; [−72.29, −47.82]
IWM	check 2025–26	20,177	444 / 457	40.2	+84.5, +62.36	−85.8, −54.97	−55.3, −12.04	**−17.28**, −7.79	[−19.60, −15.65]; [−9.42, −6.58]
QQQ	check 2025–26	48,342	444 / 457	39.3	+82.8, +141.26	−84.6, −124.45	−56.0, −27.04	**−18.87**, −20.11	[−20.90, −17.12]; [−23.03, −17.99]
SPXW	check 2025–26	106,236	444 / 457	41.1	+81.3, +1,167.90	−84.3, −982.35	−42.5, −106.13	**−16.20**, −98.09	[−18.05, −14.46]; [−123.60, −77.87]
SPY	check 2025–26	53,096	444 / 457	39.0	+80.3, +121.01	−83.2, −101.77	−50.9, −17.04	**−19.40**, −14.81	[−21.33, −17.68]; [−17.73, −12.68]
all names	check 2025–26	227,851	444 / 457	40.2	+81.7, +619.80	−84.3, −506.49	−48.3, −30.04	**−17.61**, −54.14	[−19.37, −16.02]; [−67.43, −43.81]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	pick 2022–24	+376.89, +100.0%, 2024-08-05, IWM 197.0C m1	−336.04, −100.0%, 2024-07-24, IWM 220.0C m30	0.68	68,223	180	total ≤ 0 (best 5 days +2,838; without them −70,686)
QQQ	pick 2022–24	+761.87, +100.0%, 2024-08-05, QQQ 423.0C m1	−700.04, −100.0%, 2024-08-05, QQQ 426.0P m1	0.70	631,261	551	total ≤ 0 (best 5 days +21,148; without them −651,979)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	pick 2022–24	+7,667.74, +100.0%, 2024-08-05, SPXW 5095.0C m1	−6,891.13, −100.0%, 2024-08-05, SPXW 5145.0P m1	0.79	10,776,052	683	total ≤ 0 (best 5 days +720,720; without them −11,496,772)
SPY	pick 2022–24	+709.88, +100.0%, 2024-08-05, SPY 509.0C m1	−678.04, −100.0%, 2024-08-05, SPY 513.0P m1	0.73	585,645	553	total ≤ 0 (best 5 days +27,423; without them −613,067)
all names	pick 2022–24	+7,667.74, +100.0%, 2024-08-05, SPXW 5095.0C m1	−6,891.13, −100.0%, 2024-08-05, SPXW 5145.0P m1	0.79	12,060,375	683	total ≤ 0 (best 5 days +757,864; without them −12,818,240)
IWM	check 2025–26	+344.89, +100.0%, 2025-04-07, IWM 173.0C m1	−340.04, −100.0%, 2025-04-07, IWM 179.0P m60	0.76	157,234	457	total ≤ 0 (best 5 days +6,839; without them −163,930)
QQQ	check 2025–26	+787.87, +100.0%, 2025-04-09, QQQ 415.0C m1	−792.04, −100.0%, 2025-04-09, QQQ 418.0P m1	0.73	972,278	457	total ≤ 0 (best 5 days +53,908; without them −1,026,186)
SPXW	check 2025–26	+9,037.74, +100.0%, 2025-04-09, SPXW 4940.0C m1	−18,091.13, −100.0%, 2025-11-28, SPXW 6845.0P m1	0.83	10,437,628	457	total ≤ 0 (best 5 days +910,540; without them −11,331,229)
SPY	check 2025–26	+931.87, +100.0%, 2025-04-09, SPY 492.0C m1	−880.04, −100.0%, 2025-04-09, SPY 496.0P m1	0.76	786,680	457	total ≤ 0 (best 5 days +46,040; without them −832,367)
all names	check 2025–26	+9,037.74, +100.0%, 2025-04-09, SPXW 4940.0C m1	−18,091.13, −100.0%, 2025-11-28, SPXW 6845.0P m1	0.82	12,352,616	457	total ≤ 0 (best 5 days +1,010,478; without them −13,346,862)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.02 a trade) and Robinhood's charged instead (mean $0.76). The study's fees are given beside them below. Re-pricing: exact (430,531 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	pick 2022–24	6,960	**−20.47**, −9.75	−24.05, −10.71	38.0 / 37.5	[−24.27, −17.67] / [−28.15, −21.15]
QQQ	pick 2022–24	39,574	**−19.94**, −15.94	−21.90, −16.90	38.6 / 38.5	[−21.42, −18.71] / [−23.39, −20.69]
SPXW	pick 2022–24	109,721	**−15.83**, −98.21	−15.61, −97.77	41.3 / 41.3	[−17.16, −14.68] / [−16.94, −14.47]
SPY	pick 2022–24	46,425	**−19.43**, −12.61	−21.94, −13.58	39.2 / 39.0	[−20.88, −18.21] / [−23.48, −20.59]
all names	pick 2022–24	202,680	**−17.62**, −59.50	−18.58, −59.71	40.2 / 40.1	[−18.93, −16.54] / [−19.90, −17.44]
IWM	check 2025–26	20,177	**−17.28**, −7.79	−20.51, −8.74	40.2 / 40.0	[−19.60, −15.65] / [−22.95, −18.65]
QQQ	check 2025–26	48,342	**−18.87**, −20.11	−20.52, −21.07	39.3 / 39.2	[−20.90, −17.12] / [−22.70, −18.63]
SPXW	check 2025–26	106,236	**−16.20**, −98.09	−15.96, −97.67	41.1 / 41.1	[−18.05, −14.46] / [−17.79, −14.25]
SPY	check 2025–26	53,096	**−19.40**, −14.81	−21.68, −15.78	39.0 / 38.9	[−21.33, −17.68] / [−23.83, −19.76]
all names	check 2025–26	227,851	**−17.61**, −54.14	−18.66, −54.46	40.2 / 40.1	[−19.37, −16.02] / [−20.55, −16.99]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

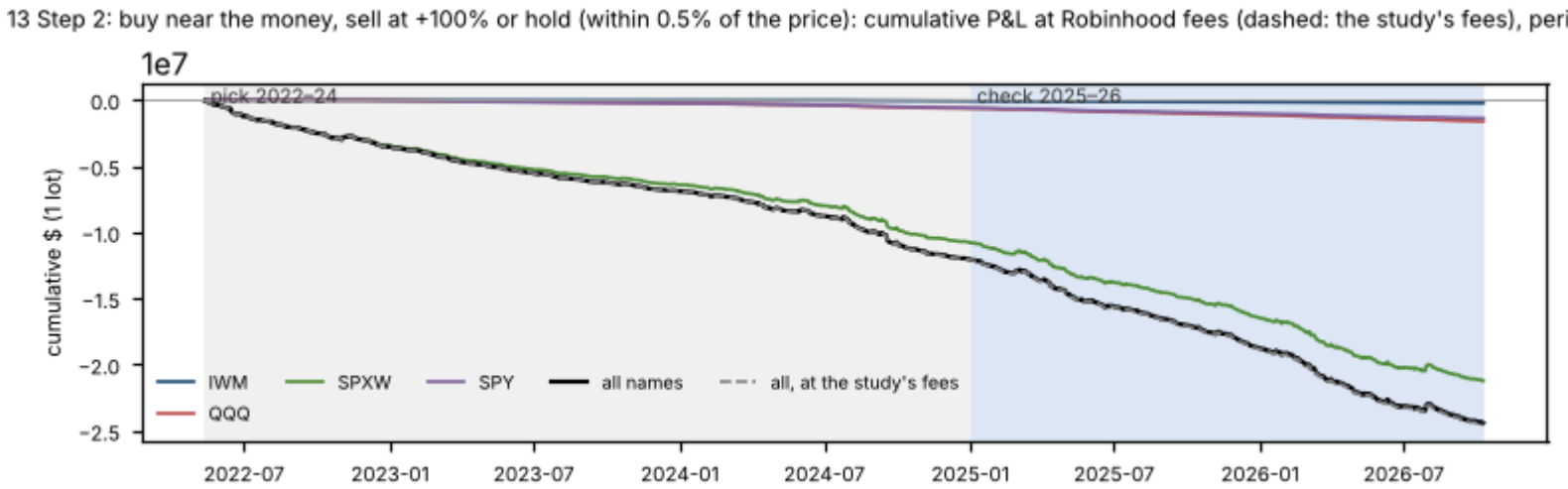
Exit reasons (share of trades):

- pick 2022–24: held to the close 70.8%; +100% target 29.2%

- check 2025–26: held to the close 71.1%; +100% target 28.9%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	6,960	38.0	−20.47	−9.75	−67,848
2025	IWM	10,251	40.2	−17.59	−7.93	−81,303
2026	IWM	9,926	40.2	−16.95	−7.64	−75,789
2022	QQQ	1,720	36.8	−21.99	−18.60	−31,989
2023	QQQ	15,750	39.3	−19.06	−13.07	−205,813
2024	QQQ	22,104	38.3	−20.41	−17.78	−393,029
2025	QQQ	24,779	38.4	−20.50	−20.25	−501,665
2026	QQQ	23,563	40.2	−17.15	−19.97	−470,613
2022	SPXW	22,444	40.9	−15.65	−153.67	−3,448,980
2023	SPXW	38,347	41.6	−15.60	−77.07	−2,955,493
2024	SPXW	48,930	41.2	−16.10	−89.34	−4,371,579
2025	SPXW	55,548	41.2	−16.17	−102.24	−5,679,112
2026	SPXW	50,688	41.1	−16.23	−93.54	−4,741,576
2022	SPY	2,386	38.0	−20.32	−17.85	−42,597
2023	SPY	19,406	39.7	−18.76	−10.60	−205,722
2024	SPY	24,633	38.8	−19.87	−13.69	−337,326
2025	SPY	27,804	39.1	−19.49	−15.35	−426,889
2026	SPY	25,292	39.0	−19.30	−14.21	−359,438
2022	all names	26,550	40.4	−16.48	−132.71	−3,523,566
2023	all names	73,503	40.6	−17.18	−45.81	−3,367,027
2024	all names	102,627	39.8	−18.23	−50.37	−5,169,782
2025	all names	118,382	40.0	−17.98	−56.50	−6,688,968
2026	all names	109,469	40.3	−17.20	−51.59	−5,647,417



Amendments, deviations and integrity checks. Formulas: step2/summarize.py (tp100, fees), the committed loader step2.double.load.

What it showed. Near-the-money 0DTE buys lose 16–22% of premium a trade.

#14 Step 3: 108 exits × 9 conditions

Folder: `step3/`. **Commits:** — / — / 09885b6. **Dates:** 2026-10-16.

Hypothesis. A target, stop or time exit, or a condition known at the buy, makes step 2’s trades pay.

Rule. 108 exits × 9 entry conditions on step 2’s trades.

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). $0.65 a contract each way.

Data. As step 2.

Periods and holdouts. discovery 2022–24 / check 2025–26.

Pass rule (as pre-registered). None pre-registered.

Verdict. Nothing made money in a way that held up; a −50% stop shrank losses; targets raised the win rate and lowered the average.

Metrics. not available — committed as per-rule summaries (exits.csv, exits_best.csv); the per-trade paths are committed but the exit grid was not re-run here.

Amendments, deviations and integrity checks. —

What it showed. No exit rule rescues the buys.

#15 Step 4: turn-up buy (1.5× the low) + 20% trailing stop, near the money

Folder: `step4/`. **Commits:** — / — / 6c0e60b. **Dates:** 2026-10-16.

Hypothesis. Buying the turn up off the low and selling near the high with a trailing stop finds an edge.

Rule. The first minute (9:36 → 15 minutes before the close) a contract's ask reaches 1.5× its low so far → buy at the ask; sell at the bid once it falls 20% from the highest bid since buying, else hold to the close. Strikes within 0.5% of the price.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.65 a contract each way (none when worthless).

Data. SPY, QQQ, IWM, SPX 0DTE 2022–2026 (this slice of 12,149,834 trades).

Periods and holdouts. pick 2022–24 / check 2025–26.

Pass rule (as pre-registered). None pre-registered; 126 rules positive in both periods, the 11 biggest failed the stress test.

Verdict. A trailing stop beats holding but still loses.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	pick 2022–24	807	177 / 182	24.5	+37.0, +28.50	−20.4, −17.31	−18.0, −10.10	**−6.33**, −6.07	[−9.52, −3.39]; [−9.07, −3.39]
QQQ	pick 2022–24	4,603	536 / 553	28.1	+39.6, +52.59	−18.8, −25.65	−15.3, −14.10	**−2.39**, −3.69	[−3.63, −0.43]; [−5.49, −0.73]
SPXW	pick 2022–24	12,787	662 / 683	26.8	+36.7, +472.57	−19.5, −244.41	−16.7, −117.26	**−4.44**, −52.48	[−5.58, −3.06]; [−69.56, −31.31]
SPY	pick 2022–24	5,388	536 / 553	27.5	+36.6, +46.54	−18.8, −22.43	−15.7, −10.10	**−3.58**, −3.46	[−4.81, −1.85]; [−5.05, −0.84]
all names	pick 2022–24	23,585	662 / 683	27.1	+37.2, +275.24	−19.2, −143.79	−16.3, −25.10	**−3.91**, −30.17	[−4.90, −2.70]; [−41.22, −17.54]
IWM	check 2025–26	2,347	444 / 457	26.0	+41.0, +37.89	−19.9, −18.58	−16.8, −10.10	**−4.05**, −3.88	[−6.62, −0.91]; [−6.61, −0.87]
QQQ	check 2025–26	5,595	444 / 457	28.9	+41.4, +86.76	−18.9, −38.41	−14.9, −20.10	**−1.42**, −2.21	[−3.82, +0.97]; [−6.43, +2.76]
SPXW	check 2025–26	12,238	444 / 457	30.0	+40.9, +721.76	−19.0, −312.11	−14.7, −122.26	**−1.07**, −2.40	[−2.69, +1.07]; [−43.29, +51.45]
SPY	check 2025–26	6,078	444 / 457	30.4	+39.4, +70.71	−18.5, −31.69	−14.0, −11.10	**−0.87**, −0.52	[−2.54, +1.14]; [−4.16, +3.71]
all names	check 2025–26	26,258	444 / 457	29.5	+40.7, +379.64	−18.9, −161.74	−14.8, −23.10	**−1.37**, −2.06	[−2.96, +0.56]; [−22.49, +24.58]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	pick 2022–24	+196.90, +161.4%, 2024-07-11, IWM 207.0C m14	−240.10, −73.9%, 2024-09-03, IWM 217.0C m30	0.54	5,042	179	total ≤ 0 (best 5 days +1,063; without them −5,960)
QQQ	pick 2022–24	+604.96, +360.1%, 2023-08-29, QQQ 367.0C m13	−270.10, −52.5%, 2023-03-22, QQQ 313.0P m270	0.80	16,971	553	total ≤ 0 (best 5 days +7,189; without them −24,160)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	pick 2022–24	+7,302.87, +268.5%, 2024-05-31, SPXW 5180.0C m211	−4,492.26, −81.8%, 2023-09-21, SPXW 4350.0C m6	0.71	673,616	673	total ≤ 0 (best 5 days +145,866; without them −816,949)
SPY	pick 2022–24	+796.88, +249.0%, 2024-08-01, SPY 555.0P m36	−330.10, −68.1%, 2024-09-18, SPY 565.0P m270	0.79	18,648	553	total ≤ 0 (best 5 days +6,787; without them −25,435)
all names	pick 2022–24	+7,302.87, +268.5%, 2024-05-31, SPXW 5180.0C m211	−4,492.26, −81.8%, 2023-09-21, SPXW 4350.0C m6	0.71	714,131	673	total ≤ 0 (best 5 days +154,196; without them −865,795)
IWM	check 2025–26	+640.96, +754.1%, 2025-02-21, IWM 225.0P m6	−245.10, −66.1%, 2025-08-01, IWM 212.5C m30	0.72	10,434	418	total ≤ 0 (best 5 days +3,653; without them −12,759)
QQQ	check 2025–26	+1,033.96, +984.7%, 2026-09-21, QQQ 730.0C m8	−274.09, −100.0%, 2025-07-30, QQQ 570.0P m104	0.92	26,331	323	total ≤ 0 (best 5 days +21,516; without them −33,895)
SPXW	check 2025–26	+21,607.74, +266.1%, 2025-04-07, SPXW 4875.0C m19	−39,942.26, −80.0%, 2025-04-09, SPXW 5340.0P m238	0.99	466,377	390	total ≤ 0 (best 5 days +421,711; without them −451,093)
SPY	check 2025–26	+1,876.85, +247.0%, 2025-04-07, SPY 490.0C m23	−2,362.10, −85.9%, 2025-04-09, SPY 521.0P m231	0.98	25,145	390	total ≤ 0 (best 5 days +20,869; without them −24,052)
all names	check 2025–26	+21,607.74, +266.1%, 2025-04-07, SPXW 4875.0C m19	−39,942.26, −80.0%, 2025-04-09, SPXW 5340.0P m238	0.98	521,558	390	total ≤ 0 (best 5 days +459,576; without them −513,627)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.30 a trade) and Robinhood's charged instead (mean $1.18). The study's fees are given beside them below. Re-pricing: exact (49,843 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	pick 2022–24	807	**−6.33**, −6.07	−8.30, −7.27	24.5 / 23.3	[−9.52, −3.39] / [−11.58, −5.37]
QQQ	pick 2022–24	4,603	**−2.39**, −3.69	−3.66, −4.89	28.1 / 27.4	[−3.63, −0.43] / [−4.88, −1.77]
SPXW	pick 2022–24	12,787	**−4.44**, −52.48	−4.29, −51.53	26.8 / 26.8	[−5.58, −3.06] / [−5.43, −2.91]
SPY	pick 2022–24	5,388	**−3.58**, −3.46	−5.57, −4.66	27.5 / 26.2	[−4.81, −1.85] / [−6.91, −3.68]
all names	pick 2022–24	23,585	**−3.91**, −30.17	−4.59, −30.20	27.1 / 26.6	[−4.90, −2.70] / [−5.59, −3.37]
IWM	check 2025–26	2,347	**−4.05**, −3.88	−5.82, −5.08	26.0 / 24.8	[−6.62, −0.91] / [−8.42, −2.59]
QQQ	check 2025–26	5,595	**−1.42**, −2.21	−2.38, −3.41	28.9 / 28.3	[−3.82, +0.97] / [−4.82, +0.08]
SPXW	check 2025–26	12,238	**−1.07**, −2.40	−0.94, −1.45	30.0 / 30.0	[−2.69, +1.07] / [−2.55, +1.19]
SPY	check 2025–26	6,078	**−0.87**, −0.52	−2.41, −1.72	30.4 / 29.3	[−2.54, +1.14] / [−4.16, −0.30]
all names	check 2025–26	26,258	**−1.37**, −2.06	−2.02, −2.26	29.5 / 29.0	[−2.96, +0.56] / [−3.63, −0.05]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

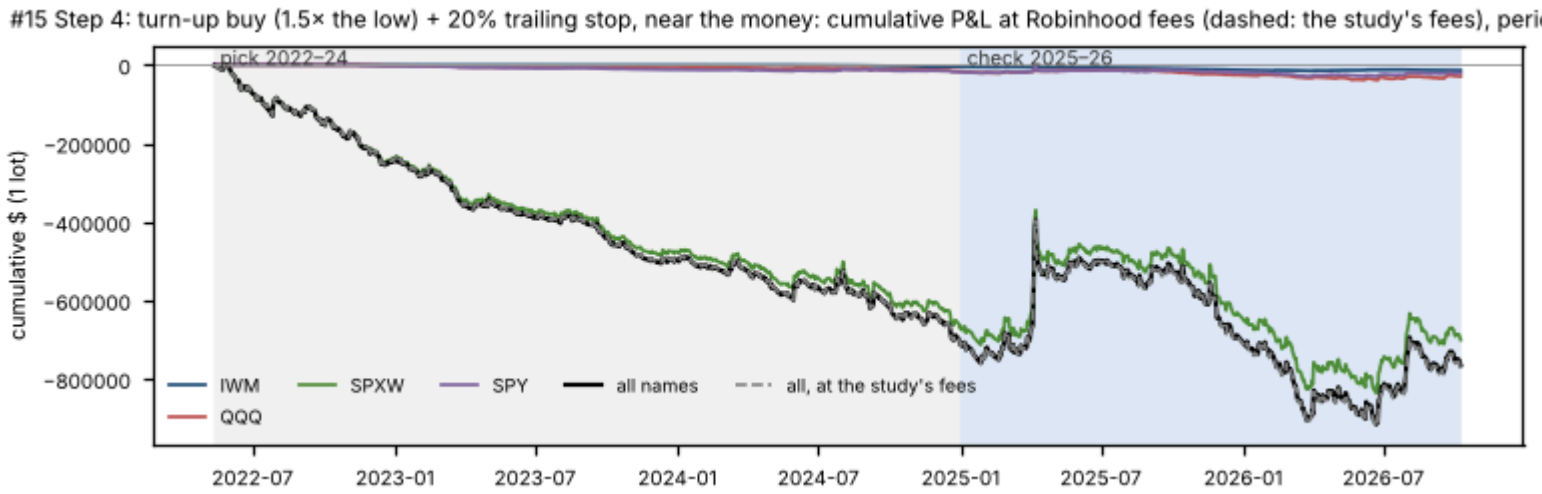
Exit reasons (share of trades):

- pick 2022–24: 20% trailing stop 99.5%; held to the close 0.5%

- check 2025–26: 20% trailing stop 99.4%; held to the close 0.6%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	807	24.5	−6.33	−6.07	−4,897
2025	IWM	1,182	24.5	−6.01	−6.99	−8,258
2026	IWM	1,165	27.6	−2.07	−0.73	−848
2022	QQQ	202	19.8	−6.67	−10.70	−2,162
2023	QQQ	1,842	28.5	−2.28	−3.25	−5,993
2024	QQQ	2,559	28.4	−2.14	−3.45	−8,816
2025	QQQ	2,874	27.7	−2.52	−3.10	−8,896
2026	QQQ	2,721	30.2	−0.27	−1.28	−3,483
2022	SPXW	2,622	25.2	−5.38	−91.83	−240,767
2023	SPXW	4,479	25.9	−5.14	−52.32	−234,351
2024	SPXW	5,686	28.2	−3.45	−34.46	−195,964
2025	SPXW	6,428	29.9	−1.07	+3.97	+25,507
2026	SPXW	5,810	30.0	−1.07	−9.45	−54,889
2022	SPY	300	22.7	−5.94	−9.47	−2,840
2023	SPY	2,240	26.7	−4.57	−4.19	−9,376
2024	SPY	2,848	28.6	−2.56	−2.26	−6,432
2025	SPY	3,194	30.4	−1.10	−0.51	−1,626
2026	SPY	2,884	30.5	−0.61	−0.54	−1,556
2022	all names	3,124	24.6	−5.52	−78.67	−245,770
2023	all names	8,561	26.7	−4.38	−29.17	−249,720
2024	all names	11,900	28.1	−3.15	−18.16	−216,109
2025	all names	13,678	29.1	−1.81	+0.49	+6,727
2026	all names	12,580	29.9	−0.88	−4.83	−60,777



Amendments, deviations and integrity checks. Formulas copied from step4/summarize.py prep().

What it showed. The near-the-money turn-up buy with a 20% trail loses 1–8% of premium a trade.

#16 Step 5: the runner model (model A/B) and its top 1% picks

Folder: `step5/`. **Commits:** — / — / ae3c63d, 1227394, f9f2746, 491a0c7. **Dates:** 2026-10-16 → 17.

Hypothesis. What was visible at the turn tells runners from duds well enough to trade.

Rule. Gradient-boosted model A (P(bid ≥ 3× paid)) and B; trade the top 1% / 0.5% / 0.1%, three exits.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.65 a contract each way.

Data. 1,602,401 turn-up and range-break buys; trained 2022–24, scored once on 2025–26.

Periods and holdouts. train 2022–24 / scored 2025–26.

Pass rule (as pre-registered). None pre-registered.

Verdict. Runners partly separable (top 1% reach 3× 26% vs 6%) but do not pay; every pick total made money almost all on 2025-11-20, and without its best 5 days it lost $620,114.

Metrics. not available — the picks are committed as daily sums (top1_daily.csv), not trades.

Amendments, deviations and integrity checks. Scored once.

What it showed. The model ranks runners but its picks do not pay without a handful of days.

#17 Step 6: the six pre-registered runner rules × three exits

Folder: `step6/`. **Commits:** 30babaa / — / 44b5547. **Dates:** 2026-10-23.

Hypothesis. A condition visible at the buy raises the odds of a +40% (family A) or +500% (family B) run enough to trade.

Rule. The top 3 rules of each family by Pick lift (one or two Part 2 buckets, ≥ 300 Pick trades on ≥ 60 dates), each buying every near-the-money population trade that meets it; exits (a) +40% / −30% / 60 minutes or 3:30, (b) 50% trail armed at +40%, out 3:30, (c) half at +40%, trail the rest 50%, out 3:30.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.65 a contract each way (SPY/QQQ/IWM), SPXW per schedule; no sale fee at a $0 bid.

Data. 240,493 near-the-money 0DTE buys 2022-05 → 2026-10-09 (the 28 held-out months never read).

Periods and holdouts. pick 2022-05→2024 / check 2025→2026-10-09; the 28 holdout-B months UNOPENED.

Pass rule (as pre-registered). Check: average r > 0, its 20-session block lower end > 0, and above the 97.5th percentile of 500 cell-matched random draws.

Verdict. FAIL: 0 of 18 pass; all lose on Check (−3.3% to −21.9% a trade); 14 of 18 below the 20th random percentile.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
A1a	pick 2022-05→2024	665	80 / 681	49.9	+36.4, +235.02	−32.7, −297.48	−0.0, −0.10	**+1.78**, −31.63	[−5.55, +10.85]; [−84.59, +65.12]
A1b	pick 2022-05→2024	665	80 / 681	32.0	+79.9, +608.96	−49.9, −418.89	−22.6, −40.10	**−8.32**, −89.67	[−24.75, +16.03]; [−197.94, +180.18]
A1c	pick 2022-05→2024	665	80 / 681	66.9	+66.1, +503.74	−156.0, −1,350.48	+15.3, +35.48	**−7.38**, −109.68	[−33.39, +30.57]; [−282.38, +320.89]
A2a	pick 2022-05→2024	1,081	186 / 678	43.1	+38.4, +324.15	−33.7, −270.27	−30.6, −19.10	**−2.60**, −14.02	[−10.00, +1.96]; [−93.03, +32.48]
A2b	pick 2022-05→2024	1,081	186 / 678	33.3	+139.4, +938.60	−50.2, −420.27	−21.8, −45.10	**+12.97**, +32.27	[−7.98, +40.79]; [−123.81, +239.18]
A2c	pick 2022-05→2024	1,081	186 / 678	64.8	+100.1, +706.13	−148.1, −1,215.00	+14.4, +40.00	**+12.62**, +29.03	[−13.99, +44.01]; [−186.99, +294.84]
A3a	pick 2022-05→2024	1,239	127 / 683	47.4	+38.3, +323.94	−33.1, −287.55	−14.7, −14.10	**+0.74**, +2.16	[−6.33, +7.31]; [−80.72, +74.27]
A3b	pick 2022-05→2024	1,239	127 / 683	29.1	+95.0, +884.01	−45.2, −357.95	−20.4, −50.10	**−4.46**, +2.91	[−26.01, +2.77]; [−269.93, +76.52]
A3c	pick 2022-05→2024	1,239	127 / 683	66.3	+69.8, +650.54	−145.1, −1,132.38	+16.4, +38.40	**−2.68**, +49.04	[−35.01, +13.09]; [−383.88, +201.66]
B1a	pick 2022-05→2024	2,426	121 / 443	42.3	+31.5, +631.01	−29.8, −598.26	−18.6, −382.26	**−3.87**, −77.87	[−5.30, −2.88]; [−104.62, −55.49]
B1b	pick 2022-05→2024	2,426	121 / 443	34.5	+106.5, +2,149.80	−52.9, −1,062.29	−21.6, −412.26	**+2.03**, +44.59	[−10.50, +14.05]; [−210.37, +298.18]
B1c	pick 2022-05→2024	2,426	121 / 443	58.4	+93.3, +1,886.36	−142.2, −2,855.75	+13.8, +253.48	**−4.66**, −85.93	[−16.86, +8.36]; [−327.73, +192.82]
B2a	pick 2022-05→2024	2,483	137 / 621	38.6	+30.0, +579.64	−27.6, −533.57	−15.5, −312.26	**−5.33**, −103.62	[−6.43, −4.58]; [−126.68, −88.57]
B2b	pick 2022-05→2024	2,483	137 / 621	34.8	+95.3, +1,817.59	−55.5, −1,073.71	−23.8, −432.26	**−2.97**, −66.47	[−11.43, +6.46]; [−246.90, +130.43]
B2c	pick 2022-05→2024	2,483	137 / 621	55.8	+90.1, +1,733.67	−142.5, −2,744.13	+10.8, +207.48	**−12.65**, −244.64	[−21.37, −3.57]; [−429.23, −53.76]
B3a	pick 2022-05→2024	752	60 / 85	43.9	+36.3, +320.14	−32.3, −203.18	−30.3, −22.10	**−2.20**, +26.47	[−10.14, −1.77]; [−74.91, +34.62]
B3b	pick 2022-05→2024	752	60 / 85	32.6	+104.2, +928.10	−50.3, −338.87	−21.9, −48.60	**+0.06**, +73.91	[−27.16, +6.00]; [−229.79, +153.45]
B3c	pick 2022-05→2024	752	60 / 85	56.0	+91.8, +801.36	−133.0, −858.15	+12.9, +26.50	**−7.11**, +70.91	[−47.60, −0.43]; [−400.77, +171.45]
A1a	check 2025→2026-10-09	1,035	101 / 451	36.6	+38.1, +430.92	−32.6, −286.49	−30.9, −32.10	**−6.72**, −23.79	[−14.64, +0.78]; [−158.27, +105.75]
A1b	check 2025→2026-10-09	1,035	101 / 451	27.4	+83.5, +329.73	−60.2, −662.52	−29.4, −72.10	**−20.74**, −390.25	[−41.61, +1.31]; [−621.10, −124.77]

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
A1c	check 2025→2026-10-09	1,035	101 / 451	54.0	+71.1, +416.03	−163.2, −1,635.74	+8.7, +8.80	**−36.67**, −527.59	[−78.56, +0.61]; [−822.26, −207.25]
A2a	check 2025→2026-10-09	1,502	155 / 456	32.4	+36.8, +377.42	−33.0, −318.01	−31.1, −39.60	**−10.43**, −92.99	[−14.25, −6.42]; [−187.40, +2.57]
A2b	check 2025→2026-10-09	1,502	155 / 456	36.0	+81.5, +878.55	−63.4, −597.62	−23.9, −45.10	**−11.19**, −65.92	[−29.74, +11.07]; [−317.65, +162.85]
A2c	check 2025→2026-10-09	1,502	155 / 456	55.9	+84.3, +927.09	−165.7, −1,585.56	+12.5, +21.70	**−25.90**, −180.35	[−58.12, +11.34]; [−619.37, +241.04]
A3a	check 2025→2026-10-09	2,435	146 / 451	37.1	+37.8, +365.85	−33.3, −304.89	−30.9, −34.10	**−6.90**, −56.15	[−11.17, −2.58]; [−116.06, +1.64]
A3b	check 2025→2026-10-09	2,435	146 / 451	29.0	+75.2, +511.39	−60.5, −623.93	−27.5, −59.10	**−21.21**, −295.22	[−31.60, −6.77]; [−507.83, −4.66]
A3c	check 2025→2026-10-09	2,435	146 / 451	54.5	+70.4, +543.60	−166.6, −1,690.55	+10.1, +13.40	**−37.46**, −473.01	[−55.26, −14.39]; [−823.39, −4.40]
B1a	check 2025→2026-10-09	6,350	197 / 457	41.6	+31.8, +686.04	−29.5, −642.27	−17.7, −432.26	**−3.96**, −89.19	[−4.78, −3.40]; [−107.28, −75.90]
B1b	check 2025→2026-10-09	6,350	197 / 457	35.0	+93.5, +2,028.67	−55.6, −1,238.59	−22.1, −442.26	**−3.33**, −93.77	[−8.44, +2.82]; [−197.96, +34.83]
B1c	check 2025→2026-10-09	6,350	197 / 457	59.0	+85.8, +1,874.77	−150.5, −3,356.40	+14.0, +293.48	**−11.00**, −268.77	[−16.67, −5.04]; [−397.73, −126.84]
B2a	check 2025→2026-10-09	5,406	136 / 446	39.7	+31.7, +703.98	−29.2, −655.49	−19.9, −452.26	**−5.05**, −116.08	[−6.43, −4.01]; [−152.33, −88.09]
B2b	check 2025→2026-10-09	5,406	136 / 446	35.4	+93.0, +2,087.91	−57.7, −1,324.10	−22.0, −462.26	**−4.28**, −114.81	[−9.47, +3.42]; [−237.38, +59.95]
B2c	check 2025→2026-10-09	5,406	136 / 446	57.3	+88.7, +2,007.76	−152.2, −3,497.85	+13.2, +277.48	**−14.04**, −340.73	[−21.01, −5.68]; [−527.97, −136.71]
B3a	check 2025→2026-10-09	5,454	321 / 457	37.5	+36.7, +376.44	−31.4, −325.41	−30.5, −40.10	**−5.86**, −62.38	[−7.25, −4.50]; [−119.96, −21.06]
B3b	check 2025→2026-10-09	5,454	321 / 457	31.6	+95.2, +1,028.90	−54.5, −557.24	−25.1, −64.10	**−7.17**, −55.57	[−11.78, −2.80]; [−130.11, +14.09]
B3c	check 2025→2026-10-09	5,454	321 / 457	53.7	+87.1, +955.61	−141.7, −1,427.58	+9.7, +19.50	**−18.89**, −148.59	[−26.18, −12.17]; [−285.84, −41.36]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
A1a	pick 2022-05→2024	+1,237.74, +39.9%, 2022-09-01, SPX 3955.0P m60	−1,412.26, −31.6%, 2022-05-19, SPX 3935.0P m90	0.79	36,376	681	total ≤ 0 (best 5 days +31,082; without them −52,115)
A1b	pick 2022-05→2024	+6,137.74, +204.6%, 2022-10-20, SPX 3755.0P m90	−3,402.26, −94.2%, 2022-05-17, SPX 4075.0P m90	0.69	82,581	681	total ≤ 0 (best 5 days +92,021; without them −151,650)
A1c	pick 2022-05→2024	+7,519.48, +200.0%, 2022-07-05, SPX 3725.0C m60	−6,804.52, −188.5%, 2022-05-17, SPX 4075.0P m90	0.75	115,672	681	total ≤ 0 (best 5 days +125,595; without them −198,535)
A2a	pick 2022-05→2024	+1,689.74, +39.9%, 2022-06-13, SPX 3750.0C m150	−1,322.26, −36.8%, 2022-05-19, SPX 3935.0P m210	0.91	65,254	660	total ≤ 0 (best 5 days +53,888; without them −69,049)
A2b	pick 2022-05→2024	+7,777.74, +437.0%, 2024-09-11, SPX 5445.0C m150	−3,622.26, −86.0%, 2022-05-24, SPX 3930.0P m60	1.12	86,433	596	468%
A2c	pick 2022-05→2024	+8,487.48, +476.8%, 2024-09-11, SPX 5445.0C m150	−7,244.52, −172.1%, 2022-05-24, SPX 3930.0P m60	1.07	112,670	543	621%
A3a	pick 2022-05→2024	+1,813.74, +40.0%, 2022-05-18, SPX 4050.0P m60	−1,562.26, −40.7%, 2022-10-21, SPX 3710.0P m90	1.01	59,262	587	1962%
A3b	pick 2022-05→2024	+8,687.74, +191.4%, 2022-05-18, SPX 4050.0P m60	−3,837.17, −99.9%, 2022-10-21, SPX 3710.0P m90	1.01	170,610	679	6266%
A3c	pick 2022-05→2024	+10,501.48, +231.3%, 2022-05-18, SPX 4050.0P m60	−7,674.34, −199.9%, 2022-10-21, SPX 3710.0P m90	1.13	187,074	633	482%
B1a	pick 2022-05→2024	+1,625.74, +39.9%, 2024-12-31, SPX 5955.0P m60	−2,557.17, −99.5%, 2023-09-21, SPX 4355.0C m1	0.77	189,080	442	total ≤ 0 (best 5 days +11,243; without them −200,162)
B1b	pick 2022-05→2024	+12,317.74, +443.1%, 2024-12-18, SPX 6090.0P m90	−3,431.13, −100.0%, 2024-04-15, SPX 5135.0C m1	1.06	233,154	324	521%

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
B1c	pick 2022-05→2024	+13,427.48, +483.0%, 2024-12-18, SPX 6090.0P m90	−6,862.26, −200.1%, 2024-04-15, SPX 5135.0C m1	0.93	391,350	443	total ≤ 0 (best 5 days +559,855; without them −768,330)
B2a	pick 2022-05→2024	+1,533.74, +39.9%, 2024-10-04, SPX 5775.0P m1	−1,382.26, −37.8%, 2022-08-10, SPX 4165.0C m1	0.68	259,220	618	total ≤ 0 (best 5 days +9,032; without them −266,317)
B2b	pick 2022-05→2024	+12,317.74, +443.1%, 2024-12-18, SPX 6090.0P m90	−3,322.26, −94.1%, 2024-10-31, SPX 5740.0C m1	0.90	456,313	621	total ≤ 0 (best 5 days +395,764; without them −560,812)
B2c	pick 2022-05→2024	+13,427.48, +483.0%, 2024-12-18, SPX 6090.0P m90	−6,644.52, −188.2%, 2024-10-31, SPX 5740.0C m1	0.80	889,284	621	total ≤ 0 (best 5 days +390,675; without them −998,112)
B3a	pick 2022-05→2024	+1,281.74, +39.9%, 2024-09-06, SPX 5410.0C m90	−1,112.26, −35.2%, 2024-09-06, SPX 5435.0C m60	1.23	18,770	61	197%
B3b	pick 2022-05→2024	+11,797.74, +655.4%, 2024-12-18, SPX 6075.0P m150	−2,932.26, −92.8%, 2024-09-06, SPX 5435.0C m60	1.32	83,638	79	265%
B3c	pick 2022-05→2024	+12,515.48, +695.3%, 2024-12-18, SPX 6075.0P m150	−5,864.52, −185.6%, 2024-09-06, SPX 5435.0C m60	1.19	137,832	79	337%
A1a	check 2025→2026-10-09	+4,161.74, +40.0%, 2025-04-09, SPX 5085.0P m60	−3,532.26, −37.5%, 2025-04-04, SPX 5255.0P m90	0.87	74,759	368	total ≤ 0 (best 5 days +140,582; without them −165,205)
A1b	check 2025→2026-10-09	+3,477.74, +37.0%, 2025-04-04, SPX 5255.0P m90	−13,347.17, −100.0%, 2025-04-09, SPX 5125.0P m60	0.19	404,331	446	total ≤ 0 (best 5 days +40,801; without them −444,708)
A1c	check 2025→2026-10-09	+7,239.48, +76.9%, 2025-04-04, SPX 5255.0P m90	−26,694.34, −200.0%, 2025-04-09, SPX 5125.0P m60	0.30	590,220	384	total ≤ 0 (best 5 days +136,512; without them −682,564)
A2a	check 2025→2026-10-09	+2,647.74, +23.2%, 2025-04-11, SPX 5155.0C m90	−4,132.26, −31.1%, 2025-04-07, SPX 4900.0C m150	0.57	155,224	408	total ≤ 0 (best 5 days +96,325; without them −235,994)
A2b	check 2025→2026-10-09	+9,607.74, +84.2%, 2025-04-11, SPX 5155.0C m90	−6,502.26, −88.7%, 2025-04-08, SPX 5025.0C m270	0.83	388,282	409	total ≤ 0 (best 5 days +308,386; without them −407,402)
A2c	check 2025→2026-10-09	+14,169.48, +124.2%, 2025-04-11, SPX 5155.0C m90	−13,004.52, −177.4%, 2025-04-08, SPX 5025.0C m270	0.74	695,495	408	total ≤ 0 (best 5 days +431,141; without them −702,028)
A3a	check 2025→2026-10-09	+4,161.74, +40.0%, 2025-04-09, SPX 5085.0P m60	−3,532.26, −37.5%, 2025-04-04, SPX 5255.0P m90	0.71	217,405	383	total ≤ 0 (best 5 days +177,528; without them −314,258)
A3b	check 2025→2026-10-09	+5,297.74, +118.5%, 2025-10-10, SPX 6680.0P m150	−13,347.17, −100.0%, 2025-04-09, SPX 5125.0P m60	0.33	754,262	451	total ≤ 0 (best 5 days +152,747; without them −871,616)
A3c	check 2025→2026-10-09	+7,239.48, +76.9%, 2025-04-04, SPX 5255.0P m90	−26,694.34, −200.0%, 2025-04-09, SPX 5125.0P m60	0.39	1,198,123	451	total ≤ 0 (best 5 days +252,347; without them −1,404,122)
B1a	check 2025→2026-10-09	+1,957.74, +40.0%, 2026-06-17, SPX 7570.0P m60	−2,891.13, −100.0%, 2025-07-30, SPX 6360.0C m60	0.76	569,189	449	total ≤ 0 (best 5 days +22,718; without them −589,061)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
B1b	check 2025→2026-10-09	+16,707.74, +843.8%, 2025-10-10, SPX 6770.0P m30	−5,517.26, −98.2%, 2026-09-16, SPX 7560.0C m1	0.88	697,999	401	total ≤ 0 (best 5 days +624,115; without them −1,219,527)
B1c	check 2025→2026-10-09	+17,497.48, +883.7%, 2025-10-10, SPX 6770.0P m30	−11,034.52, −196.3%, 2026-09-16, SPX 7560.0C m1	0.80	1,789,686	434	total ≤ 0 (best 5 days +576,212; without them −2,282,931)
B2a	check 2025→2026-10-09	+2,301.74, +40.0%, 2026-02-20, SPX 6790.0C m1	−2,891.13, −100.0%, 2025-07-30, SPX 6360.0C m60	0.71	627,925	446	total ≤ 0 (best 5 days +22,686; without them −650,206)
B2b	check 2025→2026-10-09	+16,707.74, +843.8%, 2025-10-10, SPX 6770.0P m30	−5,762.26, −97.0%, 2026-02-20, SPX 6890.0P m1	0.87	915,097	251	total ≤ 0 (best 5 days +550,915; without them −1,171,573)
B2c	check 2025→2026-10-09	+17,497.48, +883.7%, 2025-10-10, SPX 6770.0P m30	−11,524.52, −194.0%, 2026-02-20, SPX 6890.0P m1	0.77	1,932,579	251	total ≤ 0 (best 5 days +521,569; without them −2,363,566)
B3a	check 2025→2026-10-09	+2,985.74, +40.0%, 2025-04-07, SPX 4925.0C m210	−4,022.26, −33.5%, 2025-04-07, SPX 4820.0C m1	0.69	343,898	456	total ≤ 0 (best 5 days +72,225; without them −412,431)
B3b	check 2025→2026-10-09	+23,857.74, +381.1%, 2025-04-08, SPX 5295.0P m60	−7,837.17, −99.2%, 2025-04-04, SPX 5215.0C m1	0.85	482,980	390	total ≤ 0 (best 5 days +678,847; without them −981,925)
B3c	check 2025→2026-10-09	+26,359.48, +421.1%, 2025-04-08, SPX 5295.0P m60	−15,674.34, −198.4%, 2025-04-04, SPX 5215.0C m1	0.78	969,168	412	total ≤ 0 (best 5 days +852,845; without them −1,663,254)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.92 a trade) and Robinhood's charged instead (mean $2.22). The study's fees are given beside them below. Re-pricing: exact (92,484 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
A1a	pick 2022-05→2024	665	**+1.78**, −31.63	+1.07, −32.35	49.9 / 49.8	[−5.55, +10.85] / [−6.21, +10.10]
A1b	pick 2022-05→2024	665	**−8.32**, −89.67	−9.02, −90.37	32.0 / 31.9	[−24.75, +16.03] / [−25.54, +15.35]
A1c	pick 2022-05→2024	665	**−7.38**, −109.68	−8.77, −111.10	66.9 / 66.2	[−33.39, +30.57] / [−34.77, +29.18]
A2a	pick 2022-05→2024	1,081	**−2.60**, −14.02	−3.29, −14.71	43.1 / 43.1	[−10.00, +1.96] / [−10.74, +1.18]
A2b	pick 2022-05→2024	1,081	**+12.97**, +32.27	+12.30, +31.60	33.3 / 32.7	[−7.98, +40.79] / [−8.79, +40.13]
A2c	pick 2022-05→2024	1,081	**+12.62**, +29.03	+11.28, +27.68	64.8 / 64.0	[−13.99, +44.01] / [−15.39, +42.35]
A3a	pick 2022-05→2024	1,239	**+0.74**, +2.16	+0.07, +1.42	47.4 / 47.4	[−6.33, +7.31] / [−7.06, +6.68]
A3b	pick 2022-05→2024	1,239	**−4.46**, +2.91	−5.11, +2.19	29.1 / 28.6	[−26.01, +2.77] / [−26.65, +2.15]
A3c	pick 2022-05→2024	1,239	**−2.68**, +49.04	−3.97, +47.59	66.3 / 66.0	[−35.01, +13.09] / [−36.20, +11.73]
B1a	pick 2022-05→2024	2,426	**−3.87**, −77.87	−3.88, −78.17	42.3 / 42.3	[−5.30, −2.88] / [−5.32, −2.90]
B1b	pick 2022-05→2024	2,426	**+2.03**, +44.59	+2.01, +44.30	34.5 / 34.5	[−10.50, +14.05] / [−10.52, +14.04]
B1c	pick 2022-05→2024	2,426	**−4.66**, −85.93	−4.69, −86.51	58.4 / 58.4	[−16.86, +8.36] / [−16.90, +8.33]
B2a	pick 2022-05→2024	2,483	**−5.33**, −103.62	−5.35, −103.91	38.6 / 38.6	[−6.43, −4.58] / [−6.45, −4.60]
B2b	pick 2022-05→2024	2,483	**−2.97**, −66.47	−2.99, −66.76	34.8 / 34.8	[−11.43, +6.46] / [−11.44, +6.44]
B2c	pick 2022-05→2024	2,483	**−12.65**, −244.64	−12.68, −245.22	55.8 / 55.8	[−21.37, −3.57] / [−21.40, −3.60]

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
B3a	pick 2022-05→2024	752	**−2.20**, +26.47	−2.80, +25.68	43.9 / 43.8	[−10.14, −1.77] / [−10.87, −2.38]
B3b	pick 2022-05→2024	752	**+0.06**, +73.91	−0.53, +73.13	32.6 / 32.3	[−27.16, +6.00] / [−27.90, +5.39]
B3c	pick 2022-05→2024	752	**−7.11**, +70.91	−8.30, +69.36	56.0 / 55.9	[−47.60, −0.43] / [−49.01, −1.58]
A1a	check 2025→2026-10-09	1,035	**−6.72**, −23.79	−7.54, −24.69	36.6 / 36.6	[−14.64, +0.78] / [−15.50, −0.10]
A1b	check 2025→2026-10-09	1,035	**−20.74**, −390.25	−21.53, −391.13	27.4 / 27.1	[−41.61, +1.31] / [−42.34, +0.40]
A1c	check 2025→2026-10-09	1,035	**−36.67**, −527.59	−38.24, −529.36	54.0 / 53.4	[−78.56, +0.61] / [−80.04, −0.96]
A2a	check 2025→2026-10-09	1,502	**−10.43**, −92.99	−11.17, −93.85	32.4 / 32.2	[−14.25, −6.42] / [−15.01, −7.26]
A2b	check 2025→2026-10-09	1,502	**−11.19**, −65.92	−11.89, −66.75	36.0 / 35.8	[−29.74, +11.07] / [−30.37, +10.29]
A2c	check 2025→2026-10-09	1,502	**−25.90**, −180.35	−27.31, −182.01	55.9 / 55.3	[−58.12, +11.34] / [−59.32, +9.59]
A3a	check 2025→2026-10-09	2,435	**−6.90**, −56.15	−7.62, −57.06	37.1 / 37.1	[−11.17, −2.58] / [−11.88, −3.35]
A3b	check 2025→2026-10-09	2,435	**−21.21**, −295.22	−21.90, −296.11	29.0 / 28.7	[−31.60, −6.77] / [−32.27, −7.52]
A3c	check 2025→2026-10-09	2,435	**−37.46**, −473.01	−38.83, −474.77	54.5 / 54.0	[−55.26, −14.39] / [−56.61, −15.99]
B1a	check 2025→2026-10-09	6,350	**−3.96**, −89.19	−3.97, −89.48	41.6 / 41.6	[−4.78, −3.40] / [−4.79, −3.41]
B1b	check 2025→2026-10-09	6,350	**−3.33**, −93.77	−3.34, −94.06	35.0 / 35.0	[−8.44, +2.82] / [−8.46, +2.81]
B1c	check 2025→2026-10-09	6,350	**−11.00**, −268.77	−11.03, −269.36	59.0 / 59.0	[−16.67, −5.04] / [−16.70, −5.06]
B2a	check 2025→2026-10-09	5,406	**−5.05**, −116.08	−5.07, −116.37	39.7 / 39.7	[−6.43, −4.01] / [−6.44, −4.02]
B2b	check 2025→2026-10-09	5,406	**−4.28**, −114.81	−4.29, −115.10	35.4 / 35.4	[−9.47, +3.42] / [−9.49, +3.40]
B2c	check 2025→2026-10-09	5,406	**−14.04**, −340.73	−14.07, −341.32	57.3 / 57.3	[−21.01, −5.68] / [−21.04, −5.71]
B3a	check 2025→2026-10-09	5,454	**−5.86**, −62.38	−6.35, −63.19	37.5 / 37.4	[−7.25, −4.50] / [−7.74, −5.02]
B3b	check 2025→2026-10-09	5,454	**−7.17**, −55.57	−7.66, −56.38	31.6 / 31.5	[−11.78, −2.80] / [−12.26, −3.30]
B3c	check 2025→2026-10-09	5,454	**−18.89**, −148.59	−19.87, −150.21	53.7 / 53.6	[−26.18, −12.17] / [−27.20, −13.13]

Random baseline (as the study computed it, at its own fees): 500 cell-matched random draws per rule and exit (step6/out/results.csv): the rule's percentile is printed there; on Check 14 of 18 rule × exit cells sit below the 20th.

Exit reasons (share of trades):

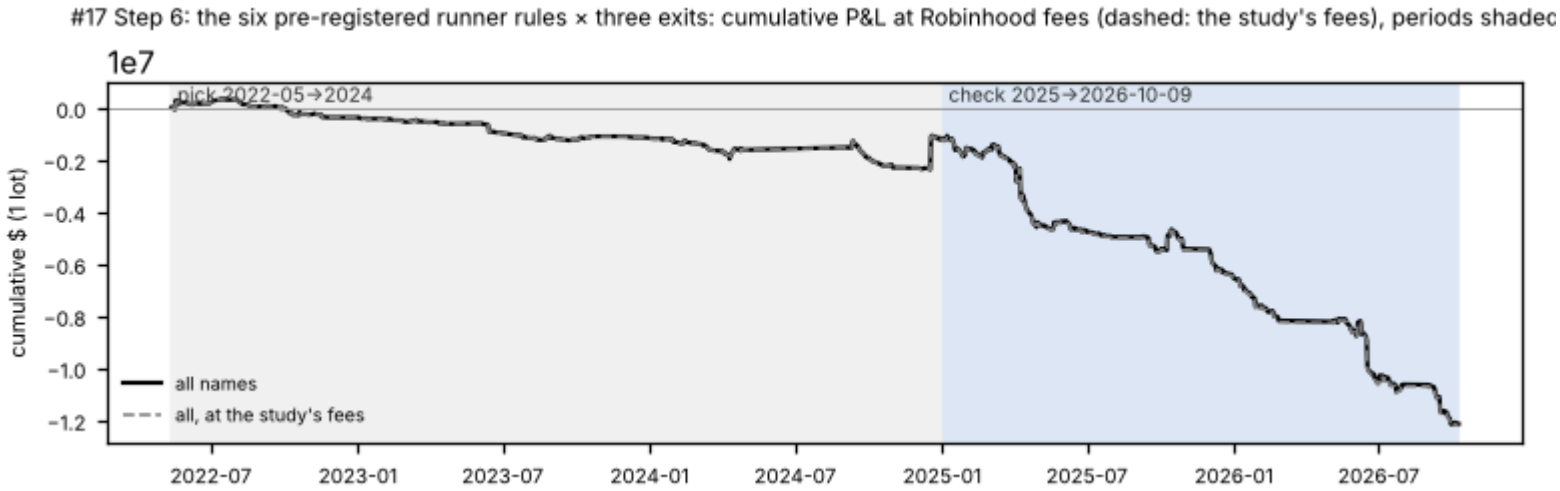
- pick 2022-05→2024: half at +40%, trail the rest 50%, out at 3:30 33.3%; time 20.4%; stop 15.9%; trail 11.9%; target 10.9%; time_armed 7.6%
- check 2025→2026-10-09: half at +40%, trail the rest 50%, out at 3:30 33.3%; time 21.0%; stop 17.5%; trail 10.7%; target 10.0%; time_armed 7.6%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	A1a	218	40.4	−1.06	−90.49	−19,727
2023	A1a	249	59.4	+7.33	+5.46	+1,359
2024	A1a	198	48.5	−2.05	−13.46	−2,665
2025	A1a	615	44.1	−1.07	+32.65	+20,082
2026	A1a	420	25.7	−14.99	−106.44	−44,704
2022	A1b	218	22.5	−9.04	−137.71	−30,020
2023	A1b	249	43.8	+6.75	−8.77	−2,183
2024	A1b	198	27.8	−26.49	−138.51	−27,425
2025	A1b	615	33.7	−3.02	−353.37	−217,325

year	name	trades	win %	mean %	mean $	total $
2026	A1b	420	18.3	−46.69	−444.24	−186,581
2022	A1c	218	61.9	−10.30	−183.45	−39,991
2023	A1c	249	75.5	+17.04	−4.26	−1,060
2024	A1c	198	61.6	−34.88	−161.05	−31,889
2025	A1c	615	64.9	−3.33	−363.00	−223,243
2026	A1c	420	38.1	−85.48	−768.59	−322,809
2022	A2a	341	37.5	−6.84	−59.24	−20,201
2023	A2a	403	46.9	−0.68	−9.28	−3,740
2024	A2a	337	44.2	−0.61	+26.06	+8,781
2025	A2a	1,058	31.1	−11.14	−122.47	−129,574
2026	A2a	444	35.4	−8.74	−22.73	−10,094
2022	A2b	341	27.3	−5.78	−71.82	−24,492
2023	A2b	403	33.3	+18.53	+48.86	+19,691
2024	A2b	337	39.5	+25.30	+117.76	+39,686
2025	A2b	1,058	35.7	−8.12	−68.40	−72,363
2026	A2b	444	36.7	−18.50	−60.03	−26,653
2022	A2c	341	62.8	−5.91	−36.06	−12,296
2023	A2c	403	70.0	+23.79	+57.73	+23,265
2024	A2c	337	60.5	+18.00	+60.56	+20,408
2025	A2c	1,058	57.1	−22.02	−212.45	−224,768
2026	A2c	444	53.2	−35.16	−103.87	−46,118
2022	A3a	493	48.5	+2.75	+48.91	+24,112
2023	A3a	440	48.2	+0.72	−23.36	−10,276
2024	A3a	306	44.4	−2.47	−36.48	−11,164
2025	A3a	1,516	36.7	−7.38	−73.51	−111,435
2026	A3a	919	37.8	−6.10	−27.52	−25,294
2022	A3b	493	27.2	−1.70	+56.95	+28,076
2023	A3b	440	30.2	−2.25	−3.95	−1,737
2024	A3b	306	30.4	−12.08	−74.28	−22,731
2025	A3b	1,516	27.9	−22.64	−401.46	−608,621
2026	A3b	919	30.7	−18.85	−119.97	−110,248
2022	A3c	493	65.9	−0.02	+89.54	+44,143
2023	A3c	440	74.5	+8.16	+115.29	+50,728
2024	A3c	306	54.9	−22.56	−111.48	−34,112
2025	A3c	1,516	52.4	−40.91	−640.99	−971,742
2026	A3c	919	58.0	−31.75	−195.90	−180,033
2023	B1a	911	43.4	−3.69	−71.84	−65,442
2024	B1a	1,515	41.7	−3.97	−81.50	−123,477
2025	B1a	3,411	42.0	−3.87	−85.14	−290,404
2026	B1a	2,939	41.3	−4.06	−93.89	−275,938

year	name	trades	win %	mean %	mean $	total $
2023	B1b	911	36.2	−3.34	−78.31	−71,343
2024	B1b	1,515	33.4	+5.26	+118.50	+179,525
2025	B1b	3,411	36.1	−1.44	−50.85	−173,447
2026	B1b	2,939	33.9	−5.51	−143.57	−421,965
2023	B1c	911	59.9	−8.59	−171.21	−155,975
2024	B1c	1,515	57.5	−2.29	−34.65	−52,500
2025	B1c	3,411	60.1	−7.57	−175.77	−599,566
2026	B1c	2,939	57.8	−14.98	−376.71	−1,107,153
2022	B2a	247	39.7	−5.50	−117.31	−28,974
2023	B2a	775	37.5	−5.42	−104.41	−80,916
2024	B2a	1,461	39.0	−5.25	−100.89	−147,395
2025	B2a	2,755	40.3	−4.72	−102.86	−283,385
2026	B2a	2,651	39.0	−5.41	−129.81	−344,135
2022	B2b	247	33.2	−17.66	−395.41	−97,667
2023	B2b	775	34.1	−10.26	−223.55	−173,248
2024	B2b	1,461	35.5	+3.38	+72.46	+105,867
2025	B2b	2,755	36.9	+1.06	+22.17	+61,092
2026	B2b	2,651	33.9	−9.82	−257.17	−681,750
2022	B2c	247	50.2	−33.33	−708.26	−174,939
2023	B2c	775	54.7	−20.22	−396.09	−306,968
2024	B2c	1,461	57.4	−5.14	−85.92	−125,530
2025	B2c	2,755	58.3	−7.49	−151.34	−416,933
2026	B2c	2,651	56.4	−20.84	−537.56	−1,425,064
2024	B3a	752	43.9	−2.20	+26.47	+19,903
2025	B3a	3,157	36.8	−6.42	−86.42	−272,818
2026	B3a	2,297	38.4	−5.10	−29.34	−67,388
2024	B3b	752	32.6	+0.06	+73.91	+55,578
2025	B3b	3,157	31.5	−6.89	−60.70	−191,615
2026	B3b	2,297	31.8	−7.56	−48.53	−111,463
2024	B3c	752	56.0	−7.11	+70.91	+53,325
2025	B3c	3,157	52.7	−19.32	−180.17	−568,806
2026	B3c	2,297	55.0	−18.30	−105.18	−241,604
2022	all names	3,897	43.9	−6.21	−90.32	−351,977
2023	all names	8,334	47.7	−1.75	−93.33	−777,846
2024	all names	13,707	44.4	−2.06	−6.99	−95,814
2025	all names	37,536	43.2	−8.88	−140.53	−5,274,871
2026	all names	29,010	42.3	−13.12	−194.04	−5,628,994



Amendments, deviations and integrity checks. The rules are chosen by the committed step6/trade.py choose() on the scratchpad population (39 MB, not committed); verify.py 0 differences in 480 figures; lookahead.py 0 differences in 104,748 values.

What it showed. No rule chosen on Pick made money on Check.

#18 Compass engine audit (six module groups)

Folder: `research_notes/Compass engine audit/, reports/Compass engine audit.md`. **Commits:** — / — / 9cf8110, 59e1a1a, d724ae4, 18cf86a, b84fa3c, 09799af; report 138ff0e.
Dates: 2026-10-21.

Hypothesis. The owner’s 16-module Compass specification picks direction and opens trades that pay after costs.

Rule. Each module tested under its own definition (exits and sizing, expected move/ATR/flow, clock and power hour, greeks and books, horizons, sweep and grade).

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). Ask in, bid out, $0.65 each way.

Data. 0DTE SPY/QQQ/IWM/SPXW 980/980/621/1,106 sessions; daily bars for horizons.

Periods and holdouts. fit 2022–24 / check 2025–26.

Pass rule (as pre-registered). Picked on 2022–24, read on 2025–26, friction-adjusted.

Verdict. No directional rule survives after fees. What holds: the 15:58 flat, the −32% backstop, a breakeven lock at +0% with a target on, a trailed remainder, the r,q pricer from parity, Method A flips with three books, net/gross confidence, the MOC magnitude as a volatility gauge, an ATR > 1.5 block on counter-trend fades.

Metrics. not available — the audit committed row-level paths and events (paths_*.parquet, events_*.parquet) and per-cell tables; its exit stacks were scored by cell, and no single trade list per rule exists to recompute (spec stack A −4.3% / B −4.5% of debit a trade, exits_sizing.md line 14).

Amendments, deviations and integrity checks. Six blockers stated (no OPRA tape, no strike-level OI history before 2026-10, …).

What it showed. Frictions, not edges.

#19 Phase 2 hyp A: the low-moc_mag closing pin (SPXW short call spread, 15:30, held)

Folder: `research_notes/Compass phase 2/hyp_a_closing_pin`. **Commits:** prereg in-doc before the OOS read / — / 5deb793. **Dates:** 2026-10-21.

Hypothesis. On quiet closes (low moc_mag) the 15:30 short credit spread pins and pays.

Rule. SPXW short call spread, short strike at 1.00 × the move rule, 25 wide, 15:30 at the NBBO, held to settlement, on low-moc_mag sessions (the pre-registered cell).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.65 a contract (leg) to open, nothing at expiry; bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPXW 2022-05-13 → 2026-10-09.

Periods and holdouts. fit 2022–24 / OOS 2025–26 (read once).

Pass rule (as pre-registered). The pick fixed on fit; read OOS for a positive mean with an interval clear of zero.

Verdict. NO EDGE: fit +$26 a spread, OOS −$33 [−84, +13].

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	fit 2022–24	252	252 / 656	91.7	+2.3, +55.08	−12.2, −299.66	+1.9, +45.42	**+1.07**, +25.52	[+0.42, +1.63]; [+9.66, +38.99]
SPXW	OOS 2025–26	177	177 / 456	87.6	+2.6, +62.08	−29.6, −709.56	+2.0, +47.92	**−1.43**, −33.83	[−3.57, +0.26]; [−85.45, +6.92]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	fit 2022–24	+197.83, +8.6%, 2022-09-16, SPXW 15:30 3880.0/3905.0C	−688.08, −28.1%, 2023-03-08, SPXW 15:30 3985.0/4010.0C	2.02	919	99	13%
SPXW	OOS 2025–26	+172.83, +7.4%, 2025-04-14, SPXW 15:30 5450.0/5475.0C	−2,312.17, −100.1%, 2025-04-30, SPXW 15:30 5535.0/5560.0C	0.62	8,138	426	total ≤ 0 (best 5 days +789; without them −6,778)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.30 a trade) and Robinhood's charged instead (mean $2.10). The study's fees are given beside them below. Re-pricing: exact (429 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPXW	fit 2022–24	252	**+1.07**, +25.52	+1.10, +26.31	91.7 / 91.7	[+0.42, +1.63] / [+0.45, +1.66]
SPXW	OOS 2025–26	177	**−1.43**, −33.83	−1.39, −33.03	87.6 / 87.6	[−3.57, +0.26] / [−3.53, +0.30]

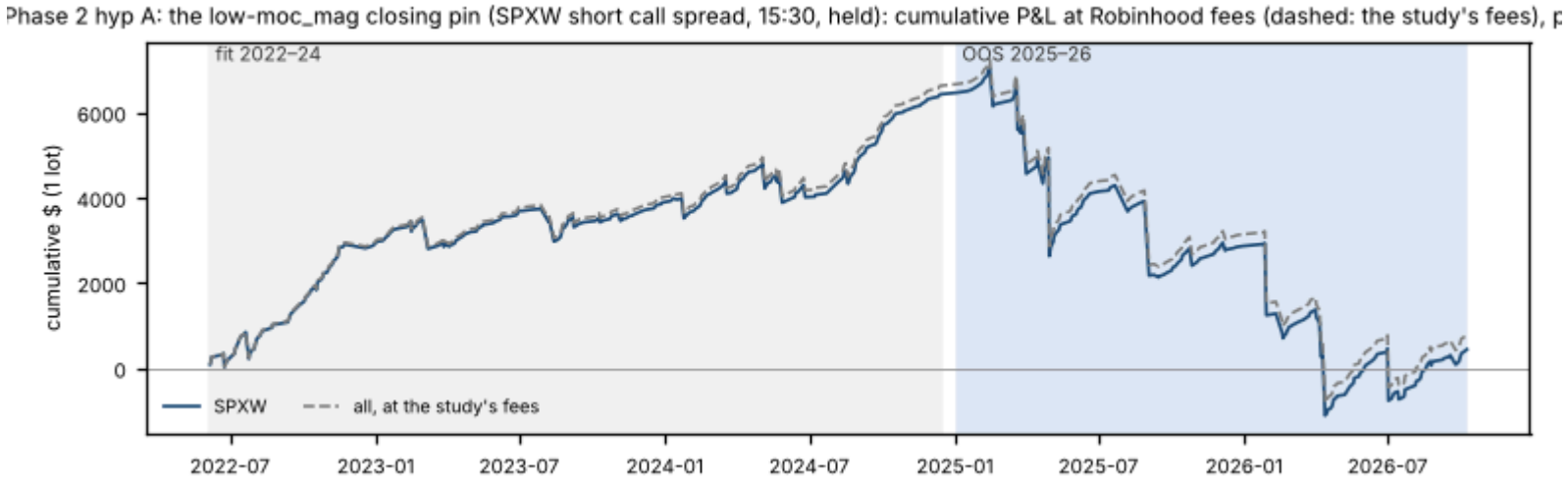
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- fit 2022–24: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	55	92.7	+2.19	+52.38	+2,881
2023	SPXW	95	90.5	+0.43	+10.56	+1,003
2024	SPXW	102	92.2	+1.05	+24.96	+2,546
2025	SPXW	94	88.3	−1.62	−37.93	−3,565
2026	SPXW	83	86.7	−1.20	−29.19	−2,423



Amendments, deviations and integrity checks. Cells and the pick written before the OOS run (s3_report.py fit, then oos).

What it showed. The pin made money on fit and lost out of sample.

#20a Phase 2 hyp B: boundary touch — SPY 2-wide 25-delta credit spread, held (the touch pick)

Folder: `research_notes/Compass phase 2/hyp_b_boundary_theta`. **Commits:** out/prereg.txt before the OOS read / — / 2c241f4. **Dates:** 2026-10-21.

Hypothesis. When price touches the straddle boundary, the far side's credit spread decays faster.

Rule. At a touch: sell the 2-wide spread on the touched side, short strike at 25 delta, held to settlement.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.65 a contract (leg) to open, nothing at expiry; bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPY 2022-11 → 2026-10.

Periods and holdouts. fit 2022–24 / OOS 2025–26.

Pass rule (as pre-registered). The highest fit lower bound (N ≥ 300, ≥ 150 sessions, positive without the best 5); its lower bound was negative ("least bad").

Verdict. No edge: the touch adds nothing over the unconditional sell at the same minute (the increment flips sign between periods).

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	fit 2022–24	2,890	512 / 553	83.7	+11.3, +19.81	−47.3, −83.52	+9.4, +16.90	**+1.74**, +3.01	[−0.14, +3.51]; [−0.35, +6.15]
SPY	OOS 2025–26	2,600	432 / 457	88.2	+12.6, +21.86	−54.9, −95.16	+11.1, +19.90	**+4.59**, +8.04	[+2.24, +6.37]; [+3.91, +11.15]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	fit 2022–24	+44.90, +29.0%, 2024-08-28, SPY 554.0/552.0 down	−194.10, −100.1%, 2024-05-31, SPY 525.0/527.0 up	1.22	3,489	165	16%
SPY	OOS 2025–26	+45.90, +29.8%, 2025-11-14, SPY 673.0/675.0 up	−190.10, −99.5%, 2026-04-13, SPY 684.0/686.0 up	1.72	4,370	59	8%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.30 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (5,490 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPY	fit 2022–24	2,890	**+1.74**, +3.01	+1.07, +1.81	83.7 / 83.5	[−0.14, +3.51] / [−0.80, +2.84]
SPY	OOS 2025–26	2,600	**+4.59**, +8.04	+3.91, +6.84	88.2 / 88.1	[+2.24, +6.37] / [+1.56, +5.68]

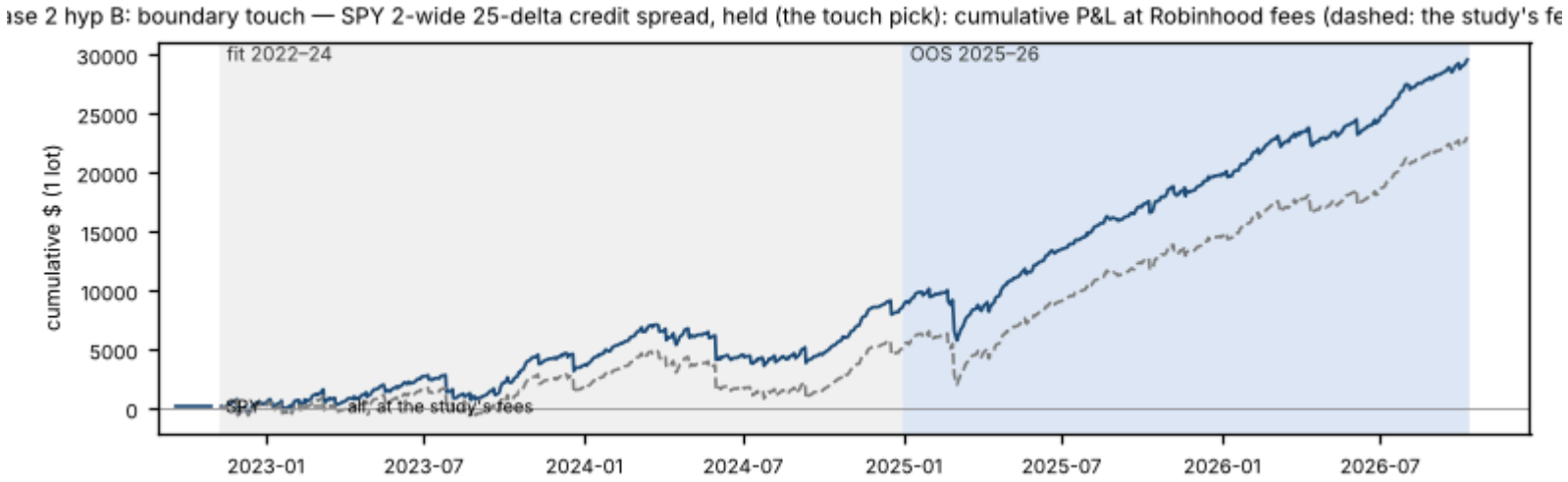
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- fit 2022–24: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPY	188	80.9	+1.43	+2.26	+425
2023	SPY	1,366	82.8	+1.38	+2.35	+3,205
2024	SPY	1,336	85.1	+2.15	+3.79	+5,068
2025	SPY	1,457	87.7	+4.36	+7.60	+11,067
2026	SPY	1,143	88.8	+4.89	+8.60	+9,836



Amendments, deviations and integrity checks. Deduplicated as analyze.py (one touch a session, side, e, mult, rule, width).

What it showed. Positive in both periods but no better than selling unconditionally.

#20b Phase 2 hyp B: the midday SPY iron condor (em1, mid-VIX), the midday pick

Folder: `research_notes/Compass phase 2/hyp_b_boundary_theta`. **Commits:** out/prereg.txt / — / 2c241f4. **Dates:** 2026-10-21.

Hypothesis. Midday iron condors on SPY are unconditional short premium that pays.

Rule. 11:30–13:30 every half hour: SPY 2-wide iron condor, shorts at the first listed strikes outside spot ± the ATM straddle, held to settlement, VIX in the fit's middle tercile.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.65 a contract (leg) to open, nothing at expiry; bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPY 2022-11 → 2026-10.

Periods and holdouts. fit 2022–24 / OOS 2025–26.

Pass rule (as pre-registered). As 20a.

Verdict. "Needs X" (a forward paper run): +4.6% of risk fit, +3.3% OOS (CI $0.03–$10.69). Became the tracker's condor_em1 (all-VIX parent, #26).

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	fit 2022–24	748	178 / 510	80.7	+17.0, +28.34	−40.0, −68.04	+14.2, +24.30	**+6.02**, +9.78	[+2.67, +9.18]; [+4.23, +15.06]
SPY	OOS 2025–26	1,202	280 / 456	81.9	+16.8, +28.18	−50.4, −83.87	+14.2, +24.80	**+4.67**, +7.95	[+1.62, +7.46]; [+2.98, +12.41]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	fit 2022–24	+64.80, +48.0%, 2024-09-18, None	−173.20, −100.1%, 2024-09-11, None	1.75	1,387	57	18%
SPY	OOS 2025–26	+58.80, +41.7%, 2026-04-29, None	−176.20, −100.1%, 2026-07-24, None	1.53	1,696	92	13%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.60 a trade) and Robinhood's charged instead (mean $0.20). The study's fees are given beside them below. Re-pricing: exact (1,950 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPY	fit 2022–24	748	**+6.02**, +9.78	+4.61, +7.39	80.7 / 80.5	[+2.67, +9.18] / [+1.27, +7.77]
SPY	OOS 2025–26	1,202	**+4.67**, +7.95	+3.25, +5.55	81.9 / 81.6	[+1.62, +7.46] / [+0.21, +6.04]

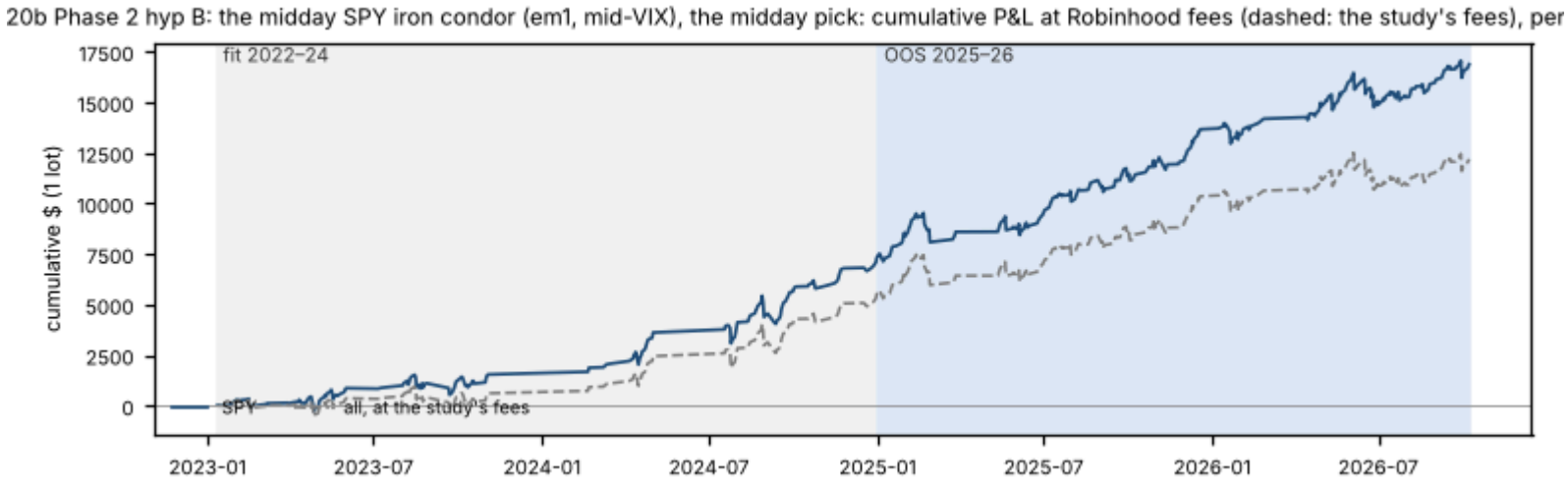
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- fit 2022–24: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2023	SPY	388	74.7	+2.43	+3.99	+1,550
2024	SPY	360	87.2	+9.88	+16.02	+5,769
2025	SPY	631	83.5	+6.05	+10.06	+6,350
2026	SPY	571	80.2	+3.13	+5.62	+3,210



Amendments, deviations and integrity checks. The VIX terciles recomputed from the fit d15 condors exactly as analyze.py.

What it showed. Small positive in both periods; $5 a condor on a four-leg structure.

#21a Phase 2 hyp C: 15:00 SPY/QQQ P1 iron ±2 butterfly, held to settlement

Folder: `research_notes/Compass phase 2/hyp_c_debit_flies`. **Commits:** preregistration.md before the OOS run / — / faea67c. **Dates:** 2026-10-21.

Hypothesis. Late-day ATM butterflies on quiet closes pin.

Rule. 15:00, SPY and QQQ, P1 iron ±2 at the 4-leg NBBO, held to settlement, unconditional.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.65 a contract (leg) to open, nothing at expiry (four legs); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPY and QQQ 2022-11 → 2026-10.

Periods and holdouts. fit 2022–24 / OOS 2025–26.

Pass rule (as pre-registered). Edge only if the OOS mean is positive with its interval clear of zero AND positive without the best five sessions AND both OOS years positive.

Verdict. Not confirmed: the picks fell to +2 to +4.5% of risk OOS with intervals through zero and die under $0.01 a leg; the un-picked iron ±5 read +5.0/+6.0% (→ the tracker's fly5).

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	fit 2022–24	521	521 / 553	64.5	+54.4, +50.94	−51.8, −49.88	+23.4, +25.80	**+16.71**, +15.14	[+11.64, +23.15]; [+10.82, +20.41]
SPY	fit 2022–24	525	525 / 553	63.4	+50.9, +49.19	−51.8, −51.01	+18.3, +21.80	**+13.33**, +12.54	[+8.43, +18.36]; [+8.36, +16.78]
all names	fit 2022–24	1,046	529 / 553	64.0	+52.7, +50.07	−51.8, −50.46	+21.5, +23.80	**+15.01**, +13.84	[+10.61, +20.36]; [+10.06, +18.07]
QQQ	OOS 2025–26	440	440 / 457	51.4	+75.9, +57.13	−69.3, −52.57	+2.3, +2.55	**+5.29**, +3.78	[−2.50, +13.57]; [−2.32, +10.23]
SPY	OOS 2025–26	438	438 / 457	55.3	+65.6, +53.87	−59.4, −48.97	+10.7, +11.05	**+9.65**, +7.85	[+2.22, +18.07]; [+1.83, +15.04]
all names	OOS 2025–26	878	440 / 457	53.3	+70.6, +55.45	−64.6, −50.85	+6.8, +6.80	**+7.46**, +5.81	[+0.15, +15.30]; [+0.44, +12.00]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	fit 2022–24	+138.80, +261.9%, 2024-02-01, QQQ 1500 iron K0 422.0 ±2.0	−142.20, −100.1%, 2023-07-07, QQQ 1500 iron K0 369.0 ±2.0	1.85	592	65	8%
SPY	fit 2022–24	+132.80, +201.2%, 2024-12-31, SPY 1500 iron K0 586.0 ±2.0	−144.20, −100.1%, 2024-03-26, SPY 1500 iron K0 521.0 ±2.0	1.67	381	52	10%
all names	fit 2022–24	+138.80, +261.9%, 2024-02-01, QQQ 1500 iron K0 422.0 ±2.0	−144.20, −100.1%, 2024-03-26, SPY 1500 iron K0 521.0 ±2.0	1.76	829	34	8%
QQQ	OOS 2025–26	+155.80, +380.0%, 2026-01-28, QQQ 1500 iron K0 633.0 ±2.0	−118.20, −100.2%, 2026-09-14, QQQ 1500 iron K0 712.0 ±2.0	1.15	923	142	42%
SPY	OOS 2025–26	+149.30, +364.1%, 2026-09-16, SPY 1500 iron K0 754.0 ±2.0	−130.20, −100.2%, 2026-01-07, SPY 1500 iron K0 692.0 ±2.0	1.36	725	85	20%

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
all names	OOS 2025–26	+155.80, +380.0%, 2026-01-28, QQQ 1500 iron K0 633.0 ±2.0	−130.20, −100.2%, 2026-01-07, SPY 1500 iron K0 692.0 ±2.0	1.24	1,552	131	26%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.60 a trade) and Robinhood's charged instead (mean $0.20). The study's fees are given beside them below. Re-pricing: exact (1,924 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	fit 2022–24	521	**+16.71**, +15.14	+14.26, +12.74	64.5 / 63.1	[+11.64, +23.15] / [+9.21, +20.69]
SPY	fit 2022–24	525	**+13.33**, +12.54	+10.95, +10.14	63.4 / 62.1	[+8.43, +18.36] / [+6.16, +15.93]
all names	fit 2022–24	1,046	**+15.01**, +13.84	+12.60, +11.44	64.0 / 62.6	[+10.61, +20.36] / [+8.22, +17.89]
QQQ	OOS 2025–26	440	**+5.29**, +3.78	+2.16, +1.38	51.4 / 50.0	[−2.50, +13.57] / [−5.77, +10.53]
SPY	OOS 2025–26	438	**+9.65**, +7.85	+6.84, +5.45	55.3 / 54.6	[+2.22, +18.07] / [−0.71, +15.31]
all names	OOS 2025–26	878	**+7.46**, +5.81	+4.49, +3.41	53.3 / 52.3	[+0.15, +15.30] / [−2.96, +12.33]

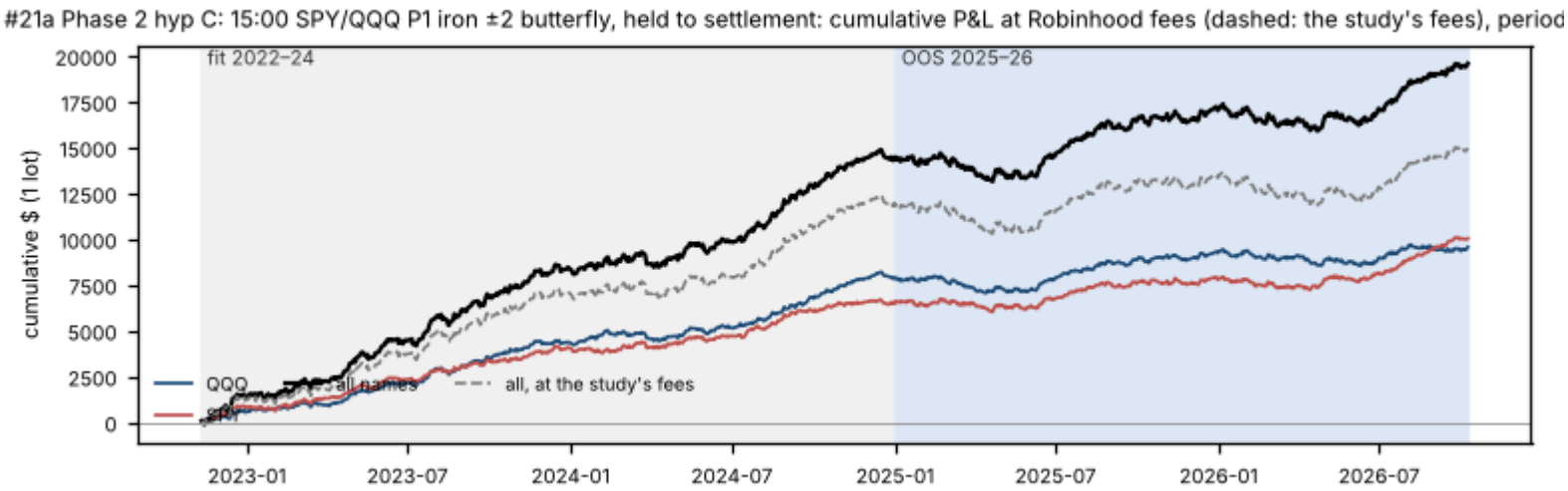
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- fit 2022–24: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	QQQ	33	75.8	+21.33	+16.12	+532
2023	QQQ	247	65.2	+16.18	+15.55	+3,842
2024	QQQ	241	62.2	+16.62	+14.59	+3,515
2025	QQQ	246	54.1	+4.09	+5.47	+1,345
2026	QQQ	194	47.9	+6.80	+1.64	+318
2022	SPY	32	65.6	+33.20	+24.89	+797
2023	SPY	246	65.4	+12.72	+13.36	+3,287
2024	SPY	247	61.1	+11.35	+10.13	+2,501
2025	SPY	244	53.7	+4.74	+4.87	+1,189
2026	SPY	194	57.2	+15.84	+11.60	+2,250
2022	all names	65	70.8	+27.18	+20.44	+1,328
2023	all names	493	65.3	+14.46	+14.46	+7,128
2024	all names	488	61.7	+13.96	+12.33	+6,017
2025	all names	490	53.9	+4.41	+5.17	+2,534
2026	all names	388	52.6	+11.32	+6.62	+2,568



Amendments, deviations and integrity checks. Fill stress ($0.01 a leg) read on both periods as a robustness check.

What it showed. Positive on fit, shrunk to noise out of sample.

#21b Phase 2 hyp C: 15:00 SPY/QQQ P2 call ±2 butterfly, held to settlement

Folder: `research_notes/Compass phase 2/hyp_c_debit_flies`. **Commits:** preregistration.md before the OOS run / — / faea67c. **Dates:** 2026-10-21.

Hypothesis. Late-day ATM butterflies on quiet closes pin.

Rule. 15:00, SPY and QQQ, P2 call ±2 at the 4-leg NBBO, held to settlement, unconditional.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.65 a contract (leg) to open, nothing at expiry (four legs); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPY and QQQ 2022-11 → 2026-10.

Periods and holdouts. fit 2022–24 / OOS 2025–26.

Pass rule (as pre-registered). Edge only if the OOS mean is positive with its interval clear of zero AND positive without the best five sessions AND both OOS years positive.

Verdict. Not confirmed: the picks fell to +2 to +4.5% of risk OOS with intervals through zero and die under $0.01 a leg; the un-picked iron ±5 read +5.0/+6.0% (→ the tracker's fly5).

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	fit 2022–24	522	522 / 553	62.3	+51.6, +49.62	−50.2, −50.30	+20.0, +22.82	**+13.21**, +11.91	[+8.31, +19.56]; [+7.53, +17.25]
SPY	fit 2022–24	525	525 / 553	61.5	+48.4, +47.92	−50.4, −51.29	+16.7, +18.82	**+10.40**, +9.74	[+5.72, +15.41]; [+5.43, +14.06]
all names	fit 2022–24	1,047	530 / 553	61.9	+50.0, +48.77	−50.3, −50.80	+18.4, +20.82	**+11.80**, +10.83	[+7.47, +17.08]; [+6.96, +15.13]
QQQ	OOS 2025–26	441	441 / 457	49.9	+73.0, +56.05	−68.4, −53.62	−0.2, −0.18	**+2.15**, +1.09	[−5.56, +10.35]; [−5.02, +7.63]
SPY	OOS 2025–26	439	439 / 457	54.7	+63.7, +53.10	−59.4, −49.86	+8.6, +8.82	**+7.90**, +6.43	[+0.28, +16.44]; [+0.32, +13.68]
all names	OOS 2025–26	880	441 / 457	52.3	+68.2, +54.51	−64.1, −51.84	+4.4, +4.82	**+5.02**, +3.75	[−2.27, +12.91]; [−1.67, +9.95]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	fit 2022–24	+137.82, +255.2%, 2024-02-01, QQQ 1500 call K0 422.0 ±2.0	−146.18, −100.1%, 2023-07-07, QQQ 1500 call K0 369.0 ±2.0	1.63	696	92	10%
SPY	fit 2022–24	+131.82, +376.6%, 2024-08-28, SPY 1500 call K0 558.0 ±2.0	−149.18, −100.1%, 2024-03-26, SPY 1500 call K0 521.0 ±2.0	1.49	495	62	12%
all names	fit 2022–24	+137.82, +255.2%, 2024-02-01, QQQ 1500 call K0 422.0 ±2.0	−149.18, −100.1%, 2024-03-26, SPY 1500 call K0 521.0 ±2.0	1.56	877	44	11%
QQQ	OOS 2025–26	+153.82, +357.7%, 2026-01-28, QQQ 1500 call K0 633.0 ±2.0	−130.18, −100.1%, 2025-11-05, QQQ 1500 call K0 626.0 ±2.0	1.04	1,152	197	144%
SPY	OOS 2025–26	+149.32, +364.2%, 2026-09-16, SPY 1500 call K0 754.0 ±2.0	−131.18, −100.1%, 2026-01-07, SPY 1500 call K0 692.0 ±2.0	1.28	823	120	25%

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
all names	OOS 2025–26	+153.82, +357.7%, 2026-01-28, QQQ 1500 call K0 633.0 ±2.0	−131.18, −100.1%, 2026-01-07, SPY 1500 call K0 692.0 ±2.0	1.15	1,781	141	39%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.60 a trade) and Robinhood's charged instead (mean $0.18). The study's fees are given beside them below. Re-pricing: exact (1,927 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	fit 2022–24	522	**+13.21**, +11.91	+10.81, +9.49	62.3 / 61.5	[+8.31, +19.56] / [+5.90, +17.14]
SPY	fit 2022–24	525	**+10.40**, +9.74	+8.07, +7.33	61.5 / 59.6	[+5.72, +15.41] / [+3.41, +13.02]
all names	fit 2022–24	1,047	**+11.80**, +10.83	+9.44, +8.41	61.9 / 60.6	[+7.47, +17.08] / [+5.18, +14.71]
QQQ	OOS 2025–26	441	**+2.15**, +1.09	−0.92, −1.33	49.9 / 48.3	[−5.56, +10.35] / [−8.71, +7.43]
SPY	OOS 2025–26	439	**+7.90**, +6.43	+5.11, +4.01	54.7 / 54.2	[+0.28, +16.44] / [−2.63, +13.75]
all names	OOS 2025–26	880	**+5.02**, +3.75	+2.09, +1.34	52.3 / 51.2	[−2.27, +12.91] / [−5.35, +10.06]

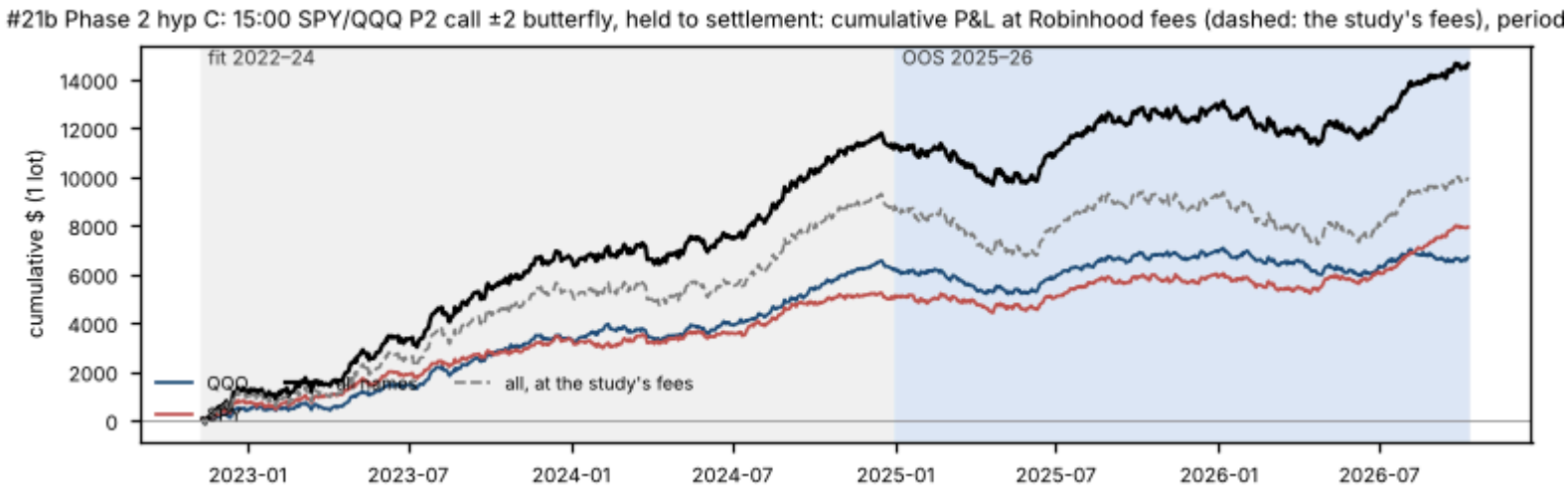
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- fit 2022–24: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	QQQ	33	72.7	+17.18	+11.71	+387
2023	QQQ	247	62.8	+12.66	+12.16	+3,004
2024	QQQ	242	60.3	+13.22	+11.69	+2,828
2025	QQQ	247	52.2	+0.89	+2.73	+675
2026	QQQ	194	46.9	+3.75	−0.99	−193
2022	SPY	32	65.6	+29.27	+21.79	+697
2023	SPY	246	63.4	+10.24	+10.86	+2,671
2024	SPY	247	59.1	+8.11	+7.07	+1,747
2025	SPY	245	52.7	+2.49	+3.08	+754
2026	SPY	194	57.2	+14.73	+10.66	+2,068
2022	all names	65	69.2	+23.14	+16.67	+1,084
2023	all names	493	63.1	+11.45	+11.51	+5,675
2024	all names	489	59.7	+10.64	+9.36	+4,575
2025	all names	492	52.4	+1.69	+2.90	+1,428
2026	all names	388	52.1	+9.24	+4.83	+1,876



Amendments, deviations and integrity checks. Fill stress ($0.01 a leg) read on both periods as a robustness check.

What it showed. Positive on fit, shrunk to noise out of sample.

#21c Phase 2 hyp C: 15:00 SPY/QQQ P3 call ±5 butterfly, held to settlement

Folder: `research_notes/Compass phase 2/hyp_c_debit_flies`. **Commits:** preregistration.md before the OOS run / — / faea67c. **Dates:** 2026-10-21.

Hypothesis. Late-day ATM butterflies on quiet closes pin.

Rule. 15:00, SPY and QQQ, P3 call ±5 at the 4-leg NBBO, held to settlement, unconditional.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.65 a contract (leg) to open, nothing at expiry (four legs); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPY and QQQ 2022-11 → 2026-10.

Periods and holdouts. fit 2022–24 / OOS 2025–26.

Pass rule (as pre-registered). Edge only if the OOS mean is positive with its interval clear of zero AND positive without the best five sessions AND both OOS years positive.

Verdict. Not confirmed: the picks fell to +2 to +4.5% of risk OOS with intervals through zero and die under $0.01 a leg; the un-picked iron ±5 read +5.0/+6.0% (→ the tracker's fly5).

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	fit 2022–24	523	523 / 553	67.7	+16.3, +58.86	−20.1, −75.05	+7.5, +30.82	**+4.53**, +15.59	[+2.84, +6.89]; [+10.26, +22.80]
SPY	fit 2022–24	524	524 / 553	66.2	+15.1, +55.82	−19.4, −74.67	+5.9, +24.82	**+3.43**, +11.74	[+1.76, +5.46]; [+5.76, +18.73]
all names	fit 2022–24	1,047	530 / 553	67.0	+15.7, +57.36	−19.7, −74.85	+7.0, +27.82	**+3.98**, +13.67	[+2.47, +6.05]; [+8.57, +20.25]
QQQ	OOS 2025–26	438	438 / 457	56.2	+25.4, +77.79	−28.9, −97.59	+5.1, +19.82	**+1.60**, +0.91	[−1.59, +4.74]; [−9.97, +11.22]
SPY	OOS 2025–26	440	440 / 457	62.3	+21.8, +71.33	−25.2, −83.42	+7.3, +25.82	**+4.08**, +12.95	[+1.51, +6.61]; [+4.72, +21.89]
all names	OOS 2025–26	878	441 / 457	59.2	+23.5, +74.39	−27.2, −91.02	+5.6, +21.82	**+2.84**, +6.94	[+0.10, +5.46]; [−2.06, +16.24]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	fit 2022–24	+288.81, +207.8%, 2024-08-28, QQQ 1500 call K0 472.0 ±5.0	−367.18, −100.0%, 2024-05-31, QQQ 1500 call K0 446.0 ±5.0	1.64	738	64	14%
SPY	fit 2022–24	+270.82, +138.2%, 2024-08-28, SPY 1500 call K0 558.0 ±5.0	−405.18, −100.0%, 2024-05-31, SPY 1500 call K0 521.0 ±5.0	1.47	851	113	17%
all names	fit 2022–24	+288.81, +207.8%, 2024-08-28, QQQ 1500 call K0 472.0 ±5.0	−405.18, −100.0%, 2024-05-31, SPY 1500 call K0 521.0 ±5.0	1.55	1,296	39	14%
QQQ	OOS 2025–26	+276.81, +170.9%, 2026-04-29, QQQ 1500 call K0 661.0 ±5.0	−468.18, −100.0%, 2026-05-12, QQQ 1500 call K0 702.0 ±5.0	1.02	2,030	214	318%
SPY	OOS 2025–26	+267.32, +119.9%, 2026-09-16, SPY 1500 call K0 754.0 ±5.0	−342.18, −100.1%, 2026-03-09, SPY 1500 call K0 670.0 ±5.0	1.41	1,171	81	20%

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
all names	OOS 2025–26	+276.81, +170.9%, 2026-04-29, QQQ 1500 call K0 661.0 ±5.0	−468.18, −100.0%, 2026-05-12, QQQ 1500 call K0 702.0 ±5.0	1.19	3,102	128	38%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.60 a trade) and Robinhood's charged instead (mean $0.18). The study's fees are given beside them below. Re-pricing: exact (1,925 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	fit 2022–24	523	**+4.53**, +15.59	+3.91, +13.17	67.7 / 66.3	[+2.84, +6.89] / [+2.23, +6.26]
SPY	fit 2022–24	524	**+3.43**, +11.74	+2.81, +9.32	66.2 / 64.9	[+1.76, +5.46] / [+1.17, +4.83]
all names	fit 2022–24	1,047	**+3.98**, +13.67	+3.36, +11.25	67.0 / 65.6	[+2.47, +6.05] / [+1.85, +5.44]
QQQ	OOS 2025–26	438	**+1.60**, +0.91	+0.90, −1.51	56.2 / 55.5	[−1.59, +4.74] / [−2.31, +4.05]
SPY	OOS 2025–26	440	**+4.08**, +12.95	+3.40, +10.53	62.3 / 61.6	[+1.51, +6.61] / [+0.80, +5.95]
all names	OOS 2025–26	878	**+2.84**, +6.94	+2.15, +4.52	59.2 / 58.5	[+0.10, +5.46] / [−0.59, +4.78]

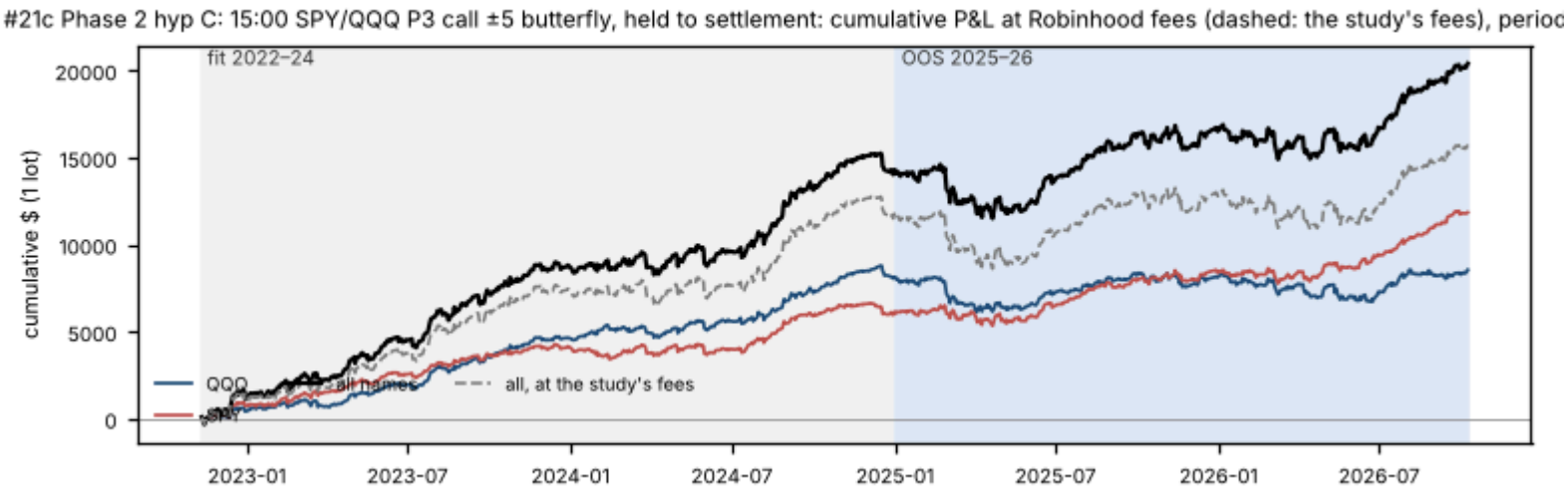
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- fit 2022–24: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	QQQ	32	75.0	+5.41	+13.08	+419
2023	QQQ	247	66.0	+4.43	+17.19	+4,246
2024	QQQ	244	68.4	+4.52	+14.31	+3,491
2025	QQQ	246	56.9	+0.95	−0.14	−34
2026	QQQ	192	55.2	+2.42	+2.26	+434
2022	SPY	33	72.7	+7.77	+22.26	+735
2023	SPY	247	67.6	+3.54	+13.88	+3,429
2024	SPY	244	63.9	+2.73	+8.15	+1,990
2025	SPY	247	60.3	+2.78	+9.04	+2,232
2026	SPY	193	64.8	+5.74	+17.95	+3,465
2022	all names	65	73.8	+6.61	+17.74	+1,153
2023	all names	494	66.8	+3.99	+15.54	+7,675
2024	all names	488	66.2	+3.62	+11.23	+5,480
2025	all names	493	58.6	+1.87	+4.46	+2,198
2026	all names	385	60.0	+4.08	+10.13	+3,898



Amendments, deviations and integrity checks. Fill stress ($0.01 a leg) read on both periods as a robustness check.

What it showed. Positive on fit, shrunk to noise out of sample.

#21d Phase 2 hyp C: debit vertical 0.70/0.30, universe B, held (the vertical pick)

Folder: `research_notes/Compass phase 2/hyp_c_debit_flies`. **Commits:** preregistration.md / — / faea67c. **Dates:** 2026-10-21.

Hypothesis. A debit vertical keeps the directional premium at lower cost.

Rule. The audit's universe B entries, 0.70Δ/0.30Δ debit vertical, held to settlement.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.65 a contract (leg) to open, nothing at expiry; bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPY, QQQ, IWM, SPXW 2022–2026.

Periods and holdouts. fit / OOS.

Pass rule (as pre-registered). Pre-registered expectation: negative OOS.

Verdict. Loses less than the naked contract but still loses (−6.7% of debit fit).

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	fit 2022–24	173	173 / 182	54.9	+60.6, +44.65	−69.8, −49.86	+10.0, +10.90	**+1.83**, +2.04	[−10.09, +9.51]; [−7.68, +8.86]
QQQ	fit 2022–24	530	530 / 553	50.2	+70.2, +93.93	−69.9, −89.38	+1.1, +1.90	**+0.42**, +2.62	[−6.74, +8.85]; [−8.28, +15.78]
SPXW	fit 2022–24	656	656 / 683	45.4	+74.7, +883.66	−77.7, −953.83	−14.5, −156.76	**−8.47**, −119.12	[−13.29, −1.93]; [−183.46, −32.45]
SPY	fit 2022–24	530	530 / 553	44.0	+69.7, +81.90	−75.6, −90.73	−20.9, −24.10	**−11.74**, −14.84	[−17.85, −4.31]; [−22.97, −4.02]
all names	fit 2022–24	1,889	656 / 683	47.2	+70.5, +349.37	−74.4, −397.10	−9.3, −14.10	**−5.95**, −44.61	[−9.17, −2.36]; [−66.16, −15.55]
IWM	OOS 2025–26	439	439 / 457	50.6	+67.0, +60.94	−73.1, −61.82	+0.9, +0.90	**−2.25**, +0.26	[−9.15, +4.45]; [−7.00, +7.58]
QQQ	OOS 2025–26	441	441 / 457	51.9	+71.6, +135.52	−70.9, −133.29	+9.4, +14.90	**+3.07**, +6.30	[−4.60, +9.89]; [−9.53, +21.99]
SPXW	OOS 2025–26	441	441 / 457	51.7	+71.7, +1,074.83	−78.1, −1,140.55	+5.4, +69.74	**−0.63**, +4.82	[−9.01, +7.72]; [−103.56, +117.40]
SPY	OOS 2025–26	440	440 / 457	53.6	+73.9, +115.70	−67.8, −110.02	+12.6, +18.90	**+8.17**, +11.05	[+0.57, +15.59]; [−2.69, +24.29]
all names	OOS 2025–26	1,761	441 / 457	52.0	+71.1, +346.37	−72.5, −362.95	+6.7, +10.90	**+2.09**, +5.61	[−2.01, +5.45]; [−21.65, +33.23]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	fit 2022–24	+137.90, +93.2%, 2024-08-08, IWM CALL 0.70/0.30 m56	−212.10, −100.0%, 2024-06-12, IWM CALL 0.70/0.30 m146	1.09	660	77	168%
QQQ	fit 2022–24	+370.90, +95.8%, 2022-12-14, QQQ PUT 0.70/0.30 m54	−321.10, −100.0%, 2024-11-20, QQQ PUT 0.70/0.30 m66	1.06	2,897	229	110%
SPXW	fit 2022–24	+4,407.74, +107.8%, 2022-09-21, SPXW PUT 0.70/0.30 m59	−4,492.26, −100.1%, 2022-12-14, SPXW CALL 0.70/0.30 m90	0.77	80,420	678	total ≤ 0 (best 5 days +14,715; without them −92,857)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	fit 2022–24	+338.90, +93.9%, 2023-03-22, SPY PUT 0.70/0.30 m96	−454.10, −100.0%, 2022-12-14, SPY CALL 0.70/0.30 m160	0.71	8,263	553	total ≤ 0 (best 5 days +1,317; without them −9,182)
all names	fit 2022–24	+4,407.74, +107.8%, 2022-09-21, SPXW PUT 0.70/0.30 m59	−4,492.26, −100.1%, 2022-12-14, SPXW CALL 0.70/0.30 m90	0.79	86,543	678	total ≤ 0 (best 5 days +13,891; without them −98,156)
IWM	OOS 2025–26	+218.90, +120.9%, 2026-09-16, IWM PUT 0.70/0.30 m108	−394.10, −94.3%, 2025-06-23, IWM PUT 0.70/0.30 m30	1.01	1,967	351	930%
QQQ	OOS 2025–26	+594.90, +117.8%, 2026-06-10, QQQ PUT 0.70/0.30 m62	−619.10, −67.0%, 2025-04-07, QQQ PUT 0.70/0.30 m66	1.10	2,749	185	92%
SPXW	OOS 2025–26	+6,047.74, +87.0%, 2025-04-11, SPXW CALL 0.70/0.30 m99	−7,092.26, −100.0%, 2025-04-10, SPXW PUT 0.70/0.30 m176	1.01	34,735	286	1149%
SPY	OOS 2025–26	+690.90, +85.4%, 2025-04-11, SPY CALL 0.70/0.30 m52	−558.10, −100.0%, 2025-04-08, SPY CALL 0.70/0.30 m197	1.22	2,148	225	51%
all names	OOS 2025–26	+6,047.74, +87.0%, 2025-04-11, SPXW CALL 0.70/0.30 m99	−7,092.26, −100.0%, 2025-04-10, SPXW PUT 0.70/0.30 m176	1.03	32,160	261	268%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.30 a trade) and Robinhood's charged instead (mean $0.75). The study's fees are given beside them below. Re-pricing: exact (2,553 trades); Robinhood fees inferred from the rule: SPXW legs' price tier ($1.00 and up) (1,097 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	fit 2022–24	173	**+1.83**, +2.04	−0.05, +0.84	54.9 / 54.9	[−10.09, +9.51] / [−12.07, +7.72]
QQQ	fit 2022–24	530	**+0.42**, +2.62	−0.66, +1.42	50.2 / 50.2	[−6.74, +8.85] / [−7.85, +7.82]
SPXW	fit 2022–24	656	**−8.47**, −119.12	−8.38, −118.16	45.4 / 45.4	[−13.29, −1.93] / [−13.19, −1.84]
SPY	fit 2022–24	530	**−11.74**, −14.84	−12.94, −16.04	44.0 / 44.0	[−17.85, −4.31] / [−19.06, −5.52]
all names	fit 2022–24	1,889	**−5.95**, −44.61	−6.73, −45.06	47.2 / 47.2	[−9.17, −2.36] / [−9.96, −3.11]
IWM	OOS 2025–26	439	**−2.25**, +0.26	−3.93, −0.94	50.6 / 49.7	[−9.15, +4.45] / [−10.90, +2.84]
QQQ	OOS 2025–26	441	**+3.07**, +6.30	+2.26, +5.10	51.9 / 51.5	[−4.60, +9.89] / [−5.49, +9.09]
SPXW	OOS 2025–26	441	**−0.63**, +4.82	−0.55, +5.78	51.7 / 51.7	[−9.01, +7.72] / [−8.93, +7.80]
SPY	OOS 2025–26	440	**+8.17**, +11.05	+7.22, +9.85	53.6 / 53.6	[+0.57, +15.59] / [−0.41, +14.64]
all names	OOS 2025–26	1,761	**+2.09**, +5.61	+1.25, +4.95	52.0 / 51.6	[−2.01, +5.45] / [−2.84, +4.65]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

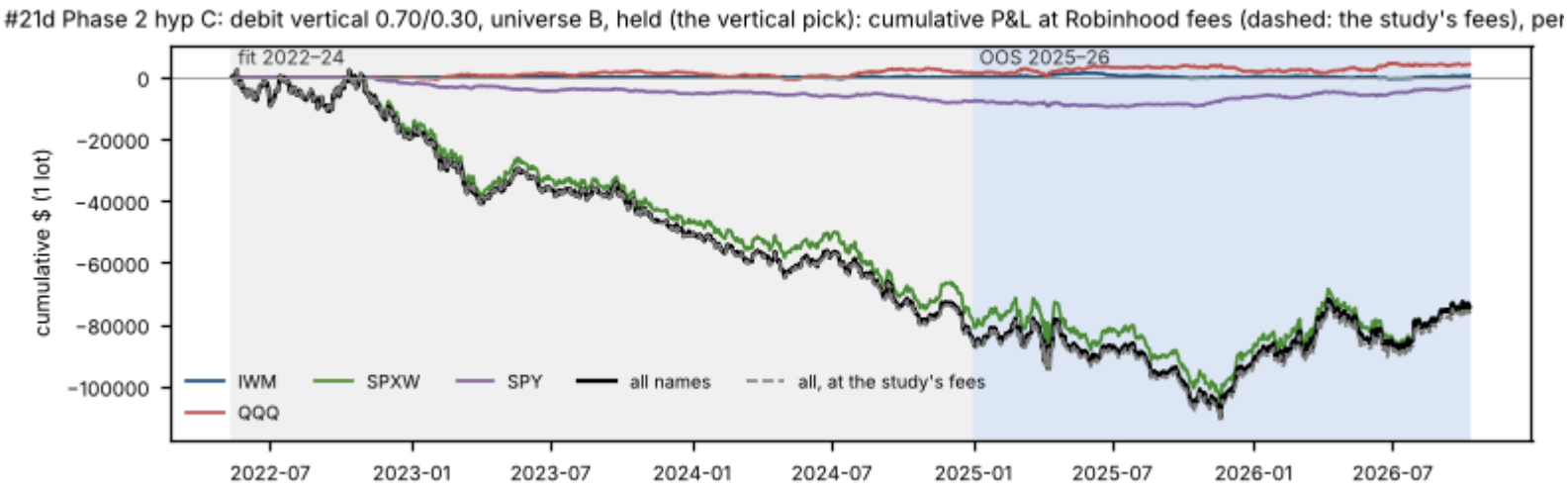
Exit reasons (share of trades):

- fit 2022–24: settle 100.0%
- OOS 2025–26: settle 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	173	54.9	+1.83	+2.04	+353

year	name	trades	win %	mean %	mean $	total $
2025	IWM	245	48.6	−6.41	−0.55	−134
2026	IWM	194	53.1	+3.00	+1.28	+247
2022	QQQ	33	42.4	−1.42	−3.18	−105
2023	QQQ	248	51.6	+1.55	+6.07	+1,507
2024	QQQ	249	49.8	−0.46	−0.05	−13
2025	QQQ	247	51.4	+0.09	+2.10	+520
2026	QQQ	194	52.6	+6.87	+11.64	+2,257
2022	SPXW	159	47.2	−6.24	−111.25	−17,688
2023	SPXW	248	46.0	−8.93	−119.26	−29,577
2024	SPXW	249	43.8	−9.45	−124.00	−30,877
2025	SPXW	247	49.8	−2.50	−39.58	−9,776
2026	SPXW	194	54.1	+1.76	+61.34	+11,900
2022	SPY	33	30.3	−30.24	−64.56	−2,130
2023	SPY	248	42.7	−13.29	−13.19	−3,272
2024	SPY	249	47.0	−7.74	−9.89	−2,463
2025	SPY	246	51.6	+3.63	+5.73	+1,410
2026	SPY	194	56.2	+13.94	+17.79	+3,452
2022	all names	225	44.0	−9.05	−88.55	−19,924
2023	all names	744	46.8	−6.89	−42.13	−31,342
2024	all names	920	48.4	−4.43	−35.87	−32,999
2025	all names	985	50.4	−1.29	−8.10	−7,981
2026	all names	776	54.0	+6.39	+23.01	+17,857



Amendments, deviations and integrity checks. —

What it showed. Verticals lose less than calls and puts, and still lose.

#22 Phase 2 hyp D: volatility breakout (straddles on the model’s top decile)

Folder: `research_notes/Compass phase 2/hyp_d_vol_breakout`. **Commits:** out/preregistered.md / — / da02507. **Dates:** 2026-10-21.

Hypothesis. The volatility model’s top decile marks straddles worth buying.

Rule. Buy straddles/strangles on the top-decile scan minutes.

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). $0.65 a contract (leg) to open, nothing at expiry

Data. SPY/QQQ/IWM/SPXW 2022–2026.

Periods and holdouts. fit / cut / OOS (read once).

Pass rule (as pre-registered). Best-on-fit cell positive OOS.

Verdict. No edge: the best-on-fit cell flipped from +4.6% to −7.0%; the model forecasts volatility (AUC 0.71–0.73) but the straddle is already priced richer.

Metrics. not available — committed as cells by scan minute and decile (cells_*.csv, minutes_*.parquet); no trade list of the pre-registered cell.

Amendments, deviations and integrity checks. —

What it showed. Forecasting volatility is not trading it.

#24 Phase 3 hyp E: short-premium lift of the vol model’s quiet deciles

Folder: `research_notes/Compass phase 3/hyp_e_short_premium_lift`. **Commits:** out/preregistered.md / — / 02d0dcd. **Dates:** 2026-10-21.

Hypothesis. Quiet deciles lift the SPY short-premium spread over selling unconditionally.

Rule. Sell the SPY spread on the bottom deciles vs unconditionally, same minute.

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). $0.65 a contract (leg) to open, nothing at expiry

Data. SPY 2022-11 → 2026-10.

Periods and holdouts. fit / OOS.

Pass rule (as pre-registered). A pre-registrable lift on fit.

Verdict. No edge established — not pre-registrable (no fit cell clear of zero); the D1 condor read later became condor_d20 (#26).

Metrics. not available — committed as by-scan cells and pooled tables; the candidate was set after the OOS read (see #26 condor_d20 for its trades).

Amendments, deviations and integrity checks. —

What it showed. No lift on fit.

#25 Phase 3 hyp F: the 15:30 SPY/QQQ long call ±5 butterfly (cash account)

Folder: `research_notes/Compass phase 3/hyp_f_cash_butterfly`. **Commits:** out/preregistered.md (20:44:54 UTC) / — / f2883c0; report b7ffa46. **Dates:** 2026-10-21.

Hypothesis. The cash-account call fly at 15:30 keeps hyp C's OOS edge under realistic complex-order fills.

Rule. 15:30, SPY and QQQ, long call butterfly ±5 around the ATM strike at the 4-leg NBBO package, held to settlement.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.65 a leg, four legs to open, none at expiry; package at the NBBO (k = 0).

Data. SPY, QQQ 2022-11-11 → 2026-10-09.

Periods and holdouts. fit 2022-11 → 2024 / OOS 2025 → 2026-10-09 (the same OOS C read).

Pass rule (as pre-registered). Four parts: OOS mean > 0 with its interval clear of zero at k = 0; also under $0.01 a leg worse; positive without the best five sessions; both names positive.

Verdict. Passes all four, but the cell was named after C's OOS was printed: a conditionally justified rule, to be confirmed forward (≈ 450 sessions); dies at $0.02 a leg.

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	fit 2022-11→2024	526	526 / 553	66.3	+13.7, +50.50	−16.5, −67.55	+5.7, +24.82	**+3.52**, +10.78	[+1.96, +6.01]; [+5.58, +19.00]
SPY	fit 2022-11→2024	522	522 / 553	65.5	+11.7, +45.76	−16.1, −65.07	+4.9, +20.82	**+2.15**, +7.54	[+0.69, +4.13]; [+1.88, +14.88]
all names	fit 2022-11→2024	1,048	529 / 553	65.9	+12.7, +48.15	−16.3, −66.30	+5.1, +21.82	**+2.84**, +9.17	[+1.40, +4.94]; [+4.22, +16.39]
QQQ	OOS 2025–26	439	439 / 457	61.3	+21.3, +67.68	−23.1, −83.58	+5.4, +20.82	**+4.14**, +9.11	[+1.31, +6.99]; [+0.01, +17.59]
SPY	OOS 2025–26	440	440 / 457	64.8	+17.6, +60.78	−18.6, −67.76	+6.4, +26.82	**+4.86**, +15.50	[+2.74, +7.06]; [+8.63, +22.33]
all names	OOS 2025–26	879	441 / 457	63.0	+19.4, +64.13	−20.9, −76.03	+5.7, +24.32	**+4.50**, +12.31	[+2.14, +6.84]; [+4.78, +19.17]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	fit 2022-11→2024	+289.81, +210.0%, 2024-08-28, QQQ 1530 call fly K0 472.0 ±5	−405.18, −100.0%, 2024-12-18, QQQ 1530 call fly K0 522.0 ±5	1.47	1,263	48	19%
SPY	fit 2022-11→2024	+228.82, +112.2%, 2024-08-28, SPY 1530 call fly K0 559.0 ±5	−420.18, −100.0%, 2024-05-31, SPY 1530 call fly K0 522.0 ±5	1.34	1,281	106	23%
all names	fit 2022-11→2024	+289.81, +210.0%, 2024-08-28, QQQ 1530 call fly K0 472.0 ±5	−420.18, −100.0%, 2024-05-31, SPY 1530 call fly K0 522.0 ±5	1.41	1,999	88	19%
QQQ	OOS 2025–26	+261.82, +147.9%, 2026-04-29, QQQ 1530 call fly K0 661.0 ±5	−332.18, −100.1%, 2025-03-04, QQQ 1530 call fly K0 502.0 ±5	1.28	1,434	153	31%
SPY	OOS 2025–26	+237.82, +104.8%, 2025-11-19, SPY 1530 call fly K0 663.0 ±5	−290.18, −73.8%, 2025-05-22, SPY 1530 call fly K0 587.0 ±5	1.65	1,080	67	16%
all names	OOS 2025–26	+261.82, +147.9%, 2026-04-29, QQQ 1530 call fly K0 661.0 ±5	−332.18, −100.1%, 2025-03-04, QQQ 1530 call fly K0 502.0 ±5	1.44	2,393	80	21%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.60 a trade) and Robinhood's charged instead (mean $0.18). The study's fees are given beside them below. Re-pricing: exact (1,927 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	fit 2022-11→2024	526	**+3.52**, +10.78	+2.93, +8.36	66.3 / 65.2	[+1.96, +6.01] / [+1.37, +5.42]
SPY	fit 2022-11→2024	522	**+2.15**, +7.54	+1.56, +5.12	65.5 / 63.8	[+0.69, +4.13] / [+0.11, +3.53]
all names	fit 2022-11→2024	1,048	**+2.84**, +9.17	+2.25, +6.75	65.9 / 64.5	[+1.40, +4.94] / [+0.83, +4.35]
QQQ	OOS 2025–26	439	**+4.14**, +9.11	+3.48, +6.69	61.3 / 60.1	[+1.31, +6.99] / [+0.65, +6.32]
SPY	OOS 2025–26	440	**+4.86**, +15.50	+4.22, +13.08	64.8 / 64.3	[+2.74, +7.06] / [+2.11, +6.41]
all names	OOS 2025–26	879	**+4.50**, +12.31	+3.85, +9.89	63.0 / 62.2	[+2.14, +6.84] / [+1.49, +6.20]

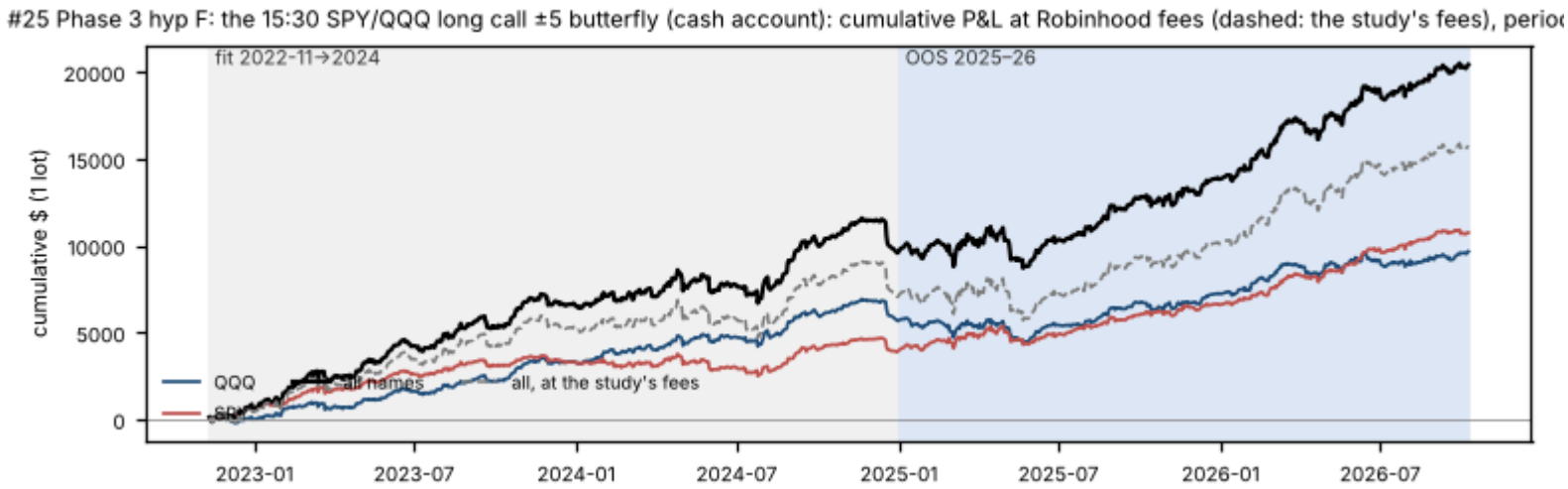
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- fit 2022-11→2024: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	QQQ	33	60.6	+0.19	−0.83	−27
2023	QQQ	248	68.1	+3.69	+13.62	+3,378
2024	QQQ	245	65.3	+3.79	+9.47	+2,320
2025	QQQ	247	59.9	+3.49	+6.41	+1,584
2026	QQQ	192	63.0	+4.97	+12.57	+2,414
2022	SPY	32	71.9	+5.26	+18.77	+601
2023	SPY	248	67.7	+2.90	+11.07	+2,744
2024	SPY	242	62.4	+0.96	+2.44	+591
2025	SPY	246	63.4	+3.60	+10.78	+2,651
2026	SPY	194	66.5	+6.45	+21.49	+4,169
2022	all names	65	66.2	+2.69	+8.82	+573
2023	all names	496	67.9	+3.30	+12.34	+6,122
2024	all names	487	63.9	+2.39	+5.98	+2,911
2025	all names	493	61.7	+3.55	+8.59	+4,235
2026	all names	386	64.8	+5.71	+17.05	+6,583



Amendments, deviations and integrity checks. The selection is not independent of the OOS it was read on (stated in the report).

What it showed. +3.85% of risk a fly OOS, carried by the same data that chose it.

#26a Paper tracker: condor_em1 (SPY 2-wide iron condor at ± the ATM straddle, 11:30–13:30)

Folder: `scripts/paper_tracker.py; research_notes/Compass phase 3/backtest_replay, backtest_audit, forward_*`. **Commits:** defaults ba117a7, 6826b98 / official-close settlement a4ebe90, 99a3d4e, 16eb3c0 / replay 94af538, 30bb73e; audit 1ca1f0e; forward d7115b6, 2a7bd3b. **Dates:** 2026-10-21 → 22.

Hypothesis. The midday SPY condor of hyp B (#20b), all-VIX form, pays forward.

Rule. 11:30, 12:00, 12:30, 13:00, 13:30: SPY 2-wide iron condor, shorts the first listed strikes outside spot ± the ATM straddle, held to the official close.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Shorts at the bid, longs at the ask; $0.65 a contract (leg) to open, nothing at expiry.

Data. The replay of every stored session through the live tracker's code, settled on the official close (data_recorder/paper_backtest_official, git-ignored, every trade).

Periods and holdouts. i development 2022-11-11 → 2024-12-31 / ii selection 2025-01-02 → 2026-10-09 (read) / iii forward 2026-10-12 → (untouched). No historical period is untouched for any candidate (AUDIT.md §5).

Pass rule (as pre-registered). Confirmation: 60 → 120 forward sessions (the candidate's threshold).

Verdict. Historical periods through zero (i +3.47 [−0.04, +6.68], ii +4.10 [−0.51, +8.15]); the 15:58 close-out turns it negative; forward pending.

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	i	2,358	525 / 553	80.5	+15.6, +26.27	−46.0, −78.29	+12.2, +21.80	**+3.56**, +5.87	[+1.48, +5.58]; [+2.36, +9.29]
SPY	ii	2,068	439 / 457	80.7	+18.8, +30.76	−58.5, −94.98	+14.8, +25.80	**+3.90**, +6.50	[+0.90, +6.46]; [+1.71, +10.55]
SPY	iii	44	9 / 9	86.4	+16.9, +28.33	−53.6, −88.53	+14.5, +25.30	**+7.25**, +12.39	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	i	+68.80, +52.5%, 2022-12-14, SPY 12:30 −C411/−C413/−P396/−P394	−189.20, −100.1%, 2023-07-07, SPY 13:30 −C444/−C446/−P441/−P439	1.38	2,344	98	10%
SPY	ii	+65.80, +49.1%, 2026-07-29, SPY 13:00 −C741/−C743/−P729/−P727	−185.20, −100.1%, 2025-04-09, SPY 13:30 −C536/−C538/−P512/−P510	1.35	2,737	67	11%
SPY	iii	+46.80, +30.6%, 2026-10-22, SPY 11:30 −C767/−C769/−P760/−P758	−168.20, −100.1%, 2026-10-16, SPY 11:30 −C772/−C774/−P766/−P764	2.03	443	4	148%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.60 a trade) and Robinhood's charged instead (mean $0.20). The study's fees are given beside them below. Re-pricing: exact (4,470 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPY	i	2,358	**+3.56**, +5.87	+2.16, +3.47	80.5 / 80.2	[+1.48, +5.58] / [+0.08, +4.18]
SPY	ii	2,068	**+3.90**, +6.50	+2.46, +4.10	80.7 / 80.4	[+0.90, +6.46] / [−0.54, +5.02]
SPY	iii	44	**+7.25**, +12.39	+5.84, +9.99	86.4 / 86.4	n/a (< 60 sessions) / n/a (< 60 sessions)

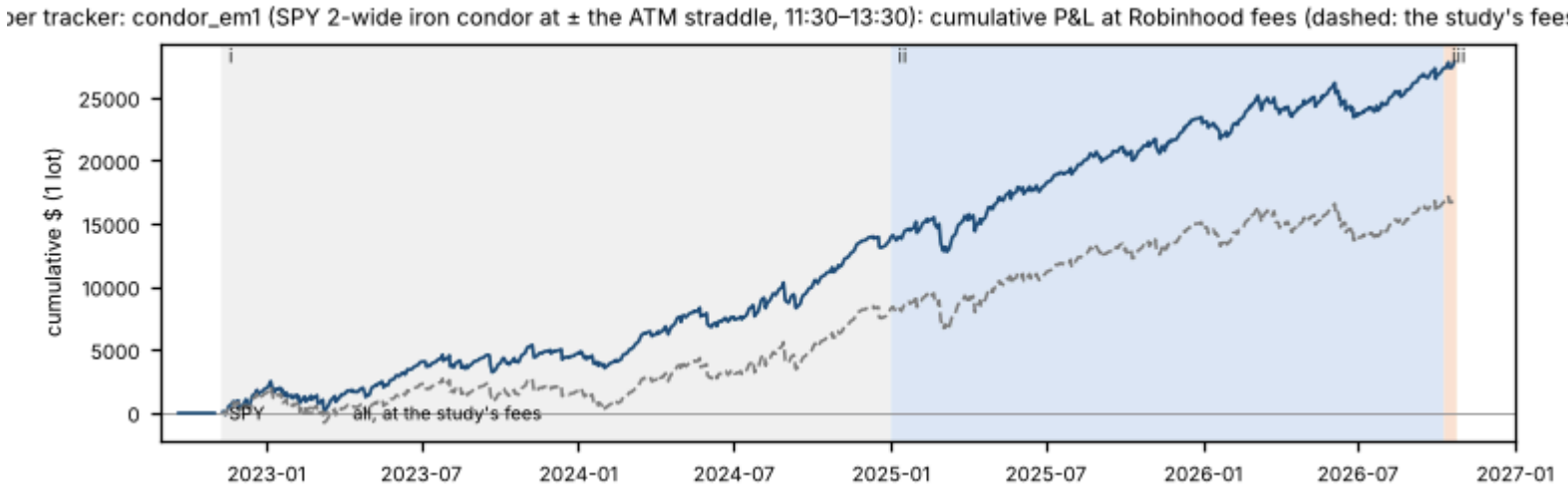
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- i: held to settlement (official close) 100.0%
- ii: held to settlement (official close) 100.0%
- iii: held to settlement (official close) 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPY	159	82.4	+6.98	+11.78	+1,873
2023	SPY	1,080	78.2	+1.59	+2.56	+2,761
2024	SPY	1,119	82.4	+4.98	+8.24	+9,216
2025	SPY	1,129	81.8	+4.97	+8.05	+9,086
2026	SPY	983	79.8	+2.82	+4.99	+4,907



Amendments, deviations and integrity checks. Deterministic (a cold replay equals the committed CSV byte for byte); 100 trades re-derived, 0 differences; re-settled on the official close (12,338 of 20,469 trades changed).

What it showed. A few dollars a condor in both historical periods, inside the noise.

#26b Paper tracker: fly5 (SPY/QQQ ±5 butterflies at 15:00 and 15:30: iron, call, put)

Folder: `scripts/paper_tracker.py; research_notes/Compass phase 3/backtest_replay, backtest_audit, forward_*`. **Commits:** defaults ba117a7, 6826b98 / official-close settlement a4ebe90, 99a3d4e, 16eb3c0 / replay 94af538, 30bb73e; audit 1ca1f0e; forward d7115b6, 2a7bd3b. **Dates:** 2026-10-21 → 22.

Hypothesis. The late-day ±5 fly of hyp C/F pays forward.

Rule. 15:00 and 15:30: SPY and QQQ ±5 iron, call and put butterflies (one trade seen three ways), held to the official close.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Shorts at the bid, longs at the ask; $0.65 a contract (leg) to open, nothing at expiry.

Data. The replay of every stored session through the live tracker's code, settled on the official close (data_recorder/paper_backtest_official, git-ignored, every trade).

Periods and holdouts. i development 2022-11-11 → 2024-12-31 / ii selection 2025-01-02 → 2026-10-09 (read) / iii forward 2026-10-12 → (untouched). No historical period is untouched for any candidate (AUDIT.md §5).

Pass rule (as pre-registered). Confirmation: 250 forward sessions a name.

Verdict. Iron clear of zero in i and ii (+16.59, +16.20 a trade) but chosen on ii; hold-to-expiry leaves 100 shares a fly; the after-hours rule (#27) DROPS the iron as a SPY/QQQ hold-to-expiry strategy.

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ call	i	1,049	528 / 553	67.8	+15.0, +54.56	−18.1, −70.70	+6.6, +27.82	**+4.29**, +14.20	[+2.86, +6.62]; [+9.41, +21.45]
QQQ iron	i	1,049	528 / 553	70.4	+16.2, +58.48	−18.2, −68.71	+8.2, +33.80	**+6.01**, +20.78	[+4.42, +8.44]; [+16.04, +27.95]
QQQ put	i	1,051	529 / 553	66.9	+14.8, +54.17	−17.7, −68.90	+6.6, +26.82	**+4.06**, +13.42	[+2.63, +6.39]; [+8.84, +20.65]
SPY call	i	1,046	525 / 553	65.2	+13.6, +51.58	−17.3, −68.37	+5.5, +22.82	**+2.84**, +9.84	[+1.44, +4.70]; [+4.49, +16.39]
SPY iron	i	1,046	525 / 553	70.1	+14.5, +54.41	−18.3, −69.95	+7.5, +30.80	**+4.68**, +17.20	[+3.38, +6.49]; [+12.45, +23.13]
SPY put	i	1,046	525 / 553	61.7	+13.5, +50.90	−16.4, −65.52	+4.4, +17.82	**+2.00**, +6.27	[+0.43, +3.98]; [+0.20, +13.46]
all names	i	6,287	530 / 553	67.0	+14.6, +54.11	−17.6, −68.58	+6.4, +26.80	**+3.98**, +13.62	[+2.65, +6.02]; [+9.09, +20.10]
QQQ call	ii	877	439 / 457	58.0	+23.6, +73.52	−25.8, −89.43	+4.9, +20.82	**+2.88**, +5.14	[+0.52, +5.22]; [−3.06, +13.07]
QQQ iron	ii	877	439 / 457	63.7	+25.3, +77.91	−26.8, −87.08	+8.4, +33.80	**+6.41**, +18.09	[+4.18, +8.62]; [+10.35, +25.33]
QQQ put	ii	877	439 / 457	57.2	+23.3, +72.48	−25.7, −89.48	+4.3, +17.82	**+2.36**, +3.23	[−0.03, +4.65]; [−5.04, +10.84]
SPY call	ii	880	441 / 457	62.5	+20.0, +67.05	−21.8, −74.57	+6.5, +26.82	**+4.31**, +13.94	[+2.58, +6.09]; [+8.02, +20.31]
SPY iron	ii	880	441 / 457	65.0	+20.8, +69.13	−22.2, −73.79	+7.8, +30.80	**+5.78**, +19.11	[+4.08, +7.55]; [+13.03, +25.40]
SPY put	ii	880	441 / 457	62.0	+19.9, +66.54	−21.9, −75.18	+6.2, +24.82	**+4.02**, +12.75	[+2.30, +5.86]; [+6.59, +19.20]
all names	ii	5,271	441 / 457	61.4	+22.1, +71.07	−24.1, −81.95	+6.2, +24.82	**+4.29**, +12.05	[+2.45, +6.13]; [+5.53, +18.35]
QQQ call	iii	18	9 / 9	77.8	+36.8, +110.68	−19.9, −74.68	+27.9, +82.32	**+24.16**, +69.49	n/a (< 60 sessions); n/a (< 60 sessions)
QQQ iron	iii	18	9 / 9	77.8	+38.7, +115.16	−18.4, −67.20	+29.9, +89.30	**+26.03**, +74.63	n/a (< 60 sessions); n/a (< 60 sessions)
QQQ put	iii	18	9 / 9	77.8	+36.9, +110.82	−19.1, −70.68	+28.4, +80.82	**+24.49**, +70.49	n/a (< 60 sessions); n/a (< 60 sessions)
SPY call	iii	18	9 / 9	61.1	+22.6, +81.91	−9.2, −37.47	+14.5, +49.32	**+10.27**, +35.49	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY iron	iii	18	9 / 9	61.1	+22.9, +82.44	−9.0, −36.92	+14.5, +54.30	**+10.47**, +36.02	n/a (< 60 sessions); n/a (< 60 sessions)
SPY put	iii	18	9 / 9	61.1	+22.7, +82.09	−9.1, −37.18	+14.4, +53.82	**+10.36**, +35.71	n/a (< 60 sessions); n/a (< 60 sessions)
all names	iii	108	9 / 9	69.4	+31.0, +98.99	−12.7, −49.43	+18.2, +67.30	**+17.63**, +53.64	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ call	i	+296.81, +215.1%, 2024-08-28, QQQ 15:30 −C467/−C472/−C477	−405.18, −100.0%, 2024-12-18, QQQ 15:30 −C517/−C522/−C527	1.62	1,830	36	14%
QQQ iron	i	+298.80, +219.7%, 2024-08-28, QQQ 15:00 −C472/−P472/−P467/−C477	−349.20, −100.1%, 2024-07-25, QQQ 15:30 −C464/−P464/−P459/−C469	2.02	1,370	31	10%
QQQ put	i	+295.81, +212.8%, 2024-08-28, QQQ 15:00 −P467/−P472/−P477	−372.18, −100.0%, 2024-07-25, QQQ 15:30 −P459/−P464/−P469	1.59	1,633	37	15%
SPY call	i	+273.82, +139.7%, 2024-08-28, SPY 15:00 −C553/−C558/−C563	−420.18, −100.0%, 2024-05-31, SPY 15:30 −C517/−C522/−C527	1.41	1,480	106	18%
SPY iron	i	+275.80, +142.2%, 2024-08-28, SPY 15:00 −C558/−P558/−P553/−C563	−399.20, −100.1%, 2024-05-31, SPY 15:30 −C522/−P522/−P517/−C527	1.82	1,046	30	10%
SPY put	i	+273.82, +139.7%, 2024-08-28, SPY 15:00 −P553/−P558/−P563	−405.18, −100.0%, 2024-05-31, SPY 15:00 −P516/−P521/−P526	1.25	3,410	207	27%
all names	i	+298.80, +219.7%, 2024-08-28, QQQ 15:00 −C472/−P472/−P467/−C477	−420.18, −100.0%, 2024-05-31, SPY 15:30 −C517/−C522/−C527	1.60	8,141	44	13%
QQQ call	ii	+280.82, +173.3%, 2026-04-29, QQQ 15:00 −C656/−C661/−C666	−468.18, −100.0%, 2026-05-12, QQQ 15:00 −C697/−C702/−C707	1.14	3,185	130	51%
QQQ iron	ii	+282.80, +176.7%, 2026-04-29, QQQ 15:00 −C661/−P661/−P656/−C666	−347.20, −100.1%, 2026-03-09, QQQ 15:00 −C600/−P600/−P595/−C605	1.57	1,714	50	15%
QQQ put	ii	+280.81, +173.3%, 2026-04-29, QQQ 15:00 −P656/−P661/−P666	−363.18, −100.0%, 2026-03-09, QQQ 15:00 −P595/−P600/−P605	1.08	3,215	180	82%
SPY call	ii	+271.82, +121.9%, 2026-09-16, SPY 15:00 −C749/−C754/−C759	−342.18, −100.1%, 2026-03-09, SPY 15:00 −C665/−C670/−C675	1.50	1,480	57	15%
SPY iron	ii	+271.80, +121.9%, 2026-09-16, SPY 15:00 −C754/−P754/−P749/−C759	−341.20, −100.1%, 2026-03-09, SPY 15:00 −C670/−P670/−P665/−C675	1.74	1,248	34	11%
SPY put	ii	+270.82, +120.9%, 2026-09-16, SPY 15:00 −P749/−P754/−P759	−346.18, −100.1%, 2026-03-09, SPY 15:00 −P665/−P670/−P675	1.45	1,569	63	17%
all names	ii	+282.80, +176.7%, 2026-04-29, QQQ 15:00 −C661/−P661/−P656/−C666	−468.18, −100.0%, 2026-05-12, QQQ 15:00 −C697/−C702/−C707	1.38	10,116	86	19%
QQQ call	iii	+177.82, +73.5%, 2026-10-21, QQQ 15:30 −C721/−C726/−C731	−151.18, −43.1%, 2026-10-15, QQQ 15:00 −C720/−C725/−C730	5.19	38	1	98%
QQQ iron	iii	+181.80, +76.4%, 2026-10-21, QQQ 15:30 −C726/−P726/−P721/−C731	−141.20, −41.4%, 2026-10-15, QQQ 15:00 −C725/−P725/−P720/−C730	6.00	27	1	95%
QQQ put	iii	+178.82, +74.2%, 2026-10-21, QQQ 15:30 −P721/−P726/−P731	−145.18, −42.1%, 2026-10-15, QQQ 15:00 −P720/−P725/−P730	5.49	37	1	98%
SPY call	iii	+158.82, +47.8%, 2026-10-21, SPY 15:30 −C768/−C773/−C778	−69.18, −16.6%, 2026-10-19, SPY 15:30 −C763/−C768/−C773	3.44	116	2	121%
SPY iron	iii	+158.80, +47.8%, 2026-10-21, SPY 15:30 −C773/−P773/−P768/−C778	−71.20, −17.0%, 2026-10-19, SPY 15:30 −C768/−P768/−P763/−C773	3.51	119	2	121%

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY put	iii	+156.82, +47.0%, 2026-10-21, SPY 15:30 −P768/−P773/−P778	−72.18, −17.2%, 2026-10-19, SPY 15:30 −P763/−P768/−P773	3.47	120	2	121%
all names	iii	+181.80, +76.4%, 2026-10-21, QQQ 15:30 −C726/−P726/−P721/−C731	−151.18, −43.1%, 2026-10-15, QQQ 15:00 −C720/−C725/−C730	4.55	295	1	96%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.60 a trade) and Robinhood's charged instead (mean $0.19). The study's fees are given beside them below. Re-pricing: exact (11,666 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ call	i	1,049	**+4.29**, +14.20	+3.69, +11.78	67.8 / 66.7	[+2.86, +6.62] / [+2.25, +6.00]
QQQ iron	i	1,049	**+6.01**, +20.78	+5.39, +18.38	70.4 / 69.5	[+4.42, +8.44] / [+3.81, +7.82]
QQQ put	i	1,051	**+4.06**, +13.42	+3.45, +11.00	66.9 / 65.8	[+2.63, +6.39] / [+2.03, +5.76]
SPY call	i	1,046	**+2.84**, +9.84	+2.24, +7.42	65.2 / 64.1	[+1.44, +4.70] / [+0.85, +4.08]
SPY iron	i	1,046	**+4.68**, +17.20	+4.08, +14.80	70.1 / 69.0	[+3.38, +6.49] / [+2.77, +5.88]
SPY put	i	1,046	**+2.00**, +6.27	+1.41, +3.85	61.7 / 60.7	[+0.43, +3.98] / [−0.16, +3.37]
all names	i	6,287	**+3.98**, +13.62	+3.38, +11.21	67.0 / 66.0	[+2.65, +6.02] / [+2.05, +5.40]
QQQ call	ii	877	**+2.88**, +5.14	+2.20, +2.72	58.0 / 57.5	[+0.52, +5.22] / [−0.17, +4.54]
QQQ iron	ii	877	**+6.41**, +18.09	+5.71, +15.69	63.7 / 62.7	[+4.18, +8.62] / [+3.47, +7.94]
QQQ put	ii	877	**+2.36**, +3.23	+1.68, +0.81	57.2 / 56.4	[−0.03, +4.65] / [−0.72, +3.99]
SPY call	ii	880	**+4.31**, +13.94	+3.65, +11.52	62.5 / 61.6	[+2.58, +6.09] / [+1.91, +5.44]
SPY iron	ii	880	**+5.78**, +19.11	+5.12, +16.71	65.0 / 64.2	[+4.08, +7.55] / [+3.41, +6.91]
SPY put	ii	880	**+4.02**, +12.75	+3.37, +10.33	62.0 / 60.8	[+2.30, +5.86] / [+1.63, +5.21]
all names	ii	5,271	**+4.29**, +12.05	+3.62, +9.64	61.4 / 60.5	[+2.45, +6.13] / [+1.77, +5.47]
QQQ call	iii	18	**+24.16**, +69.49	+23.42, +67.07	77.8 / 77.8	n/a (< 60 sessions) / n/a (< 60 sessions)
QQQ iron	iii	18	**+26.03**, +74.63	+25.29, +72.23	77.8 / 77.8	n/a (< 60 sessions) / n/a (< 60 sessions)
QQQ put	iii	18	**+24.49**, +70.49	+23.75, +68.07	77.8 / 77.8	n/a (< 60 sessions) / n/a (< 60 sessions)
SPY call	iii	18	**+10.27**, +35.49	+9.63, +33.07	61.1 / 61.1	n/a (< 60 sessions) / n/a (< 60 sessions)
SPY iron	iii	18	**+10.47**, +36.02	+9.83, +33.62	61.1 / 61.1	n/a (< 60 sessions) / n/a (< 60 sessions)
SPY put	iii	18	**+10.36**, +35.71	+9.72, +33.29	61.1 / 61.1	n/a (< 60 sessions) / n/a (< 60 sessions)
all names	iii	108	**+17.63**, +53.64	+16.94, +51.22	69.4 / 69.4	n/a (< 60 sessions) / n/a (< 60 sessions)

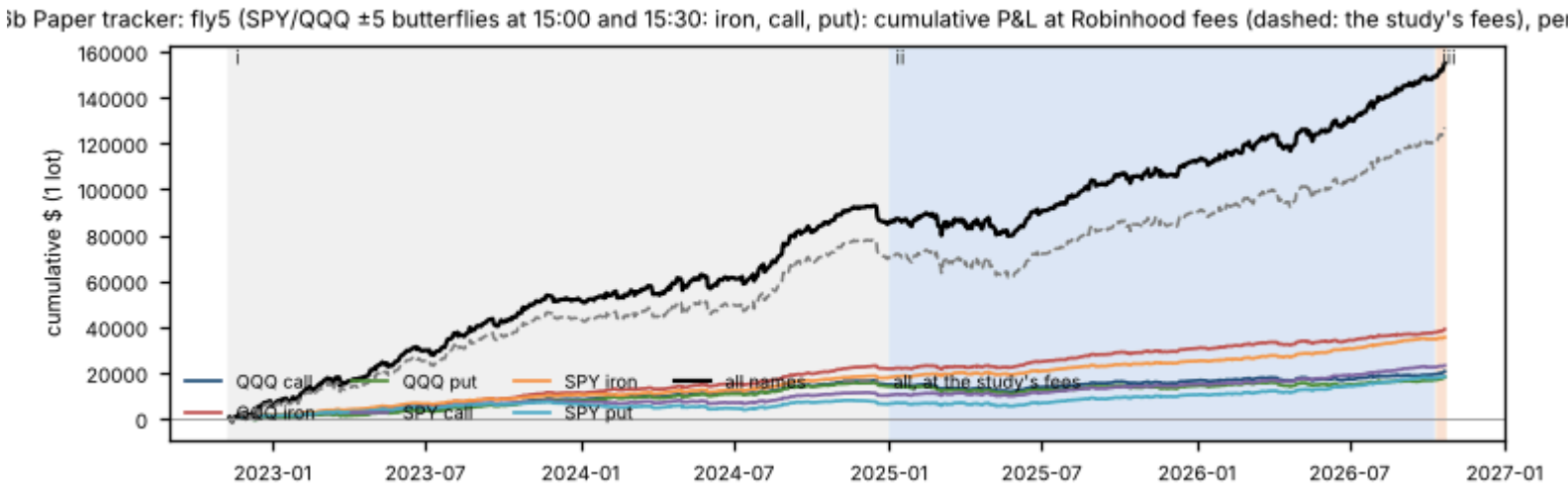
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- i: held to settlement (official close) 100.0%
- ii: held to settlement (official close) 100.0%
- iii: held to settlement (official close) 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	QQQ call	65	67.7	+2.76	+6.17	+401
2023	QQQ call	495	68.7	+4.34	+16.49	+8,162
2024	QQQ call	489	66.9	+4.46	+12.95	+6,332
2025	QQQ call	493	57.6	+2.24	+3.37	+1,663
2026	QQQ call	402	59.5	+4.62	+10.19	+4,098
2022	QQQ iron	65	72.3	+4.59	+13.69	+890
2023	QQQ iron	495	70.5	+5.57	+21.23	+10,507
2024	QQQ iron	489	69.9	+6.64	+21.26	+10,397
2025	QQQ iron	493	64.9	+6.15	+17.93	+8,842
2026	QQQ iron	402	62.9	+7.61	+20.80	+8,363
2022	QQQ put	65	69.2	+2.60	+5.50	+357
2023	QQQ put	495	68.1	+4.10	+15.53	+7,689
2024	QQQ put	491	65.4	+4.20	+12.34	+6,060
2025	QQQ put	493	55.4	+1.43	+0.36	+180
2026	QQQ put	402	60.4	+4.49	+9.75	+3,920
2022	SPY call	65	69.2	+6.33	+19.73	+1,282
2023	SPY call	495	67.9	+3.26	+12.65	+6,262
2024	SPY call	486	61.9	+1.94	+5.64	+2,743
2025	SPY call	493	61.1	+2.89	+9.42	+4,646
2026	SPY call	405	64.2	+6.30	+20.40	+8,264
2022	SPY iron	65	72.3	+7.71	+24.26	+1,577
2023	SPY iron	495	71.9	+4.65	+18.17	+8,993
2024	SPY iron	486	67.9	+4.31	+15.26	+7,416
2025	SPY iron	493	62.5	+4.32	+14.32	+7,060
2026	SPY iron	405	67.9	+7.77	+25.69	+10,405
2022	SPY put	65	70.8	+6.22	+19.23	+1,250
2023	SPY put	495	65.1	+2.73	+10.33	+5,115
2024	SPY put	486	57.0	+0.70	+0.39	+191
2025	SPY put	493	60.0	+2.48	+7.72	+3,805
2026	SPY put	405	64.4	+6.18	+19.89	+8,057
2022	all names	390	70.3	+5.04	+14.76	+5,758
2023	all names	2,970	68.7	+4.11	+15.73	+46,730
2024	all names	2,927	64.8	+3.71	+11.32	+33,139
2025	all names	2,958	60.2	+3.25	+8.86	+26,194
2026	all names	2,421	63.2	+6.17	+17.81	+43,106



Amendments, deviations and integrity checks. As 26a.

What it showed. Positive historically, but on the period that chose it, and not operationally clean.

#26c Paper tracker: condor_d20 (SPY 2-wide 20-delta iron condor, 13:00–15:00)

Folder: `scripts/paper_tracker.py; research_notes/Compass phase 3/backtest_replay, backtest_audit, forward_*`. **Commits:** defaults ba117a7, 6826b98 / official-close settlement a4ebe90, 99a3d4e, 16eb3c0 / replay 94af538, 30bb73e; audit 1ca1f0e; forward d7115b6, 2a7bd3b; candidate 3 99a3d4e. **Dates:** 2026-10-21 → 22.

Hypothesis. The 20-delta SPY condor (from hyp E's D1 read) pays forward.

Rule. 13:00, 13:30, 14:00, 14:30, 15:00: SPY 2-wide iron condor at 20 delta, held to the official close.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Shorts at the bid, longs at the ask; $0.65 a contract (leg) to open, nothing at expiry.

Data. The replay of every stored session through the live tracker's code, settled on the official close (data_recorder/paper_backtest_official, git-ignored, every trade).

Periods and holdouts. i-a model training 2022-11 → 2024-06 (in-sample) / i-b decile cuts 2024-07 → 12 / ii selection 2025 → 2026-10-09 / iii forward.

Pass rule (as pre-registered). Confirmation: 150 forward sessions for the D1 re-test.

Verdict. Positive in ii (+7.67 [+3.87, +10.89]) but set after ii was read; the after-hours rule DROPS it as a hold-to-expiry strategy.

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	i-a	1,823	403 / 422	77.4	+16.2, +27.08	−42.4, −71.43	+12.9, +22.80	**+2.94**, +4.82	[+1.04, +4.52]; [+1.61, +7.40]
SPY	i-b	572	125 / 131	83.9	+17.9, +29.46	−51.9, −84.65	+14.5, +24.80	**+6.68**, +11.11	[+3.50, +10.97]; [+5.84, +18.09]
SPY	ii	2,090	440 / 457	82.8	+18.2, +29.85	−52.1, −84.97	+14.8, +25.80	**+6.11**, +10.07	[+3.87, +8.01]; [+6.37, +13.18]
SPY	iii	41	9 / 9	95.1	+17.6, +29.34	−26.2, −43.70	+17.5, +29.80	**+15.43**, +25.77	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	i-a	+71.80, +56.1%, 2022-12-14, SPY 13:30 −C411/−C413/−P397/−P395	−189.20, −100.1%, 2023-07-07, SPY 13:30 −C444/−C446/−P441/−P439	1.30	1,349	68	15%
SPY	i-b	+65.80, +49.1%, 2024-09-18, SPY 13:00 −C569/−C571/−P557/−P555	−183.20, −100.1%, 2024-07-12, SPY 15:00 −C564/−C566/−P562/−P560	1.82	1,517	20	21%
SPY	ii	+62.80, +45.8%, 2025-04-10, SPY 13:30 −C530/−C532/−P509/−P507	−187.20, −100.1%, 2026-04-13, SPY 14:00 −C684/−C686/−P680/−P678	1.69	2,139	34	7%
SPY	iii	+43.80, +28.1%, 2026-10-22, SPY 13:00 −C767/−C769/−P761/−P759	−45.20, −27.2%, 2026-10-15, SPY 14:30 −C771/−C773/−P768/−P766	13.09	0	0	72%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.60 a trade) and Robinhood's charged instead (mean $0.20). The study's fees are given beside them below. Re-pricing: exact (4,526 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPY	i-a	1,823	**+2.94**, +4.82	+1.53, +2.42	77.4 / 76.9	[+1.04, +4.52] / [−0.37, +3.11]

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPY	i-b	572	**+6.68**, +11.11	+5.25, +8.71	83.9 / 83.9	[+3.50, +10.97] / [+2.06, +9.54]
SPY	ii	2,090	**+6.11**, +10.07	+4.68, +7.67	82.8 / 82.1	[+3.87, +8.01] / [+2.43, +6.59]
SPY	iii	41	**+15.43**, +25.77	+14.02, +23.38	95.1 / 95.1	n/a (< 60 sessions) / n/a (< 60 sessions)

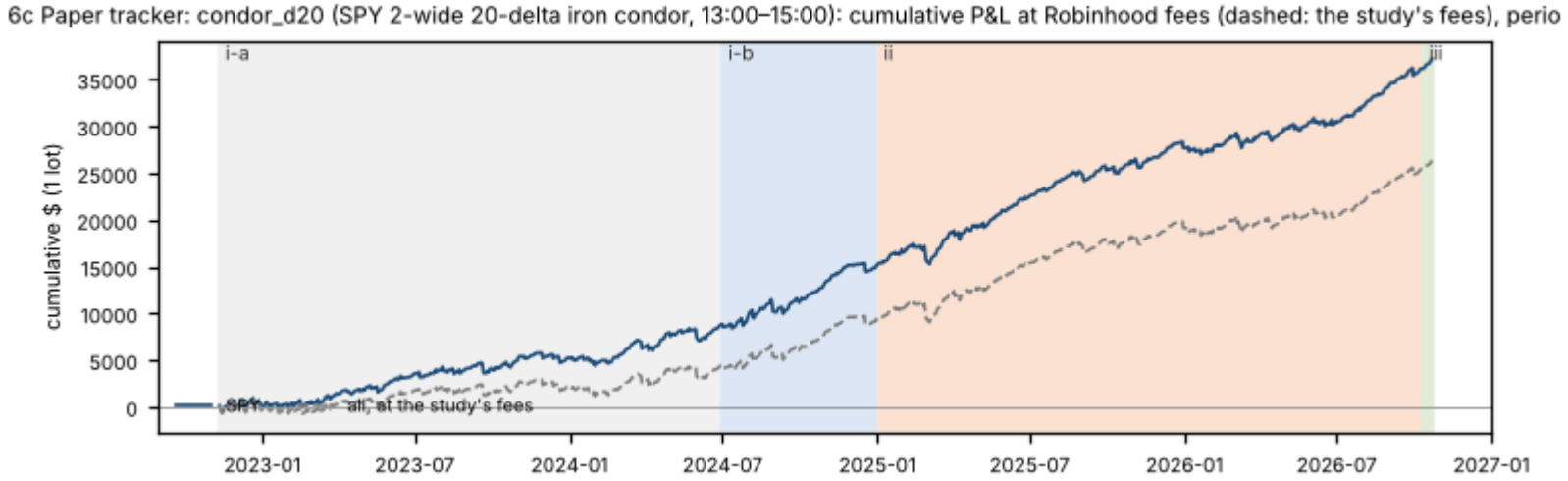
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- i-a: held to settlement (official close) 100.0%
- i-b: held to settlement (official close) 100.0%
- ii: held to settlement (official close) 100.0%
- iii: held to settlement (official close) 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPY	163	77.9	+0.37	+0.14	+22
2023	SPY	1,102	76.7	+2.84	+4.73	+5,212
2024	SPY	1,130	81.3	+5.30	+8.76	+9,901
2025	SPY	1,156	82.7	+6.70	+10.86	+12,557
2026	SPY	975	83.4	+5.81	+9.79	+9,547



Amendments, deviations and integrity checks. As 26a.

What it showed. Positive on the period it was chosen on.

#26d Paper tracker: fly_spxw (SPX ±5 / ±25 flies)

Folder: `scripts/paper_tracker.py; research_notes/Compass phase 3/backtest_replay, backtest_audit, forward_*`. **Commits:** scripts/paper_backtest.py (off unless listed). **Dates:** 2026-10-21.

Hypothesis. The SPXW analogue of fly5.

Rule. 15:00 and 15:30 SPXW ±5 and ±25 flies.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC)

Data. One week, 2026-10-05 → 09.

Periods and holdouts. —

Pass rule (as pre-registered). —

Verdict. Run only as a one-week sanity run outside the repository; replaced by the pre-registered SPXW test (#28).

Metrics. not available — a one-week sanity run outside the repository; no rows were kept.

Amendments, deviations and integrity checks. —

What it showed. No result.

#27a After-hours exercise study: fly5 iron ±5, case (b) the post-market decision

Folder: `research_notes/Compass phase 3/backtest_audit/AFTERHOURS.md`. **Commits:** owner's rule fixed before the read / — / 633fbff. **Dates:** 2026-10-22.

Hypothesis. The SPY/QQQ hold-to-expiry candidates survive after-hours exercise.

Rule. fly5 iron held to expiry; case (b): the counterparty exercises on the 17:30 post-market price, the position re-marked at the next 09:30 open.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). As #26 plus the exercise/assignment fee.

Data. SPY/QQQ, period ii (2025-10-13 → 2026-10-09 overlap) and iii (2026-10-12 → 21).

Periods and holdouts. ii / iii (7 sessions with a next open).

Pass rule (as pre-registered). "If a candidate's mean under (b) is ≤ 0 or its block-20 interval includes zero, drop it as a SPY/QQQ hold-to-expiry strategy."

Verdict. DROP: (b) +22.52 a trade [−20.07, +52.89] in ii.

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	ii	492	246 / 258	55.7	+119.6, +393.62	−131.8, −428.94	+16.4, +61.30	**+8.20**, +29.15	[−7.17, +18.66]; [−21.23, +63.21]
SPY	ii	495	248 / 258	56.0	+72.4, +258.78	−80.1, −285.08	+9.2, +37.36	**+5.26**, +19.26	[−6.09, +14.29]; [−18.43, +50.48]
all names	ii	987	248 / 258	55.8	+95.9, +325.83	−106.0, −357.01	+11.1, +43.36	**+6.72**, +24.19	[−5.36, +15.74]; [−17.22, +53.53]
QQQ	iii	14	7 / 7	42.9	+142.6, +506.30	−156.5, −511.84	−51.5, −168.96	**−28.36**, −75.49	n/a (< 60 sessions); n/a (< 60 sessions)
SPY	iii	14	7 / 7	42.9	+56.5, +226.86	−49.2, −177.93	−27.7, −107.20	**−3.94**, −4.45	n/a (< 60 sessions); n/a (< 60 sessions)
all names	iii	28	7 / 7	42.9	+99.5, +366.58	−102.9, −344.88	−27.7, −107.20	**−16.15**, −39.97	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	ii	+2,198.80, +672.4%, 2026-06-22, QQQ 15:00 [["C", 736.0, "short", 1], ["P", 736.0, "short", 1], ["P", 731.0, "long", 1], ["C", 741.0, "long", 1]]	−2,193.20, −681.1%, 2026-04-07, QQQ 15:30 [["C", 585.0, "short", 1], ["P", 585.0, "short", 1], ["P", 580.0, "long", 1], ["C", 590.0, "long", 1]]	1.15	10,269	68	88%
SPY	ii	+1,155.80, +318.4%, 2026-06-22, SPY 15:00 [["C", 744.0, "short", 1], ["P", 744.0, "short", 1], ["P", 739.0, "long", 1], ["C", 749.0, "long", 1]]	−1,759.20, −549.8%, 2026-04-07, SPY 15:30 [["C", 657.0, "short", 1], ["P", 657.0, "short", 1], ["P", 652.0, "long", 1], ["C", 662.0, "long", 1]]	1.15	7,172	88	83%
all names	ii	+2,198.80, +672.4%, 2026-06-22, QQQ 15:00 [["C", 736.0, "short", 1], ["P", 736.0, "short", 1], ["P", 731.0, "long", 1], ["C", 741.0, "long", 1]]	−2,193.20, −681.1%, 2026-04-07, QQQ 15:30 [["C", 585.0, "short", 1], ["P", 585.0, "short", 1], ["P", 580.0, "long", 1], ["C", 590.0, "long", 1]]	1.15	15,447	73	83%
QQQ	iii	+1,067.29, +365.5%, 2026-10-16, QQQ 15:30 [["C", 711.0, "short", 1], ["P", 711.0, "short", 1], ["P", 706.0, "long", 1], ["C", 716.0, "long", 1]]	−874.71, −266.7%, 2026-10-14, QQQ 15:30 [["C", 729.0, "short", 1], ["P", 729.0, "short", 1], ["P", 724.0, "long", 1], ["C", 734.0, "long", 1]]	0.74	2,203	5	total ≤ 0 (best 5 days +1,170; without them −2,227)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	iii	+375.80, +87.0%, 2026-10-13, SPY 15:00 [["C", 778.0, "short", 1], ["P", 778.0, "short", 1], ["P", 773.0, "long", 1], ["C", 783.0, "long", 1]]	−340.20, −107.3%, 2026-10-16, SPY 15:00 [["C", 762.0, "short", 1], ["P", 762.0, "short", 1], ["P", 757.0, "long", 1], ["C", 767.0, "long", 1]]	0.96	1,171	3	total ≤ 0 (best 5 days +863; without them −925)
all names	iii	+1,067.29, +365.5%, 2026-10-16, QQQ 15:30 [["C", 711.0, "short", 1], ["P", 711.0, "short", 1], ["P", 706.0, "long", 1], ["C", 716.0, "long", 1]]	−874.71, −266.7%, 2026-10-14, QQQ 15:30 [["C", 729.0, "short", 1], ["P", 729.0, "short", 1], ["P", 724.0, "long", 1], ["C", 734.0, "long", 1]]	0.80	3,063	5	total ≤ 0 (best 5 days +1,259; without them −2,378)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.60 a trade) and Robinhood's charged instead (mean $0.92). The study's fees are given beside them below. Re-pricing: exact (1,015 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	ii	492	**+8.20**, +29.15	+7.69, +27.42	55.7 / 55.5	[−7.17, +18.66] / [−7.67, +18.13]
SPY	ii	495	**+5.26**, +19.26	+4.82, +17.65	56.0 / 56.0	[−6.09, +14.29] / [−6.54, +13.87]
all names	ii	987	**+6.72**, +24.19	+6.25, +22.52	55.8 / 55.7	[−5.36, +15.74] / [−5.84, +15.27]
QQQ	iii	14	**−28.36**, −75.49	−28.86, −77.24	42.9 / 42.9	n/a (< 60 sessions) / n/a (< 60 sessions)
SPY	iii	14	**−3.94**, −4.45	−4.44, −6.39	42.9 / 42.9	n/a (< 60 sessions) / n/a (< 60 sessions)
all names	iii	28	**−16.15**, −39.97	−16.65, −41.81	42.9 / 42.9	n/a (< 60 sessions) / n/a (< 60 sessions)

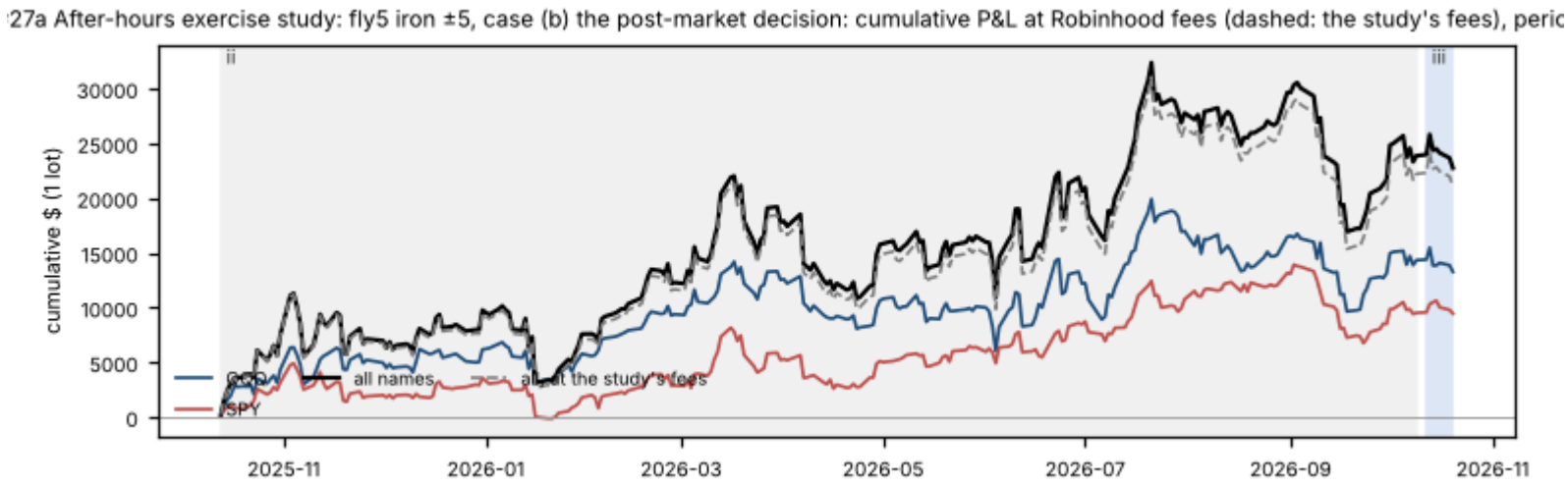
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- ii: held; after-hours decision (b) on the 17:30 price 100.0%
- iii: held; after-hours decision (b) on the 17:30 price 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	QQQ	108	57.4	+11.35	+57.33	+6,191
2026	QQQ	398	54.8	+6.05	+17.83	+7,095
2025	SPY	108	59.3	+5.69	+33.24	+3,590
2026	SPY	401	54.6	+4.82	+14.67	+5,882
2025	all names	216	58.3	+8.52	+45.28	+9,781
2026	all names	799	54.7	+5.43	+16.24	+12,977



Amendments, deviations and integrity checks. 16 name-sessions of UW 1-minute post-market bars fetched once with the owner’s leave; SPY 2025-04-09 official close flagged.

What it showed. Counting after-hours exercise, the iron fly’s mean is inside the noise.

#27b After-hours exercise study: condor_d20, case (b)

Folder: `research_notes/Compass phase 3/backtest_audit/AFTERHOURS.md`. **Commits:** — / — / 633fbff. **Dates:** 2026-10-22.

Hypothesis. As 27a.

Rule. condor_d20 held to expiry, case (b).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). As 27a.

Data. As 27a.

Periods and holdouts. ii / iii.

Pass rule (as pre-registered). As 27a.

Verdict. DROP: (b) +3.84 [−19.94, +22.74] in ii; iii undecided (7 sessions).

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	ii	1,176	247 / 258	83.3	+38.3, +62.32	−169.4, −275.19	+16.2, +27.80	**+3.70**, +6.07	[−11.28, +15.87]; [−17.97, +25.12]
SPY	iii	36	8 / 8	94.4	+31.6, +54.28	−84.0, −143.70	+17.5, +29.80	**+25.14**, +43.28	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	ii	+868.20, +598.8%, 2026-09-16, SPY 14:00 [["C", 765.0, "short", 1], ["C", 767.0, "long", 1], ["P", 755.0, "short", 1], ["P", 753.0, "long", 1]]	−1,796.20, −1,144.1%, 2026-04-07, SPY 14:30 [["C", 658.0, "short", 1], ["C", 660.0, "long", 1], ["P", 654.0, "short", 1], ["P", 652.0, "long", 1]]	1.13	11,417	130	156%
SPY	iii	+354.19, +212.1%, 2026-10-16, SPY 13:00 [["C", 769.0, "short", 1], ["C", 771.0, "long", 1], ["P", 764.0, "short", 1], ["P", 762.0, "long", 1]]	−190.20, −114.6%, 2026-10-15, SPY 14:30 [["C", 771.0, "short", 1], ["C", 773.0, "long", 1], ["P", 768.0, "short", 1], ["P", 766.0, "long", 1]]	6.42	59	1	105%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.60 a trade) and Robinhood's charged instead (mean $0.38). The study's fees are given beside them below. Re-pricing: exact (1,212 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPY	ii	1,176	**+3.70**, +6.07	+2.37, +3.84	83.3 / 83.3	[−11.28, +15.87] / [−12.63, +14.57]
SPY	iii	36	**+25.14**, +43.28	+23.77, +40.93	94.4 / 94.4	n/a (< 60 sessions) / n/a (< 60 sessions)

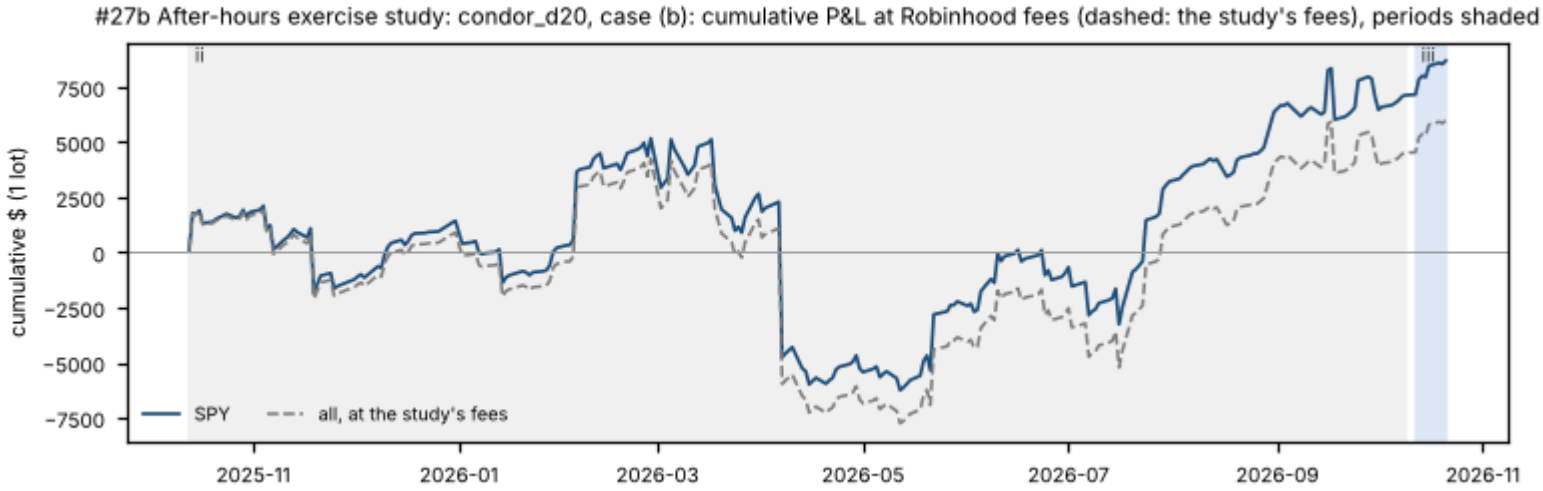
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- ii: held; after-hours decision (b) on the 17:30 price 100.0%
- iii: held; after-hours decision (b) on the 17:30 price 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	SPY	242	84.3	+2.92	+5.92	+1,432
2026	SPY	970	83.5	+4.68	+7.48	+7,258



Amendments, deviations and integrity checks. As 27a.

What it showed. Inside the noise once exercise is counted.

#28a SPXW analogue A: iron fly ±5×ratio at 15:00 and 15:30 (fly5 iron)

Folder: `research_notes/Compass phase 3/spxw_prereg`. **Commits:** 2820bb4 / — / 6bb230e. **Dates:** 2026-10-22.

Hypothesis. The ETF fly5 iron result transfers to cash-settled SPXW.

Rule. SPXW iron fly, wings ±5 × the SPX/SPY ratio, 15:00 and 15:30, package fill at the NBBO rounded against us to $0.05, held to cash settlement on the official SPX close.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); about $4.93 to open a four-leg structure.

Data. SPXW 1,105 full sessions 2022-05-13 → 2026-10-21.

Periods and holdouts. development 2022-05 → 2024 / selection 2025 → 2026-10-09 / forward 8 sessions.

Pass rule (as pre-registered). (1) selection mean > 0 and block-20 lower bound > 0; (2) selection mean > 0 at +$0.05 package stress; (3) development mean > 0.

Verdict. FAIL: selection −$39.64 a session [−173.06, +87.48].

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	development	1,312	656 / 683	62.6	+12.7, +488.32	−18.3, −708.36	+4.6, +195.16	**+1.08**, +40.48	[−0.12, +2.36]; [−3.76, +88.63]
SPXW	selection	882	441 / 457	58.8	+15.0, +535.02	−22.4, −811.67	+3.5, +150.16	**−0.40**, −19.23	[−2.27, +1.43]; [−86.71, +46.19]
SPXW	forward	16	8 / 8	62.5	+10.9, +433.56	−7.7, −324.67	+1.4, +54.66	**+3.92**, +149.22	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	development	+2,463.48, +123.2%, 2022-12-14, None	−4,034.34, −100.1%, 2024-05-31, None	1.15	29,822	272	28%
SPXW	selection	+2,569.48, +114.5%, 2026-09-16, None	−3,549.34, −100.1%, 2026-03-09, None	0.94	39,040	422	total ≤ 0 (best 5 days +15,024; without them −31,986)
SPXW	forward	+921.66, +25.4%, 2026-10-16, None	−746.34, −17.3%, 2026-10-19, None	2.23	813	2	141%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $4.93 a trade) and Robinhood's charged instead (mean $4.35). The study's fees are given beside them below. Re-pricing: exact (2,210 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPXW	development	1,312	**+1.08**, +40.48	+1.07, +39.89	62.6 / 62.6	[−0.12, +2.36] / [−0.13, +2.35]
SPXW	selection	882	**−0.40**, −19.23	−0.41, −19.82	58.8 / 58.8	[−2.27, +1.43] / [−2.28, +1.41]
SPXW	forward	16	**+3.92**, +149.22	+3.91, +148.63	62.5 / 62.5	n/a (< 60 sessions) / n/a (< 60 sessions)

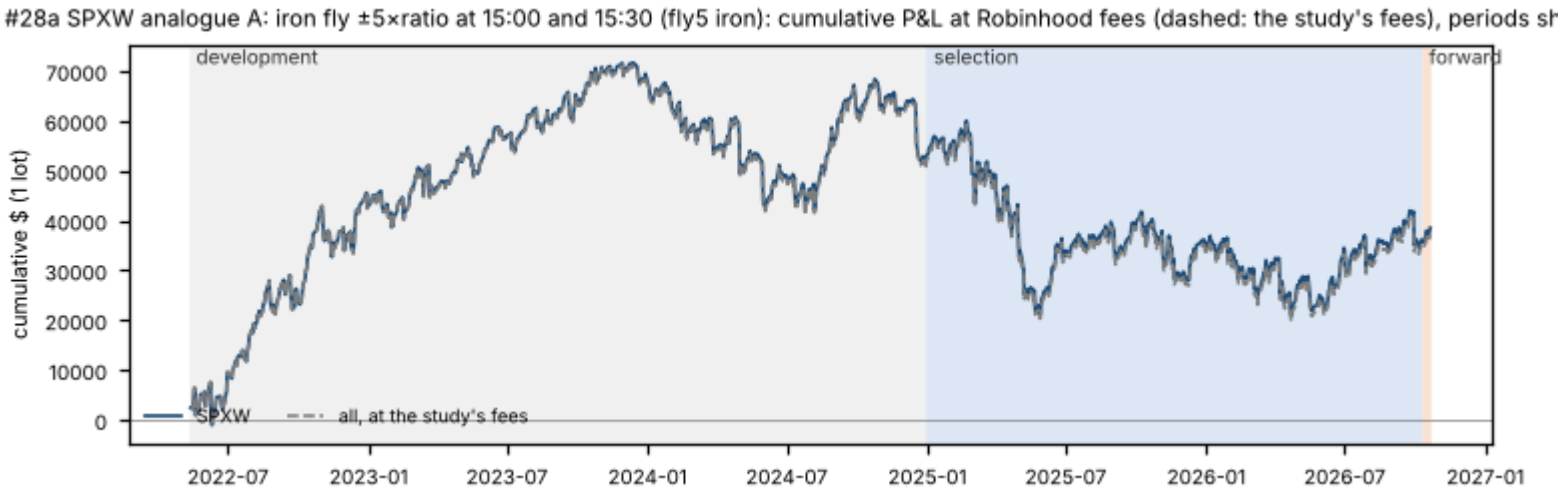
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- development: held to cash settlement (official SPX close) 100.0%
- selection: held to cash settlement (official SPX close) 100.0%
- forward: held to cash settlement (official SPX close) 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	318	63.5	+3.79	+133.85	+42,565
2023	SPXW	496	64.5	+1.29	+54.21	+26,888
2024	SPXW	498	60.0	−0.85	−32.82	−16,344
2025	SPXW	494	58.9	−0.88	−38.33	−18,934
2026	SPXW	404	58.9	+0.36	+10.79	+4,360



Amendments, deviations and integrity checks. The package rule cost nothing: the summed leg NBBO is always a multiple of $0.05.

What it showed. SPXW did not reproduce the ETF fly.

#28b SPXW analogue B: 20-delta iron condor, wings 2×ratio, 13:00–15:00 (condor_d20)

Folder: `research_notes/Compass phase 3/spxw_prereg`. **Commits:** 2820bb4 / — / 6bb230e. **Dates:** 2026-10-22.

Hypothesis. As 28a for condor_d20.

Rule. SPXW 20-delta iron condor, wings 2 × ratio, five entries 13:00–15:00, package fill, held to cash settlement.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC)

Data. As 28a.

Periods and holdouts. As 28a.

Pass rule (as pre-registered). As 28a.

Verdict. FAIL: selection +$120.76 a session [−95.03, +286.78] — positive but the interval crosses zero.

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	development	3,193	656 / 683	74.4	+16.9, +281.25	−47.0, −784.69	+13.1, +230.66	**+0.52**, +7.84	[−1.29, +2.21]; [−21.92, +35.82]
SPXW	selection	2,157	441 / 457	76.8	+17.5, +289.67	−51.4, −848.63	+13.4, +235.66	**+1.53**, +25.28	[−1.08, +3.50]; [−18.27, +58.23]
SPXW	forward	38	8 / 8	89.5	+15.4, +262.42	−15.0, −260.32	+14.4, +250.66	**+12.23**, +207.39	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	development	+705.48, +54.7%, 2022-12-14, SPXW 13:00 3945.0/3965.0P 4105.0/4125.0C	−1,879.16, −100.2%, 2023-12-20, SPXW 13:30 4745.0/4765.0P 4785.0/4805.0C	1.04	55,785	560	51%
SPXW	selection	+605.48, +43.6%, 2026-07-29, SPXW 13:00 7290.0/7310.0P 7430.0/7450.0C	−1,849.34, −100.2%, 2026-04-13, SPXW 14:30 6815.0/6835.0P 6860.0/6880.0C	1.13	34,778	171	24%
SPXW	forward	+435.48, +27.9%, 2026-10-16, SPXW 13:30 7620.0/7640.0P 7695.0/7715.0C	−645.34, −37.2%, 2026-10-15, SPXW 14:30 7685.0/7705.0P 7735.0/7755.0C	8.57	0	0	80%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $4.95 a trade) and Robinhood's charged instead (mean $4.36). The study's fees are given beside them below. Re-pricing: exact (5,388 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPXW	development	3,193	**+0.52**, +7.84	+0.49, +7.25	74.4 / 74.4	[−1.29, +2.21] / [−1.33, +2.18]
SPXW	selection	2,157	**+1.53**, +25.28	+1.49, +24.69	76.8 / 76.8	[−1.08, +3.50] / [−1.12, +3.46]
SPXW	forward	38	**+12.23**, +207.39	+12.19, +206.80	89.5 / 89.5	n/a (< 60 sessions) / n/a (< 60 sessions)

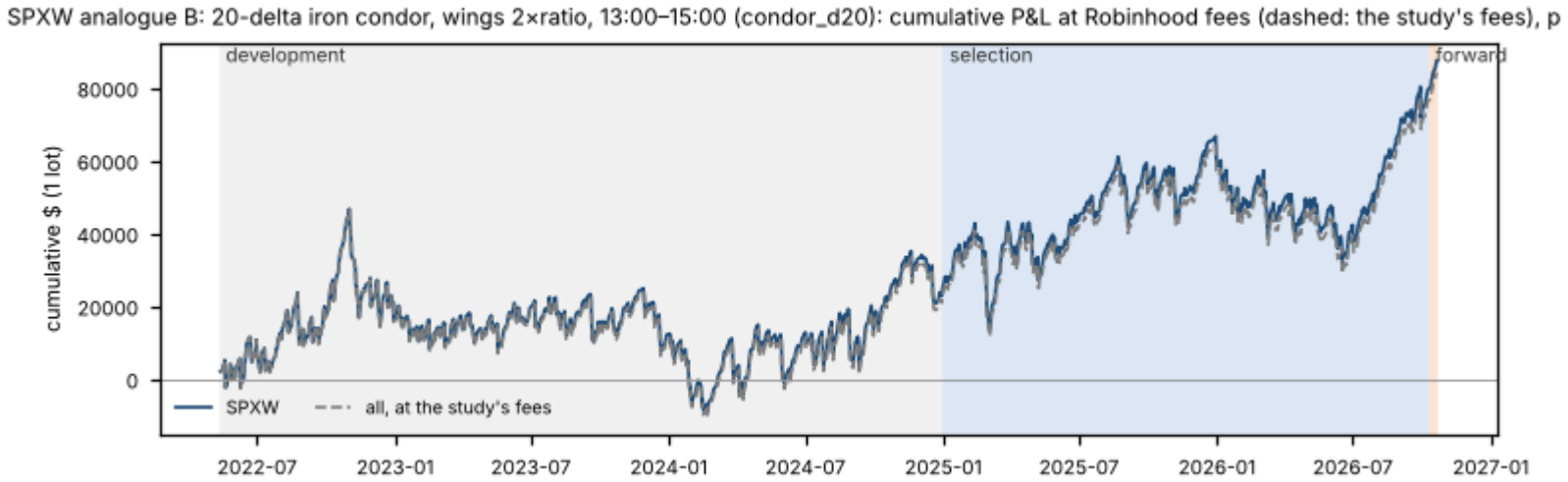
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- development: held to cash settlement (official SPX close) 100.0%
- selection: held to cash settlement (official SPX close) 100.0%
- forward: held to cash settlement (official SPX close) 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	794	73.8	+1.34	+20.24	+16,067
2023	SPXW	1,190	73.4	−0.23	−3.36	−3,994
2024	SPXW	1,209	75.6	+0.73	+10.72	+12,955
2025	SPXW	1,203	77.1	+1.83	+28.80	+34,645
2026	SPXW	992	76.9	+1.57	+27.98	+27,761



Amendments, deviations and integrity checks. As 28a; the ETF−SPXW late-day extrinsic gap measured beside it.

What it showed. Positive in both periods, inside the noise.

#29a Picker (GeorgeFi rule), PRIMARY: opening-range fade, stop 0.68, target 1.25, 5-minute stop, flat 15:58

Folder: `research_notes/Compass phase 3/picker_prereg`. **Commits:** c502122 / — / 1e10053. **Dates:** 2026-10-22.

Hypothesis. The opening-range sweep fade the GeorgeFi cards describe makes money as written.

Rule. Opening range = the 09:30–09:59 bars; 10:00–11:59: the first bar with high ≥ ORH × (1 + 2 bp) and close < ORH → PUT (the low side → CALL); buy the nearest OTM 0DTE contract at the ask one bar later; exits: stop at 0.68 × ask, target 1.25 × ask (resting), out at 5 minutes unless +6% reached, flat 15:58.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData); a target at its resting limit

Data. SPY, QQQ (846 sessions), IWM (484), SPXW (972), 2022 → 2026-10-21, the 134 GeorgeFi card dates excluded.

Periods and holdouts. The whole tested span (verdict) and the never-read line 2026-10-14 → 21 (no verdict). The 2022–2026 NBBO had been read by the engine audit for the same universe (disclosed).

Pass rule (as pre-registered). "Pass: mean > 0 with the interval clear of zero on each name, judged only on the span with the excluded dates removed" (PRIMARY only; mean % of premium, 20-session block interval; every name must pass).

Verdict. FAIL on every name: SPY −6.28% [−8.18, −4.41], QQQ −5.21%, IWM −9.47%, SPXW −4.96%.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	whole span (card dates out)	438	394 / 634	36.1	+23.8, +13.92	−23.7, −13.18	−10.6, −5.10	**−6.54**, −3.41	[−9.08, −4.31]; [−5.25, −1.95]
QQQ	whole span (card dates out)	814	698 / 1,005	37.3	+22.8, +28.93	−20.2, −26.03	−7.3, −7.60	**−4.12**, −5.50	[−5.86, −2.43]; [−8.41, −2.71]
SPXW	whole span (card dates out)	902	765 / 1,135	36.9	+21.9, +269.51	−20.7, −252.86	−7.9, −82.26	**−5.00**, −60.01	[−6.67, −3.48]; [−84.67, −36.93]
SPY	whole span (card dates out)	771	660 / 1,005	36.2	+22.0, +24.93	−20.2, −22.79	−7.7, −6.10	**−4.97**, −5.52	[−6.71, −3.28]; [−8.49, −2.91]
all names	whole span (card dates out)	2,925	898 / 1,135	36.7	+22.4, +100.28	−20.9, −92.95	−7.8, −9.10	**−4.98**, −22.00	[−6.17, −3.69]; [−30.20, −14.23]
QQQ	never-read 2026-10-14→21	6	5 / 6	66.7	+25.1, +50.40	−10.0, −19.10	+25.0, +45.90	**+13.42**, +27.23	n/a (< 60 sessions); n/a (< 60 sessions)
SPXW	never-read 2026-10-14→21	6	5 / 6	50.0	+24.9, +291.07	−12.5, −145.59	+10.1, +87.74	**+6.24**, +72.74	n/a (< 60 sessions); n/a (< 60 sessions)
SPY	never-read 2026-10-14→21	6	5 / 6	50.0	+25.4, +26.57	−11.5, −13.10	+10.0, +5.90	**+6.98**, +6.73	n/a (< 60 sessions); n/a (< 60 sessions)
all names	never-read 2026-10-14→21	18	5 / 6	55.6	+25.2, +115.45	−11.5, −64.29	+24.8, +24.40	**+8.88**, +35.57	n/a (< 60 sessions); n/a (< 60 sessions)
IWM	whole span (verdict)	438	394 / 634	36.1	+23.8, +13.92	−23.7, −13.18	−10.6, −5.10	**−6.54**, −3.41	[−9.08, −4.31]; [−5.25, −1.95]

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	whole span (verdict)	820	703 / 1,018	37.6	+22.8, +29.20	−20.1, −26.00	−7.1, −7.10	**−3.99**, −5.27	[−5.98, −2.44]; [−8.47, −2.91]
SPXW	whole span (verdict)	908	770 / 1,148	37.0	+21.9, +269.70	−20.7, −252.30	−7.8, −77.26	**−4.93**, −59.13	[−6.70, −3.46]; [−85.43, −36.69]
SPY	whole span (verdict)	777	665 / 1,018	36.3	+22.0, +24.95	−20.2, −22.73	−7.7, −6.10	**−4.87**, −5.42	[−6.71, −3.20]; [−8.52, −2.98]
all names	whole span (verdict)	2,943	903 / 1,148	36.8	+22.5, +100.42	−20.8, −92.83	−7.8, −9.10	**−4.89**, −21.65	[−6.21, −3.68]; [−30.26, −13.55]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	whole span (card dates out)	+60.90, +25.1%, 2025-04-09, IWM 2025-04-09 176.0P	−119.10, −57.8%, 2025-04-07, IWM 2025-04-07 176.0P	0.60	1,533	624	total ≤ 0 (best 5 days +292; without them −1,784)
QQQ	whole span (card dates out)	+141.89, +25.1%, 2025-04-09, QQQ 2025-04-09 423.0P	−299.10, −51.1%, 2025-04-07, QQQ 2025-04-07 416.0P	0.66	4,558	990	total ≤ 0 (best 5 days +691; without them −5,172)
SPXW	whole span (card dates out)	+1,387.74, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−2,952.26, −45.6%, 2025-04-07, SPXW 2025-04-07 4970.0P	0.62	54,455	1,135	total ≤ 0 (best 5 days +6,120; without them −60,249)
SPY	whole span (card dates out)	+128.89, +25.1%, 2025-04-09, SPY 2025-04-09 495.0C	−294.10, −45.2%, 2025-04-07, SPY 2025-04-07 495.0P	0.62	4,290	1,005	total ≤ 0 (best 5 days +572; without them −4,828)
all names	whole span (card dates out)	+1,387.74, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−2,952.26, −45.6%, 2025-04-07, SPXW 2025-04-07 4970.0P	0.63	64,706	1,135	total ≤ 0 (best 5 days +7,330; without them −71,688)
QQQ	never-read 2026-10-14→21	+54.90, +25.1%, 2026-10-15, QQQ 2026-10-15 720.0P	−34.10, −17.9%, 2026-10-19, QQQ 2026-10-19 720.0P	5.28	38	2	100%
SPXW	never-read 2026-10-14→21	+347.74, +25.0%, 2026-10-19, SPXW 2026-10-19 7690.0C	−222.26, −17.8%, 2026-10-14, SPXW 2026-10-14 7760.0C	2.00	295	3	100%
SPY	never-read 2026-10-14→21	+30.90, +24.9%, 2026-10-21, SPY 2026-10-21 770.0C	−17.10, −17.8%, 2026-10-14, SPY 2026-10-14 774.0C	2.03	25	3	100%
all names	never-read 2026-10-14→21	+347.74, +25.0%, 2026-10-19, SPXW 2026-10-19 7690.0C	−222.26, −17.8%, 2026-10-14, SPXW 2026-10-14 7760.0C	2.24	218	2	100%
IWM	whole span (verdict)	+60.90, +25.1%, 2025-04-09, IWM 2025-04-09 176.0P	−119.10, −57.8%, 2025-04-07, IWM 2025-04-07 176.0P	0.60	1,533	624	total ≤ 0 (best 5 days +292; without them −1,784)
QQQ	whole span (verdict)	+141.89, +25.1%, 2025-04-09, QQQ 2025-04-09 423.0P	−299.10, −51.1%, 2025-04-07, QQQ 2025-04-07 416.0P	0.68	4,558	1,003	total ≤ 0 (best 5 days +691; without them −5,009)
SPXW	whole span (verdict)	+1,387.74, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−2,952.26, −45.6%, 2025-04-07, SPXW 2025-04-07 4970.0P	0.63	54,455	1,148	total ≤ 0 (best 5 days +6,120; without them −59,812)
SPY	whole span (verdict)	+128.89, +25.1%, 2025-04-09, SPY 2025-04-09 495.0C	−294.10, −45.2%, 2025-04-07, SPY 2025-04-07 495.0P	0.63	4,290	1,018	total ≤ 0 (best 5 days +572; without them −4,787)
all names	whole span (verdict)	+1,387.74, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−2,952.26, −45.6%, 2025-04-07, SPXW 2025-04-07 4970.0P	0.63	64,706	1,148	total ≤ 0 (best 5 days +7,330; without them −71,048)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.74 a trade) and Robinhood's charged instead (mean $0.77). The study's fees are given beside them below. Re-pricing: exact (2,943 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	whole span (card dates out)	438	**−6.54**, −3.41	−9.47, −4.68	36.1 / 34.9	[−9.08, −4.31] / [−12.02, −7.26]
QQQ	whole span (card dates out)	814	**−4.12**, −5.50	−5.34, −6.78	37.3 / 36.4	[−5.86, −2.43] / [−7.05, −3.69]
SPXW	whole span (card dates out)	902	**−5.00**, −60.01	−5.03, −60.30	36.9 / 36.9	[−6.67, −3.48] / [−6.69, −3.51]
SPY	whole span (card dates out)	771	**−4.97**, −5.52	−6.37, −6.79	36.2 / 34.4	[−6.71, −3.28] / [−8.12, −4.67]
all names	whole span (card dates out)	2,925	**−4.98**, −22.00	−6.14, −22.97	36.7 / 35.8	[−6.17, −3.69] / [−7.36, −4.81]
QQQ	never-read 2026-10-14→21	6	**+13.42**, +27.23	+12.77, +25.96	66.7 / 66.7	n/a (< 60 sessions) / n/a (< 60 sessions)
SPXW	never-read 2026-10-14→21	6	**+6.24**, +72.74	+6.21, +72.44	50.0 / 50.0	n/a (< 60 sessions) / n/a (< 60 sessions)
SPY	never-read 2026-10-14→21	6	**+6.98**, +6.73	+5.82, +5.46	50.0 / 50.0	n/a (< 60 sessions) / n/a (< 60 sessions)
all names	never-read 2026-10-14→21	18	**+8.88**, +35.57	+8.27, +34.62	55.6 / 55.6	n/a (< 60 sessions) / n/a (< 60 sessions)
IWM	whole span (verdict)	438	**−6.54**, −3.41	−9.47, −4.68	36.1 / 34.9	[−9.08, −4.31] / [−12.02, −7.26]
QQQ	whole span (verdict)	820	**−3.99**, −5.27	−5.21, −6.54	37.6 / 36.6	[−5.98, −2.44] / [−7.13, −3.67]
SPXW	whole span (verdict)	908	**−4.93**, −59.13	−4.96, −59.43	37.0 / 37.0	[−6.70, −3.46] / [−6.73, −3.49]
SPY	whole span (verdict)	777	**−4.87**, −5.42	−6.28, −6.70	36.3 / 34.5	[−6.71, −3.20] / [−8.12, −4.61]
all names	whole span (verdict)	2,943	**−4.89**, −21.65	−6.05, −22.62	36.8 / 35.9	[−6.21, −3.68] / [−7.41, −4.82]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

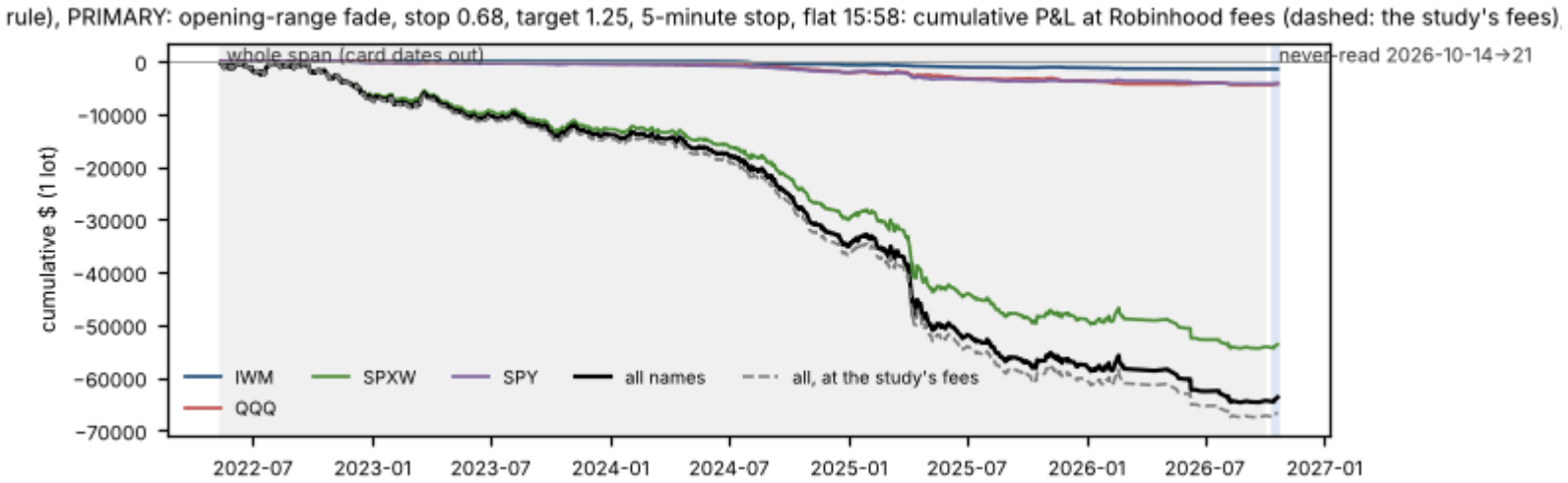
Exit reasons (share of trades):

- whole span (card dates out): time5 47.2%; target 32.1%; stop 20.7%
- never-read 2026-10-14→21: target 55.6%; time5 44.4%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	152	30.9	−8.85	−4.73	−719
2025	IWM	232	38.8	−5.08	−2.29	−531
2026	IWM	54	38.9	−6.33	−4.47	−241
2022	QQQ	30	36.7	−2.88	−1.93	−58
2023	QQQ	248	39.5	−2.10	−1.50	−371
2024	QQQ	244	34.8	−5.34	−7.19	−1,756
2025	QQQ	233	39.1	−3.80	−6.37	−1,483
2026	QQQ	65	35.4	−7.39	−9.99	−650
2022	SPXW	164	40.9	−3.26	−41.41	−6,791
2023	SPXW	240	41.7	−3.14	−25.47	−6,113
2024	SPXW	228	28.5	−7.29	−75.33	−17,175
2025	SPXW	213	36.2	−5.61	−88.18	−18,782
2026	SPXW	63	42.9	−5.23	−76.71	−4,832
2022	SPY	33	30.3	−9.79	−10.31	−340
2023	SPY	240	42.5	−2.30	−1.08	−259
2024	SPY	228	28.1	−7.19	−6.86	−1,564
2025	SPY	213	36.6	−4.88	−7.78	−1,657

year	name	trades	win %	mean %	mean $	total $
2026	SPY	63	44.4	−3.70	−6.26	−394
2022	all names	227	38.8	−4.15	−31.67	−7,189
2023	all names	728	41.2	−2.51	−9.26	−6,743
2024	all names	852	30.6	−6.98	−24.90	−21,214
2025	all names	891	37.7	−4.82	−25.20	−22,454
2026	all names	245	40.4	−5.65	−24.97	−6,118



Amendments, deviations and integrity checks. verify_picker.py re-derives trades from the raw files; the flow gate had one session of data (0 trades gated).

What it showed. The fade loses 5–9% of premium a trade on every name.

#29b Picker, SECONDARY: the narrative’s +6% breakeven lock and VIX trail

Folder: `research_notes/Compass phase 3/picker_prereg`. **Commits:** c502122 / — / 1e10053. **Dates:** 2026-10-22.

Hypothesis. The vendor’s narrative exits rescue the rule.

Rule. Opening range = the 09:30–09:59 bars; 10:00–11:59: the first bar with high ≥ ORH × (1 + 2 bp) and close < ORH → PUT (the low side → CALL); buy the nearest OTM 0DTE contract at the ask one bar later; exits: stop, target, breakeven lock at +6%, VIX-scaled trail, flat 15:58.

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData); a target at its resting limit

Data. SPY, QQQ (846 sessions), IWM (484), SPXW (972), 2022 → 2026-10-21, the 134 GeorgeFi card dates excluded.

Periods and holdouts. The whole tested span (verdict) and the never-read line 2026-10-14 → 21 (no verdict). The 2022–2026 NBBO had been read by the engine audit for the same universe (disclosed).

Pass rule (as pre-registered). No verdict of its own (labelled).

Verdict. Labelled: SPY −6.29%, QQQ −5.44%, IWM −9.91%, SPXW −5.01%.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib’s 20-session moving-block bootstrap over the period’s calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	whole span (card dates out)	438	394 / 634	38.1	+18.5, +10.62	−22.7, −11.83	−3.9, −2.10	**−6.98**, −3.27	[−9.29, −4.90]; [−4.67, −2.02]
QQQ	whole span (card dates out)	814	698 / 1,005	44.3	+17.2, +21.82	−21.6, −26.43	−0.9, −1.10	**−4.37**, −5.03	[−5.68, −2.80]; [−7.18, −2.60]
SPXW	whole span (card dates out)	902	765 / 1,135	41.4	+16.6, +210.90	−20.1, −242.81	−1.6, −12.26	**−4.95**, −55.19	[−6.42, −3.35]; [−72.29, −35.44]
SPY	whole span (card dates out)	771	660 / 1,005	41.8	+16.6, +19.26	−20.2, −22.35	−1.7, −1.10	**−4.85**, −4.97	[−6.60, −3.28]; [−6.93, −3.16]
all names	whole span (card dates out)	2,925	898 / 1,135	41.8	+17.0, +77.29	−20.9, −90.28	−1.7, −2.10	**−5.07**, −20.22	[−6.13, −3.84]; [−25.53, −13.49]
QQQ	never-read 2026-10-14→21	6	5 / 6	83.3	+25.1, +50.30	−32.5, −62.10	+25.1, +48.40	**+15.52**, +31.57	n/a (< 60 sessions); n/a (< 60 sessions)
SPXW	never-read 2026-10-14→21	6	5 / 6	33.3	+25.0, +312.74	−26.9, −334.76	−17.7, −187.26	**−9.57**, −118.93	n/a (< 60 sessions); n/a (< 60 sessions)
SPY	never-read 2026-10-14→21	6	5 / 6	33.3	+25.2, +29.90	−27.3, −34.35	−18.2, −17.10	**−9.79**, −12.93	n/a (< 60 sessions); n/a (< 60 sessions)
all names	never-read 2026-10-14→21	18	5 / 6	50.0	+25.1, +104.09	−27.7, −170.95	+10.8, +12.90	**−1.28**, −33.43	n/a (< 60 sessions); n/a (< 60 sessions)
IWM	whole span (verdict)	438	394 / 634	38.1	+18.5, +10.62	−22.7, −11.83	−3.9, −2.10	**−6.98**, −3.27	[−9.29, −4.90]; [−4.67, −2.02]
QQQ	whole span (verdict)	820	703 / 1,018	44.6	+17.3, +22.21	−21.6, −26.50	−0.9, −1.10	**−4.22**, −4.76	[−5.88, −2.87]; [−7.35, −2.80]

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	whole span (verdict)	908	770 / 1,148	41.3	+16.6, +211.45	−20.2, −243.50	−1.6, −17.26	**−4.98**, −55.61	[−6.55, −3.44]; [−73.51, −36.97]
SPY	whole span (verdict)	777	665 / 1,018	41.7	+16.7, +19.33	−20.3, −22.46	−1.7, −1.10	**−4.89**, −5.04	[−6.62, −3.34]; [−7.10, −3.27]
all names	whole span (verdict)	2,943	903 / 1,148	41.9	+17.1, +77.48	−21.0, −90.71	−1.7, −2.10	**−5.04**, −20.30	[−6.20, −3.90]; [−25.95, −13.96]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	whole span (card dates out)	+60.90, +25.1%, 2025-04-09, IWM 2025-04-09 176.0P	−59.10, −39.1%, 2024-08-05, IWM 2024-08-05 199.0P	0.55	1,453	632	total ≤ 0 (best 5 days +225; without them −1,658)
QQQ	whole span (card dates out)	+141.89, +25.1%, 2025-04-09, QQQ 2025-04-09 423.0P	−139.10, −34.6%, 2025-04-08, QQQ 2025-04-08 440.0P	0.66	4,142	1,001	total ≤ 0 (best 5 days +616; without them −4,709)
SPXW	whole span (card dates out)	+1,387.74, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−1,182.26, −34.4%, 2025-04-23, SPXW 2025-04-23 5445.0P	0.61	49,779	1,135	total ≤ 0 (best 5 days +5,592; without them −55,371)
SPY	whole span (card dates out)	+128.89, +25.1%, 2025-04-09, SPY 2025-04-09 495.0C	−109.10, −34.7%, 2025-04-23, SPY 2025-04-23 542.0P	0.62	4,064	913	total ≤ 0 (best 5 days +531; without them −4,366)
all names	whole span (card dates out)	+1,387.74, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−1,182.26, −34.4%, 2025-04-23, SPXW 2025-04-23 5445.0P	0.62	59,140	1,135	total ≤ 0 (best 5 days +6,267; without them −65,407)
QQQ	never-read 2026-10-14→21	+54.90, +25.1%, 2026-10-15, QQQ 2026-10-15 720.0P	−62.10, −32.5%, 2026-10-19, QQQ 2026-10-19 720.0P	4.05	12	2	100%
SPXW	never-read 2026-10-14→21	+347.74, +25.0%, 2026-10-19, SPXW 2026-10-19 7690.0C	−562.26, −36.0%, 2026-10-15, SPXW 2026-10-15 7695.0P	0.47	997	6	total ≤ 0 (best 5 days −714; without them +0)
SPY	never-read 2026-10-14→21	+30.90, +24.9%, 2026-10-21, SPY 2026-10-21 770.0C	−57.10, −34.4%, 2026-10-15, SPY 2026-10-15 767.0P	0.44	91	6	total ≤ 0 (best 5 days −78; without them +0)
all names	never-read 2026-10-14→21	+347.74, +25.0%, 2026-10-19, SPXW 2026-10-19 7690.0C	−562.26, −36.0%, 2026-10-15, SPXW 2026-10-15 7695.0P	0.61	951	6	total ≤ 0 (best 5 days −602; without them +0)
IWM	whole span (verdict)	+60.90, +25.1%, 2025-04-09, IWM 2025-04-09 176.0P	−59.10, −39.1%, 2024-08-05, IWM 2024-08-05 199.0P	0.55	1,453	632	total ≤ 0 (best 5 days +225; without them −1,658)
QQQ	whole span (verdict)	+141.89, +25.1%, 2025-04-09, QQQ 2025-04-09 423.0P	−139.10, −34.6%, 2025-04-08, QQQ 2025-04-08 440.0P	0.68	4,142	1,014	total ≤ 0 (best 5 days +616; without them −4,520)
SPXW	whole span (verdict)	+1,387.74, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−1,182.26, −34.4%, 2025-04-23, SPXW 2025-04-23 5445.0P	0.61	50,776	1,148	total ≤ 0 (best 5 days +5,592; without them −56,085)
SPY	whole span (verdict)	+128.89, +25.1%, 2025-04-09, SPY 2025-04-09 495.0C	−109.10, −34.7%, 2025-04-23, SPY 2025-04-23 542.0P	0.62	4,155	926	total ≤ 0 (best 5 days +531; without them −4,444)
all names	whole span (verdict)	+1,387.74, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−1,182.26, −34.4%, 2025-04-23, SPXW 2025-04-23 5445.0P	0.62	60,091	1,148	total ≤ 0 (best 5 days +6,267; without them −66,009)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.74 a trade) and Robinhood's charged instead (mean $0.77). The study's fees are given beside them below. Re-pricing: exact (2,943 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	whole span (card dates out)	438	**−6.98**, −3.27	−9.91, −4.55	38.1 / 33.8	[−9.29, −4.90] / [−12.27, −7.73]

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	whole span (card dates out)	814	**−4.37**, −5.03	−5.59, −6.30	44.3 / 42.3	[−5.68, −2.80] / [−6.92, −4.02]
SPXW	whole span (card dates out)	902	**−4.95**, −55.19	−4.98, −55.48	41.4 / 41.4	[−6.42, −3.35] / [−6.45, −3.38]
SPY	whole span (card dates out)	771	**−4.85**, −4.97	−6.25, −6.25	41.8 / 38.4	[−6.60, −3.28] / [−8.05, −4.65]
all names	whole span (card dates out)	2,925	**−5.07**, −20.22	−6.22, −21.19	41.8 / 39.7	[−6.13, −3.84] / [−7.32, −4.97]
QQQ	never-read 2026-10-14→21	6	**+15.52**, +31.57	+14.87, +30.29	83.3 / 83.3	n/a (< 60 sessions) / n/a (< 60 sessions)
SPXW	never-read 2026-10-14→21	6	**−9.57**, −118.93	−9.60, −119.22	33.3 / 33.3	n/a (< 60 sessions) / n/a (< 60 sessions)
SPY	never-read 2026-10-14→21	6	**−9.79**, −12.93	−10.95, −14.21	33.3 / 33.3	n/a (< 60 sessions) / n/a (< 60 sessions)
all names	never-read 2026-10-14→21	18	**−1.28**, −33.43	−1.89, −34.38	50.0 / 50.0	n/a (< 60 sessions) / n/a (< 60 sessions)
IWM	whole span (verdict)	438	**−6.98**, −3.27	−9.91, −4.55	38.1 / 33.8	[−9.29, −4.90] / [−12.27, −7.73]
QQQ	whole span (verdict)	820	**−4.22**, −4.76	−5.44, −6.03	44.6 / 42.6	[−5.88, −2.87] / [−7.06, −4.10]
SPXW	whole span (verdict)	908	**−4.98**, −55.61	−5.01, −55.90	41.3 / 41.3	[−6.55, −3.44] / [−6.58, −3.47]
SPY	whole span (verdict)	777	**−4.89**, −5.04	−6.29, −6.31	41.7 / 38.4	[−6.62, −3.34] / [−8.04, −4.75]
all names	whole span (verdict)	2,943	**−5.04**, −20.30	−6.20, −21.27	41.9 / 39.8	[−6.20, −3.90] / [−7.38, −5.02]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

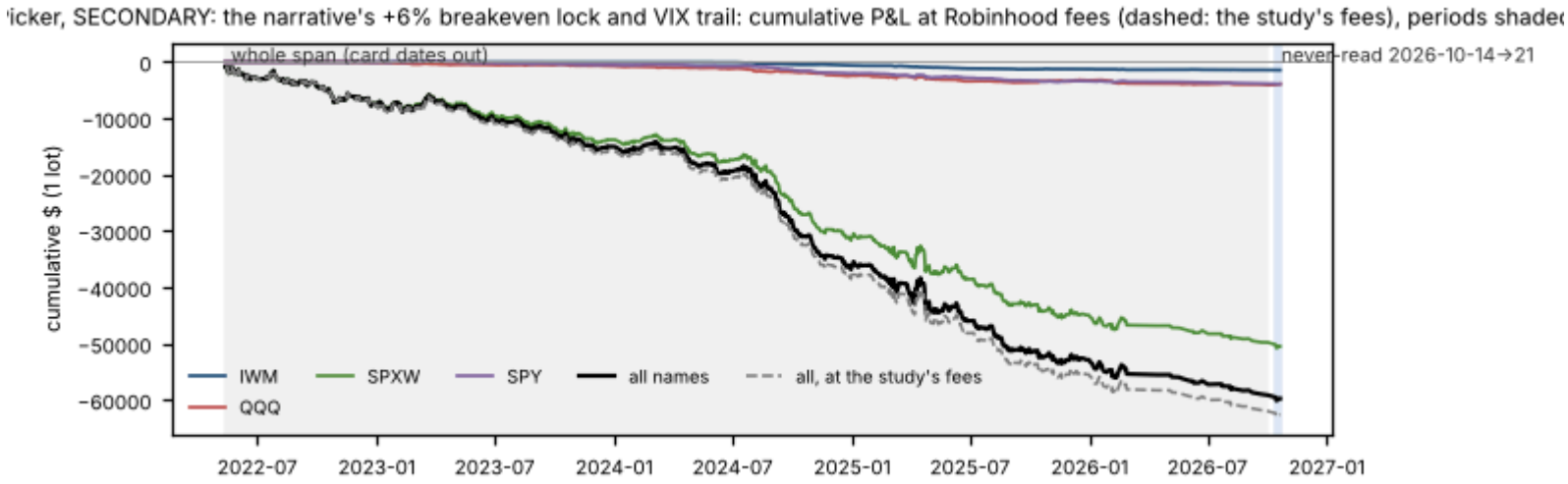
Exit reasons (share of trades):

- whole span (card dates out): stop 30.8%; lock 27.4%; target 25.0%; trail 16.9%
- never-read 2026-10-14→21: target 50.0%; stop 38.9%; lock 11.1%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	152	35.5	−7.99	−4.01	−609
2025	IWM	232	37.9	−6.81	−2.91	−674
2026	IWM	54	46.3	−4.91	−2.77	−149
2022	QQQ	30	40.0	−3.25	−1.43	−43
2023	QQQ	248	48.8	−3.31	−2.68	−664
2024	QQQ	244	41.8	−6.02	−7.03	−1,716
2025	QQQ	233	45.9	−2.90	−3.65	−851
2026	QQQ	65	36.9	−6.19	−9.69	−630
2022	SPXW	164	36.6	−3.36	−44.70	−7,331
2023	SPXW	240	46.2	−3.50	−27.01	−6,483
2024	SPXW	228	41.2	−6.27	−76.38	−17,415
2025	SPXW	213	41.3	−5.50	−64.33	−13,702
2026	SPXW	63	34.9	−8.39	−88.29	−5,562
2022	SPY	33	36.4	−1.51	−1.13	−37
2023	SPY	240	44.6	−3.28	−1.88	−452
2024	SPY	228	41.7	−5.84	−6.62	−1,510
2025	SPY	213	42.3	−5.23	−6.44	−1,371
2026	SPY	63	31.7	−8.15	−8.59	−541

year	name	trades	win %	mean %	mean $	total $
2022	all names	227	37.0	−3.08	−32.65	−7,411
2023	all names	728	46.6	−3.36	−10.44	−7,599
2024	all names	852	40.5	−6.39	−24.94	−21,250
2025	all names	891	41.9	−5.09	−18.63	−16,599
2026	all names	245	37.1	−6.98	−28.09	−6,883



Amendments, deviations and integrity checks. As 29a.

What it showed. The narrative exits change nothing.

#29c Picker, S2: “5 minutes” as a hard exit

Folder: `research_notes/Compass phase 3/picker_prereg`. **Commits:** c502122 / — / 1e10053. **Dates:** 2026-10-22.

Hypothesis. Reading the 5-minute rule as a hard exit helps.

Rule. Opening range = the 09:30–09:59 bars; 10:00–11:59: the first bar with high ≥ ORH × (1 + 2 bp) and close < ORH → PUT (the low side → CALL); buy the nearest OTM 0DTE contract at the ask one bar later; exit at 5 minutes, or the stop/target first.

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData); a target at its resting limit

Data. SPY, QQQ (846 sessions), IWM (484), SPXW (972), 2022 → 2026-10-21, the 134 GeorgeFi card dates excluded.

Periods and holdouts. The whole tested span (verdict) and the never-read line 2026-10-14 → 21 (no verdict). The 2022–2026 NBBO had been read by the engine audit for the same universe (disclosed).

Pass rule (as pre-registered). Sensitivity, no verdict.

Verdict. Labelled: SPY −4.20%, QQQ −3.64%, IWM −7.42%, SPXW −3.10%.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib’s 20-session moving-block bootstrap over the period’s calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	whole span (card dates out)	438	394 / 634	39.3	+16.3, +9.27	−17.9, −9.57	−5.6, −2.10	**−4.49**, −2.17	[−6.50, −2.52]; [−3.66, −0.83]
QQQ	whole span (card dates out)	814	698 / 1,005	40.7	+15.1, +19.23	−14.6, −18.54	−3.6, −4.10	**−2.54**, −3.18	[−3.72, −1.21]; [−5.00, −1.09]
SPXW	whole span (card dates out)	902	765 / 1,135	40.4	+13.6, +172.88	−14.4, −172.45	−3.9, −42.26	**−3.08**, −33.09	[−4.30, −1.84]; [−47.58, −17.29]
SPY	whole span (card dates out)	771	660 / 1,005	39.4	+14.0, +16.08	−13.8, −15.10	−4.0, −3.10	**−2.81**, −2.81	[−4.17, −1.52]; [−4.37, −1.17]
all names	whole span (card dates out)	2,925	898 / 1,135	40.0	+14.5, +64.71	−14.8, −63.47	−4.1, −4.10	**−3.07**, −12.16	[−4.02, −2.06]; [−16.90, −6.70]
QQQ	never-read 2026-10-14→21	6	5 / 6	66.7	+25.1, +50.40	−10.0, −19.10	+25.0, +45.90	**+13.42**, +27.23	n/a (< 60 sessions); n/a (< 60 sessions)
SPXW	never-read 2026-10-14→21	6	5 / 6	33.3	+15.0, +177.74	−10.2, −117.26	−3.9, −52.26	**−1.76**, −18.93	n/a (< 60 sessions); n/a (< 60 sessions)
SPY	never-read 2026-10-14→21	6	5 / 6	33.3	+14.8, +17.90	−9.6, −10.60	−4.5, −5.60	**−1.47**, −1.10	n/a (< 60 sessions); n/a (< 60 sessions)
all names	never-read 2026-10-14→21	18	5 / 6	44.4	+20.0, +74.11	−9.9, −54.96	−2.6, −3.60	**+3.40**, +2.40	n/a (< 60 sessions); n/a (< 60 sessions)
IWM	whole span (verdict)	438	394 / 634	39.3	+16.3, +9.27	−17.9, −9.57	−5.6, −2.10	**−4.49**, −2.17	[−6.50, −2.52]; [−3.66, −0.83]
QQQ	whole span (verdict)	820	703 / 1,018	40.9	+15.2, +19.61	−14.6, −18.55	−3.5, −4.10	**−2.42**, −2.96	[−3.76, −1.27]; [−5.12, −1.14]

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	whole span (verdict)	908	770 / 1,148	40.3	+13.7, +172.90	−14.4, −172.04	−3.9, −42.26	**−3.07**, −33.00	[−4.43, −1.78]; [−48.54, −16.94]
SPY	whole span (verdict)	777	665 / 1,018	39.4	+14.0, +16.09	−13.7, −15.07	−4.0, −3.10	**−2.80**, −2.80	[−4.16, −1.48]; [−4.41, −1.16]
all names	whole span (verdict)	2,943	903 / 1,148	40.1	+14.6, +64.78	−14.8, −63.42	−4.0, −4.10	**−3.03**, −12.07	[−4.12, −2.02]; [−17.44, −6.50]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	whole span (card dates out)	+33.90, +14.0%, 2025-04-09, IWM 2025-04-09 176.0P	−47.10, −39.6%, 2026-02-13, IWM 2026-02-13 261.0P	0.63	967	629	total ≤ 0 (best 5 days +186; without them −1,136)
QQQ	whole span (card dates out)	+83.90, +25.2%, 2025-03-11, QQQ 2025-03-11 472.0C	−100.10, −39.4%, 2026-02-20, QQQ 2026-02-20 604.0P	0.71	2,590	1,005	total ≤ 0 (best 5 days +523; without them −3,113)
SPXW	whole span (card dates out)	+907.74, +16.4%, 2025-04-09, SPXW 2025-04-09 4960.0C	−992.26, −34.1%, 2026-02-05, SPXW 2026-02-05 6810.0C	0.68	30,354	1,038	total ≤ 0 (best 5 days +4,440; without them −34,289)
SPY	whole span (card dates out)	+88.89, +17.3%, 2025-04-09, SPY 2025-04-09 495.0C	−97.10, −32.5%, 2026-02-05, SPY 2026-02-05 679.0C	0.69	2,417	914	total ≤ 0 (best 5 days +411; without them −2,577)
all names	whole span (card dates out)	+907.74, +16.4%, 2025-04-09, SPXW 2025-04-09 4960.0C	−992.26, −34.1%, 2026-02-05, SPXW 2026-02-05 6810.0C	0.68	36,007	1,038	total ≤ 0 (best 5 days +4,952; without them −40,507)
QQQ	never-read 2026-10-14→21	+54.90, +25.1%, 2026-10-15, QQQ 2026-10-15 720.0P	−34.10, −17.9%, 2026-10-19, QQQ 2026-10-19 720.0P	5.28	38	2	100%
SPXW	never-read 2026-10-14→21	+247.74, +22.3%, 2026-10-21, SPXW 2026-10-21 7725.0C	−222.26, −17.8%, 2026-10-14, SPXW 2026-10-14 7760.0C	0.76	327	6	total ≤ 0 (best 5 days −114; without them +0)
SPY	never-read 2026-10-14→21	+25.90, +20.9%, 2026-10-21, SPY 2026-10-21 770.0C	−17.10, −17.8%, 2026-10-14, SPY 2026-10-14 774.0C	0.84	28	6	total ≤ 0 (best 5 days −7; without them +0)
all names	never-read 2026-10-14→21	+247.74, +22.3%, 2026-10-21, SPXW 2026-10-21 7725.0C	−222.26, −17.8%, 2026-10-14, SPXW 2026-10-14 7760.0C	1.08	218	5	100%
IWM	whole span (verdict)	+33.90, +14.0%, 2025-04-09, IWM 2025-04-09 176.0P	−47.10, −39.6%, 2026-02-13, IWM 2026-02-13 261.0P	0.63	967	629	total ≤ 0 (best 5 days +186; without them −1,136)
QQQ	whole span (verdict)	+83.90, +25.2%, 2025-03-11, QQQ 2025-03-11 472.0C	−100.10, −39.4%, 2026-02-20, QQQ 2026-02-20 604.0P	0.73	2,590	1,018	total ≤ 0 (best 5 days +523; without them −2,950)
SPXW	whole span (verdict)	+907.74, +16.4%, 2025-04-09, SPXW 2025-04-09 4960.0C	−992.26, −34.1%, 2026-02-05, SPXW 2026-02-05 6810.0C	0.68	30,575	1,051	total ≤ 0 (best 5 days +4,440; without them −34,402)
SPY	whole span (verdict)	+88.89, +17.3%, 2025-04-09, SPY 2025-04-09 495.0C	−97.10, −32.5%, 2026-02-05, SPY 2026-02-05 679.0C	0.69	2,434	927	total ≤ 0 (best 5 days +411; without them −2,583)
all names	whole span (verdict)	+907.74, +16.4%, 2025-04-09, SPXW 2025-04-09 4960.0C	−992.26, −34.1%, 2026-02-05, SPXW 2026-02-05 6810.0C	0.68	36,172	1,051	total ≤ 0 (best 5 days +4,952; without them −40,464)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.74 a trade) and Robinhood's charged instead (mean $0.77). The study's fees are given beside them below. Re-pricing: exact (2,943 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	whole span (card dates out)	438	**−4.49**, −2.17	−7.42, −3.44	39.3 / 35.8	[−6.50, −2.52] / [−9.45, −5.42]

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	whole span (card dates out)	814	**−2.54**, −3.18	−3.76, −4.46	40.7 / 38.9	[−3.72, −1.21] / [−4.96, −2.44]
SPXW	whole span (card dates out)	902	**−3.08**, −33.09	−3.11, −33.39	40.4 / 40.4	[−4.30, −1.84] / [−4.33, −1.87]
SPY	whole span (card dates out)	771	**−2.81**, −2.81	−4.21, −4.08	39.4 / 36.4	[−4.17, −1.52] / [−5.64, −2.91]
all names	whole span (card dates out)	2,925	**−3.07**, −12.16	−4.23, −13.13	40.0 / 38.3	[−4.02, −2.06] / [−5.22, −3.18]
QQQ	never-read 2026-10-14→21	6	**+13.42**, +27.23	+12.77, +25.96	66.7 / 66.7	n/a (< 60 sessions) / n/a (< 60 sessions)
SPXW	never-read 2026-10-14→21	6	**−1.76**, −18.93	−1.78, −19.22	33.3 / 33.3	n/a (< 60 sessions) / n/a (< 60 sessions)
SPY	never-read 2026-10-14→21	6	**−1.47**, −1.10	−2.63, −2.38	33.3 / 33.3	n/a (< 60 sessions) / n/a (< 60 sessions)
all names	never-read 2026-10-14→21	18	**+3.40**, +2.40	+2.79, +1.45	44.4 / 44.4	n/a (< 60 sessions) / n/a (< 60 sessions)
IWM	whole span (verdict)	438	**−4.49**, −2.17	−7.42, −3.44	39.3 / 35.8	[−6.50, −2.52] / [−9.45, −5.42]
QQQ	whole span (verdict)	820	**−2.42**, −2.96	−3.64, −4.23	40.9 / 39.1	[−3.76, −1.27] / [−4.99, −2.47]
SPXW	whole span (verdict)	908	**−3.07**, −33.00	−3.10, −33.29	40.3 / 40.3	[−4.43, −1.78] / [−4.46, −1.80]
SPY	whole span (verdict)	777	**−2.80**, −2.80	−4.20, −4.07	39.4 / 36.4	[−4.16, −1.48] / [−5.61, −2.85]
all names	whole span (verdict)	2,943	**−3.03**, −12.07	−4.18, −13.04	40.1 / 38.3	[−4.12, −2.02] / [−5.32, −3.14]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

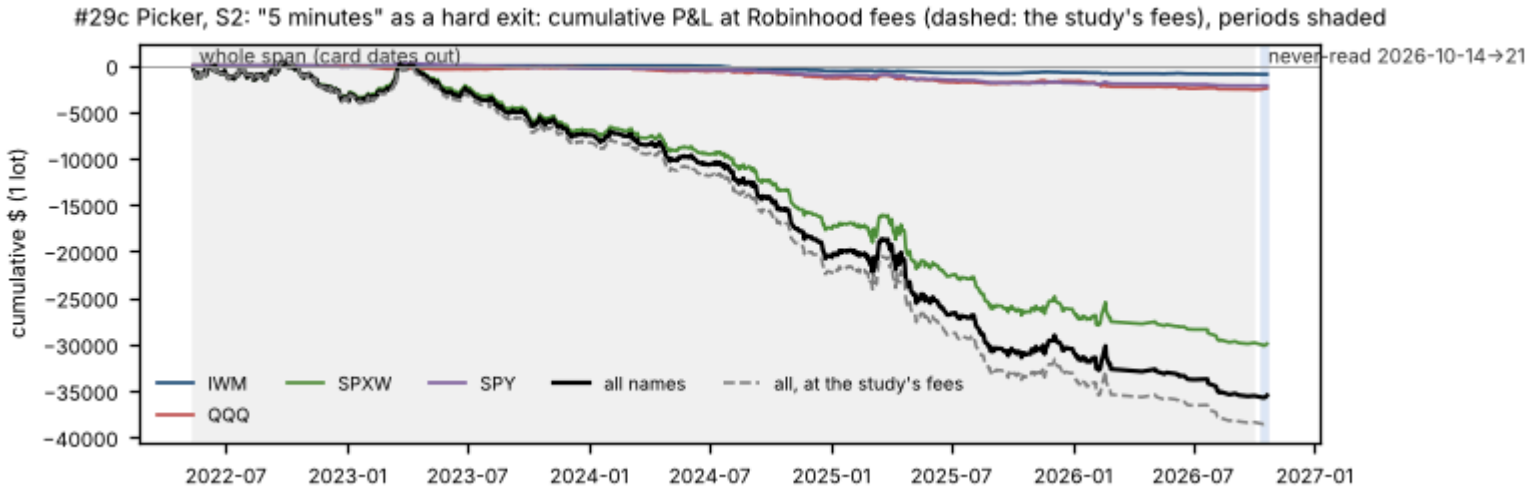
Exit reasons (share of trades):

- whole span (card dates out): time5 80.8%; target 13.2%; stop 6.0%
- never-read 2026-10-14→21: time5 77.8%; target 22.2%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	152	32.2	−6.92	−3.95	−600
2025	IWM	232	44.0	−2.52	−0.60	−140
2026	IWM	54	38.9	−6.13	−3.88	−209
2022	QQQ	30	36.7	−0.96	−0.13	−4
2023	QQQ	248	43.5	−1.14	−0.83	−205
2024	QQQ	244	38.5	−3.67	−4.65	−1,135
2025	QQQ	233	43.3	−1.52	−1.46	−339
2026	QQQ	65	32.3	−6.52	−11.44	−744
2022	SPXW	164	45.7	−1.47	−19.88	−3,261
2023	SPXW	240	42.9	−2.70	−15.51	−3,723
2024	SPXW	228	34.2	−3.87	−46.08	−10,505
2025	SPXW	213	39.0	−3.43	−40.99	−8,732
2026	SPXW	63	42.9	−4.60	−59.40	−3,742
2022	SPY	33	42.4	−4.09	−4.49	−148
2023	SPY	240	44.2	−1.80	−0.52	−124
2024	SPY	228	34.2	−3.26	−3.60	−820
2025	SPY	213	38.5	−2.84	−3.55	−755
2026	SPY	63	41.3	−4.09	−5.15	−324

year	name	trades	win %	mean %	mean $	total $
2022	all names	227	44.1	−1.79	−15.04	−3,413
2023	all names	728	43.5	−1.87	−5.57	−4,052
2024	all names	852	35.1	−4.19	−15.33	−13,060
2025	all names	891	41.3	−2.55	−11.19	−9,967
2026	all names	245	38.8	−5.32	−20.49	−5,020



Amendments, deviations and integrity checks. As 29a.

What it showed. Shorter holds lose less, still lose.

#29d Picker, S3: SPXW rides to the official close when in the money at 15:58

Folder: `research_notes/Compass phase 3/picker_prereg`. **Commits:** c502122 / — / 1e10053. **Dates:** 2026-10-22.

Hypothesis. SPXW's cash settlement adds value.

Rule. Opening range = the 09:30–09:59 bars; 10:00–11:59: the first bar with high ≥ ORH × (1 + 2 bp) and close < ORH → PUT (the low side → CALL); buy the nearest OTM 0DTE contract at the ask one bar later; SPXW only; ITM at 15:58 → settle.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData); a target at its resting limit

Data. SPY, QQQ (846 sessions), IWM (484), SPXW (972), 2022 → 2026-10-21, the 134 GeorgeFi card dates excluded.

Periods and holdouts. The whole tested span (verdict) and the never-read line 2026-10-14 → 21 (no verdict). The 2022–2026 NBBO had been read by the engine audit for the same universe (disclosed).

Pass rule (as pre-registered). Sensitivity, no verdict.

Verdict. Labelled: identical to PRIMARY (no trade was open at 15:58).

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	whole span (card dates out)	902	765 / 1,135	36.9	+21.9, +269.51	−20.7, −252.86	−7.9, −82.26	**−5.00**, −60.01	[−6.67, −3.48]; [−84.67, −36.93]
SPXW	never-read 2026-10-14→21	6	5 / 6	50.0	+24.9, +291.07	−12.5, −145.59	+10.1, +87.74	**+6.24**, +72.74	n/a (< 60 sessions); n/a (< 60 sessions)
SPXW	whole span (verdict)	908	770 / 1,148	37.0	+21.9, +269.70	−20.7, −252.30	−7.8, −77.26	**−4.93**, −59.13	[−6.70, −3.46]; [−85.43, −36.69]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	whole span (card dates out)	+1,387.74, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−2,952.26, −45.6%, 2025-04-07, SPXW 2025-04-07 4970.0P	0.62	54,455	1,135	total ≤ 0 (best 5 days +6,120; without them −60,249)
SPXW	never-read 2026-10-14→21	+347.74, +25.0%, 2026-10-19, SPXW 2026-10-19 7690.0C	−222.26, −17.8%, 2026-10-14, SPXW 2026-10-14 7760.0C	2.00	295	3	100%
SPXW	whole span (verdict)	+1,387.74, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−2,952.26, −45.6%, 2025-04-07, SPXW 2025-04-07 4970.0P	0.63	54,455	1,148	total ≤ 0 (best 5 days +6,120; without them −59,812)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.55 a trade) and Robinhood's charged instead (mean $2.26). The study's fees are given beside them below. Re-pricing: exact (908 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
SPXW	whole span (card dates out)	902	**−5.00**, −60.01	−5.03, −60.30	36.9 / 36.9	[−6.67, −3.48] / [−6.69, −3.51]
SPXW	never-read 2026-10-14→21	6	**+6.24**, +72.74	+6.21, +72.44	50.0 / 50.0	n/a (< 60 sessions) / n/a (< 60 sessions)
SPXW	whole span (verdict)	908	**−4.93**, −59.13	−4.96, −59.43	37.0 / 37.0	[−6.70, −3.46] / [−6.73, −3.49]

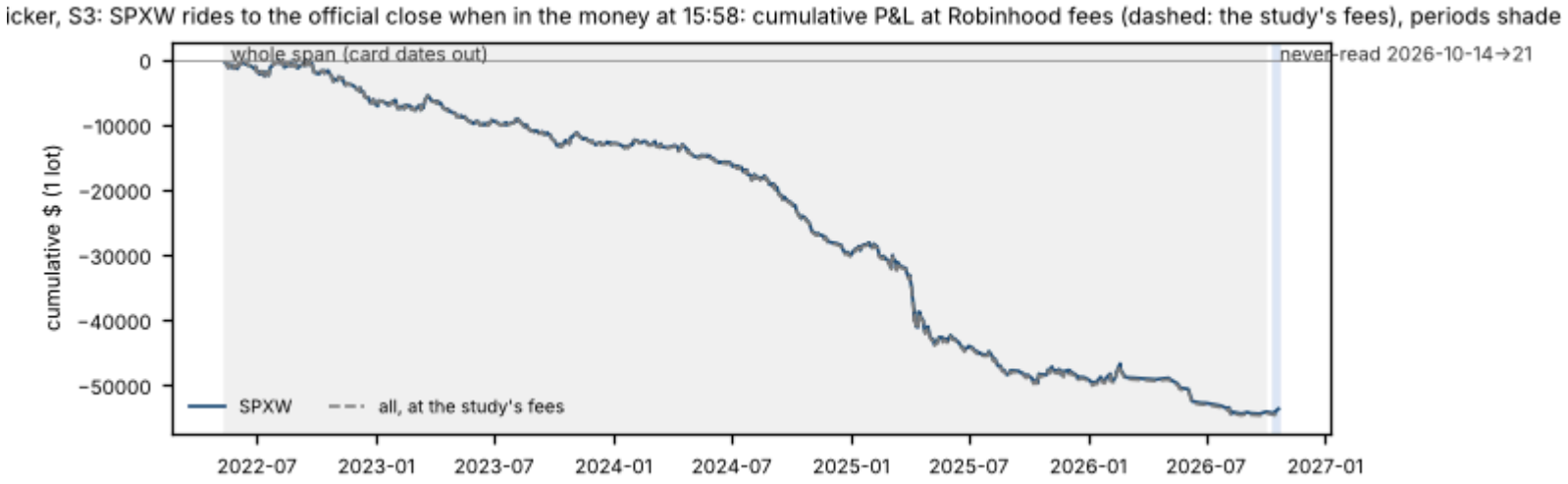
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- whole span (card dates out): time5 48.9%; target 31.5%; stop 19.6%
- never-read 2026-10-14→21: time5 50.0%; target 50.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	164	40.9	−3.26	−41.41	−6,791
2023	SPXW	240	41.7	−3.14	−25.47	−6,113
2024	SPXW	228	28.5	−7.29	−75.33	−17,175
2025	SPXW	213	36.2	−5.61	−88.18	−18,782
2026	SPXW	63	42.9	−5.23	−76.71	−4,832



Amendments, deviations and integrity checks. As 29a.

What it showed. No trade reached 15:58.

#29e Picker, IMPROVED (picked on the cards): PDH sweep → put, 10:00–12:00, +25% or 5 minutes

Folder: `research_notes/Compass phase 3/picker_prereg`. **Commits:** c502122 / — / 1e10053. **Dates:** 2026-10-22.

Hypothesis. The owner's "improved" parameters, traced to the cards, work.

Rule. SPXW/SPY/IWM: prior-day-high sweep → put, 10:00–12:00, sell at +25% or at 5 minutes.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData); a target at its resting limit

Data. SPY, QQQ (846 sessions), IWM (484), SPXW (972), 2022 → 2026-10-21, the 134 GeorgeFi card dates excluded.

Periods and holdouts. The whole tested span (verdict) and the never-read line 2026-10-14 → 21 (no verdict). The 2022–2026 NBBO had been read by the engine audit for the same universe (disclosed).

Pass rule (as pre-registered). Labelled, never pooled; the owner's rule shown for reading only.

Verdict. FAIL on every name (read only): SPXW −3.81% [−6.76, −0.95], SPY −4.77%, IWM −10.38%.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	whole span (card dates out)	106	106 / 615	35.8	+16.1, +9.00	−20.6, −11.22	−8.3, −4.10	**−7.45**, −3.97	[−11.56, −3.37]; [−7.45, −0.87]
SPXW	whole span (card dates out)	217	217 / 1,105	40.1	+16.9, +196.93	−17.9, −197.72	−6.2, −62.26	**−3.92**, −39.50	[−6.82, −1.00]; [−77.20, −2.81]
SPY	whole span (card dates out)	191	191 / 977	40.3	+17.4, +18.54	−17.6, −17.83	−6.6, −5.10	**−3.48**, −3.17	[−6.09, +0.27]; [−6.48, +1.24]
all names	whole span (card dates out)	514	275 / 1,125	39.3	+17.0, +93.58	−18.4, −91.34	−7.0, −7.10	**−4.48**, −18.67	[−7.09, −1.91]; [−36.18, −2.07]
SPXW	never-read 2026-10-14→21	1	1 / 1	100.0	+25.2, +337.74	—, —	+25.2, +337.74	**+25.20**, +337.74	n/a (< 60 sessions); n/a (< 60 sessions)
SPY	never-read 2026-10-14→21	1	1 / 1	100.0	+24.9, +35.90	—, —	+24.9, +35.90	**+24.93**, +35.90	n/a (< 60 sessions); n/a (< 60 sessions)
all names	never-read 2026-10-14→21	2	1 / 1	100.0	+25.1, +186.82	—, —	+25.1, +186.82	**+25.07**, +186.82	n/a (< 60 sessions); n/a (< 60 sessions)
IWM	whole span (verdict)	106	106 / 615	35.8	+16.1, +9.00	−20.6, −11.22	−8.3, −4.10	**−7.45**, −3.97	[−11.56, −3.37]; [−7.45, −0.87]
SPXW	whole span (verdict)	218	218 / 1,145	40.4	+17.0, +198.53	−17.9, −197.72	−6.2, −62.26	**−3.78**, −37.77	[−6.90, −1.07]; [−80.50, −2.94]
SPY	whole span (verdict)	192	192 / 1,017	40.6	+17.5, +18.76	−17.6, −17.83	−6.6, −5.10	**−3.33**, −2.97	[−6.21, +0.07]; [−6.54, +1.13]
all names	whole span (verdict)	516	276 / 1,145	39.5	+17.0, +94.49	−18.4, −91.34	−6.9, −7.10	**−4.37**, −17.87	[−7.23, −1.84]; [−37.83, −1.65]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	whole span (card dates out)	+28.90, +24.9%, 2024-07-17, IWM 2024-07-17 224.0P	−129.10, −60.3%, 2025-04-07, IWM 2025-04-07 182.0P	0.45	516	512	total ≤ 0 (best 5 days +117; without them −538)
SPXW	whole span (card dates out)	+477.74, +25.0%, 2024-08-07, SPXW 2024-08-07 5310.0P	−1,172.26, −49.7%, 2026-02-20, SPXW 2026-02-20 6870.0P	0.67	9,095	1,084	total ≤ 0 (best 5 days +2,259; without them −10,829)
SPY	whole span (card dates out)	+50.90, +22.5%, 2025-11-20, SPY 2025-11-20 667.0P	−108.10, −48.9%, 2026-02-20, SPY 2026-02-20 685.0P	0.70	642	977	total ≤ 0 (best 5 days +224; without them −830)
all names	whole span (card dates out)	+477.74, +25.0%, 2024-08-07, SPXW 2024-08-07 5310.0P	−1,172.26, −49.7%, 2026-02-20, SPXW 2026-02-20 6870.0P	0.66	10,121	1,104	total ≤ 0 (best 5 days +2,494; without them −12,090)
SPXW	never-read 2026-10-14→21	+337.74, +25.2%, 2026-10-20, SPXW 2026-10-20 7715.0P	+337.74, +25.2%, 2026-10-20, SPXW 2026-10-20 7715.0P	∞	0	0	100%
SPY	never-read 2026-10-14→21	+35.90, +24.9%, 2026-10-20, SPY 2026-10-20 769.0P	+35.90, +24.9%, 2026-10-20, SPY 2026-10-20 769.0P	∞	0	0	100%
all names	never-read 2026-10-14→21	+337.74, +25.2%, 2026-10-20, SPXW 2026-10-20 7715.0P	+35.90, +24.9%, 2026-10-20, SPY 2026-10-20 769.0P	∞	0	0	100%
IWM	whole span (verdict)	+28.90, +24.9%, 2024-07-17, IWM 2024-07-17 224.0P	−129.10, −60.3%, 2025-04-07, IWM 2025-04-07 182.0P	0.45	516	512	total ≤ 0 (best 5 days +117; without them −538)
SPXW	whole span (verdict)	+477.74, +25.0%, 2024-08-07, SPXW 2024-08-07 5310.0P	−1,172.26, −49.7%, 2026-02-20, SPXW 2026-02-20 6870.0P	0.68	9,095	1,124	total ≤ 0 (best 5 days +2,259; without them −10,492)
SPY	whole span (verdict)	+50.90, +22.5%, 2025-11-20, SPY 2025-11-20 667.0P	−108.10, −48.9%, 2026-02-20, SPY 2026-02-20 685.0P	0.72	642	1,017	total ≤ 0 (best 5 days +224; without them −794)
all names	whole span (verdict)	+477.74, +25.0%, 2024-08-07, SPXW 2024-08-07 5310.0P	−1,172.26, −49.7%, 2026-02-20, SPXW 2026-02-20 6870.0P	0.68	10,121	1,124	total ≤ 0 (best 5 days +2,494; without them −11,717)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.87 a trade) and Robinhood's charged instead (mean $1.01). The study's fees are given beside them below. Re-pricing: exact (516 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
IWM	whole span (card dates out)	106	**−7.45**, −3.97	−10.38, −5.24	35.8 / 34.9	[−11.56, −3.37] / [−14.45, −6.32]
SPXW	whole span (card dates out)	217	**−3.92**, −39.50	−3.95, −39.79	40.1 / 40.1	[−6.82, −1.00] / [−6.85, −1.03]
SPY	whole span (card dates out)	191	**−3.48**, −3.17	−4.92, −4.44	40.3 / 38.7	[−6.09, +0.27] / [−7.55, −1.17]
all names	whole span (card dates out)	514	**−4.48**, −18.67	−5.64, −19.53	39.3 / 38.5	[−7.09, −1.91] / [−8.26, −3.06]
SPXW	never-read 2026-10-14→21	1	**+25.20**, +337.74	+25.18, +337.44	100.0 / 100.0	n/a (< 60 sessions) / n/a (< 60 sessions)
SPY	never-read 2026-10-14→21	1	**+24.93**, +35.90	+24.05, +34.62	100.0 / 100.0	n/a (< 60 sessions) / n/a (< 60 sessions)
all names	never-read 2026-10-14→21	2	**+25.07**, +186.82	+24.61, +186.03	100.0 / 100.0	n/a (< 60 sessions) / n/a (< 60 sessions)
IWM	whole span (verdict)	106	**−7.45**, −3.97	−10.38, −5.24	35.8 / 34.9	[−11.56, −3.37] / [−14.45, −6.32]
SPXW	whole span (verdict)	218	**−3.78**, −37.77	−3.81, −38.06	40.4 / 40.4	[−6.90, −1.07] / [−6.93, −1.10]
SPY	whole span (verdict)	192	**−3.33**, −2.97	−4.77, −4.24	40.6 / 39.1	[−6.21, +0.07] / [−7.69, −1.37]
all names	whole span (verdict)	516	**−4.37**, −17.87	−5.52, −18.73	39.5 / 38.8	[−7.23, −1.84] / [−8.38, −2.95]

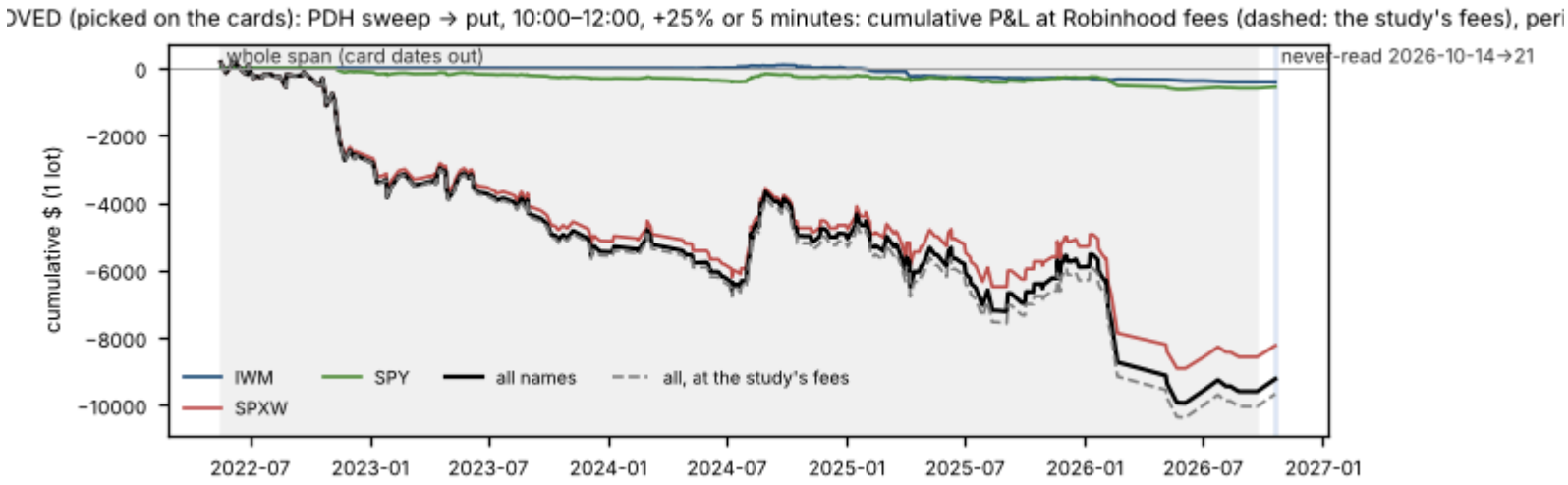
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- whole span (card dates out): time5 81.9%; target 18.1%
- never-read 2026-10-14→21: target 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	37	45.9	−2.88	−0.24	−9
2025	IWM	52	34.6	−9.24	−5.89	−306
2026	IWM	17	17.6	−11.89	−6.22	−106
2022	SPXW	33	42.4	−5.28	−74.99	−2,475
2023	SPXW	61	37.7	−5.03	−43.57	−2,658
2024	SPXW	50	38.0	−1.08	+9.34	+467
2025	SPXW	57	47.4	−2.03	−11.03	−629
2026	SPXW	17	29.4	−10.22	−172.85	−2,938
2022	SPY	7	28.6	−14.54	−15.39	−108
2023	SPY	61	37.7	−4.59	−3.46	−211
2024	SPY	50	42.0	−0.41	+1.14	+57
2025	SPY	57	47.4	−1.23	−0.61	−35
2026	SPY	17	29.4	−9.88	−16.04	−273
2022	all names	40	40.0	−6.90	−64.56	−2,582
2023	all names	122	37.7	−4.81	−23.52	−2,869
2024	all names	137	41.6	−1.32	+3.76	+515
2025	all names	166	43.4	−4.01	−5.84	−970
2026	all names	51	25.5	−10.67	−65.04	−3,317



Amendments, deviations and integrity checks. Parameters picked from the cards, whose dates are excluded.

What it showed. Negative on every name.

#29f Picker: flow gate and grade check

Folder: `research_notes/Compass phase 3/picker_prereg`. **Commits:** c502122 / — / 1e10053. **Dates:** 2026-10-22.

Hypothesis. UW sweep flow on the trade's side, and the vendor's grades, separate winners.

Rule. Gate: trade-side UW sweep ask-premium over the 5 minutes before entry > the other side. Grade: A+/A/B+ on the vendor cards.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). —

Data. Flow: one session (2026-10-21). Grades: the D1 ledger (489 cards).

Periods and holdouts. —

Pass rule (as pre-registered). Descriptive only.

Verdict. Flow gate: 0 of 5 trades gated (one session). Grades (vendor-recorded, not re-marked): A+ +8.09% [+1.81, +19.44], A +3.12%, B+ −3.23%; A+ − A +4.97 [−3.79, +15.09].

Metrics. not available — one session of flow (5 trades) and the vendor's own recorded returns (see #1, vendor-recorded).

Amendments, deviations and integrity checks. —

What it showed. No usable gate; the grades rank the vendor's own record.

#30a Compass features: the base trades (every feature is judged against them)

Folder: `research_notes/Compass features/`. **Commits:** inventory 04ce685; prereg a2e14e4 / Phase B prereg b6737c7; forward rule b101ae1 / Phase A 2a49049; Phase B e5dd8c6; forward wired 850268e. **Dates:** 2026-10-22.

Hypothesis. (The reference set.)

Rule. Every base trade, fixed exit: 0DTE sold at the 15:58 bid (SPXW held to settlement), multi-day at a fixed horizon.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData) (0DTE at 13 fixed minutes; 1–60 DTE at the EOD closing ask)

Data. The base trades: 0DTE at 13 fixed minutes (SPY/QQQ/SPXW and the singles' expiry days) and 1–7, 8–30, 31–60 DTE at the EOD close, 15 names, five strike buckets, 2012-06 → 2026-10 (by span).

Periods and holdouts. exploration / holdout — the holdout (43 months + AVGO, MU, QQQ) was OPENED: read twice in Phase A (the first on a table with duplicated rows) and once in Phase B; forward from 2026-10-22 (0 sessions).

Pass rule (as pre-registered). —

Verdict. Descriptive: −13.35% a trade (exploration), −11.03% (holdout).

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
0DTE→60 DTE, 15 names	exploration	651,232	2,596 / 3,671	27.0	+168.5, +741.48	−78.4, −330.47	−73.4, −56.10	**−11.66**, −40.85	[−14.67, −8.76]; [−57.10, −24.10]
0DTE→60 DTE, 15 names	holdout (opened)	580,234	3,577 / 3,686	28.3	+159.0, +669.32	−75.9, −306.57	−66.0, −47.10	**−9.39**, −30.17	[−12.26, −6.22]; [−43.06, −9.06]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
0DTE→60 DTE, 15 names	exploration	+73,267.74, +453.4%, 2025-03-25, SPXW 31-60 5900.0P	−27,252.26, −77.6%, 2025-04-08, SPXW 31-60 5175.0P	0.83	26,826,744	3,671	total ≤ 0 (best 5 days +4,030,356; without them −30,633,237)
0DTE→60 DTE, 15 names	holdout (opened)	+45,477.74, +143.5%, 2026-03-30, SPXW 31-60 6220.0C	−32,491.13, −100.0%, 2026-10-19, SPXW 1-7 7740.0P	0.86	17,504,031	3,686	total ≤ 0 (best 5 days +2,437,620; without them −19,941,651)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.29 a trade) and Robinhood's charged instead (mean $0.24). The study's fees are given beside them below. Re-pricing: exact (1,231,466 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
0DTE→60 DTE, 15 names	exploration	651,232	**−11.66**, −40.85	−13.35, −41.86	27.0 / 26.9	[−14.67, −8.76] / [−16.39, −10.41]
0DTE→60 DTE, 15 names	holdout (opened)	580,234	**−9.39**, −30.17	−11.03, −31.26	28.3 / 28.2	[−12.26, −6.22] / [−13.95, −7.83]

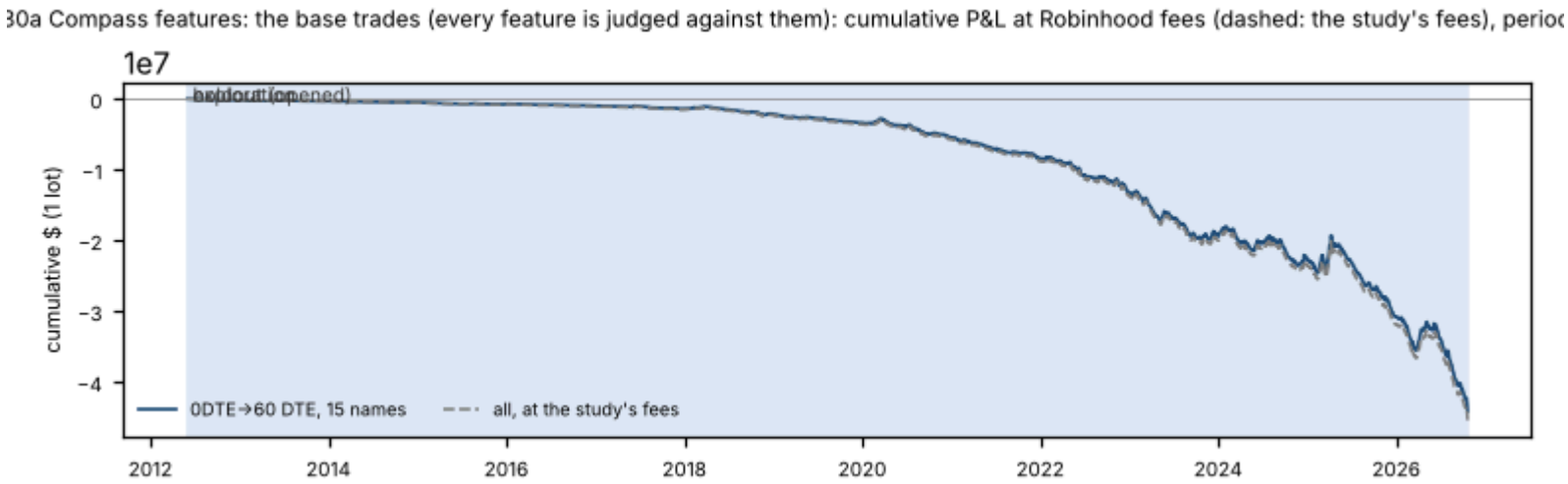
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- exploration: fixed exit 100.0%
- holdout (opened): fixed exit 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	0DTE→60 DTE, 15 names	7,483	33.4	−14.57	−19.13	−143,128
2013	0DTE→60 DTE, 15 names	13,829	33.8	−8.36	−12.16	−168,121
2014	0DTE→60 DTE, 15 names	14,053	35.3	−7.14	−10.83	−152,153
2015	0DTE→60 DTE, 15 names	14,318	32.3	−8.75	−15.94	−228,164
2016	0DTE→60 DTE, 15 names	14,750	32.3	−8.71	−17.81	−262,733
2017	0DTE→60 DTE, 15 names	26,450	26.4	−11.80	−15.92	−421,047
2018	0DTE→60 DTE, 15 names	45,006	27.0	−15.94	−17.01	−765,774
2019	0DTE→60 DTE, 15 names	41,854	25.6	−23.61	−28.81	−1,206,001
2020	0DTE→60 DTE, 15 names	52,798	26.3	−14.46	−39.88	−2,105,827
2021	0DTE→60 DTE, 15 names	50,593	24.5	−24.22	−58.86	−2,977,693
2022	0DTE→60 DTE, 15 names	90,557	27.2	−12.55	−53.30	−4,826,972
2023	0DTE→60 DTE, 15 names	194,773	28.3	−8.15	−31.31	−6,098,522
2024	0DTE→60 DTE, 15 names	210,416	27.9	−6.83	−15.66	−3,294,940
2025	0DTE→60 DTE, 15 names	223,997	27.7	−9.02	−36.18	−8,103,173
2026	0DTE→60 DTE, 15 names	230,589	26.9	−9.86	−57.91	−13,352,664



Amendments, deviations and integrity checks. The holdout join duplicated rows on the first read (777,571 rows for 580,274 trades); fixed with no feature, threshold or rule changed.

What it showed. The base trades lose 11–13% of the debit a trade.

#30b Compass features V1: veto a spread over 10% of mid (PASSED)

Folder: `research_notes/Compass features/`. **Commits:** inventory 04ce685; prereg a2e14e4 / Phase B prereg b6737c7; forward rule b101ae1 / Phase A 2a49049; Phase B e5dd8c6; forward wired 850268e. **Dates:** 2026-10-22.

Hypothesis. Skipping wide spreads makes a losing set lose less.

Rule. The base trades with (ask − bid) ÷ mid ≤ 10% at the entry quote.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData) (0DTE at 13 fixed minutes; 1–60 DTE at the EOD closing ask)

Data. The base trades: 0DTE at 13 fixed minutes (SPY/QQQ/SPXW and the singles' expiry days) and 1–7, 8–30, 31–60 DTE at the EOD close, 15 names, five strike buckets, 2012-06 → 2026-10 (by span).

Periods and holdouts. exploration / holdout — the holdout (43 months + AVGO, MU, QQQ) was OPENED: read twice in Phase A (the first on a table with duplicated rows) and once in Phase B; forward from 2026-10-22 (0 sessions).

Pass rule (as pre-registered). Holdout Δ (with − without, % of debit) > 0 with its 20-session block 95% interval clear of zero.

Verdict. PASS: holdout Δ +2.61 [+1.76, +3.46] (−8.43 with vs −11.03 without).

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
0DTE→60 DTE, 15 names	exploration	538,468	2,594 / 3,671	28.1	+163.5, +774.06	−77.5, −355.59	−69.2, −61.04	**−9.72**, −37.88	[−12.93, −6.46]; [−56.55, −18.77]
0DTE→60 DTE, 15 names	holdout (opened)	474,526	3,572 / 3,686	29.7	+153.4, +673.84	−74.9, −321.38	−61.4, −49.04	**−7.14**, −26.08	[−9.97, −3.99]; [−39.67, −4.48]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
0DTE→60 DTE, 15 names	exploration	+73,267.74, +453.4%, 2025-03-25, SPXW 31-60 5900.0P	−27,252.26, −77.6%, 2025-04-08, SPXW 31-60 5175.0P	0.85	20,616,811	3,671	total ≤ 0 (best 5 days +4,132,210; without them −24,528,078)
0DTE→60 DTE, 15 names	holdout (opened)	+45,477.74, +143.5%, 2026-03-30, SPXW 31-60 6220.0C	−29,000.04, −100.0%, 2026-10-19, MU 1-7 1000.0C	0.88	12,374,254	3,686	total ≤ 0 (best 5 days +2,369,409; without them −14,743,663)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.31 a trade) and Robinhood's charged instead (mean $0.27). The study's fees are given beside them below. Re-pricing: exact (1,012,994 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
0DTE→60 DTE, 15 names	exploration	538,468	**−9.72**, −37.88	−10.89, −38.87	28.1 / 28.0	[−12.93, −6.46] / [−14.09, −7.65]
0DTE→60 DTE, 15 names	holdout (opened)	474,526	**−7.14**, −26.08	−8.43, −27.16	29.7 / 29.5	[−9.97, −3.99] / [−11.25, −5.25]

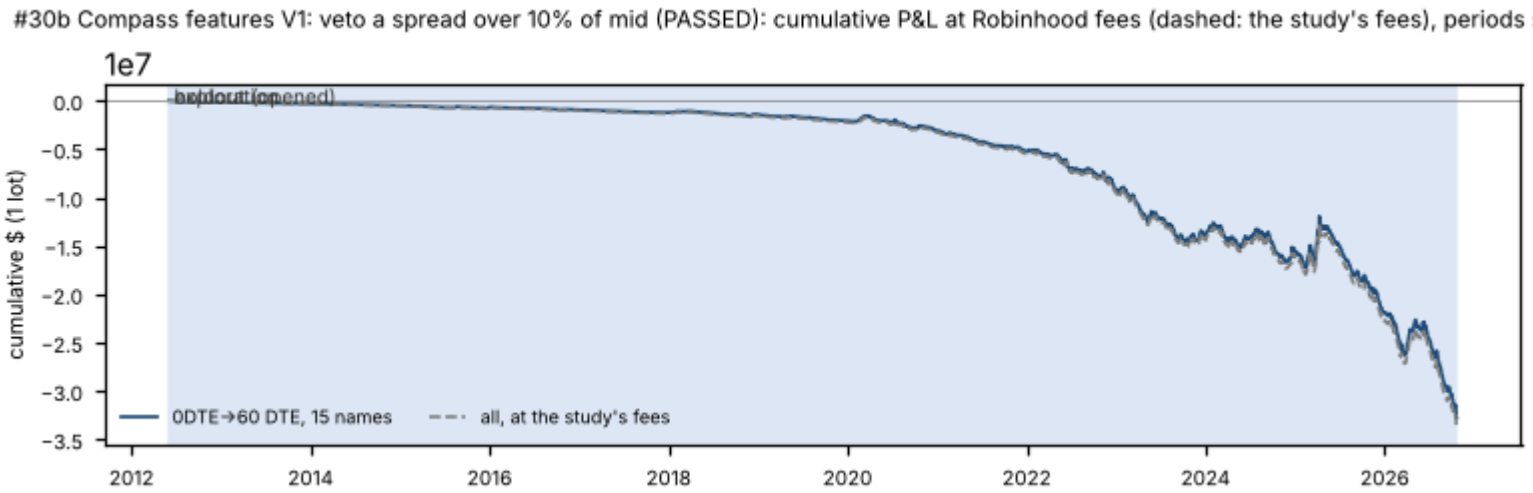
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- exploration: fixed exit 100.0%
- holdout (opened): fixed exit 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	0DTE→60 DTE, 15 names	6,826	33.4	−15.13	−20.70	−141,305
2013	0DTE→60 DTE, 15 names	12,329	33.9	−9.51	−13.44	−165,709
2014	0DTE→60 DTE, 15 names	12,822	36.1	−6.59	−11.18	−143,386
2015	0DTE→60 DTE, 15 names	13,158	33.2	−6.84	−15.85	−208,518
2016	0DTE→60 DTE, 15 names	14,153	32.4	−9.19	−18.67	−264,175
2017	0DTE→60 DTE, 15 names	15,517	30.3	−11.23	−19.48	−302,248
2018	0DTE→60 DTE, 15 names	23,379	32.6	−6.95	−8.56	−200,046
2019	0DTE→60 DTE, 15 names	25,570	31.3	−14.22	−25.13	−642,470
2020	0DTE→60 DTE, 15 names	32,249	29.9	−7.61	−31.64	−1,020,510
2021	0DTE→60 DTE, 15 names	35,164	27.0	−21.52	−60.05	−2,111,715
2022	0DTE→60 DTE, 15 names	76,404	28.1	−11.42	−51.57	−3,940,490
2023	0DTE→60 DTE, 15 names	169,119	29.2	−6.98	−28.25	−4,778,061
2024	0DTE→60 DTE, 15 names	179,301	28.7	−5.20	−9.23	−1,654,092
2025	0DTE→60 DTE, 15 names	201,814	28.1	−7.93	−31.15	−6,286,096
2026	0DTE→60 DTE, 15 names	195,189	27.5	−9.28	−55.90	−10,911,300



Amendments, deviations and integrity checks. As 30a.

What it showed. A cost filter: the with-set still loses 8.4% a trade.

#30c Compass features X3: sell a same-day long option at the 14:00 bid (PASSED)

Folder: `research_notes/Compass features/`. **Commits:** inventory 04ce685; prereg a2e14e4 / Phase B prereg b6737c7; forward rule b101ae1 / Phase A 2a49049; Phase B e5dd8c6; forward wired 850268e. **Dates:** 2026-10-22.

Hypothesis. Selling 0DTE longs at 14:00 beats holding to 15:58/settlement.

Rule. 0DTE base trades entered before 14:00, sold at the 14:00 bid.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData) (0DTE at 13 fixed minutes; 1–60 DTE at the EOD closing ask)

Data. The base trades: 0DTE at 13 fixed minutes (SPY/QQQ/SPXW and the singles' expiry days) and 1–7, 8–30, 31–60 DTE at the EOD close, 15 names, five strike buckets, 2012-06 → 2026-10 (by span).

Periods and holdouts. exploration / holdout — the holdout (43 months + AVGO, MU, QQQ) was OPENED: read twice in Phase A (the first on a table with duplicated rows) and once in Phase B; forward from 2026-10-22 (0 sessions).

Pass rule (as pre-registered). As 30b.

Verdict. PASS: holdout Δ +7.78 [+4.14, +10.95] (−9.51 with vs −17.29 without).

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
0DTE→60 DTE, 15 names	exploration	291,090	964 / 2,388	28.9	+109.1, +374.81	−60.5, −184.85	−42.6, −25.10	**−11.48**, −23.17	[−13.72, −9.24]; [−32.52, −11.51]
0DTE→60 DTE, 15 names	holdout (opened)	233,454	1,290 / 2,428	29.6	+121.7, +266.96	−61.2, −127.43	−42.2, −21.10	**−6.97**, −10.58	[−11.22, −4.91]; [−18.52, −4.17]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
0DTE→60 DTE, 15 names	exploration	+36,117.74, +538.3%, 2025-04-09, SPXW 0DTE 4940.0C	−9,452.26, −96.9%, 2025-04-09, SPXW 0DTE 5040.0P	0.82	6,745,277	2,388	total ≤ 0 (best 5 days +1,951,222; without them −8,695,714)
0DTE→60 DTE, 15 names	holdout (opened)	+19,167.74, +891.5%, 2025-11-20, SPXW 0DTE 6780.0P	−4,252.26, −93.3%, 2025-11-21, SPXW 0DTE 6580.0P	0.88	2,806,725	2,418	total ≤ 0 (best 5 days +1,179,475; without them −3,649,568)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.58 a trade) and Robinhood's charged instead (mean $0.49). The study's fees are given beside them below. Re-pricing: exact (524,544 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
0DTE→60 DTE, 15 names	exploration	291,090	**−11.48**, −23.17	−13.88, −24.21	28.9 / 28.3	[−13.72, −9.24] / [−16.20, −11.58]
0DTE→60 DTE, 15 names	holdout (opened)	233,454	**−6.97**, −10.58	−9.51, −11.72	29.6 / 29.0	[−11.22, −4.91] / [−13.87, −7.42]

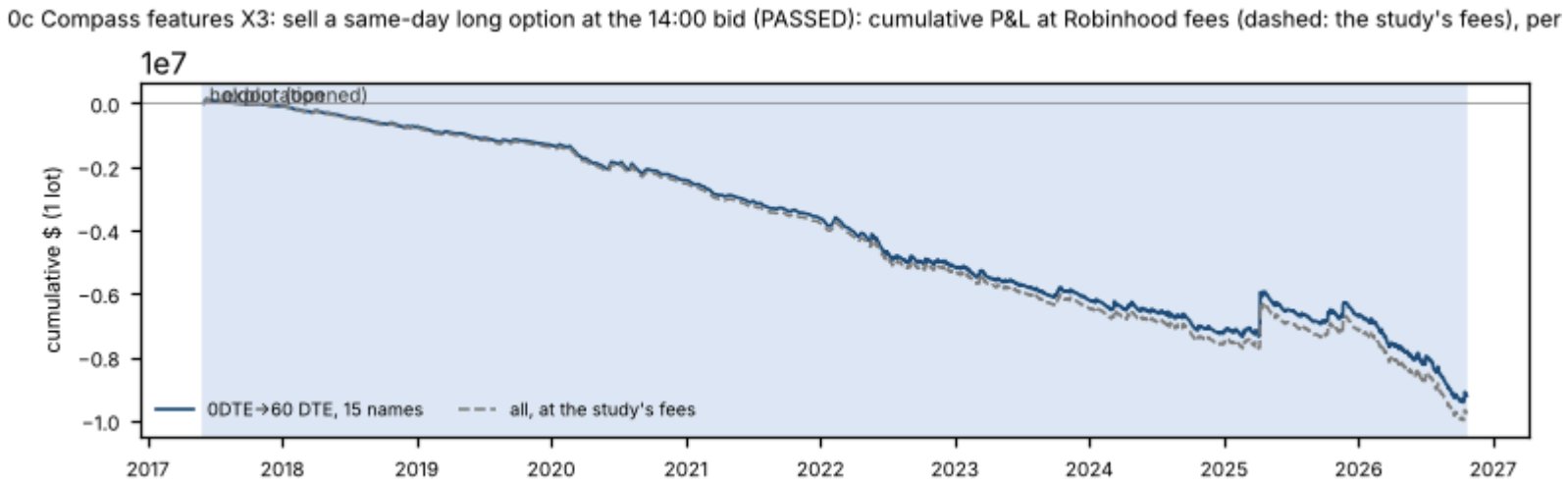
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- exploration: sold at the 14:00 bid 100.0%
- holdout (opened): sold at the 14:00 bid 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2017	0DTE→60 DTE, 15 names	9,741	21.3	−6.70	−9.41	−91,695
2018	0DTE→60 DTE, 15 names	23,860	25.2	−19.64	−27.75	−662,143
2019	0DTE→60 DTE, 15 names	21,472	23.8	−19.62	−25.50	−547,548
2020	0DTE→60 DTE, 15 names	29,501	27.2	−13.55	−37.94	−1,119,259
2021	0DTE→60 DTE, 15 names	27,934	26.5	−15.96	−43.34	−1,210,556
2022	0DTE→60 DTE, 15 names	47,279	30.3	−9.69	−32.09	−1,517,213
2023	0DTE→60 DTE, 15 names	72,946	31.4	−6.40	−14.26	−1,040,264
2024	0DTE→60 DTE, 15 names	84,765	29.9	−8.98	−12.43	−1,053,383
2025	0DTE→60 DTE, 15 names	93,284	31.3	−4.32	+5.61	+523,027
2026	0DTE→60 DTE, 15 names	113,762	28.9	−9.47	−21.94	−2,495,551



Amendments, deviations and integrity checks. As 30a.

What it showed. Selling at 14:00 halves the loss; the with-set still loses 9.5% a trade.

#30d Compass features BV1b: the V1 veto at 3% (PASSED, Phase B)

Folder: `research_notes/Compass features/`. **Commits:** inventory 04ce685; prereg a2e14e4 / Phase B prereg b6737c7; forward rule b101ae1 / Phase A 2a49049; Phase B e5dd8c6; forward wired 850268e. **Dates:** 2026-10-22.

Hypothesis. A stricter spread veto.

Rule. The base trades with spread ≤ 3% of mid.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData) (0DTE at 13 fixed minutes; 1–60 DTE at the EOD closing ask)

Data. The base trades: 0DTE at 13 fixed minutes (SPY/QQQ/SPXW and the singles' expiry days) and 1–7, 8–30, 31–60 DTE at the EOD close, 15 names, five strike buckets, 2012-06 → 2026-10 (by span).

Periods and holdouts. exploration / holdout — the holdout (43 months + AVGO, MU, QQQ) was OPENED: read twice in Phase A (the first on a table with duplicated rows) and once in Phase B; forward from 2026-10-22 (0 sessions).

Pass rule (as pre-registered). As 30b (Phase B, read once).

Verdict. PASS: holdout Δ +4.30 [+2.66, +5.85] (−6.73 with vs −11.03) — the holdout was not virgin for it.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
0DTE→60 DTE, 15 names	exploration	265,416	2,576 / 3,671	30.9	+140.3, +922.74	−73.4, −478.52	−55.0, −81.10	**−7.31**, −45.05	[−10.79, −3.54]; [−74.18, −13.33]
0DTE→60 DTE, 15 names	holdout (opened)	251,808	3,437 / 3,686	32.0	+135.1, +690.58	−72.3, −362.73	−52.4, −64.10	**−5.93**, −25.68	[−8.57, −3.08]; [−39.04, −8.60]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
0DTE→60 DTE, 15 names	exploration	+73,267.74, +453.4%, 2025-03-25, SPXW 31-60 5900.0P	−27,252.26, −77.6%, 2025-04-08, SPXW 31-60 5175.0P	0.86	12,142,678	3,671	total ≤ 0 (best 5 days +3,638,743; without them −15,595,427)
0DTE→60 DTE, 15 names	holdout (opened)	+45,477.74, +143.5%, 2026-03-30, SPXW 31-60 6220.0C	−26,112.26, −90.4%, 2026-03-30, SPXW 31-60 6510.0P	0.90	6,467,637	3,686	total ≤ 0 (best 5 days +1,736,209; without them −8,203,846)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.33 a trade) and Robinhood's charged instead (mean $0.32). The study's fees are given beside them below. Re-pricing: exact (517,224 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
0DTE→60 DTE, 15 names	exploration	265,416	**−7.31**, −45.05	−7.95, −46.00	30.9 / 30.8	[−10.79, −3.54] / [−11.44, −4.19]
0DTE→60 DTE, 15 names	holdout (opened)	251,808	**−5.93**, −25.68	−6.73, −26.77	32.0 / 31.9	[−8.57, −3.08] / [−9.38, −3.86]

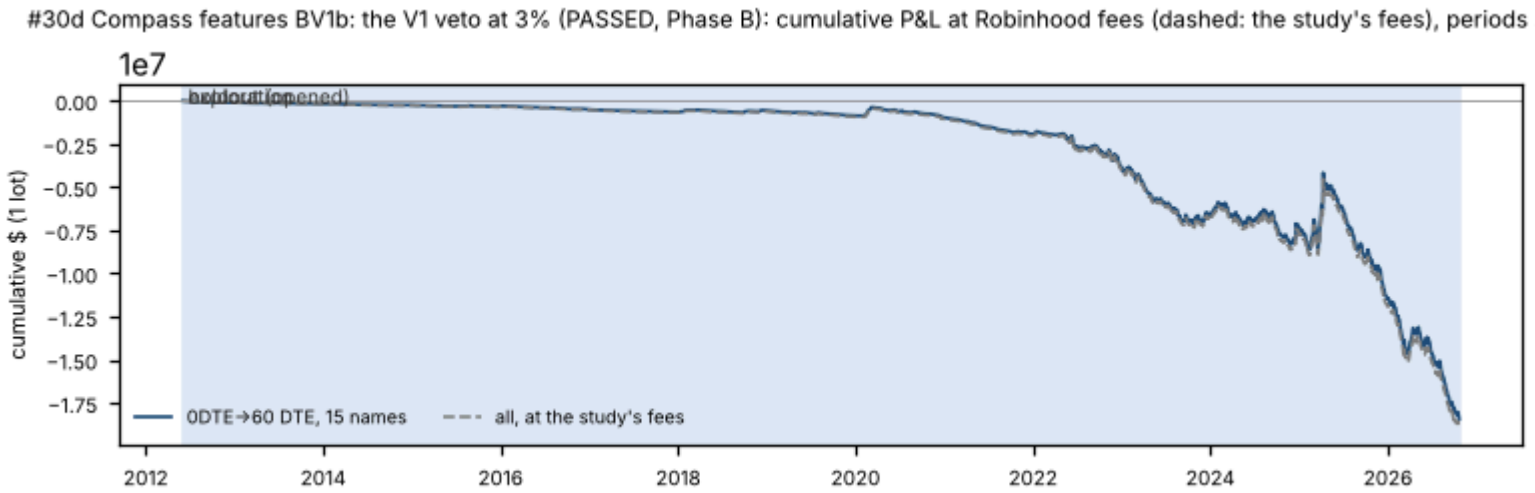
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- exploration: fixed exit 100.0%
- holdout (opened): fixed exit 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	0DTE→60 DTE, 15 names	3,539	34.8	−14.38	−27.47	−97,221
2013	0DTE→60 DTE, 15 names	5,476	35.5	−9.22	−15.17	−83,074
2014	0DTE→60 DTE, 15 names	5,572	37.3	−5.08	−10.09	−56,242
2015	0DTE→60 DTE, 15 names	4,958	35.4	−4.12	−15.29	−75,801
2016	0DTE→60 DTE, 15 names	8,797	33.3	−10.26	−21.22	−186,698
2017	0DTE→60 DTE, 15 names	5,822	31.5	−13.12	−25.73	−149,817
2018	0DTE→60 DTE, 15 names	8,205	38.2	+4.31	+9.81	+80,509
2019	0DTE→60 DTE, 15 names	11,814	36.3	−9.23	−24.84	−293,426
2020	0DTE→60 DTE, 15 names	12,459	35.3	+2.21	−8.66	−107,908
2021	0DTE→60 DTE, 15 names	14,837	31.8	−14.67	−64.48	−956,640
2022	0DTE→60 DTE, 15 names	36,317	32.2	−6.88	−53.37	−1,938,310
2023	0DTE→60 DTE, 15 names	85,208	32.4	−6.14	−33.26	−2,834,113
2024	0DTE→60 DTE, 15 names	93,592	31.4	−4.08	−7.83	−732,520
2025	0DTE→60 DTE, 15 names	120,615	30.2	−6.40	−33.05	−3,985,784
2026	0DTE→60 DTE, 15 names	100,013	29.3	−9.26	−70.06	−7,007,275



Amendments, deviations and integrity checks. 33 variants tried for 3 candidates.

What it showed. Still −6.7% a trade.

#30e Compass features BX3e: the 14:00 exit on the physically settled names only (PASSED, Phase B)

Folder: `research_notes/Compass features/`. **Commits:** inventory 04ce685; prereg a2e14e4 / Phase B prereg b6737c7; forward rule b101ae1 / Phase A 2a49049; Phase B e5dd8c6; forward wired 850268e. **Dates:** 2026-10-22.

Hypothesis. X3 without SPXW.

Rule. X3 on SPY/QQQ/singles (physical settlement).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData) (0DTE at 13 fixed minutes; 1–60 DTE at the EOD closing ask)

Data. The base trades: 0DTE at 13 fixed minutes (SPY/QQQ/SPXW and the singles' expiry days) and 1–7, 8–30, 31–60 DTE at the EOD close, 15 names, five strike buckets, 2012-06 → 2026-10 (by span).

Periods and holdouts. exploration / holdout — the holdout (43 months + AVGO, MU, QQQ) was OPENED: read twice in Phase A (the first on a table with duplicated rows) and once in Phase B; forward from 2026-10-22 (0 sessions).

Pass rule (as pre-registered). As 30d.

Verdict. PASS: holdout Δ +7.67 [+4.23, +10.69] (−11.03 with vs −18.71).

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
0DTE→60 DTE, 15 names	exploration	225,023	897 / 2,388	28.0	+110.3, +202.19	−60.9, −107.45	−44.3, −19.10	**−12.88**, −20.65	[−15.20, −10.62]; [−27.07, −15.33]
0DTE→60 DTE, 15 names	holdout (opened)	203,289	1,257 / 2,428	29.1	+121.7, +149.87	−61.4, −79.55	−43.3, −19.10	**−8.12**, −12.83	[−12.22, −6.21]; [−18.15, −9.08]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
0DTE→60 DTE, 15 names	exploration	+10,634.57, +181.8%, 2022-02-04, AMZN 0DTE 3045.0C	−6,108.10, −97.0%, 2020-09-04, AMZN 0DTE 3150.0P	0.73	4,646,111	2,388	total ≤ 0 (best 5 days +870,133; without them −5,516,244)
0DTE→60 DTE, 15 names	holdout (opened)	+7,779.71, +421.7%, 2026-07-02, MU 0DTE 1055.0P	−3,457.10, −99.3%, 2021-12-17, AMZN 0DTE 3330.0P	0.77	2,772,055	2,418	total ≤ 0 (best 5 days +485,048; without them −3,093,089)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.36 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (428,312 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
0DTE→60 DTE, 15 names	exploration	225,023	**−12.88**, −20.65	−15.96, −21.91	28.0 / 27.3	[−15.20, −10.62] / [−18.40, −13.67]
0DTE→60 DTE, 15 names	holdout (opened)	203,289	**−8.12**, −12.83	−11.03, −14.09	29.1 / 28.4	[−12.22, −6.21] / [−15.23, −9.12]

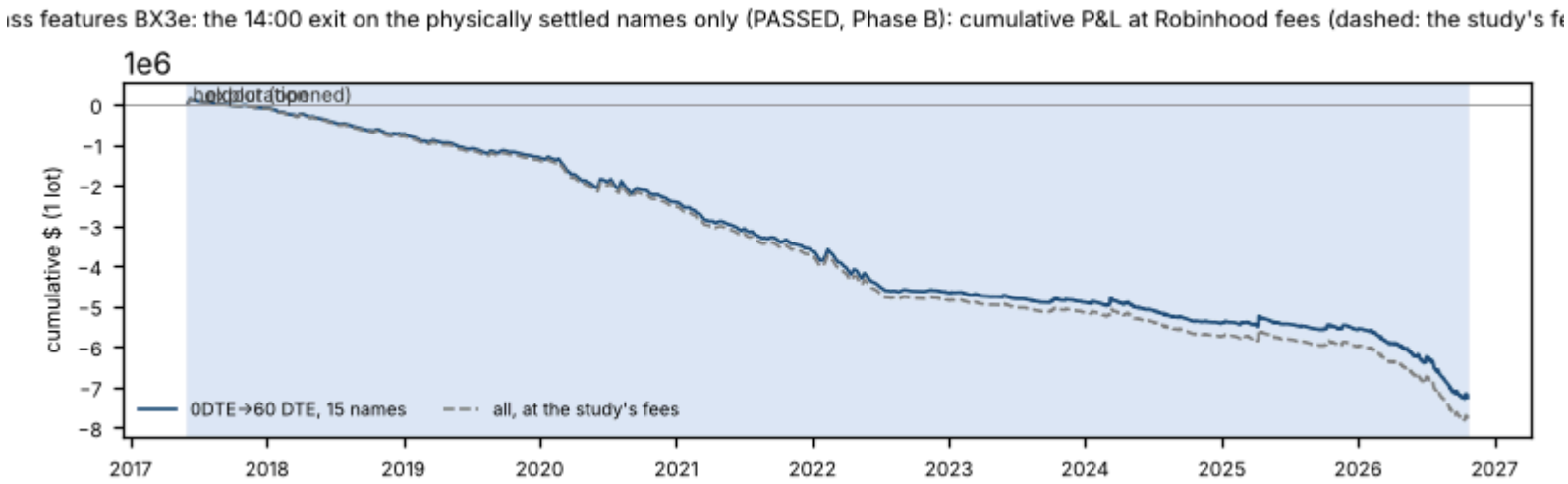
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- exploration: sold at the 14:00 bid 100.0%
- holdout (opened): sold at the 14:00 bid 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2017	0DTE→60 DTE, 15 names	9,741	21.3	−6.70	−9.41	−91,695
2018	0DTE→60 DTE, 15 names	23,860	25.2	−19.64	−27.75	−662,143
2019	0DTE→60 DTE, 15 names	21,472	23.8	−19.62	−25.50	−547,548
2020	0DTE→60 DTE, 15 names	29,501	27.2	−13.55	−37.94	−1,119,259
2021	0DTE→60 DTE, 15 names	27,934	26.5	−15.96	−43.34	−1,210,556
2022	0DTE→60 DTE, 15 names	33,013	28.8	−12.57	−31.38	−1,035,845
2023	0DTE→60 DTE, 15 names	51,821	30.8	−6.42	−4.53	−234,964
2024	0DTE→60 DTE, 15 names	63,319	29.1	−10.42	−8.25	−522,496
2025	0DTE→60 DTE, 15 names	71,666	30.9	−5.03	−2.24	−160,403
2026	0DTE→60 DTE, 15 names	95,985	28.6	−10.21	−17.39	−1,669,244



Amendments, deviations and integrity checks. As 30d.

What it showed. Still −11% a trade.

#30f Compass features VN1 (venue SPY vs SPXW) and BC3 (V1 + X3 + D6)

Folder: `research_notes/Compass features/`. **Commits:** inventory 04ce685; prereg a2e14e4 / Phase B prereg b6737c7; forward rule b101ae1 / Phase A 2a49049; Phase B e5dd8c6; forward wired 850268e. **Dates:** 2026-10-22.

Hypothesis. The cheaper venue, and the stacked model, lose less.

Rule. VN1: keep the SPXW/SPY venue with the lower ((ask − mid) + fee ÷ 100) ÷ mid. BC3: V1 veto + X3 exit + D6 direction.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData) (0DTE at 13 fixed minutes; 1–60 DTE at the EOD closing ask)

Data. The base trades: 0DTE at 13 fixed minutes (SPY/QQQ/SPXW and the singles' expiry days) and 1–7, 8–30, 31–60 DTE at the EOD close, 15 names, five strike buckets, 2012-06 → 2026-10 (by span).

Periods and holdouts. exploration / holdout — the holdout (43 months + AVGO, MU, QQQ) was OPENED: read twice in Phase A (the first on a table with duplicated rows) and once in Phase B; forward from 2026-10-22 (0 sessions).

Pass rule (as pre-registered). As 30b.

Verdict. PASS: VN1 holdout Δ +1.18 [+0.50, +2.20] (failed on the duplicated table); BC3 +6.77 [+4.82, +8.19] (−4.26 with vs −11.03).

Metrics. not available — VN1 pairs venues and BC3 needs the D6 direction feature; both are computed inside the study's screen code, and their per-trade arms are not on disk.

Amendments, deviations and integrity checks. VN1 failed on the first (duplicated) holdout read and passes on the corrected one.

What it showed. Cost and timing filters; no direction feature passed.

#31 Compass direction: post-earnings drift in 31–60 DTE options (rerun)

Folder: `research_notes/Compass direction/`. **Commits:** 38b11bf / ledger code 043d723 / 94fb69d. **Dates:** 2026-10-22.

Hypothesis. After a report, options on the side of the earnings-day move drift in its direction (PEAD).

Rule. At the close after the report: the 31–60 DTE expiry nearest 45 days; buy five strike buckets on the picked side (the sign of the report-day return) at the closing ask; sell at the closing bid 10 full sessions later ($0 if no bid).

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; the closing NBBO (EOD).

Data. Exploration set X: 1,601 names, 3,129 events (12,114 picked-side trades) 2019-04 → 2026-09.

Periods and holdouts. Exploration screen only; the 1,600 holdout names (2019-01 → 2022-12) UNOPENED.

Pass rule (as pre-registered). Screen: Δ (picked − both sides) > 0 with its block-20 interval clear of zero; then the holdout once.

Verdict. FAIL: Δ −0.90 pp [−3.53, +1.71]; the holdout is not fetched and not read.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
single names	exploration (holdout unopened)	12,114	893 / 1,920	26.2	+79.9, +471.35	−60.2, −312.80	−46.4, −85.10	**−23.52**, −107.74	[−27.45, −19.03]; [−135.54, −77.96]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
single names	exploration (holdout unopened)	+18,029.34, +195.5%, 2023-10-30, CMG 2023-12-15 1880.0C	−13,865.21, −72.0%, 2026-06-26, MU 2026-08-07 1040.0C	0.53	1,312,036	1,840	total ≤ 0 (best 5 days +160,683; without them −1,465,799)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.34 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (12,114 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
single names	exploration (holdout unopened)	12,114	**−23.52**, −107.74	−24.29, −108.97	26.2 / 26.1	[−27.45, −19.03] / [−28.19, −19.82]

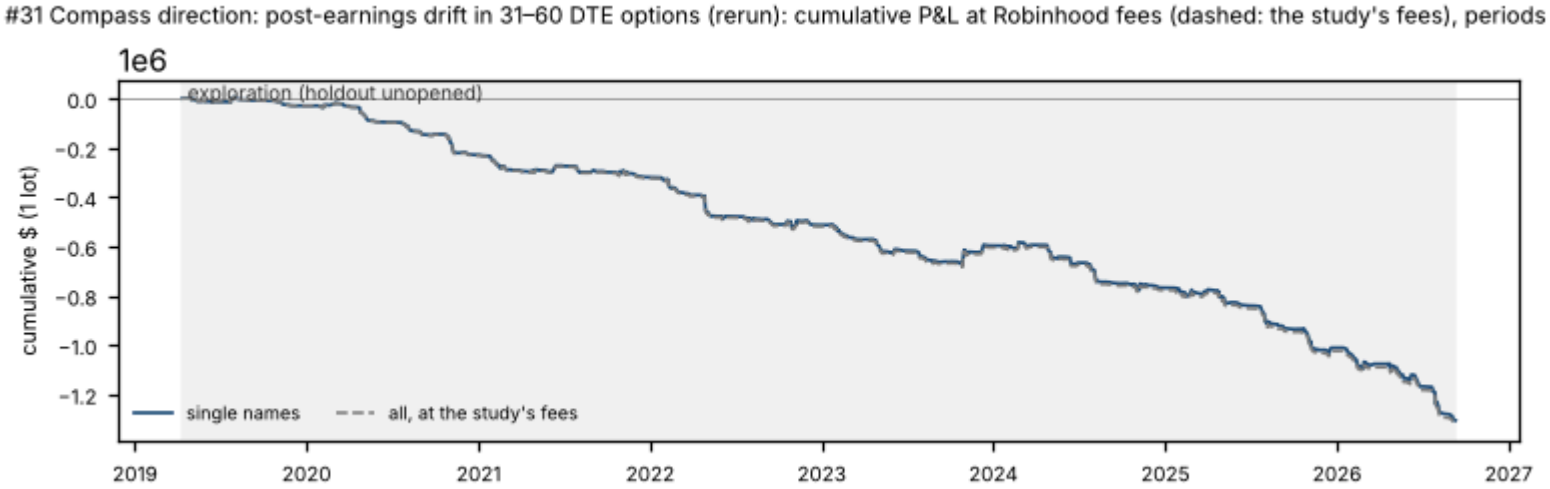
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- exploration (holdout unopened): closing bid 10 sessions later 94.5%; no bid at exit ($0) 5.5%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2019	single names	702	29.8	−9.07	−40.75	−28,604
2020	single names	1,187	19.5	−27.14	−168.04	−199,461
2021	single names	1,485	28.7	−19.96	−60.81	−90,303
2022	single names	1,549	28.3	−18.58	−125.40	−194,237
2023	single names	1,520	25.2	−24.32	−54.94	−83,511
2024	single names	1,604	28.3	−23.90	−106.55	−170,913
2025	single names	2,170	27.1	−23.81	−112.42	−243,953
2026	single names	1,897	23.0	−32.14	−155.05	−294,133



Amendments, deviations and integrity checks. Rerun for this report at 94fb69d from the stored files: screen.json equal to the committed one to the last float digit (summation order); the trade table expl_trades.parquet rewritten in the scratchpad.

What it showed. The picked side loses 24% of the debit a trade, no better than the other side.

#32 Compass H1 / edge H1: overnight reversal of the late-day move (SPY, QQQ stock)

Folder: `research_notes/Compass H1/`. **Commits:** 38aca66 / — / 4a02144. **Dates:** 2026-10-23.

Hypothesis. The 15:00 → close move reverses overnight.

Rule. Short 100 shares at the official close if the close > the 15:00 price, long if below; exit at the next 09:30 open.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.01 a share each way ($2.00 a trade at 1×); 2× shown in the study.

Data. SPY 2017-01-03 → 2025-10-10, QQQ 2012-06-01 → 2025-10-10.

Periods and holdouts. Single test; no holdout.

Pass rule (as pre-registered). In each name: mean $ a trade after costs > 0 and the 20-session block interval clear of zero (both names).

Verdict. FAIL: SPY +$11.09 [−4.42, +28.72], QQQ +$9.42 [−0.06, +18.61].

Metrics at Robinhood fees (return = % of notional ($ = per 100 sh); win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	single test	3,299	3,299 / 3,420	51.6	+0.534, +132.20	−0.491, −118.32	+0.020, +2.67	**+0.038**, +10.92	[+0.005, +0.074]; [+1.33, +20.38]
SPY	single test	2,177	2,177 / 2,266	51.2	+0.479, +181.19	−0.431, −165.10	+0.014, +5.37	**+0.035**, +12.26	[−0.009, +0.088]; [−3.39, +28.97]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	single test	+2,401.93, +5.4%, 2024-08-02, 100 QQQ short	−1,448.25, −3.4%, 2025-04-07, 100 QQQ short	1.19	8,286	585	28%
SPY	single test	+3,028.28, +11.2%, 2020-03-13, 100 SPY short	−1,749.48, −3.5%, 2025-04-07, 100 SPY short	1.15	10,312	410	42%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.00 a trade) and Robinhood's charged instead (mean $0.63). The study's fees are given beside them below. Re-pricing: exact (5,476 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	single test	3,299	**+0.038**, +10.92	+0.027, +9.42	51.6 / 50.5	[+0.005, +0.074] / [−0.006, +0.064]
SPY	single test	2,177	**+0.035**, +12.26	+0.032, +11.09	51.2 / 50.8	[−0.009, +0.088] / [−0.013, +0.084]

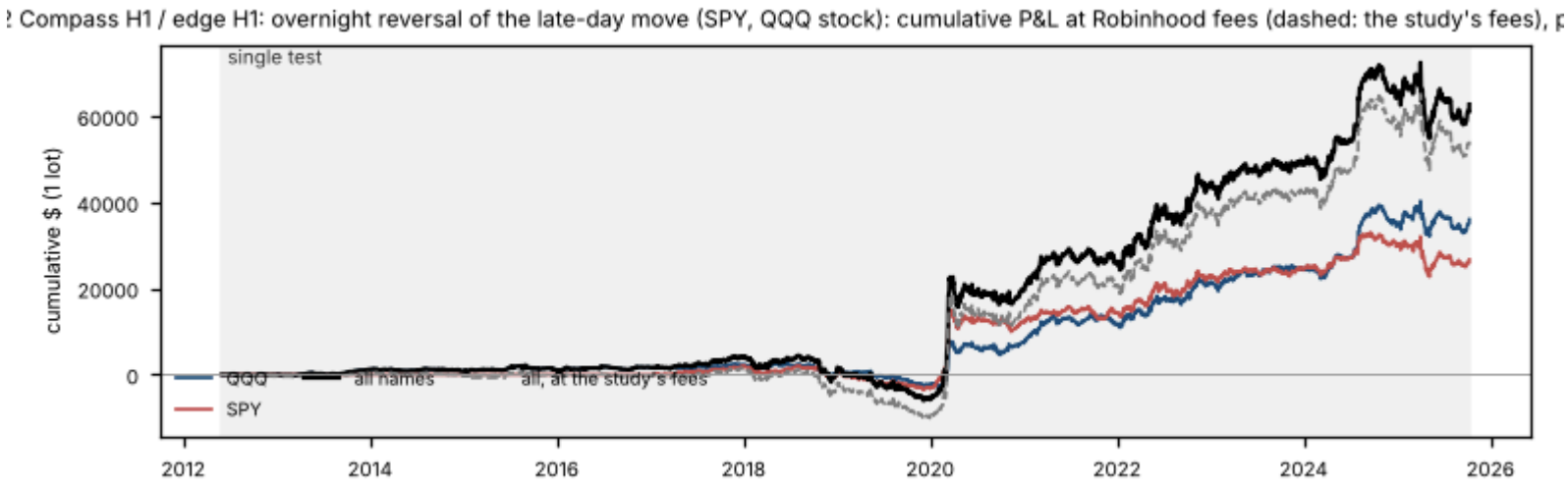
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- single test: next 09:30 open 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	QQQ	142	54.2	+0.00	+0.06	+9
2013	QQQ	247	57.1	+0.06	+4.85	+1,197
2014	QQQ	244	50.8	−0.02	−1.40	−341
2015	QQQ	246	50.4	+0.00	+0.00	+1
2016	QQQ	248	47.6	+0.03	+3.18	+788
2017	QQQ	248	52.8	+0.02	+2.62	+649
2018	QQQ	247	46.2	−0.04	−6.77	−1,673
2019	QQQ	248	44.4	−0.07	−12.60	−3,125
2020	QQQ	247	54.7	+0.19	+39.47	+9,749
2021	QQQ	247	55.9	+0.06	+18.36	+4,535
2022	QQQ	249	53.4	+0.13	+37.97	+9,454
2023	QQQ	246	52.4	+0.04	+12.36	+3,041
2024	QQQ	248	54.0	+0.10	+42.75	+10,601
2025	QQQ	192	49.0	+0.00	+6.01	+1,154
2017	SPY	245	53.9	+0.03	+7.74	+1,897
2018	SPY	246	47.6	−0.03	−7.42	−1,824
2019	SPY	248	43.1	−0.05	−13.72	−3,402
2020	SPY	251	57.4	+0.21	+61.86	+15,526
2021	SPY	251	54.2	+0.02	+9.21	+2,311
2022	SPY	247	54.3	+0.08	+32.83	+8,109
2023	SPY	247	49.0	+0.02	+7.11	+1,756
2024	SPY	249	52.6	+0.05	+23.44	+5,836
2025	SPY	193	48.2	−0.04	−18.23	−3,518
2012	all names	142	54.2	+0.00	+0.06	+9
2013	all names	247	57.1	+0.06	+4.85	+1,197
2014	all names	244	50.8	−0.02	−1.40	−341
2015	all names	246	50.4	+0.00	+0.00	+1
2016	all names	248	47.6	+0.03	+3.18	+788
2017	all names	493	53.3	+0.02	+5.17	+2,547
2018	all names	493	46.9	−0.03	−7.09	−3,498
2019	all names	496	43.8	−0.06	−13.16	−6,527
2020	all names	498	56.0	+0.20	+50.75	+25,275
2021	all names	498	55.0	+0.04	+13.75	+6,846
2022	all names	496	53.8	+0.11	+35.41	+17,564
2023	all names	493	50.7	+0.03	+9.73	+4,797
2024	all names	497	53.3	+0.07	+33.07	+16,437
2025	all names	385	48.6	−0.02	−6.14	−2,364



Amendments, deviations and integrity checks. verify_h1.py: 50 random trades re-derived, 0 differences.

What it showed. The longs carry the overnight drift; the shorts lose.

#34 Edge H3: earnings IV-crush short iron condor (single names)

Folder: `research_notes/Edge search/H3`. **Commits:** 001a136 / — / 34c8fd8. **Dates:** 2026-10-23.

Hypothesis. Implied volatility collapses across the report enough to pay a short condor.

Rule. Shorts at K0 ± 0.5 implied move, longs ± 1.5, first expiry after R within 14 days, sold at the closing quote before the report, bought back at the reaction session's close.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Bid/ask both ways, $0.68748 a contract each way.

Data. Exploration: 1,601 X names, 2,209 condors 2019-04 → 2022-12.

Periods and holdouts. Exploration screen; holdout (1,600 H names) not fetched, UNOPENED.

Pass rule (as pre-registered). Screen: mean > 0 with the block interval clear of zero.

Verdict. FAIL: −$161.54 a condor [−180.68, −142.46] (−30.2% of max risk); at the mid −$20.

Metrics at Robinhood fees (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
single names	exploration (holdout unopened)	2,209	554 / 957	31.6	+22.3, +117.80	−50.7, −283.90	−20.0, −40.34	**−27.65**, −156.97	[−29.75, −25.26]; [−176.19, −138.32]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
single names	exploration (holdout unopened)	+1,564.55, +19.3%, 2022-04-26, CMG 2022-04-29 1270.0/1385.0P 1495.0/1605.0C	−6,535.53, −111.1%, 2022-07-26, CMG 2022-07-29 1180.0/1270.0P 1365.0/1455.0C	0.19	346,757	956	total ≤ 0 (best 5 days +5,046; without them −351,794)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $4.93 a trade) and Robinhood's charged instead (mean $0.36). The study's fees are given beside them below. Re-pricing: exact (2,208 trades); Robinhood fees inferred from the rule: no exit quotes kept: four legs closed at $1.00 assumed (1 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
single names	exploration (holdout unopened)	2,209	**−27.65**, −156.97	−30.24, −161.54	31.6 / 30.3	[−29.75, −25.26] / [−32.38, −27.80]

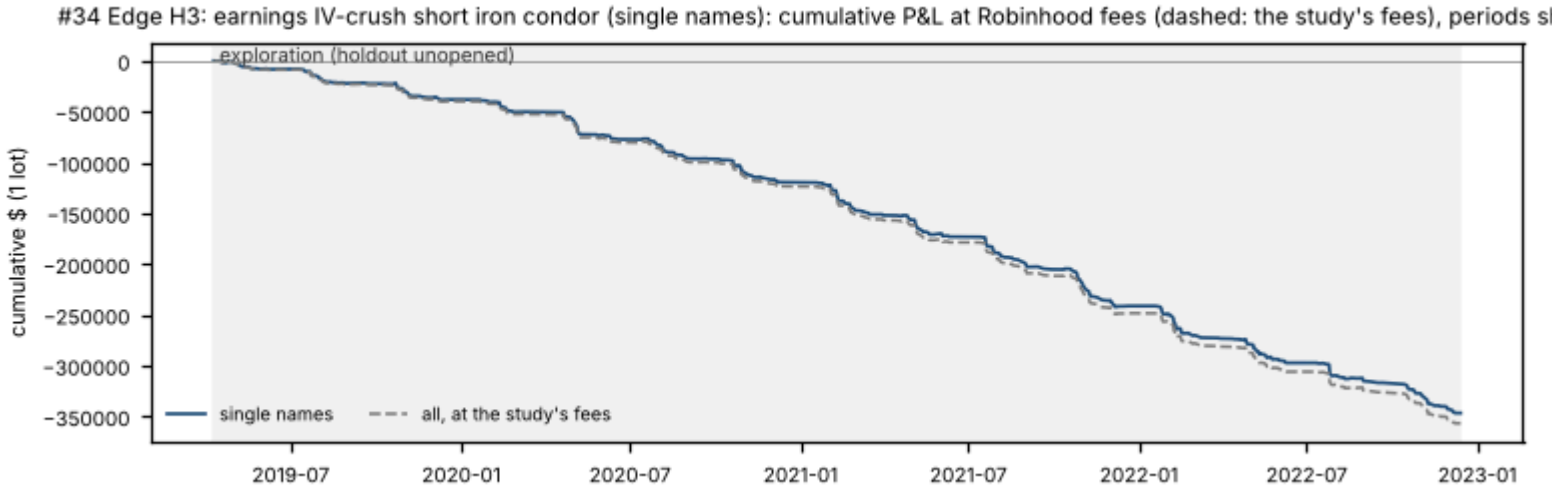
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- exploration (holdout unopened): quotes 100.0%; split-sized move: booked at max loss 0.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2019	single names	403	36.5	−24.96	−94.66	−38,148
2020	single names	504	27.0	−29.62	−161.93	−81,614
2021	single names	649	31.3	−30.20	−187.39	−121,616
2022	single names	653	32.5	−25.24	−161.36	−105,369



Amendments, deviations and integrity checks. —

What it showed. The spread and fees take the crush.

#35 Edge H4: market intraday momentum (SPY, QQQ stock)

Folder: `research_notes/Edge search/H4`. **Commits:** b14b640 / — / dfa575d. **Dates:** 2026-10-23.

Hypothesis. The first half hour's return predicts the last half hour (Gao, Han, Li & Zhou 2018).

Rule. r1 = the 09:59 bar close ÷ the previous official close − 1; long 100 at the 15:30 bar open if r1 > 0, short if < 0; exit at the official close.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.005 a share entry + $0.01 auction ($1.50 a trade).

Data. Test 2025-10-13 → 2026-10-22 (257 sessions a name); exploration (context) SPY 2017 → 2025-10-10, QQQ 2012 → .

Periods and holdouts. test (the verdict) / exploration (context only, spent by S12 and others).

Pass rule (as pre-registered). Each name: mean > 0 with the block-20 interval clear of zero.

Verdict. FAIL: SPY −$2.95 [−19.01, +15.37], QQQ +$10.32 [−8.56, +35.77].

Metrics at Robinhood fees (return = % of notional ($ = per 100 sh); win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	exploration (context)	3,319	3,319 / 3,419	49.4	+0.193, +44.97	−0.194, −47.99	−0.004, −0.66	**−0.003**, −2.03	[−0.013, +0.010]; [−4.65, +1.07]
SPY	exploration (context)	2,184	2,184 / 2,265	49.0	+0.192, +72.15	−0.190, −72.04	−0.003, −0.99	**−0.003**, −1.33	[−0.018, +0.014]; [−6.76, +4.60]
QQQ	test 2025-10-13→2026-10-22	257	257 / 267	49.8	+0.180, +119.61	−0.152, −98.91	−0.001, −0.77	**+0.013**, +9.92	[−0.017, +0.048]; [−9.77, +34.85]
SPY	test 2025-10-13→2026-10-22	257	257 / 267	45.5	+0.125, +88.87	−0.114, −80.60	−0.013, −9.11	**−0.005**, −3.45	[−0.029, +0.019]; [−20.01, +14.24]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	exploration (context)	+780.08, +4.2%, 2020-03-13, 100 QQQ long	−751.87, −4.4%, 2020-03-18, 100 QQQ short	0.92	12,257	1,401	total ≤ 0 (best 5 days +3,464; without them −10,205)
SPY	exploration (context)	+1,249.46, +4.8%, 2020-03-13, 100 SPY long	−744.13, −2.6%, 2020-02-28, 100 SPY short	0.96	9,189	1,417	total ≤ 0 (best 5 days +4,114; without them −7,020)
QQQ	test 2025-10-13→2026-10-22	+902.08, +1.3%, 2026-07-29, 100 QQQ short	−377.73, −0.6%, 2025-11-20, 100 QQQ long	1.20	2,304	142	96%
SPY	test 2025-10-13→2026-10-22	+572.95, +0.8%, 2026-07-29, 100 SPY short	−373.10, −0.5%, 2026-09-30, 100 SPY long	0.92	3,175	240	total ≤ 0 (best 5 days +1,735; without them −2,621)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.00 a trade) and Robinhood's charged instead (mean $0.70). The study's fees are given beside them below. Re-pricing: exact (6,017 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	exploration (context)	3,319	**−0.003**, −2.03	−0.007, −2.53	49.4 / 48.1	[−0.013, +0.010] / [−0.017, +0.005]
SPY	exploration (context)	2,184	**−0.003**, −1.33	−0.003, −1.51	49.0 / 48.9	[−0.018, +0.014] / [−0.018, +0.013]
QQQ	test 2025-10-13→2026-10-22	257	**+0.013**, +9.92	+0.014, +10.32	49.8 / 49.8	[−0.017, +0.048] / [−0.016, +0.048]
SPY	test 2025-10-13→2026-10-22	257	**−0.005**, −3.45	−0.004, −2.95	45.5 / 45.5	[−0.029, +0.019] / [−0.028, +0.020]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

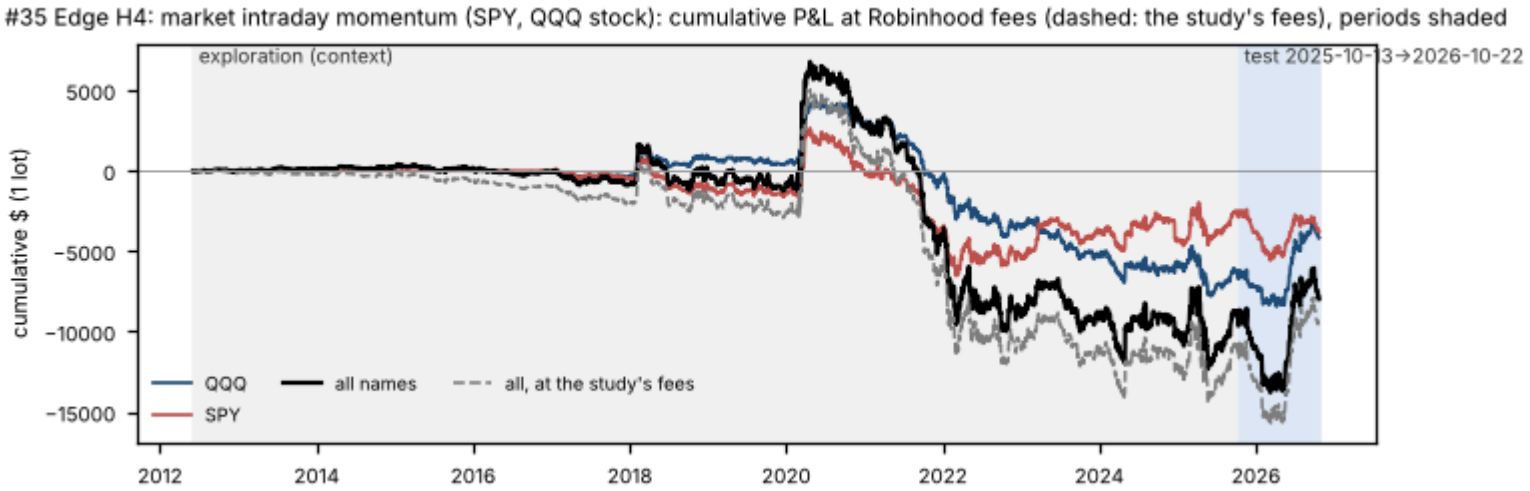
Exit reasons (share of trades):

- exploration (context): official close 100.0%
- test 2025-10-13→2026-10-22: official close 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	QQQ	142	51.4	−0.00	−0.01	−1
2013	QQQ	245	49.0	+0.01	+0.42	+102
2014	QQQ	245	49.8	+0.01	+0.46	+112
2015	QQQ	250	46.8	−0.00	−0.11	−28
2016	QQQ	251	52.2	−0.01	−1.15	−288
2017	QQQ	249	47.4	−0.01	−0.86	−214
2018	QQQ	248	56.5	+0.03	+4.60	+1,140
2019	QQQ	249	50.2	−0.01	−1.47	−366
2020	QQQ	251	52.6	+0.06	+9.25	+2,321
2021	QQQ	250	44.4	−0.04	−12.96	−3,240
2022	QQQ	250	47.6	−0.04	−12.50	−3,125
2023	QQQ	248	48.0	−0.02	−6.55	−1,626
2024	QQQ	248	48.0	−0.01	−2.60	−645
2025	QQQ	247	47.8	−0.01	−5.41	−1,337
2026	QQQ	203	51.7	+0.02	+14.79	+3,003
2017	SPY	246	45.1	−0.01	−2.24	−550
2018	SPY	248	50.8	−0.01	−1.80	−447
2019	SPY	248	51.2	−0.01	−1.69	−420
2020	SPY	251	46.6	+0.03	+5.35	+1,344
2021	SPY	251	43.4	−0.03	−14.02	−3,519
2022	SPY	250	50.0	−0.02	−7.48	−1,870
2023	SPY	248	51.2	+0.02	+7.07	+1,754
2024	SPY	249	49.4	−0.00	−2.75	−684
2025	SPY	247	52.2	+0.00	+2.51	+621
2026	SPY	203	46.3	−0.00	−0.11	−21
2012	all names	142	51.4	−0.00	−0.01	−1
2013	all names	245	49.0	+0.01	+0.42	+102

year	name	trades	win %	mean %	mean $	total $
2014	all names	245	49.8	+0.01	+0.46	+112
2015	all names	250	46.8	−0.00	−0.11	−28
2016	all names	251	52.2	−0.01	−1.15	−288
2017	all names	495	46.3	−0.01	−1.54	−764
2018	all names	496	53.6	+0.01	+1.40	+693
2019	all names	497	50.7	−0.01	−1.58	−786
2020	all names	502	49.6	+0.04	+7.30	+3,665
2021	all names	501	43.9	−0.03	−13.49	−6,759
2022	all names	500	48.8	−0.03	−9.99	−4,995
2023	all names	496	49.6	+0.00	+0.26	+129
2024	all names	497	48.7	−0.01	−2.67	−1,328
2025	all names	494	50.0	−0.00	−1.45	−716
2026	all names	406	49.0	+0.01	+7.34	+2,981



Amendments, deviations and integrity checks. A-priori MDE $28–37 a trade: weak evidence either way.

What it showed. No momentum after costs.

#37a Edge H7: the pre-FOMC drift (SPY, QQQ stock) — PASSED

Folder: `research_notes/Edge search/H7_H8_H9`. **Commits:** edca472 / forward rule fcc1f4d / 6c94bb8; wired 712b0ad; baseline 827bd5e. **Dates:** 2026-10-23.

Hypothesis. Stocks drift up in the 24 hours before a scheduled FOMC statement (Lucca & Moench 2015).

Rule. Buy 100 at the official close of the session before each scheduled 2:00 pm statement (2013-03-20 on); sell at the 13:58 bar's close on the statement day.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.01 a share auction entry + $0.005 exit ($1.50 a trade).

Data. SPY 77 statements 2017-02 → 2026-09; QQQ 108 2013-03 → 2026-09.

Periods and holdouts. Single test (no study had read the signed pre-FOMC return); forward from 2026-10-28 (N 366 SPY / 483 QQQ).

Pass rule (as pre-registered). Each name: mean > 0 with the block-20 interval (calendar sessions, 2,000 draws, seed 20261023) clear of zero at 1× and 2×; Bonferroni m = 13 for an EDGE.

Verdict. PASS both names: SPY +$70.43 [+14.64, +129.31], QQQ +$69.59 [+23.10, +122.14]; QQQ clears Bonferroni (EDGE under rule 5); not "works" before its forward N.

Metrics at Robinhood fees (return = % of notional ($ = per 100 sh); win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	single test	108	108 / 3,462	61.1	+0.649, +195.59	−0.469, −127.34	+0.115, +18.20	**+0.214**, +70.01	[+0.053, +0.358]; [+18.16, +115.30]
SPY	single test	77	77 / 2,487	55.8	+0.563, +224.77	−0.280, −124.53	+0.062, +15.94	**+0.191**, +70.54	[+0.060, +0.355]; [+11.25, +131.20]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	single test	+1,161.51, +2.5%, 2024-07-30, 100 QQQ long, FOMC 2024-07-31	−542.91, −0.8%, 2026-07-28, 100 QQQ long, FOMC 2026-07-29	2.41	1,001	1,370	56%
SPY	single test	+822.94, +1.5%, 2024-07-30, 100 SPY long, FOMC 2024-07-31	−620.30, −1.6%, 2021-01-26, 100 SPY long, FOMC 2021-01-27	2.28	1,068	295	66%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.00 a trade) and Robinhood's charged instead (mean $0.71). The study's fees are given beside them below. Re-pricing: exact (185 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	single test	108	**+0.214**, +70.01	+0.210, +69.59	61.1 / 61.1	[+0.053, +0.358] / [+0.049, +0.355]
SPY	single test	77	**+0.191**, +70.54	+0.190, +70.43	55.8 / 55.8	[+0.060, +0.355] / [+0.059, +0.355]

Random baseline (as the study computed it, at its own fees): descriptive baseline against every non-FOMC session over the same clock window (H7_BASELINE.md, 827bd5e), no new verdict.

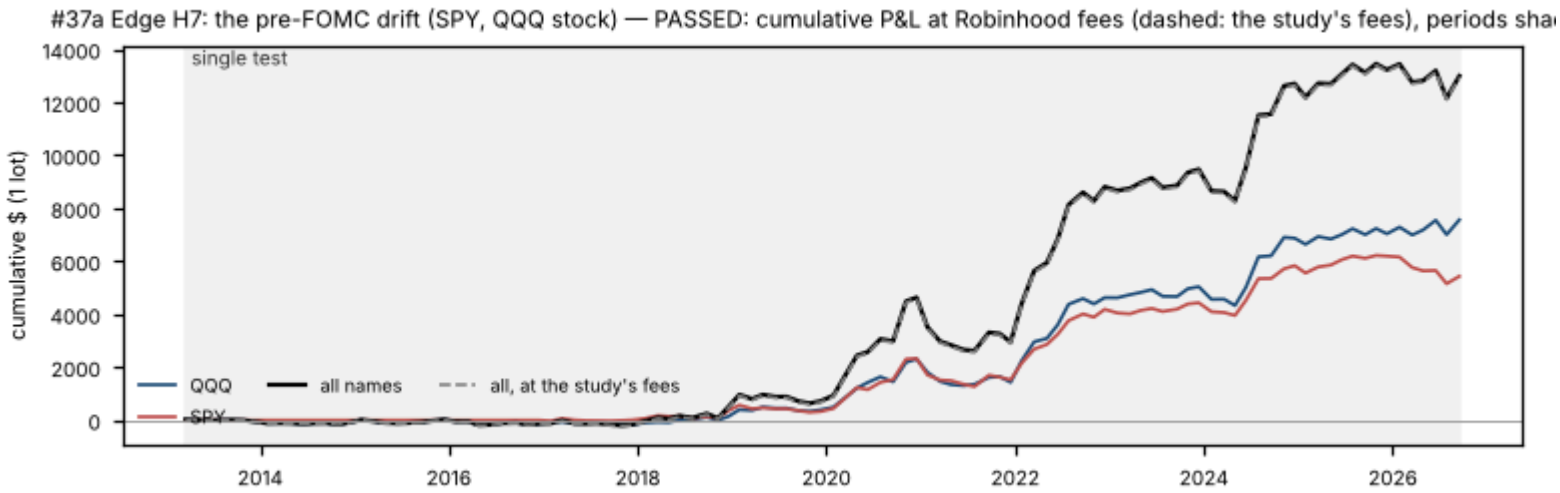
Exit reasons (share of trades):

- single test: 13:58 bar close on the statement day 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2013	QQQ	7	42.9	−0.10	−9.11	−64
2014	QQQ	8	50.0	−0.04	−2.65	−21
2015	QQQ	8	62.5	+0.13	+13.32	+107
2016	QQQ	8	62.5	−0.19	−20.01	−160
2017	QQQ	8	62.5	−0.02	−3.88	−31
2018	QQQ	8	75.0	+0.23	+38.81	+310
2019	QQQ	8	50.0	+0.19	+31.29	+250
2020	QQQ	7	85.7	+1.12	+273.44	+1,914
2021	QQQ	8	37.5	−0.34	−109.57	−877
2022	QQQ	8	87.5	+1.28	+399.64	+3,197
2023	QQQ	8	62.5	+0.16	+51.44	+412
2024	QQQ	8	62.5	+0.46	+229.10	+1,833
2025	QQQ	8	50.0	+0.04	+20.63	+165
2026	QQQ	6	66.7	+0.11	+87.60	+526
2017	SPY	8	37.5	+0.00	+0.97	+8
2018	SPY	8	62.5	+0.17	+44.34	+355
2019	SPY	8	50.0	−0.01	−4.28	−34
2020	SPY	7	85.7	+0.91	+285.49	+1,998
2021	SPY	8	12.5	−0.26	−100.50	−804
2022	SPY	8	87.5	+0.82	+332.64	+2,661
2023	SPY	8	62.5	+0.07	+30.91	+247
2024	SPY	8	62.5	+0.30	+175.36	+1,403
2025	SPY	8	62.5	+0.08	+44.94	+360
2026	SPY	6	33.3	−0.18	−127.04	−762
2013	all names	7	42.9	−0.10	−9.11	−64
2014	all names	8	50.0	−0.04	−2.65	−21
2015	all names	8	62.5	+0.13	+13.32	+107
2016	all names	8	62.5	−0.19	−20.01	−160
2017	all names	16	50.0	−0.01	−1.46	−23
2018	all names	16	68.8	+0.20	+41.58	+665
2019	all names	16	50.0	+0.09	+13.51	+216
2020	all names	14	85.7	+1.01	+279.46	+3,912
2021	all names	16	25.0	−0.30	−105.03	−1,681
2022	all names	16	87.5	+1.05	+366.14	+5,858
2023	all names	16	62.5	+0.11	+41.18	+659
2024	all names	16	62.5	+0.38	+202.23	+3,236
2025	all names	16	56.2	+0.06	+32.79	+525

year	name	trades	win %	mean %	mean $	total $
2026	all names	12	50.0	−0.03	−19.72	−237



Amendments, deviations and integrity checks. The gross is the night (close → 09:30 ≈ +$73–74); QQQ before March 2016 −$3.00.

What it showed. About +$70 a trade (0.2% of notional) in both names, carried overnight.

#37b Edge H8: turn of the month (SPY, QQQ stock)

Folder: `research_notes/Edge search/H7_H8_H9`. **Commits:** edca472 / — / 6c94bb8. **Dates:** 2026-10-23.

Hypothesis. Returns concentrate from the last trading day to the third of the new month.

Rule. Buy 100 at the official close of the second-to-last session of each month, sell at the official close of the third session of the next.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.01 a share each way ($2.00 a trade).

Data. SPY 117 turns 2017-01 → 2026-09; QQQ 172 2012-06 → 2026-09.

Periods and holdouts. Single test.

Pass rule (as pre-registered). As H7.

Verdict. SPY FAIL; QQQ PASS at 1× only (+$110.22 [+0.84, +218.26]) — not an EDGE.

Metrics at Robinhood fees (return = % of notional ($ = per 100 sh); win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	single test	172	172 / 3,651	59.3	+1.854, +505.33	−1.727, −461.97	+0.419, +82.00	**+0.397**, +111.66	[+0.016, +0.705]; [−9.31, +213.25]
SPY	single test	117	117 / 2,498	60.7	+1.482, +608.65	−1.488, −595.82	+0.417, +155.21	**+0.315**, +135.10	[−0.049, +0.646]; [−19.75, +265.54]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	single test	+3,373.50, +4.9%, 2026-07-30, 100 QQQ long, turn 2026-07	−2,369.27, −3.8%, 2026-01-29, 100 QQQ long, turn 2026-01	1.59	4,227	865	63%
SPY	single test	+2,808.39, +3.8%, 2026-07-30, 100 SPY long, turn 2026-07	−2,463.09, −4.5%, 2024-07-30, 100 SPY long, turn 2024-07	1.58	4,623	498	66%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $2.00 a trade) and Robinhood's charged instead (mean $0.70). The study's fees are given beside them below. Re-pricing: exact (289 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
QQQ	single test	172	**+0.397**, +111.66	+0.387, +110.22	59.3 / 59.3	[+0.016, +0.705] / [+0.006, +0.694]
SPY	single test	117	**+0.315**, +135.10	+0.311, +134.00	60.7 / 60.7	[−0.049, +0.646] / [−0.052, +0.642]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

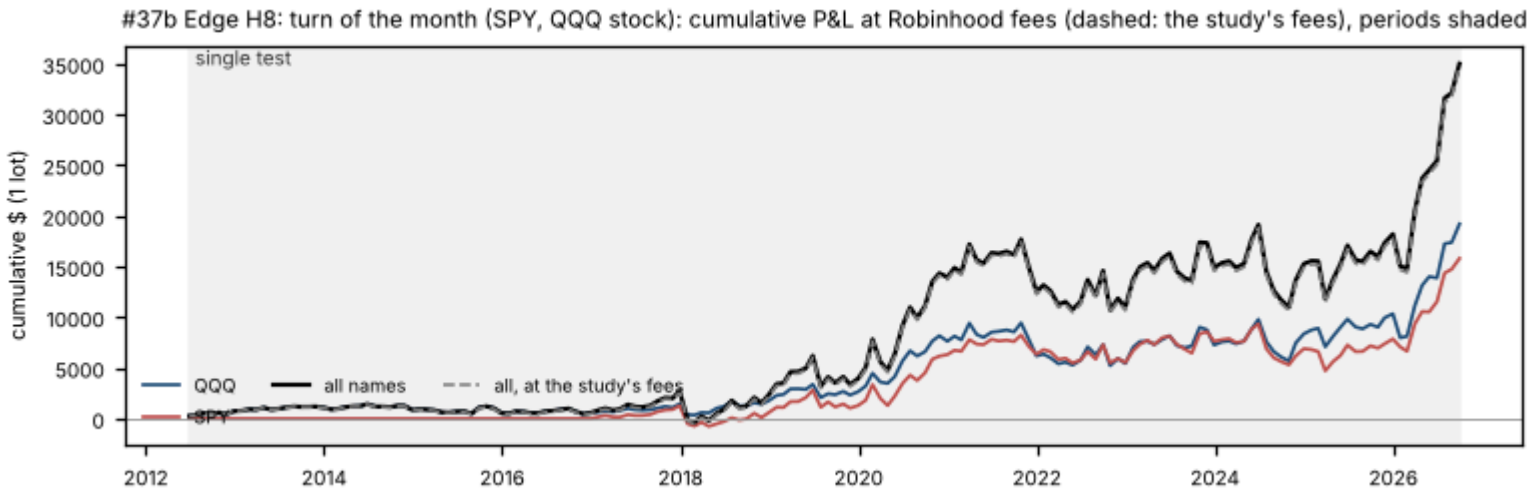
Exit reasons (share of trades):

- single test: official close, 3rd session 100.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	QQQ	7	71.4	+1.65	+105.45	+738
2013	QQQ	12	66.7	+0.44	+29.33	+352
2014	QQQ	12	50.0	−0.17	−20.55	−247
2015	QQQ	12	41.7	−0.17	−25.75	−309
2016	QQQ	12	50.0	+0.05	+3.41	+41
2017	QQQ	12	66.7	+0.50	+73.51	+882
2018	QQQ	12	58.3	+0.19	+25.29	+303
2019	QQQ	12	66.7	+0.46	+75.09	+901
2020	QQQ	12	66.7	+1.68	+414.53	+4,974
2021	QQQ	12	50.0	−0.24	−120.50	−1,446
2022	QQQ	12	50.0	−0.19	−61.99	−744
2023	QQQ	12	58.3	+0.61	+151.75	+1,821
2024	QQQ	12	58.3	+0.17	+91.68	+1,100
2025	QQQ	12	66.7	+0.27	+165.08	+1,981
2026	QQQ	9	77.8	+1.50	+984.12	+8,857
2017	SPY	12	75.0	+0.42	+106.38	+1,277
2018	SPY	12	58.3	−0.17	−58.17	−698
2019	SPY	12	66.7	+0.25	+59.99	+720
2020	SPY	12	75.0	+1.25	+418.64	+5,024
2021	SPY	12	41.7	+0.06	+2.17	+26
2022	SPY	12	50.0	−0.14	−68.36	−820
2023	SPY	12	58.3	+0.48	+175.34	+2,104
2024	SPY	12	58.3	−0.11	−60.73	−729
2025	SPY	12	58.3	+0.11	+79.77	+957
2026	SPY	9	66.7	+1.25	+882.91	+7,946
2012	all names	7	71.4	+1.65	+105.45	+738
2013	all names	12	66.7	+0.44	+29.33	+352
2014	all names	12	50.0	−0.17	−20.55	−247
2015	all names	12	41.7	−0.17	−25.75	−309
2016	all names	12	50.0	+0.05	+3.41	+41
2017	all names	24	70.8	+0.46	+89.95	+2,159
2018	all names	24	58.3	+0.01	−16.44	−395
2019	all names	24	66.7	+0.35	+67.54	+1,621
2020	all names	24	70.8	+1.47	+416.58	+9,998
2021	all names	24	45.8	−0.09	−59.17	−1,420
2022	all names	24	50.0	−0.16	−65.18	−1,564
2023	all names	24	58.3	+0.55	+163.54	+3,925
2024	all names	24	58.3	+0.03	+15.47	+371
2025	all names	24	62.5	+0.19	+122.43	+2,938

year	name	trades	win %	mean %	mean $	total $
2026	all names	18	72.2	+1.37	+933.52	+16,803



Amendments, deviations and integrity checks. The nine 2026 turns are about half of each total.

What it showed. Positive means, intervals at or through zero.

#36 Edge H5 (SPY − SPXW late straddle) and H6 (monthly put spreads), forward only

Folder: `research_notes/Edge search/`. **Commits:** e114e30 / — / wired 23c0cc3. **Dates:** 2026-10-23.

Hypothesis. (See the prereg.)

Rule. (See the prereg.)

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). —

Data. Forward only.

Periods and holdouts. Forward; no historical read.

Pass rule (as pre-registered). At N forward trades only.

Verdict. No verdict.

Metrics. not available — forward-only: no trade exists. Every metric (trades, win rate, mean, interval, drawdown, by year) is "not available" because no forward session has been counted. H5: no forward session read yet (N = 1,326 sessions); H6: entries from 2026-11-02 (N = 449 months). Their historical reads are spent (the SPXW report, S17).

Amendments, deviations and integrity checks. —

What it showed. Nothing to report yet.

#38 Edge H9: overnight drift (forward only, withdrawn)

Folder: `research_notes/Edge search/`. **Commits:** edca472 / — / wired 712b0ad, removed c68f389. **Dates:** 2026-10-23.

Hypothesis. (See the prereg.)

Rule. (See the prereg.)

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). —

Data. Forward only.

Periods and holdouts. Forward; no historical read.

Pass rule (as pre-registered). At N forward trades only.

Verdict. No verdict.

Metrics. not available — forward-only: no trade exists. Every metric (trades, win rate, mean, interval, drawdown, by year) is "not available" because no forward session has been counted. Withdrawn by the owner 2026-10-23 (c68f389) before any forward row was written; the registered rule edca472 is unchanged.

Amendments, deviations and integrity checks. —

What it showed. Nothing to report yet.

#39 Compass scanner (score ≥ 80, SPY/QQQ/IWM 0DTE)

Folder: `research_notes/Compass scanner/`. **Commits:** 142fcf4 / — / 9e3888e. **Dates:** 2026-10-23.

Hypothesis. The owner's scanner score (RSI/VWAP/RVOL/structure) gives a forward move worth an option trade.

Rule. Signal check first: the underlying's forward return after a signal vs non-signal minutes; only then the option P&L.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). —

Data. SPY/QQQ/IWM bars, Pick 2017-07-05 → 2022-11-18 and the option layer's Pick.

Periods and holdouts. Pick / Check; the 28 holdout months and AMD/AVGO UNOPENED.

Pass rule (as pre-registered). At 15 or 30 minutes: excess > 0 with its block-20 interval clear of zero and a positive signal mean.

Verdict. FAIL at the gate: 15 min −0.30 bp [−0.56, −0.07] (wrong side); 30 min −0.19 bp [−0.57, +0.21]. The study stopped.

Metrics. not available — signal-only: the gate failed before any option was priced, so trades, win rate, average win/loss, biggest win/loss, profit factor, drawdown, top-5 share and by-year are not available; no exit or option P&L was ever read.

Amendments, deviations and integrity checks. No held-out month read for an outcome.

What it showed. The scanner's signals do not move the underlying their way.

#40a Compass EMA ladder: PREVIEW 2025-03, 13 versions (rerun)

Folder: `research_notes/Compass EMA ladder/`. **Commits:** 0dd8c5d / — / cb79c23. **Dates:** 2026-10-23.

Hypothesis. The owner's EMA ladder entries on 0DTE options make money.

Rule. EMA 9 / ladder L1 (V1, the preview's base) and V2–V13 (one change each); the best-R/R execution contract; exits per PREREG_EMA_LADDER §3.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.68748 a contract each way; ask in, bid out; bid 0 → $0, no exit fee.

Data. SPY, QQQ, IWM 0DTE, 21 sessions 2025-03-03 → 31 (inside Pick, not held out).

Periods and holdouts. PREVIEW, no verdict; the full study was never run; the 28-month holdout UNOPENED for this rule.

Pass rule (as pre-registered). None for a preview (the study's rule waits for the full run).

Verdict. PREVIEW: every version loses.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
V1	preview 2025-03	2,781	21 / 21	18.6	+34.9, +12.50	−14.0, −4.68	−10.7, −3.10	**−4.89**, −1.48	n/a (< 60 sessions); n/a (< 60 sessions)
V10	preview 2025-03	1,996	21 / 21	19.9	+31.0, +11.89	−12.4, −4.74	−9.7, −3.10	**−3.79**, −1.43	n/a (< 60 sessions); n/a (< 60 sessions)
V11	preview 2025-03	2,765	21 / 21	18.7	+35.1, +12.54	−13.8, −4.71	−10.5, −3.10	**−4.67**, −1.48	n/a (< 60 sessions); n/a (< 60 sessions)
V12	preview 2025-03	2,350	21 / 21	20.7	+31.3, +15.34	−12.2, −6.01	−9.1, −4.10	**−3.21**, −1.59	n/a (< 60 sessions); n/a (< 60 sessions)
V13	preview 2025-03	2,794	21 / 21	23.8	+23.0, +31.86	−10.1, −12.08	−6.6, −7.10	**−2.23**, −1.64	n/a (< 60 sessions); n/a (< 60 sessions)
V2	preview 2025-03	768	21 / 21	13.9	+83.4, +30.95	−28.2, −10.04	−23.5, −7.10	**−12.66**, −4.33	n/a (< 60 sessions); n/a (< 60 sessions)
V3	preview 2025-03	253	21 / 21	11.5	+156.1, +57.24	−43.3, −17.44	−40.3, −13.10	**−20.41**, −8.88	n/a (< 60 sessions); n/a (< 60 sessions)
V4	preview 2025-03	2,214	21 / 21	14.1	+61.7, +22.11	−15.1, −5.08	−12.4, −4.10	**−4.22**, −1.24	n/a (< 60 sessions); n/a (< 60 sessions)
V5	preview 2025-03	511	21 / 21	11.4	+132.7, +43.61	−30.7, −11.33	−28.0, −9.10	**−12.12**, −5.09	n/a (< 60 sessions); n/a (< 60 sessions)
V6	preview 2025-03	192	21 / 21	8.9	+223.3, +90.66	−46.9, −19.72	−44.8, −16.10	**−22.97**, −9.95	n/a (< 60 sessions); n/a (< 60 sessions)
V7	preview 2025-03	2,749	21 / 21	18.6	+34.9, +12.56	−13.9, −4.69	−10.7, −3.10	**−4.86**, −1.48	n/a (< 60 sessions); n/a (< 60 sessions)
V8	preview 2025-03	2,101	21 / 21	17.8	+42.0, +15.45	−14.1, −4.68	−11.1, −3.10	**−4.16**, −1.11	n/a (< 60 sessions); n/a (< 60 sessions)
V9	preview 2025-03	2,076	21 / 21	17.8	+42.1, +15.56	−14.1, −4.69	−11.1, −3.10	**−4.09**, −1.09	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
V1	preview 2025-03	+148.90, +330.9%, 2025-03-28, QQQ 473P m34	−30.10, −38.6%, 2025-03-05, QQQ 486P m24	0.61	4,124	21	total ≤ 0 (best 5 days −132; without them −3,992)
V10	preview 2025-03	+148.90, +330.9%, 2025-03-28, QQQ 473P m34	−30.10, −38.6%, 2025-03-05, QQQ 486P m24	0.62	2,864	21	total ≤ 0 (best 5 days +243; without them −3,106)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
V11	preview 2025-03	+148.90, +330.9%, 2025-03-28, QQQ 473P m34	−30.10, −38.6%, 2025-03-05, QQQ 486P m24	0.61	4,094	21	total ≤ 0 (best 5 days −130; without them −3,965)
V12	preview 2025-03	+148.90, +330.9%, 2025-03-28, QQQ 473P m34	−40.10, −55.7%, 2025-03-07, QQQ 489P m346	0.67	3,746	21	total ≤ 0 (best 5 days +65; without them −3,812)
V13	preview 2025-03	+358.89, +212.4%, 2025-03-03, QQQ 509P m195	−97.10, −29.2%, 2025-03-05, QQQ 495P m24	0.82	4,731	21	total ≤ 0 (best 5 days +1,476; without them −6,058)
V2	preview 2025-03	+248.90, +607.1%, 2025-03-17, QQQ 482C m229	−46.10, −59.9%, 2025-03-11, QQQ 483C m18	0.50	3,322	21	total ≤ 0 (best 5 days +328; without them −3,651)
V3	preview 2025-03	+306.90, +682.0%, 2025-03-28, QQQ 473P m30	−145.10, −61.0%, 2025-03-03, QQQ 506P m30	0.43	2,769	21	total ≤ 0 (best 5 days +902; without them −3,147)
V4	preview 2025-03	+495.89, +1,549.7%, 2025-03-03, QQQ 501P m317	−34.10, −51.7%, 2025-03-07, QQQ 479P m209	0.72	3,754	19	total ≤ 0 (best 5 days +1,022; without them −3,758)
V5	preview 2025-03	+288.90, +656.6%, 2025-03-28, QQQ 473P m35	−45.10, −64.4%, 2025-03-11, QQQ 463P m90	0.49	3,302	19	total ≤ 0 (best 5 days +968; without them −3,570)
V6	preview 2025-03	+307.90, +699.8%, 2025-03-28, QQQ 473P m35	−145.10, −61.0%, 2025-03-03, QQQ 506P m30	0.45	2,887	21	total ≤ 0 (best 5 days +1,039; without them −2,949)
V7	preview 2025-03	+148.90, +330.9%, 2025-03-28, QQQ 473P m34	−30.10, −38.6%, 2025-03-05, QQQ 486P m24	0.61	4,068	21	total ≤ 0 (best 5 days −132; without them −3,935)
V8	preview 2025-03	+377.90, +944.7%, 2025-03-03, QQQ 502P m316	−27.10, −51.1%, 2025-03-19, QQQ 471P m269	0.71	3,090	20	total ≤ 0 (best 5 days +899; without them −3,221)
V9	preview 2025-03	+377.90, +944.7%, 2025-03-03, QQQ 502P m316	−27.10, −51.1%, 2025-03-19, QQQ 471P m269	0.72	3,037	20	total ≤ 0 (best 5 days +899; without them −3,168)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $1.37 a trade) and Robinhood's charged instead (mean $0.10). The study's fees are given beside them below. Re-pricing: exact (23,550 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
V1	preview 2025-03	2,781	**−4.89**, −1.48	−9.20, −2.76	18.6 / 15.5	n/a (< 60 sessions) / n/a (< 60 sessions)
V10	preview 2025-03	1,996	**−3.79**, −1.43	−7.62, −2.71	19.9 / 17.0	n/a (< 60 sessions) / n/a (< 60 sessions)
V11	preview 2025-03	2,765	**−4.67**, −1.48	−8.88, −2.76	18.7 / 15.6	n/a (< 60 sessions) / n/a (< 60 sessions)
V12	preview 2025-03	2,350	**−3.21**, −1.59	−5.91, −2.87	20.7 / 18.5	n/a (< 60 sessions) / n/a (< 60 sessions)
V13	preview 2025-03	2,794	**−2.23**, −1.64	−3.61, −2.91	23.8 / 22.3	n/a (< 60 sessions) / n/a (< 60 sessions)
V2	preview 2025-03	768	**−12.66**, −4.33	−16.78, −5.60	13.9 / 13.2	n/a (< 60 sessions) / n/a (< 60 sessions)
V3	preview 2025-03	253	**−20.41**, −8.88	−24.32, −10.15	11.5 / 10.7	n/a (< 60 sessions) / n/a (< 60 sessions)
V4	preview 2025-03	2,214	**−4.22**, −1.24	−8.50, −2.51	14.1 / 12.5	n/a (< 60 sessions) / n/a (< 60 sessions)
V5	preview 2025-03	511	**−12.12**, −5.09	−16.15, −6.37	11.4 / 11.2	n/a (< 60 sessions) / n/a (< 60 sessions)
V6	preview 2025-03	192	**−22.97**, −9.95	−26.68, −11.22	8.9 / 8.3	n/a (< 60 sessions) / n/a (< 60 sessions)
V7	preview 2025-03	2,749	**−4.86**, −1.48	−9.15, −2.75	18.6 / 15.5	n/a (< 60 sessions) / n/a (< 60 sessions)
V8	preview 2025-03	2,101	**−4.16**, −1.11	−8.50, −2.38	17.8 / 14.8	n/a (< 60 sessions) / n/a (< 60 sessions)
V9	preview 2025-03	2,076	**−4.09**, −1.09	−8.40, −2.37	17.8 / 14.8	n/a (< 60 sessions) / n/a (< 60 sessions)

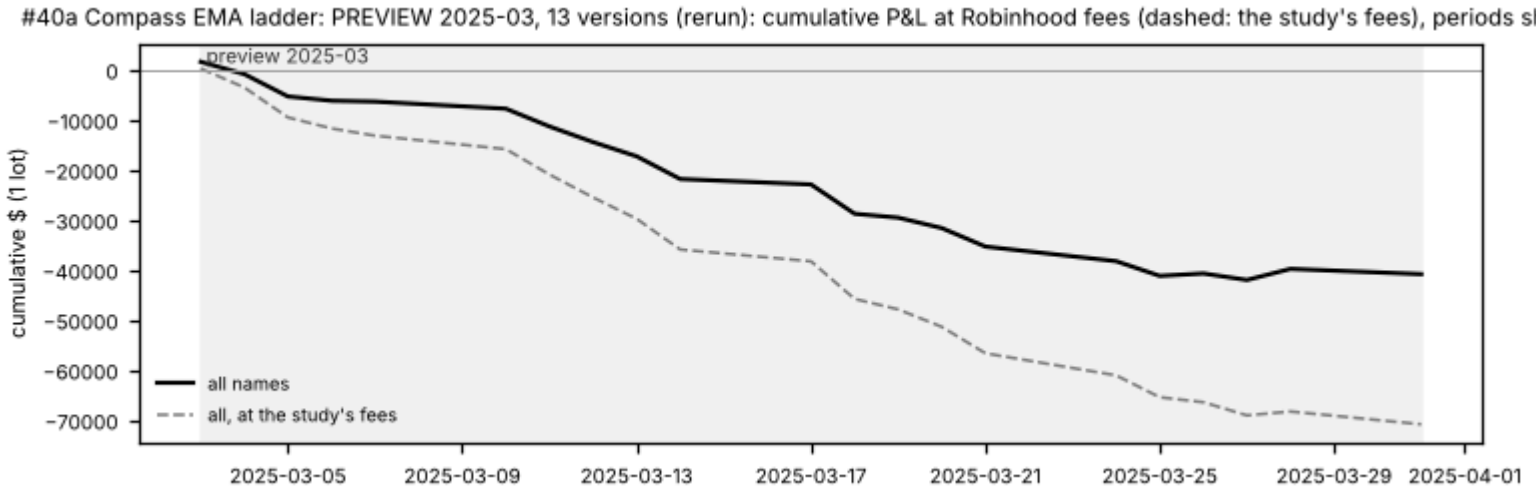
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- preview 2025-03: signal 99.6%; 15:58 0.2%; 14:00 0.2%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	V1	2,781	18.6	−4.89	−1.48	−4,124
2025	V10	1,996	19.9	−3.79	−1.43	−2,864
2025	V11	2,765	18.7	−4.67	−1.48	−4,094
2025	V12	2,350	20.7	−3.21	−1.59	−3,746
2025	V13	2,794	23.8	−2.23	−1.64	−4,582
2025	V2	768	13.9	−12.66	−4.33	−3,322
2025	V3	253	11.5	−20.41	−8.88	−2,245
2025	V4	2,214	14.1	−4.22	−1.24	−2,736
2025	V5	511	11.4	−12.12	−5.09	−2,602
2025	V6	192	8.9	−22.97	−9.95	−1,910
2025	V7	2,749	18.6	−4.86	−1.48	−4,068
2025	V8	2,101	17.8	−4.16	−1.11	−2,322
2025	V9	2,076	17.8	−4.09	−1.09	−2,269
2025	all names	23,550	18.5	−4.81	−1.74	−40,885



Amendments, deviations and integrity checks. Rerun for this report at cb79c23/e9d13c0 from the stored files: preview_2025-03.json byte-identical to the original run (sha256 81edab0c934effe5…).

What it showed. All 13 versions lose 3.6–26.7% of premium a trade in the month.

#40b Compass EMA ladder: “take profits are the nearest EMAs” PREVIEW (E1 ATM, E2 best-R/R)

Folder: `research_notes/Compass EMA ladder/`. **Commits:** rule aad56cf / — / e9d13c0. **Dates:** 2026-10-23.

Hypothesis. Targets at the nearest EMAs rescue the entries.

Rule. E1: the ATM contract (3% spread veto); E2: the best-R/R contract (3% cap); targets at the nearest EMAs, exits on the ATM chart.

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). As 40a.

Data. 2025-02 (19 sessions, fresh) and 2025-03 (21, seen before).

Periods and holdouts. PREVIEW, no verdict.

Pass rule (as pre-registered). None.

Verdict. PREVIEW: both versions lose in both months.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib’s 20-session moving-block bootstrap over the period’s calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
E1	preview 2025-02 (fresh)	1,761	19 / 20	31.8	+16.0, +16.99	−9.3, −9.71	−5.5, −4.10	**−1.25**, −1.22	n/a (< 60 sessions); n/a (< 60 sessions)
E2	preview 2025-02 (fresh)	1,544	19 / 20	29.7	+19.7, +9.11	−11.6, −5.36	−7.6, −3.10	**−2.34**, −1.07	n/a (< 60 sessions); n/a (< 60 sessions)
E1	preview 2025-03 (seen before)	2,179	21 / 21	33.2	+16.2, +23.00	−9.8, −13.06	−5.6, −6.10	**−1.14**, −1.08	n/a (< 60 sessions); n/a (< 60 sessions)
E2	preview 2025-03 (seen before)	2,044	21 / 21	31.3	+20.1, +9.86	−12.1, −5.96	−7.8, −4.10	**−2.06**, −1.01	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days’ share of profit
E1	preview 2025-02 (fresh)	+176.90, +83.1%, 2025-02-03, SPY 592C m52	−85.10, −60.4%, 2025-02-07, QQQ 527.5C m77	0.82	2,149	20	total ≤ 0 (best 5 days +571; without them −2,720)
E2	preview 2025-02 (fresh)	+80.90, +120.7%, 2025-02-25, SPY 596P m359	−30.10, −61.4%, 2025-02-07, QQQ 530C m77	0.72	1,649	20	total ≤ 0 (best 5 days +103; without them −1,752)
E1	preview 2025-03 (seen before)	+140.90, +134.2%, 2025-03-19, QQQ 484P m348	−63.10, −21.5%, 2025-03-11, QQQ 473P m120	0.88	2,888	16	total ≤ 0 (best 5 days +947; without them −3,305)
E2	preview 2025-03 (seen before)	+109.90, +144.6%, 2025-03-05, QQQ 502P m346	−23.10, −30.4%, 2025-03-13, QQQ 466P m42	0.75	2,068	21	total ≤ 0 (best 5 days +61; without them −2,129)

Fees. Every figure in this section is at **Robinhood’s fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study’s own fill prices, the study’s modelled fee added back (mean $1.37 a trade) and Robinhood’s charged instead (mean $0.10). The study’s fees are given beside them below. Re-pricing: exact (7,528 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
E1	preview 2025-02 (fresh)	1,761	**−1.25**, −1.22	−2.75, −2.49	31.8 / 30.8	n/a (< 60 sessions) / n/a (< 60 sessions)
E2	preview 2025-02 (fresh)	1,544	**−2.34**, −1.07	−5.21, −2.34	29.7 / 27.1	n/a (< 60 sessions) / n/a (< 60 sessions)
E1	preview 2025-03 (seen before)	2,179	**−1.14**, −1.08	−2.28, −2.36	33.2 / 32.5	n/a (< 60 sessions) / n/a (< 60 sessions)
E2	preview 2025-03 (seen before)	2,044	**−2.06**, −1.01	−4.76, −2.29	31.3 / 29.4	n/a (< 60 sessions) / n/a (< 60 sessions)

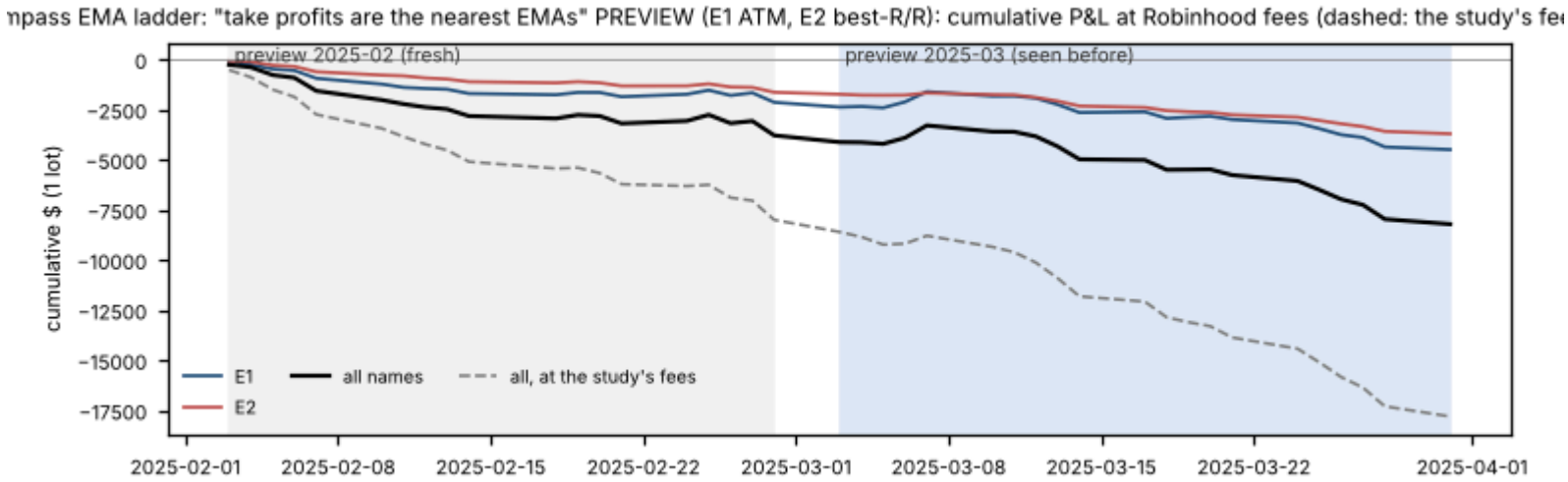
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- preview 2025-02 (fresh): stop 72.4%; target 27.6%
- preview 2025-03 (seen before): stop 70.3%; target 29.7%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	E1	3,940	32.6	−1.19	−1.14	−4,506
2025	E2	3,588	30.6	−2.18	−1.04	−3,717
2025	all names	7,528	31.6	−1.66	−1.09	−8,223



Amendments, deviations and integrity checks. Rerun byte-identical (targets_2025-02.json 6de804ea3ddfa9d5..., targets_2025-03.json 2b005e352e86e126...).

What it showed. E1 −2.3 to −2.8% and E2 −4.8 to −5.2% of premium a trade.

#41a EMA system v0: PREVIEW on 2026-10-12 → 16 (5 sessions)

Folder: `research_notes/EMA system/`. **Commits:** af32405 / — / 8765f71. **Dates:** 2026-10-23.

Hypothesis. The owner's EMA options system as specified (v0 defaults pinned).

Rule. RULEBOOK_v0.md: Config A/B entries, exits X1 (+1R), X2 (+2R), X3 (runner), X4; SPXW (SPX prints), SPY, QQQ, IWM.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). $0.68748 SPY/QQQ/IWM; SPXW $1.18748/$1.27748; ask in, bid out.

Data. 5 full sessions 2026-10-12 → 16 (a falling week: SPY −1.96%).

Periods and holdouts. PREVIEW, no verdict.

Pass rule (as pre-registered). None.

Verdict. PREVIEW: every row net positive in a falling week, 86% of trades puts.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
Config A X1	preview 2026-10-12→16	57	5 / 5	61.4	+42.2, +135.86	−36.9, −102.47	+26.5, +33.89	**+11.68**, +43.87	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X2	preview 2026-10-12→16	53	5 / 5	49.1	+76.2, +233.00	−40.8, −102.23	−13.2, −10.10	**+16.56**, +62.22	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X3	preview 2026-10-12→16	48	5 / 5	29.2	+136.9, +561.09	−33.8, −100.48	−22.8, −27.70	**+15.96**, +92.48	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X4	preview 2026-10-12→16	48	5 / 5	33.3	+90.3, +3,192.34	−31.9, −911.52	−20.4, −221.49	**+8.86**, +456.43	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X1	preview 2026-10-12→16	55	5 / 5	63.6	+39.6, +126.18	−37.9, −105.20	+26.8, +33.89	**+11.39**, +42.04	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X2	preview 2026-10-12→16	49	5 / 5	46.9	+71.3, +209.81	−41.4, −101.50	−16.9, −11.12	**+11.51**, +44.62	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X3	preview 2026-10-12→16	46	5 / 5	28.3	+125.6, +597.50	−34.3, −100.00	−23.6, −29.17	**+10.90**, +97.12	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X4	preview 2026-10-12→16	46	5 / 5	32.6	+82.5, +3,362.13	−32.7, −906.58	−21.8, −236.46	**+4.82**, +485.39	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
Config A X1	preview 2026-10-12→16	+717.74, +69.7%, 2026-10-14, SPXW 7770.0P 09:53	−772.26, −51.1%, 2026-10-15, SPXW 7665.0P 09:55	2.11	618	2	100%
Config A X2	preview 2026-10-12→16	+1,427.74, +138.6%, 2026-10-14, SPXW 7770.0P 09:53	−527.26, −65.1%, 2026-10-16, SPXW 7640.0P 14:37	2.19	724	2	100%
Config A X3	preview 2026-10-12→16	+3,277.74, +337.9%, 2026-10-16, SPXW 7715.0P 11:42	−527.26, −65.1%, 2026-10-16, SPXW 7640.0P 14:37	2.30	605	2	100%
Config A X4	preview 2026-10-12→16	+21,707.39, +210.8%, 2026-10-14, SPXW 7770.0P 09:53	−3,602.61, −44.5%, 2026-10-16, SPXW 7640.0P 14:37	1.75	5,442	2	100%
Config B X1	preview 2026-10-12→16	+717.74, +69.7%, 2026-10-14, SPXW 7770.0P 09:53	−772.26, −51.1%, 2026-10-15, SPXW 7665.0P 09:55	2.10	618	2	100%
Config B X2	preview 2026-10-12→16	+1,427.74, +138.6%, 2026-10-14, SPXW 7770.0P 09:53	−527.26, −65.1%, 2026-10-16, SPXW 7640.0P 14:37	1.83	602	2	100%
Config B X3	preview 2026-10-12→16	+3,277.74, +337.9%, 2026-10-16, SPXW 7715.0P 11:42	−527.26, −65.1%, 2026-10-16, SPXW 7640.0P 14:37	2.35	572	2	100%
Config B X4	preview 2026-10-12→16	+21,707.39, +210.8%, 2026-10-14, SPXW 7770.0P 09:53	−3,602.61, −44.5%, 2026-10-16, SPXW 7640.0P 14:37	1.79	5,442	2	100%

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $5.65 a trade) and Robinhood's charged instead (mean $2.92). The study's fees are given beside them below. Re-pricing: exact (402 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
Config A X1	preview 2026-10-12→16	57	**+11.68**, +43.87	+10.67, +43.03	61.4 / 61.4	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X2	preview 2026-10-12→16	53	**+16.56**, +62.22	+15.56, +61.38	49.1 / 49.1	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X3	preview 2026-10-12→16	48	**+15.96**, +92.48	+14.82, +91.60	29.2 / 29.2	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X4	preview 2026-10-12→16	48	**+8.86**, +456.43	+7.71, +447.58	33.3 / 33.3	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X1	preview 2026-10-12→16	55	**+11.39**, +42.04	+10.41, +41.19	63.6 / 63.6	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X2	preview 2026-10-12→16	49	**+11.51**, +44.62	+10.54, +43.78	46.9 / 46.9	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X3	preview 2026-10-12→16	46	**+10.90**, +97.12	+9.80, +96.23	28.3 / 28.3	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X4	preview 2026-10-12→16	46	**+4.82**, +485.39	+3.72, +476.48	32.6 / 32.6	n/a (< 60 sessions) / n/a (< 60 sessions)

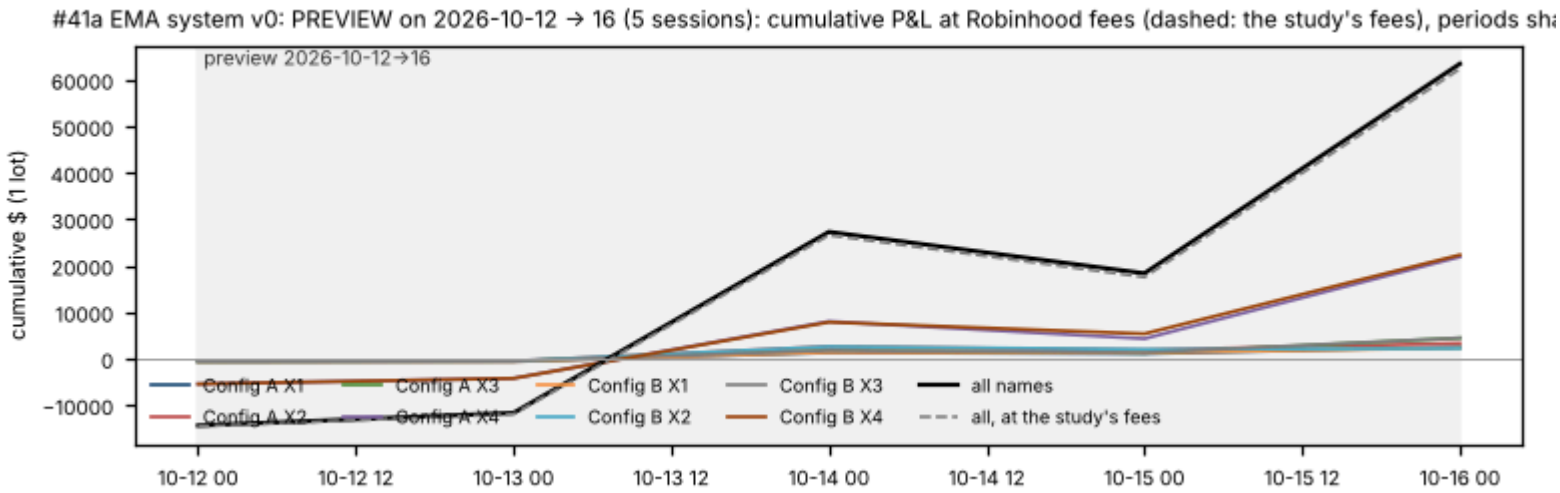
Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

Exit reasons (share of trades):

- preview 2026-10-12→16: stop 45.0%; runner 24.9%; +1R 17.4%; +2R 12.2%; flat 0.5%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2026	Config A X1	57	61.4	+11.68	+43.87	+2,501
2026	Config A X2	53	49.1	+16.56	+62.22	+3,298
2026	Config A X3	48	29.2	+15.96	+92.48	+4,439
2026	Config A X4	48	33.3	+8.86	+456.43	+21,909
2026	Config B X1	55	63.6	+11.39	+42.04	+2,312
2026	Config B X2	49	46.9	+11.51	+44.62	+2,187
2026	Config B X3	46	28.3	+10.90	+97.12	+4,467
2026	Config B X4	46	32.6	+4.82	+485.39	+22,328
2026	all names	402	44.0	+11.56	+157.81	+63,440



Amendments, deviations and integrity checks. —

What it showed. One falling week, mostly puts.

#41b EMA system v1: PREVIEW on three drawn weeks (up, flat, down)

Folder: `research_notes/EMA system/`. **Commits:** f171654 / — / f512d21. **Dates:** 2026-10-23.

Hypothesis. v0 with four changes (the three passed rules V1, X3, VN1 and a fix).

Rule. RULEBOOK_v1.md.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). As 41a.

Data. Weeks drawn by seed 2026102310: up 2025-07-21, flat 2026-02-02, down 2026-07-13.

Periods and holdouts. PREVIEW, no verdict; no day in a held-out month.

Pass rule (as pre-registered). None.

Verdict. PREVIEW: about zero pooled; the sign follows the week.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
Config A X1	preview week: up	20	5 / 5	45.0	+18.9, +28.03	−42.1, −69.72	−26.0, −17.12	**−14.67**, −25.73	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X2	preview week: up	20	5 / 5	20.0	+52.1, +158.81	−45.6, −73.60	−34.6, −22.21	**−26.04**, −27.12	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X3	preview week: up	21	5 / 5	9.5	+25.6, +10.89	−28.5, −57.92	−28.7, −20.13	**−23.34**, −51.37	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X4	preview week: up	21	5 / 5	14.3	+21.3, +92.09	−28.0, −586.46	−27.9, −201.23	**−20.93**, −489.52	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X1	preview week: up	18	5 / 5	50.0	+18.9, +28.03	−42.2, −78.63	−9.2, −5.11	**−11.65**, −25.30	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X2	preview week: up	18	5 / 5	22.2	+52.1, +158.81	−46.1, −79.88	−34.6, −22.21	**−24.28**, −26.84	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X3	preview week: up	19	5 / 5	10.5	+25.6, +10.89	−28.0, −61.95	−28.7, −17.14	**−22.31**, −54.28	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X4	preview week: up	19	5 / 5	15.8	+21.3, +92.09	−27.3, −630.20	−27.9, −171.27	**−19.65**, −516.16	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X1	preview week: flat	32	5 / 5	68.8	+44.7, +218.82	−39.8, −206.96	+28.9, +34.83	**+18.29**, +85.77	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X2	preview week: flat	27	5 / 5	55.6	+74.3, +368.70	−35.5, −244.81	+51.7, +47.81	**+25.50**, +96.03	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X3	preview week: flat	24	5 / 5	45.8	+88.1, +487.11	−30.4, −162.95	−14.0, −10.65	**+23.94**, +134.99	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X4	preview week: flat	24	5 / 5	50.0	+65.4, +3,380.28	−28.8, −1,525.87	−2.4, −47.30	**+18.29**, +927.21	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X1	preview week: flat	32	5 / 5	68.8	+44.7, +218.82	−39.8, −206.96	+28.9, +34.83	**+18.29**, +85.77	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X2	preview week: flat	27	5 / 5	55.6	+74.3, +368.70	−35.5, −244.81	+51.7, +47.81	**+25.50**, +96.03	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X3	preview week: flat	24	5 / 5	45.8	+88.1, +487.11	−30.4, −162.95	−14.0, −10.65	**+23.94**, +134.99	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X4	preview week: flat	24	5 / 5	50.0	+65.4, +3,380.28	−28.8, −1,525.87	−2.4, −47.30	**+18.29**, +927.21	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X1	preview week: down	12	5 / 5	33.3	+63.9, +202.11	−42.2, −348.48	−29.4, −93.68	**−6.80**, −164.95	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X2	preview week: down	12	5 / 5	16.7	+80.3, +531.35	−48.0, −294.91	−47.3, −138.21	**−26.64**, −157.20	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X3	preview week: down	13	5 / 5	23.1	+19.4, +61.47	−32.3, −245.24	−28.5, −82.20	**−20.34**, −174.46	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X4	preview week: down	13	5 / 5	30.8	+35.4, +1,469.19	−35.8, −2,724.57	−28.5, −821.88	**−13.90**, −1,434.19	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X1	preview week: down	12	5 / 5	33.3	+63.9, +202.11	−42.2, −348.48	−29.4, −93.68	**−6.80**, −164.95	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
Config B X2	preview week: down	12	5 / 5	16.7	+80.3, +531.35	−48.0, −294.91	−47.3, −138.21	**−26.64**, −157.20	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X3	preview week: down	13	5 / 5	23.1	+19.4, +61.47	−32.3, −245.24	−28.5, −82.20	**−20.34**, −174.46	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X4	preview week: down	13	5 / 5	30.8	+35.4, +1,469.19	−35.8, −2,724.57	−28.5, −821.88	**−13.90**, −1,434.19	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X1	three weeks pooled	64	15 / 257	54.7	+40.3, +167.85	−41.3, −193.94	+10.2, +10.81	**+3.29**, +3.91	[−13.43, +29.47]; [−70.94, +165.39]
Config A X2	three weeks pooled	59	15 / 257	35.6	+70.7, +344.21	−43.1, −185.90	−27.8, −22.20	**−2.58**, +2.78	[−20.60, +34.39]; [−59.32, +240.66]
Config A X3	three weeks pooled	58	15 / 257	27.6	+67.4, +347.78	−30.0, −135.03	−26.4, −17.18	**−3.11**, −1.84	[−21.35, +41.96]; [−96.00, +244.91]
Config A X4	three weeks pooled	58	15 / 257	32.8	+52.1, +2,458.76	−30.0, −1,368.92	−22.3, −161.32	**−3.13**, −115.02	[−18.53, +30.59]; [−542.27, +1,595.76]
Config B X1	three weeks pooled	62	15 / 257	56.5	+40.3, +167.85	−41.3, −206.12	+12.7, +12.35	**+4.74**, +5.00	[−11.65, +29.47]; [−70.94, +165.39]
Config B X2	three weeks pooled	57	15 / 257	36.8	+70.7, +344.21	−43.1, −194.59	−26.0, −22.20	**−1.20**, +3.92	[−20.60, +34.39]; [−59.32, +240.66]
Config B X3	three weeks pooled	56	15 / 257	28.6	+67.4, +347.78	−29.8, −140.60	−24.5, −16.15	**−2.03**, −1.06	[−20.52, +41.96]; [−96.00, +244.91]
Config B X4	three weeks pooled	56	15 / 257	33.9	+52.1, +2,458.76	−29.9, −1,430.13	−20.9, −146.13	**−2.05**, −110.69	[−17.89, +30.59]; [−562.97, +1,595.76]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
Config A X1	preview week: up	+97.74, +19.2%, 2025-07-23, SPXW 6340.0C 12:43	−322.26, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.33	541	4	total ≤ 0 (best 5 days −515; without them −0)
Config A X2	preview week: up	+387.74, +76.0%, 2025-07-23, SPXW 6340.0C 12:43	−322.26, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.54	806	5	total ≤ 0 (best 5 days −542; without them −0)
Config A X3	preview week: up	+18.96, +46.2%, 2025-07-25, QQQ 567.0C 12:09	−322.26, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.02	1,079	5	total ≤ 0 (best 5 days −1,079; without them +0)
Config A X4	preview week: up	+221.16, +53.9%, 2025-07-25, QQQ 567.0C 12:09	−3,222.61, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.03	10,280	5	total ≤ 0 (best 5 days −10,280; without them +0)
Config B X1	preview week: up	+97.74, +19.2%, 2025-07-23, SPXW 6340.0C 12:43	−322.26, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.36	520	4	total ≤ 0 (best 5 days −455; without them +0)
Config B X2	preview week: up	+387.74, +76.0%, 2025-07-23, SPXW 6340.0C 12:43	−322.26, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.57	768	5	total ≤ 0 (best 5 days −483; without them −0)
Config B X3	preview week: up	+18.96, +46.2%, 2025-07-25, QQQ 567.0C 12:09	−322.26, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.02	1,031	5	total ≤ 0 (best 5 days −1,031; without them +0)
Config B X4	preview week: up	+221.16, +53.9%, 2025-07-25, QQQ 567.0C 12:09	−3,222.61, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.03	9,807	5	total ≤ 0 (best 5 days −9,807; without them +0)
Config A X1	preview week: flat	+1,357.74, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−872.26, −66.1%, 2026-02-03, SPXW 6925.0P 10:25	2.33	555	2	100%
Config A X2	preview week: flat	+1,357.74, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−1,052.26, −47.6%, 2026-02-05, SPXW 6785.0P 10:46	1.88	1,198	2	100%
Config A X3	preview week: flat	+4,288.87, +376.2%, 2026-02-03, SPXW 6925.0P 11:51	−782.26, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	2.53	906	3	100%
Config A X4	preview week: flat	+28,233.05, +247.7%, 2026-02-03, SPXW 6925.0P 11:51	−7,822.61, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	2.22	4,590	3	100%
Config B X1	preview week: flat	+1,357.74, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−872.26, −66.1%, 2026-02-03, SPXW 6925.0P 10:25	2.33	555	2	100%
Config B X2	preview week: flat	+1,357.74, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−1,052.26, −47.6%, 2026-02-05, SPXW 6785.0P 10:46	1.88	1,198	2	100%
Config B X3	preview week: flat	+4,288.87, +376.2%, 2026-02-03, SPXW 6925.0P 11:51	−782.26, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	2.53	906	3	100%
Config B X4	preview week: flat	+28,233.05, +247.7%, 2026-02-03, SPXW 6925.0P 11:51	−7,822.61, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	2.22	4,590	3	100%
Config A X1	preview week: down	+627.74, +91.0%, 2026-07-13, SPXW 7530.0P 12:17	−632.26, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.29	2,607	4	total ≤ 0 (best 5 days −1,979; without them +0)
Config A X2	preview week: down	+1,047.74, +151.8%, 2026-07-13, SPXW 7530.0P 12:17	−632.26, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.36	2,934	4	total ≤ 0 (best 5 days −1,886; without them −0)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
Config A X3	preview week: down	+137.74, +20.0%, 2026-07-13, SPXW 7530.0P 12:17	−632.26, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.08	2,406	4	total ≤ 0 (best 5 days −2,268; without them +0)
Config A X4	preview week: down	+5,087.39, +73.7%, 2026-07-13, SPXW 7530.0P 12:17	−6,322.61, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.24	23,732	4	total ≤ 0 (best 5 days −18,644; without them −0)
Config B X1	preview week: down	+627.74, +91.0%, 2026-07-13, SPXW 7530.0P 12:17	−632.26, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.29	2,607	4	total ≤ 0 (best 5 days −1,979; without them +0)
Config B X2	preview week: down	+1,047.74, +151.8%, 2026-07-13, SPXW 7530.0P 12:17	−632.26, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.36	2,934	4	total ≤ 0 (best 5 days −1,886; without them −0)
Config B X3	preview week: down	+137.74, +20.0%, 2026-07-13, SPXW 7530.0P 12:17	−632.26, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.08	2,406	4	total ≤ 0 (best 5 days −2,268; without them +0)
Config B X4	preview week: down	+5,087.39, +73.7%, 2026-07-13, SPXW 7530.0P 12:17	−6,322.61, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.24	23,732	4	total ≤ 0 (best 5 days −18,644; without them −0)
Config A X1	three weeks pooled	+1,357.74, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−872.26, −66.1%, 2026-02-03, SPXW 6925.0P 10:25	1.04	2,607	136	1601%
Config A X2	three weeks pooled	+1,357.74, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−1,052.26, −47.6%, 2026-02-05, SPXW 6785.0P 10:46	1.02	2,934	137	3122%
Config A X3	three weeks pooled	+4,288.87, +376.2%, 2026-02-03, SPXW 6925.0P 11:51	−782.26, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	0.98	3,173	138	total ≤ 0 (best 5 days +4,205; without them −4,312)
Config A X4	three weeks pooled	+28,233.05, +247.7%, 2026-02-03, SPXW 6925.0P 11:51	−7,822.61, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	0.88	23,732	138	total ≤ 0 (best 5 days +31,995; without them −38,666)
Config B X1	three weeks pooled	+1,357.74, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−872.26, −66.1%, 2026-02-03, SPXW 6925.0P 10:25	1.06	2,607	136	1302%
Config B X2	three weeks pooled	+1,357.74, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−1,052.26, −47.6%, 2026-02-05, SPXW 6785.0P 10:46	1.03	2,934	137	2303%
Config B X3	three weeks pooled	+4,288.87, +376.2%, 2026-02-03, SPXW 6925.0P 11:51	−782.26, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	0.99	3,173	138	total ≤ 0 (best 5 days +4,213; without them −4,273)
Config B X4	three weeks pooled	+28,233.05, +247.7%, 2026-02-03, SPXW 6925.0P 11:51	−7,822.61, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	0.88	23,732	138	total ≤ 0 (best 5 days +32,257; without them −38,455)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $5.62 a trade) and Robinhood's charged instead (mean $2.65). The study's fees are given beside them below. Re-pricing: exact (470 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
Config A X1	preview week: up	20	**−14.67**, −25.73	−16.37, −26.72	45.0 / 45.0	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X2	preview week: up	20	**−26.04**, −27.12	−27.76, −28.17	20.0 / 20.0	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X3	preview week: up	21	**−23.34**, −51.37	−25.11, −52.37	9.5 / 9.5	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X4	preview week: up	21	**−20.93**, −489.52	−22.71, −499.61	14.3 / 9.5	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X1	preview week: up	18	**−11.65**, −25.30	−13.33, −26.26	50.0 / 50.0	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X2	preview week: up	18	**−24.28**, −26.84	−25.99, −27.87	22.2 / 22.2	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X3	preview week: up	19	**−22.31**, −54.28	−24.09, −55.26	10.5 / 10.5	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X4	preview week: up	19	**−19.65**, −516.16	−21.43, −526.01	15.8 / 10.5	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X1	preview week: flat	32	**+18.29**, +85.77	+17.52, +84.77	68.8 / 68.8	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X2	preview week: flat	27	**+25.50**, +96.03	+24.68, +94.99	55.6 / 55.6	n/a (< 60 sessions) / n/a (< 60 sessions)

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
Config A X3	preview week: flat	24	**+23.94**, +134.99	+23.21, +133.99	45.8 / 45.8	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X4	preview week: flat	24	**+18.29**, +927.21	+17.55, +917.31	50.0 / 45.8	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X1	preview week: flat	32	**+18.29**, +85.77	+17.52, +84.77	68.8 / 68.8	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X2	preview week: flat	27	**+25.50**, +96.03	+24.68, +94.99	55.6 / 55.6	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X3	preview week: flat	24	**+23.94**, +134.99	+23.21, +133.99	45.8 / 45.8	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X4	preview week: flat	24	**+18.29**, +927.21	+17.55, +917.31	50.0 / 45.8	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X1	preview week: down	12	**−6.80**, −164.95	−7.58, −165.65	33.3 / 33.3	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X2	preview week: down	12	**−26.64**, −157.20	−27.42, −157.90	16.7 / 16.7	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X3	preview week: down	13	**−20.34**, −174.46	−21.10, −175.16	23.1 / 23.1	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X4	preview week: down	13	**−13.90**, −1,434.19	−14.66, −1,441.33	30.8 / 30.8	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X1	preview week: down	12	**−6.80**, −164.95	−7.58, −165.65	33.3 / 33.3	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X2	preview week: down	12	**−26.64**, −157.20	−27.42, −157.90	16.7 / 16.7	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X3	preview week: down	13	**−20.34**, −174.46	−21.10, −175.16	23.1 / 23.1	n/a (< 60 sessions) / n/a (< 60 sessions)
Config B X4	preview week: down	13	**−13.90**, −1,434.19	−14.66, −1,441.33	30.8 / 30.8	n/a (< 60 sessions) / n/a (< 60 sessions)
Config A X1	three weeks pooled	64	**+3.29**, +3.91	+2.23, +2.98	54.7 / 54.7	[−13.43, +29.47] / [−14.91, +28.63]
Config A X2	three weeks pooled	59	**−2.58**, +2.78	−3.69, +1.80	35.6 / 35.6	[−20.60, +34.39] / [−21.32, +33.47]
Config A X3	three weeks pooled	58	**−3.11**, −1.84	−4.22, −2.78	27.6 / 27.6	[−21.35, +41.96] / [−23.21, +41.16]
Config A X4	three weeks pooled	58	**−3.13**, −115.02	−4.24, −124.37	32.8 / 29.3	[−18.53, +30.59] / [−20.19, +29.83]
Config B X1	three weeks pooled	62	**+4.74**, +5.00	+3.71, +4.07	56.5 / 56.5	[−11.65, +29.47] / [−13.33, +28.63]
Config B X2	three weeks pooled	57	**−1.20**, +3.92	−2.29, +2.95	36.8 / 36.8	[−20.60, +34.39] / [−21.32, +33.47]
Config B X3	three weeks pooled	56	**−2.03**, −1.06	−3.12, −1.99	28.6 / 28.6	[−20.52, +41.96] / [−22.39, +41.16]
Config B X4	three weeks pooled	56	**−2.05**, −110.69	−3.15, −119.93	33.9 / 30.4	[−17.89, +30.59] / [−19.76, +29.83]

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

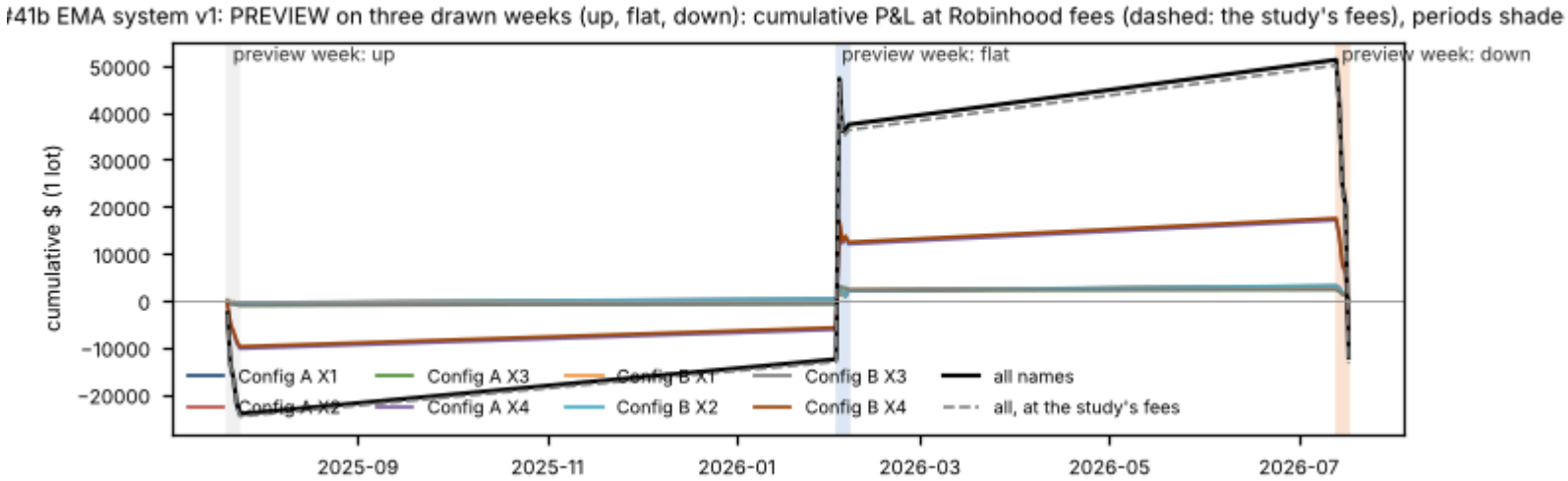
Exit reasons (share of trades):

- preview week: up: stop 46.2%; runner 34.6%; +1R 11.5%; +2R 5.1%; 14:00 2.6%
- preview week: flat: runner 31.8%; stop 25.2%; +1R 19.6%; +2R 12.1%; 14:00 11.2%
- preview week: down: stop 56.0%; runner 32.0%; +1R 6.0%; 14:00 4.0%; +2R 2.0%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	Config A X1	20	45.0	−14.67	−25.73	−515
2026	Config A X1	44	59.1	+11.45	+17.39	+765
2025	Config A X2	20	20.0	−26.04	−27.12	−542
2026	Config A X2	39	43.6	+9.46	+18.11	+706
2025	Config A X3	21	9.5	−23.34	−51.37	−1,079
2026	Config A X3	37	37.8	+8.38	+26.27	+972
2025	Config A X4	21	14.3	−20.93	−489.52	−10,280
2026	Config A X4	37	43.2	+6.98	+97.53	+3,609

year	name	trades	win %	mean %	mean $	total $
2025	Config B X1	18	50.0	−11.65	−25.30	−455
2026	Config B X1	44	59.1	+11.45	+17.39	+765
2025	Config B X2	18	22.2	−24.28	−26.84	−483
2026	Config B X2	39	43.6	+9.46	+18.11	+706
2025	Config B X3	19	10.5	−22.31	−54.28	−1,031
2026	Config B X3	37	37.8	+8.38	+26.27	+972
2025	Config B X4	19	15.8	−19.65	−516.16	−9,807
2026	Config B X4	37	43.2	+6.98	+97.53	+3,609
2025	all names	156	23.1	−20.43	−155.08	−24,192
2026	all names	314	46.5	+9.18	+38.55	+12,104



Amendments, deviations and integrity checks. —

What it showed. The sign follows the week.

#42 George A+ entry + EMA targets

Folder: `(research_notes/George A+ EMA targets/ was planned)`. **Commits:** none. **Dates:** 2026-10-23.

Hypothesis. The GeorgeFi “A+” entry with EMA price-target exits.

Rule. —

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). —

Data. —

Periods and holdouts. —

Pass rule (as pre-registered). —

Verdict. Never run.

Metrics. not available — the run was stopped before its prereg was committed; no file exists on disk or in history, so every metric is not available.

Amendments, deviations and integrity checks. —

What it showed. No result.

#43 Compass ICT: sweep → displacement/MSS → FVG on SPXW 0DTE

Folder: `research_notes/Compass ICT/`. **Commits:** 9d6c37e / Amendment 1 9dbc0bf / preview 315fe8e; exploration 0943030. **Dates:** 2026-10-23.

Hypothesis. The owner's ICT rule (v0) and its variants trade SPXW 0DTE at a profit.

Rule. HTF bias, a sweep of a level in a killzone, displacement/MSS, entry in the FVG; 2 contracts; TP1 half, TP2, stop; V1 OTE, V2 out 14:00, V3 range gate, V4 5-min; four ablations.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); ask in, bid out.

Data. SPXW 0DTE, 818 full sessions 2022-05-16 → 2026-10-09 (the 28 held-out months out); preview 2026-10-09 → 22.

Periods and holdouts. exploration / preview (10 sessions). Holdout A and holdout B UNOPENED; finalists named, never read.

Pass rule (as pre-registered). Finalists: v0 plus at most 2 candidates with exploration mean % > 0 and underlying return > 0; holdout B pass: options mean % > 0, interval clear of zero, ≥ 80 trades.

Verdict. Exploration only: v0 −0.8% [−7.7, +7.2], +$48 a trade; no holdout read.

Metrics at Robinhood fees (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
A no FVG	exploration 2022-05→2026-10-09	202	158 / 1,123	33.2	+66.1, +1,840.25	−29.6, −694.87	−4.9, −109.52	**+2.12**, +145.99	[−6.81, +12.27]; [−150.34, +476.87]
A no HTF bias	exploration 2022-05→2026-10-09	357	281 / 1,135	37.3	+37.0, +893.82	−26.3, −627.90	−4.6, −104.52	**−2.74**, −60.98	[−7.84, +2.11]; [−203.54, +88.43]
A no killzones	exploration 2022-05→2026-10-09	245	175 / 1,126	33.1	+43.7, +1,040.36	−22.7, −518.74	−4.2, −94.52	**−0.74**, −3.29	[−6.27, +6.15]; [−147.76, +185.82]
A no sweep	exploration 2022-05→2026-10-09	1,266	698 / 1,128	41.3	+30.8, +864.70	−28.7, −753.63	−2.7, −54.52	**−4.11**, −85.08	[−6.21, −1.64]; [−140.68, −24.20]
V1 OTE	exploration 2022-05→2026-10-09	94	79 / 1,117	33.0	+36.7, +833.54	−22.6, −525.71	−12.0, −214.52	**−3.04**, −77.45	[−11.14, +6.40]; [−264.40, +154.92]
V2 out 14:00	exploration 2022-05→2026-10-09	142	116 / 1,117	39.4	+34.0, +896.55	−21.2, −446.47	−3.5, −64.52	**+0.58**, +83.17	[−6.39, +8.93]; [−101.94, +310.88]
V3 range gate	exploration 2022-05→2026-10-09	84	71 / 1,105	36.9	+26.4, +713.87	−24.9, −498.63	−5.4, −84.52	**−5.95**, −51.16	[−14.25, +2.73]; [−259.00, +237.27]
V4 5-min	exploration 2022-05→2026-10-09	37	37 / 1,077	27.0	+77.6, +1,528.98	−41.1, −988.03	−5.1, −84.52	**−8.99**, −307.76	[−24.94, +18.00]; [−673.32, +279.37]
v0	exploration 2022-05→2026-10-09	142	116 / 1,117	37.3	+35.5, +955.29	−22.3, −490.95	−3.9, −74.52	**−0.73**, +48.84	[−7.74, +7.57]; [−153.60, +292.78]
A no FVG	preview 2026-10-09→22	2	2 / 6	50.0	+15.1, +175.48	−37.8, −1,004.52	−11.3, −414.52	**−11.32**, −414.52	n/a (< 60 sessions); n/a (< 60 sessions)
A no HTF bias	preview 2026-10-09→22	4	3 / 3	25.0	+98.7, +3,455.48	−32.1, −924.52	−25.5, −574.52	**+0.59**, +170.48	n/a (< 60 sessions); n/a (< 60 sessions)
A no killzones	preview 2026-10-09→22	4	3 / 6	0.0	—, —	−50.0, −721.98	−47.2, −824.52	**−49.96**, −721.98	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
A no sweep	preview 2026-10-09→22	17	9 / 9	35.3	+68.9, +1,563.81	−41.9, −1,029.07	−8.8, −204.52	**−2.84**, −113.93	n/a (< 60 sessions); n/a (< 60 sessions)
V1 OTE	preview 2026-10-09→22	1	1 / 1	0.0	—, —	−10.0, −264.52	−10.0, −264.52	**−10.02**, −264.52	n/a (< 60 sessions); n/a (< 60 sessions)
V2 out 14:00	preview 2026-10-09→22	1	1 / 1	0.0	—, —	−40.6, −884.52	−40.6, −884.52	**−40.57**, −884.52	n/a (< 60 sessions); n/a (< 60 sessions)
V3 range gate	preview 2026-10-09→22	1	1 / 1	100.0	+6.7, +175.48	—, —	+6.7, +175.48	**+6.70**, +175.48	n/a (< 60 sessions); n/a (< 60 sessions)
V4 5-min	preview 2026-10-09→22	1	1 / 1	0.0	—, —	−13.1, −104.52	−13.1, −104.52	**−13.07**, −104.52	n/a (< 60 sessions); n/a (< 60 sessions)
v0	preview 2026-10-09→22	1	1 / 1	0.0	—, —	−40.6, −884.52	−40.6, −884.52	**−40.57**, −884.52	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
A no FVG	exploration 2022-05→2026-10-09	+21,545.48, +282.7%, 2025-04-08, SPXW 5220.0- ×2 m84	−4,074.52, −91.8%, 2026-02-20, SPXW 6860.0- ×2 m47	1.31	22,163	337	193%
A no HTF bias	exploration 2022-05→2026-10-09	+9,735.48, +333.4%, 2026-07-27, SPXW 7465.0- ×2 m16	−10,804.52, −69.9%, 2025-04-07, SPXW 5075.01 ×2 m53	0.85	45,284	1,011	total ≤ 0 (best 5 days +35,068; without them −56,839)
A no killzones	exploration 2022-05→2026-10-09	+9,735.48, +333.4%, 2026-07-27, SPXW 7465.0- ×2 m16	−3,284.34, −97.7%, 2025-05-05, SPXW 5655.01 ×2 m33	0.99	16,881	518	total ≤ 0 (best 5 days +30,157; without them −30,962)
A no sweep	exploration 2022-05→2026-10-09	+24,995.48, +262.6%, 2025-04-04, SPXW 5245.0- ×2 m6	−13,624.52, −92.2%, 2025-04-07, SPXW 4895.0- ×2 m3	0.81	120,952	1,083	total ≤ 0 (best 5 days +58,768; without them −166,476)
V1 OTE	exploration 2022-05→2026-10-09	+4,615.48, +219.8%, 2023-08-10, SPXW 4510.0- ×2 m86	−2,124.52, −37.7%, 2026-06-12, SPXW 7405.0- ×2 m37	0.78	7,416	1,117	total ≤ 0 (best 5 days +16,423; without them −23,703)
V2 out 14:00	exploration 2022-05→2026-10-09	+9,735.48, +333.4%, 2026-07-27, SPXW 7465.0- ×2 m16	−3,044.52, −51.4%, 2026-06-12, SPXW 7395.0- ×2 m31	1.31	7,047	584	212%
V3 range gate	exploration 2022-05→2026-10-09	+7,705.48, +157.9%, 2022-10-14, SPXW 3675.0- ×2 m30	−3,044.52, −51.4%, 2026-06-12, SPXW 7395.0- ×2 m31	0.84	7,261	1,008	total ≤ 0 (best 5 days +16,393; without them −20,690)
V4 5-min	exploration 2022-05→2026-10-09	+4,025.48, +156.0%, 2026-06-04, SPXW 7550.01 ×2 m61	−3,764.52, −64.2%, 2026-06-12, SPXW 7380.0- ×2 m27	0.57	13,988	1,077	total ≤ 0 (best 5 days +13,137; without them −24,524)
v0	exploration 2022-05→2026-10-09	+9,735.48, +333.4%, 2026-07-27, SPXW 7465.0- ×2 m16	−3,284.34, −97.7%, 2025-05-05, SPXW 5655.01 ×2 m33	1.16	10,583	878	393%
A no FVG	preview 2026-10-09→22	+175.48, +15.1%, 2026-10-21, SPXW 7745.0- ×2 m330	−1,004.52, −37.8%, 2026-10-14, SPXW 7785.01 ×2 m18	0.17	1,005	6	total ≤ 0 (best 5 days −829; without them +0)
A no HTF bias	preview 2026-10-09→22	+3,455.48, +98.7%, 2026-10-15, SPXW 7695.01 ×2 m35	−1,624.52, −45.4%, 2026-10-15, SPXW 7695.01 ×2 m13	1.25	885	1	100%
A no killzones	preview 2026-10-09→22	−294.52, −11.0%, 2026-10-20, SPXW 7720.0- ×2 m106	−944.34, −94.4%, 2026-10-21, SPXW 7745.0- ×2 m347	0.00	2,888	6	total ≤ 0 (best 5 days −2,888; without them +0)
A no sweep	preview 2026-10-09→22	+6,545.48, +320.9%, 2026-10-16, SPXW 7740.0- ×2 m86	−2,224.52, −74.6%, 2026-10-21, SPXW 7720.0- ×2 m38	0.83	5,612	4	total ≤ 0 (best 5 days +5,139; without them −7,076)
V1 OTE	preview 2026-10-09→22	−264.52, −10.0%, 2026-10-14, SPXW 7775.01 ×2 m23	−264.52, −10.0%, 2026-10-14, SPXW 7775.01 ×2 m23	0.00	265	1	total ≤ 0 (best 5 days −265; without them +0)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
V2 out 14:00	preview 2026-10-09→22	−884.52, −40.6%, 2026-10-14, SPXW 7790.01 ×2 m20	−884.52, −40.6%, 2026-10-14, SPXW 7790.01 ×2 m20	0.00	885	1	total ≤ 0 (best 5 days −885; without them +0)
V3 range gate	preview 2026-10-09→22	+175.48, +6.7%, 2026-10-14, SPXW 7760.01 ×2 m30	+175.48, +6.7%, 2026-10-14, SPXW 7760.01 ×2 m30	∞	0	0	100%
V4 5-min	preview 2026-10-09→22	−104.52, −13.1%, 2026-10-13, SPXW 7805.01 ×2 m286	−104.52, −13.1%, 2026-10-13, SPXW 7805.01 ×2 m286	0.00	105	1	total ≤ 0 (best 5 days −105; without them +0)
v0	preview 2026-10-09→22	−884.52, −40.6%, 2026-10-14, SPXW 7790.01 ×2 m20	−884.52, −40.6%, 2026-10-14, SPXW 7790.01 ×2 m20	0.00	885	1	total ≤ 0 (best 5 days −885; without them +0)

Fees. Every figure in this section is at **Robinhood's fees** (the default cost model from 2026-10-24, `docs/FEES_ROBINHOOD.md`): the study's own fill prices, the study's modelled fee added back (mean $5.09 a trade) and Robinhood's charged instead (mean $4.50). The study's fees are given beside them below. Re-pricing: exact (2,601 trades).

name	period	trades	mean %, $ at Robinhood fees	mean %, $ at the study's fees	win % Robinhood / study	20-session block 95% (mean %) Robinhood / study
A no FVG	exploration 2022-05→2026-10-09	202	**+2.12**, +145.99	+2.10, +145.40	33.2 / 33.2	[−6.81, +12.27] / [−6.84, +12.24]
A no HTF bias	exploration 2022-05→2026-10-09	357	**−2.74**, −60.98	−2.77, −61.57	37.3 / 37.3	[−7.84, +2.11] / [−7.87, +2.08]
A no killzones	exploration 2022-05→2026-10-09	245	**−0.74**, −3.29	−0.77, −3.87	33.1 / 33.1	[−6.27, +6.15] / [−6.30, +6.13]
A no sweep	exploration 2022-05→2026-10-09	1,266	**−4.11**, −85.08	−4.14, −85.66	41.3 / 41.3	[−6.21, −1.64] / [−6.24, −1.67]
V1 OTE	exploration 2022-05→2026-10-09	94	**−3.04**, −77.45	−3.07, −78.04	33.0 / 33.0	[−11.14, +6.40] / [−11.17, +6.37]
V2 out 14:00	exploration 2022-05→2026-10-09	142	**+0.58**, +83.17	+0.55, +82.59	39.4 / 39.4	[−6.39, +8.93] / [−6.41, +8.90]
V3 range gate	exploration 2022-05→2026-10-09	84	**−5.95**, −51.16	−5.98, −51.74	36.9 / 36.9	[−14.25, +2.73] / [−14.28, +2.70]
V4 5-min	exploration 2022-05→2026-10-09	37	**−8.99**, −307.76	−9.02, −308.33	27.0 / 27.0	[−24.94, +18.00] / [−24.97, +17.97]
v0	exploration 2022-05→2026-10-09	142	**−0.73**, +48.84	−0.76, +48.26	37.3 / 37.3	[−7.74, +7.57] / [−7.77, +7.54]
A no FVG	preview 2026-10-09→22	2	**−11.32**, −414.52	−11.35, −415.11	50.0 / 50.0	n/a (< 60 sessions) / n/a (< 60 sessions)
A no HTF bias	preview 2026-10-09→22	4	**+0.59**, +170.48	+0.57, +169.89	25.0 / 25.0	n/a (< 60 sessions) / n/a (< 60 sessions)
A no killzones	preview 2026-10-09→22	4	**−49.96**, −721.98	−50.00, −722.56	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
A no sweep	preview 2026-10-09→22	17	**−2.84**, −113.93	−2.86, −114.52	35.3 / 35.3	n/a (< 60 sessions) / n/a (< 60 sessions)
V1 OTE	preview 2026-10-09→22	1	**−10.02**, −264.52	−10.04, −265.11	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
V2 out 14:00	preview 2026-10-09→22	1	**−40.57**, −884.52	−40.60, −885.11	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
V3 range gate	preview 2026-10-09→22	1	**+6.70**, +175.48	+6.68, +174.89	100.0 / 100.0	n/a (< 60 sessions) / n/a (< 60 sessions)
V4 5-min	preview 2026-10-09→22	1	**−13.07**, −104.52	−13.14, −105.11	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)
v0	preview 2026-10-09→22	1	**−40.57**, −884.52	−40.60, −885.11	0.0 / 0.0	n/a (< 60 sessions) / n/a (< 60 sessions)

Random baseline (as the study computed it, at its own fees): not available — no random baseline was computed by this study

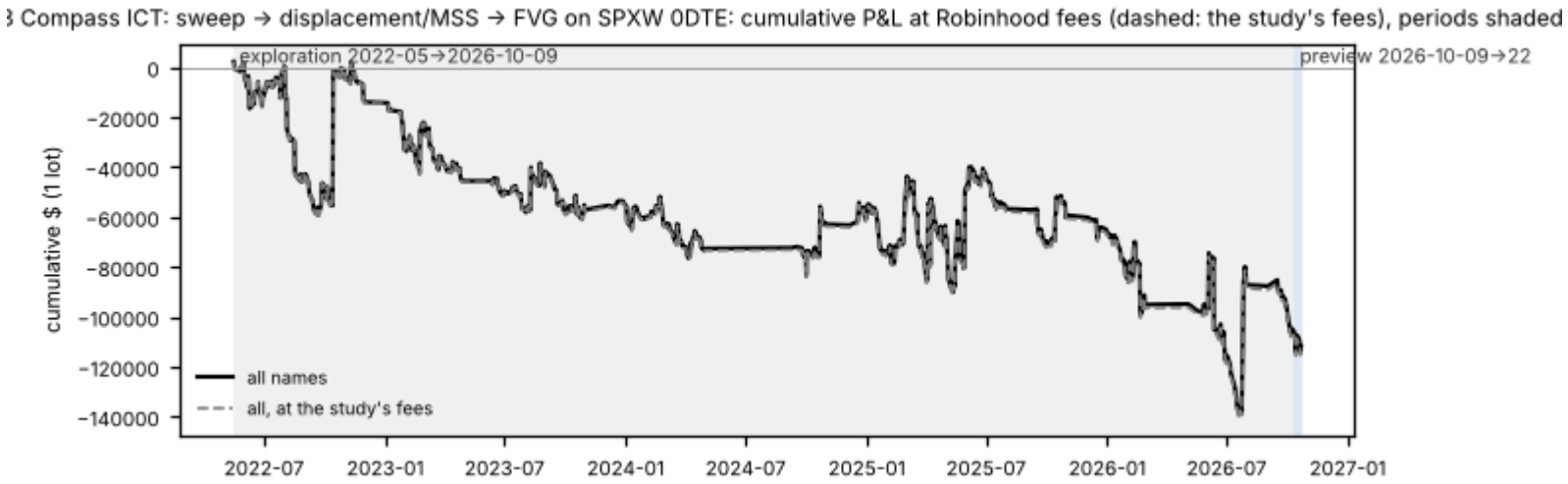
Exit reasons (share of trades):

- exploration 2022-05→2026-10-09: TP1+stop@entry 45.9%; stop 29.9%; TP1+TP2 7.3%; TP1 6.5%; TP1+flat 15:58 5.2%; flat 15:58 3.3%; TP2 1.5%; TP1+14:00 0.4%
- preview 2026-10-09→22: stop 56.2%; TP1+stop@entry 21.9%; TP1 12.5%; TP1+flat 15:58 3.1%; TP2 3.1%; TP1+TP2 3.1%

By year at Robinhood fees (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	A no FVG	40	25.0	−4.75	+70.29	+2,811
2023	A no FVG	61	39.3	−2.07	+7.29	+444
2024	A no FVG	29	37.9	+22.81	+384.79	+11,159
2025	A no FVG	43	34.9	+3.42	+484.32	+20,826
2026	A no FVG	31	25.8	−2.75	−212.26	−6,580
2022	A no HTF bias	68	39.7	−1.72	+22.06	+1,500
2023	A no HTF bias	102	37.3	−4.05	−92.65	−9,451
2024	A no HTF bias	51	31.4	−6.61	−80.89	−4,125
2025	A no HTF bias	90	36.7	−4.22	−203.41	−18,307
2026	A no HTF bias	50	40.0	+5.41	+185.88	+9,294
2022	A no killzones	52	36.5	−3.93	+35.62	+1,852
2023	A no killzones	70	31.4	−0.26	+46.91	+3,284
2024	A no killzones	35	25.7	−2.16	+12.06	+422
2025	A no killzones	60	36.7	−2.25	−137.27	−8,236
2026	A no killzones	32	28.1	+1.62	−31.70	−1,014
2022	A no sweep	233	43.8	−3.09	−42.51	−9,904
2023	A no sweep	307	41.4	−4.37	−92.89	−28,518
2024	A no sweep	201	40.3	−5.21	−69.49	−13,967
2025	A no sweep	338	40.5	−2.11	−26.64	−9,004
2026	A no sweep	204	40.2	−7.00	−236.53	−48,251
2022	V1 OTE	17	35.3	−12.61	−310.11	−5,272
2023	V1 OTE	26	38.5	+1.94	+94.13	+2,447
2024	V1 OTE	10	30.0	−4.99	−61.52	−615
2025	V1 OTE	25	32.0	+2.50	+30.48	+762
2026	V1 OTE	17	23.5	−8.46	−286.29	−4,867
2022	V2 out 14:00	30	43.3	−6.74	+19.22	+577
2023	V2 out 14:00	40	32.5	−1.79	+8.73	+349
2024	V2 out 14:00	20	35.0	−4.63	+34.49	+690
2025	V2 out 14:00	35	48.6	+4.12	+107.48	+3,762
2026	V2 out 14:00	18	33.3	+14.69	+308.26	+5,549
2022	V3 range gate	18	44.4	−9.70	+91.16	+1,641
2023	V3 range gate	20	25.0	−9.37	−152.02	−3,040
2024	V3 range gate	11	36.4	−15.00	−145.41	−1,600
2025	V3 range gate	27	44.4	+1.30	+25.48	+688
2026	V3 range gate	9	33.3	−0.15	−201.19	−1,811

year	name	trades	win %	mean %	mean $	total $
2022	V4 5-min	8	12.5	−42.21	−999.22	−7,994
2023	V4 5-min	11	18.2	−25.16	−390.65	−4,297
2024	V4 5-min	6	50.0	+24.16	+477.98	+2,868
2025	V4 5-min	5	40.0	+46.49	+619.48	+3,097
2026	V4 5-min	8	25.0	−13.59	−645.75	−5,166
2022	v0	30	40.0	−7.46	+31.39	+942
2023	v0	40	30.0	−3.79	−42.77	−1,711
2024	v0	20	35.0	−5.26	+16.49	+330
2025	v0	35	45.7	+0.75	−5.66	−198
2026	v0	18	33.3	+17.22	+371.59	+6,689
2022	all names	496	39.9	−4.81	−27.92	−13,847
2023	all names	677	37.4	−3.74	−59.81	−40,493
2024	all names	383	36.8	−2.78	−12.63	−4,839
2025	all names	658	39.8	−0.88	−10.05	−6,610
2026	all names	387	36.2	−2.25	−119.27	−46,158



Amendments, deviations and integrity checks. Amendment 1 (ablations become candidates) committed before any exploration result.

What it showed. v0 is about zero, and 398% of its profit comes from its five best days.

#44 Compass EMA: the owner’s three EMA / ORB models

Folder: `research_notes/Compass EMA/`. **Commits:** 9af93fc (prereg + HOLDOUT_LEDGER) / — / none. **Dates:** 2026-10-23.

Hypothesis. (See the prereg.)

Rule. PREREG_COMPASS_EMA.md v0.

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). —

Data. SPXW (primary), QQQ (secondary).

Periods and holdouts. Holdouts A and B UNOPENED.

Pass rule (as pre-registered). (See the prereg.)

Verdict. Never computed.

Metrics. not available — the run was stopped before computing: a partial engine and a session list exist in the scratchpad (compass_ema_partial/), no signal, trade or P&L.

Amendments, deviations and integrity checks. —

What it showed. No result.

#45 DR-GCE gates 1 and 2 (Mode 1 Gamma Vacuum Squeeze)

Folder: `research_notes/DR-GCE/`. **Commits:** 01d5c8e, 294d4f2 / A1 041baba / b20eb9c, 00eb3d5. **Dates:** 2026-10-23.

Hypothesis. Mode 1 signals touch +18% within 4 minutes far more often than random (gate 1), and in their own direction (gate 2).

Rule. Signal rows on SPXW 0DTE; the contract under the gate's rules; a touch = +18% within 4 minutes before the stop.

Fills and costs (as the study ran them; re-priced at Robinhood's fees below). Real NBBO.

Data. 504 exploration sessions 2023-12-01 → 2026-10-09; 98 Mode 1 signals.

Periods and holdouts. Exploration; the 28 held-out months UNOPENED.

Pass rule (as pre-registered). Gate 1: touch-rate lower bound above the random baseline's 97.5%; gate 2: signal − opposite side touch difference clear of zero.

Verdict. Gate 1 PASS (narrow: 32.7% [23.3, 40.2] vs 16.0% [9.2, 22.4]); gate 2 FAIL (the lift is volatility, not direction). No backtest was written.

Metrics. not available — signal-only: touch outcomes (signals.csv, gate2_pairs.csv), never traded to an exit; no P&L metric exists.

Amendments, deviations and integrity checks. Guard selftest 5 of 5.

What it showed. The lift is volatility after the impulse, not direction.

#46 Trend-day test

Folder: `–`. **Commits:** none. **Dates:** —.

Hypothesis. —

Rule. —

Fills and costs (as the study ran them; re-priced at Robinhood’s fees below). —

Data. —

Periods and holdouts. —

Pass rule (as pre-registered). —

Verdict. Never registered, never run.

Metrics. not available — never run (the trend-day share is only described in step 6).

Amendments, deviations and integrity checks. —

What it showed. No result.